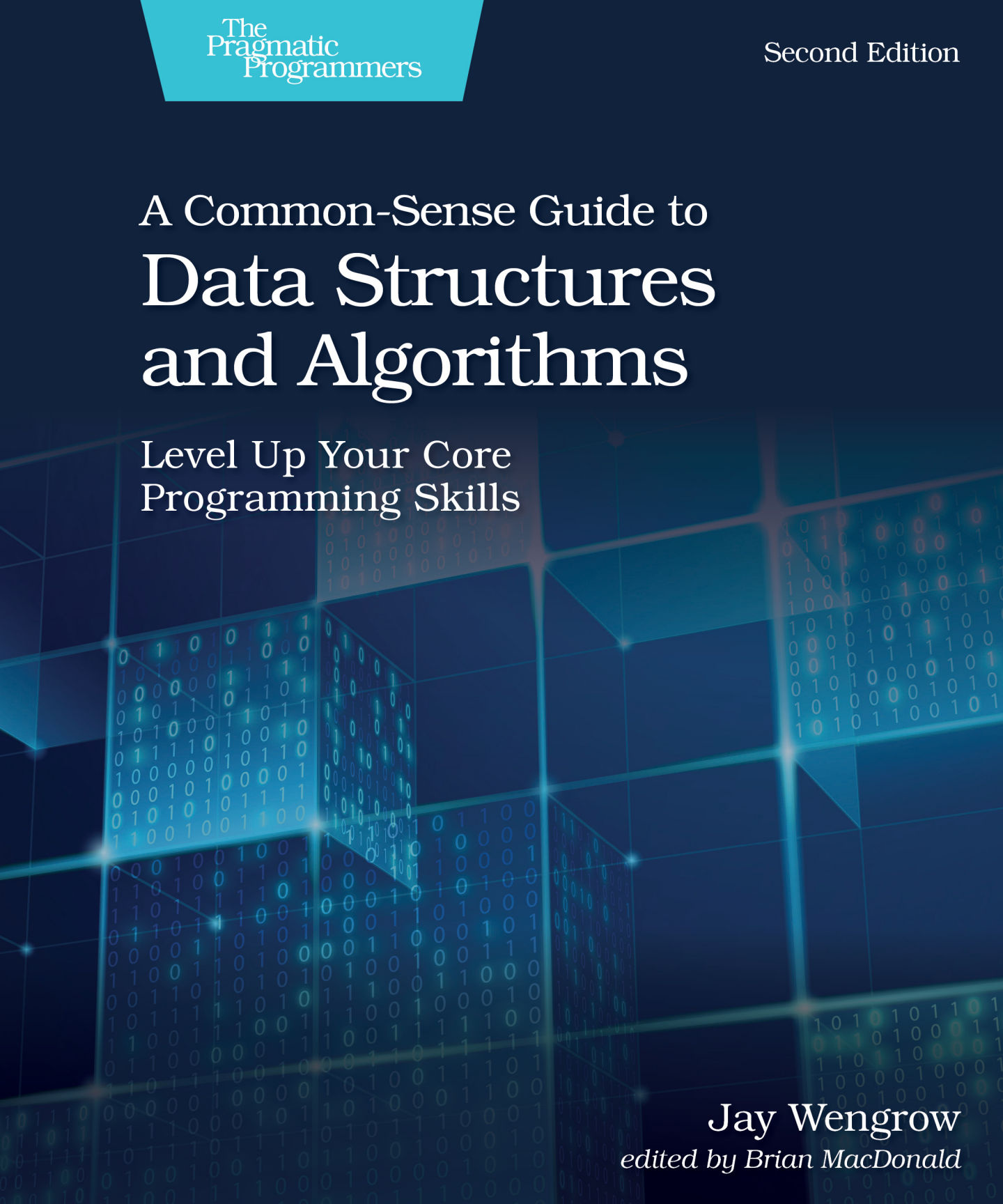

The Pragmatic Programmers
Second Edition
A Common-Sense Guide to
Data Structures and Algorithms
Level Up Your Core Programming Skills
Jay Wengrow
edited by Brian MacDonald

Early Praise for *A Common-Sense Guide to Data Structures and Algorithms, Second Edition*

If you—like me—got into coding without the benefit of a "traditional" computer science education, this book is a fantastic way to pick up the basics of how to think about algorithms, as well as how to use and implement a range of common data structures. All that, presented in plain language with a minimum of jargon, and a fun, pleasant tone! A great read for folks looking to pick up foundational programming knowledge.

➤ **John Anderson**
VP, Technology, Infinity Interactive

I've been coding for well over 30 years and one thing I've learned is to ALWAYS return to basics. *A Common-Sense Guide to Data Structures and Algorithms* has been a great way for me to re-assert all the assumptions I've made in the past, and re-cement my understanding of the core skills I'll use in the future. Don't buy this book if you think it'll help you on whiteboarding tests. Those tests suck anyway. Buy this book to continue a path to real algorithmic thinking. Whether you're early in your career or further along like me, you want a full toolbox of data structures and the common (and not so common) algorithms that complement them. Even better, you'll learn how, when, and why to optimize your code. You'll make it work, make it fast, then make it elegant—and learn the trade-offs along the way.

➤ **Scott Hanselman**
Microsoft Programmer, Professor, Blogger, Podcast

Despite being a software professional for fifteen years, I've learnt a great deal from this updated edition. This is also the book I wish I'd had twenty years ago, when learning these concepts as an undergraduate. It's like I have a new superpower tuning me in to the opportunities hash tables can provide to improve the Big O of code. Forget what it looks like, forget what it feels like, forget what you think it is, forget what habits you've accumulated: forget all that stuff; it won't give you the most efficient Big O!

➤ **Nigel Lowry**
Principal Consultant, Lemmata

The second edition of this book includes many worthy additions to an already excellent resource for learning data structures and algorithms. Additional common-sense explanations of topics like dynamic programming, and questions to check your understanding at the end of each chapter, make this book invaluable to developers with or without a computer science background.

➤ **Jason Pike**
Senior Software Engineer, KEYSYS Consulting

The perfect introduction to algorithms and structures for beginners. Highly recommended!

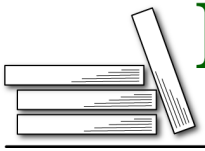
➤ **Brian Schau**
Lead Developer, Schau Consulting

A Common-Sense Guide to Data Structures and Algorithms, Second Edition

Level Up Your Core Programming Skills

Jay Wengrow

The Pragmatic Bookshelf
Raleigh, North Carolina



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Preface

Data structures and algorithms are more than abstract concepts. Mastering them enables you to write code that is *efficient,* leading to software that runs faster and consumes less memory. This is a big deal for today's software applications, which exist on increasingly mobile platforms and handle increasingly greater amounts of data.

The problem with most resources on these subjects, though, is that they are…well…obtuse. Most texts go heavy on the math jargon, and if you are not a mathematician, it can be difficult to grasp what on Earth is going on. Even books that claim to make algorithms "easy" seem to assume that the reader has an advanced math degree. Because of this, too many people shy away from these concepts, feeling that they are not "smart" enough to understand them.

The truth, though, is that everything about data structures and algorithms boils down to common sense. Mathematical notation itself is simply a particular language, and everything in math can also be explained with common-sense terminology. In this book, I use that common-sense language (plus a lot of diagrams!) to explain these concepts in simple, and dare I say, enjoyable ways.

Once you understand these concepts, you will be equipped to write code that is efficient, fast, and elegant. You will be able to weigh the pros and cons of various code alternatives, and be able to make educated decisions as to which code is best for the given situation.

In this book, I go out of my way to make these concepts real and practical with ideas that you can make use of *today*. Sure, you'll learn some really cool computer science along the way. But this book is about taking that seemingly abstract stuff and making it directly practical. You'll be writing better code and faster software by the time you're done reading this book.

Who Is This Book For?

This book is ideal for several audiences:

- You are a computer science student who wants a text that explains data structures and algorithms in plain English. This book can serve as a supplement to whatever "classic" textbook you happen to be using.
- You are a beginning developer who knows basic programming, but wants to learn the fundamentals of computer science to write better code and increase your programming knowledge and skills.
- You are a self-taught developer who has never studied formal computer science (or a developer who did but forgot everything!) and wants to leverage the power of data structures and algorithms to write more scalable and elegant code.

Whoever you may be, I tried to write this book so it can be accessed and enjoyed by people of all skill levels.

What's New in the Second Edition

Why a second edition? In the few years since the original edition was published, I've had the opportunity to teach these topics to various audiences. Over time, I continued to refine my explanations as well as discover additional topics that I thought were exciting and important. There was also quite a bit of demand for hands-on exercises that would help people gain practice with these concepts.

Accordingly, the second edition has the following new features:

1. *Revised material.* I've made significant revisions to the original chapters for the sake of clarity. While I thought that the first edition did a pretty good job in making these complex topics easy to understand, I found that there was room to make certain areas even clearer.

   Many sections of the original chapters have been entirely rewritten, and I've added brand-new sections as well. I feel that these revisions alone have enhanced the book to make it worthy of a new edition.

2. *New chapters and topics.* The second edition contains six new chapters that cover topics I'm particularly excited about.

   The book has always blended practice with theory, but I added even more material that you can take straight to the bank. The chapters *Big O in Everyday Code* and *Techniques for Code Optimization* focus exclusively

on day-to-day code, and how knowledge of data structures and algorithms can help you write more efficient software.

I went all out when it came to recursion. While the previous edition contained a chapter on this topic, I devoted an entirely new chapter, *Learning to Write in Recursive*, to teach you *how* to write recursive code, which can be confusing for beginners. I haven't seen this explained anywhere else, and I think it's a unique and valuable addition. I also added the chapter, *Dynamic Programming*, which is a popular subject, and one critical to making recursive code more efficient.

There are many data structures out there, and it's difficult to select which to include and which to leave out. However, there's been increased demand to learn about heaps and tries, and I've found them to be fascinating as well. Hence the chapters, *Keeping Your Priorities Straight with Heaps* and *It Doesn't Hurt to Trie*.

3. *Exercises and solutions*. Each chapter now has a number of exercises that allow you to gain hands-on practice with each topic within the book. I also added detailed solutions, which are available in an appendix at the back of the book. This is a significant enhancement that transforms the book into a more complete learning experience.

What's in This Book?

As you may have guessed, this book talks quite a bit about data structures and algorithms. More specifically, the book is laid out as follows:

In *Why Data Structures Matter* and *Why Algorithms Matter*, I explain what data structures and algorithms are and explore the concept of time complexity—which is used to determine how fast an algorithm is. In the process, I also talk a great deal about arrays, sets, and binary search.

In *O Yes! Big O Notation*, I unveil Big O Notation and explain it in terms that are easy to understand. We use this notation throughout the book, so this chapter is pretty important.

In *Speeding Up Your Code with Big O*, *Optimizing Code with and Without Big O*, and *Optimizing for Optimistic Scenarios*, we delve further into Big O Notation and use it to make our day-to-day code faster. Along the way, I cover various sorting algorithms, including Bubble Sort, Selection Sort, and Insertion Sort.

In *Big O in Everyday Code*, you apply all that you learned about Big O Notation and analyze the efficiency of code from the real world.

In *Blazing Fast Lookup with Hash Tables* and *Crafting Elegant Code*, I discuss a few additional data structures, including hash tables, stacks, and queues. I show how they impact the speed and elegance of our code and how we can use them to solve real-world problems.

Recursively Recurse with Recursion introduces recursion, an anchor concept in the world of computer science. We break it down in this chapter and see how it can be a great tool for certain situations. *Learning to Write in Recursive* teaches you how to write recursive code, which can otherwise be quite confusing.

Dynamic Programming shows you how to optimize recursive code and prevent it from spiraling out of control. And *Recursive Algorithms for Speed* shows you how to use recursion as the foundation for turbo-fast algorithms like Quicksort and Quickselect, and takes your algorithm-development skills up a few notches.

The following chapters, *Node-Based Data Structures*, *Speeding Up All the Things*, *Keeping Your Priorities Straight with Heaps*, *It Doesn't Hurt to Trie*, and *Connecting Everything with Graphs*, explore node-based data structures including the linked list, the binary tree, the heap, the trie, and the graph, and show how each is ideal for various applications.

Dealing with Space Constraints explores space complexity, which is important when programming for devices with relatively small amounts of disk space, or when dealing with big data.

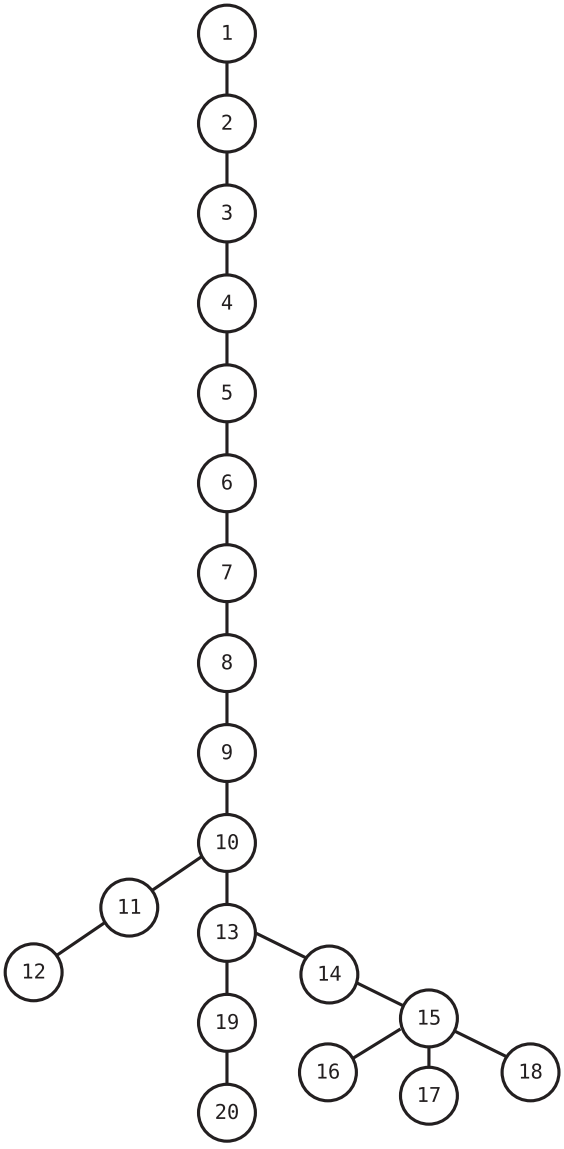
The final chapter, *Techniques for Code Optimization*, walks you through various practical techniques for optimizing the efficiency of code, and gives you new ideas for improving the code you write every day.

How to Read This Book

You've got to read this book in order. There are books out there where you can read each chapter independently and skip around a bit, but *this isn't one of them*. Each chapter assumes you've read the previous ones, and the book is carefully constructed so you can ramp up your understanding as you proceed.

That being said, there are certain chapters in the latter half of the book that don't depend entirely on each other. The diagram on page xvii depicts which chapters are prerequisites for other chapters.

For example, you could technically skip from Chapter 10 to Chapter 13 if you wanted to. (Oh! And this diagram is based on a data structure called a *tree*. You're going to learn about it in Chapter 15.)



Another important note: to make this book easy to understand, I don't always reveal everything about a particular concept when I first introduce it. Sometimes, the best way to break down a complex concept is to reveal a small piece of it, and only reveal the next piece when the first piece has sunken in. If I

define a particular term as such-and-such, don't take that as the textbook definition until you've completed the entire section on that topic.

It's a trade-off: to make the book digestible, I've chosen to oversimplify certain concepts at first and clarify them over time, rather than ensure that every sentence is completely, academically, accurate. But don't worry too much, because by the end, you'll see the entire accurate picture.

Code Examples

The concepts in this book aren't specific to one particular programming language. Because of this, I chose to use a *variety of languages* to demo the examples in this book. The examples are written in Ruby, Python, and JavaScript, so having a basic understanding of these languages would be helpful.

That being said, I've tried to write the examples in such a way that even if you're not familiar with the language of a given example, you should be able to follow along. To help with this, I don't always follow the popular idioms for each language where I feel that an idiom may confuse someone new to that particular language.

I understand that having the book switch languages frequently can come with a certain mental switching cost. However, I felt that it was important to keep the book language-agnostic. And again, I've tried to keep the code in *all* languages easy to read and straightforward.

There are some longer code snippets under the headings that read "Code Implementation." I certainly encourage you to study these code samples, but you don't necessarily have to understand every last line in order to proceed to the next section of the book. If these long pieces of code are bogging you down, just skip (or skim) them for now.

Finally, it's important to note that not every code snippet is "production ready." My greatest focus has been to clarify the concept at hand, and while I did also try to make the code generally complete, I may not have accounted for every edge case. There's certainly room for you to optimize the code further—so feel free to go crazy with that!

Online Resources

This book has its own web page[1] on which you can find more information about the book and help improve it by reporting errata, including content suggestions and typos.

1. https://pragprog.com/titles/jwdsal2

Acknowledgments

While the task of writing a book may seem like a solitary one, this book simply could not have happened without the *many* people who have supported me in my journey writing it. I'd like to personally thank *all* of you.

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When I first submitted my manuscript to the Pragmatic Bookshelf, I thought it was good. However, through the expertise, suggestions, and demands of all the wonderful people who work there, the book has become something much, much better than I could have written on my own. To my editor, Brian MacDonald—you've shown me how a book should be written, and your insights have sharpened each chapter; this book has your imprint all over it. To the original managing editor, Susannah Pfalzer and executive editor, Dave Rankin—you've given me the vision for what this book could be, taking my theory-based manuscript and transforming it into a book that can be applied to the everyday programmer. To the publishers, Andy Hunt and Dave Thomas—thank you for believing in this book and making the Pragmatic Bookshelf the most wonderful publishing company to write for.

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I've been fortunate that so many experts have reviewed this book. Your feedback has been extremely helpful and has made sure that this book can be as accurate as possible. I'd like to thank all of you for your contributions.

The reviewers for the first edition were: Alessandro Bahgat, Ivo Balbaert, Alberto Boschetti, Javier Collado, Mohamed Fouad, Derek Graham, Neil Hainer, Peter Hampton, Rod Hilton, Jeff Holland, Jessica Janiuk, Aaron Kalair, Stephan Kämper, Arun S. Kumar, Sean Lindsay, Nigel Lowry, Joy McCaffrey, Daivid Morgan, Jasdeep Narang, Stephen Orr, Kenneth Parekh, Jason Pike, Sam Rose, Frank Ruiz, Brian Schau, Tibor Simic, Matteo Vaccari, Stephen Wolff, and Peter W. A. Wood.

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Thank you all for making this book a reality.

Connecting

I enjoy connecting with my readers, and invite you to find me on LinkedIn.[2] I'd gladly accept your connection request—just send a message that you're a reader of this book. I look forward to hearing from you!

Jay Wengrow
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May, 2020

2. https://www.linkedin.com/in/jaywengrow

CHAPTER 1

Why Data Structures Matter

When people first learn to code, their focus is—and *should* be—on getting their code to run properly. Their code is measured using one simple metric: does the code actually work?

As software engineers gain more experience, though, they begin to learn about additional layers and nuances regarding the *quality* of their code. They learn that there can be two snippets of code that both accomplish the same task, but that one snippet is *better* than the other.

There are numerous measures of code quality. One important measure is code maintainability. Maintainability of code involves aspects such as the readability, organization, and modularity of one's code.

However, there's another aspect of high-quality code, and that is code *efficiency*. For example, you can have two code snippets that both achieve the same goal, but one *runs faster than the other*.

Take a look at these two functions, both of which print all the even numbers from 2 to 100:

```
def print_numbers_version_one():
  number = 2

  while number <= 100:
    # If number is even, print it:
    if number % 2 == 0:
      print(number)

    number += 1
```

```
def print_numbers_version_two():
  number = 2

  while number <= 100:
    print(number)

    # Increase number by 2, which, by definition,
    # is the next even number:
    number += 2
```

Which of these functions do you think runs faster?

If you said Version 2, you're right. This is because Version 1 ends up looping 100 times, while Version 2 only loops 50 times. The first version then, takes twice as many steps as the second version.

This book is about writing *efficient* code. Having the ability to write code that runs quickly is an important aspect of becoming a better software developer.

The first step in writing fast code is to understand what data structures are and how different data structures can affect the speed of our code. So, let's dive in.

Data Structures

Let's talk about data.

Data is a broad term that refers to all types of information, down to the most basic numbers and strings. In the simple but classic "Hello World!" program, the string "Hello World!" is a piece of data. In fact, even the most complex pieces of data usually break down into a bunch of numbers and strings.

Data structures refer to how data is *organized*. You're going to learn how the same data can be organized in a variety of ways.

Let's look at the following code:

```
x = "Hello! "
y = "How are you "
z = "today?"

print x + y + z
```

This simple program deals with three pieces of data, outputting three strings to make one coherent message. If we were to describe how the data is organized in this program, we'd say that we have three independent strings, each contained within a single variable.

However, this same data can also be stored in an array:

```
array = ["Hello! ", "How are you ", "today?"]

print array[0] + array[1] + array[2]
```

You're going to learn in this book that the organization of data doesn't just matter for organization's sake, but can significantly impact *how fast your code runs.* Depending on how you choose to organize your data, your program may run faster or slower by orders of magnitude. And if you're building a program that needs to deal with lots of data, or a web app used by thousands of people simultaneously, the data structures you select may affect whether your software runs at all, or simply conks out because it can't handle the load.

When you have a solid grasp on data structures' performance implications on the software you are creating, you will have the keys to write fast and elegant code, and your expertise as a software engineer will be greatly enhanced.

In this chapter, we're going to begin our analysis of two data structures: arrays and sets. While the two data structures may seem almost identical, you're going to learn the tools to analyze the performance implications of each choice.

The Array: The Foundational Data Structure

The *array* is one of the most basic data structures in computer science. I assume you have worked with arrays before, so you are aware that an array is a list of data elements. The array is versatile, and can serve as a useful tool in many situations, but let's take a look at one quick example.

If you are looking at the source code for an application that allows users to create and use shopping lists for the grocery store, you might find code like this:

```
array = ["apples", "bananas", "cucumbers", "dates", "elderberries"]
```

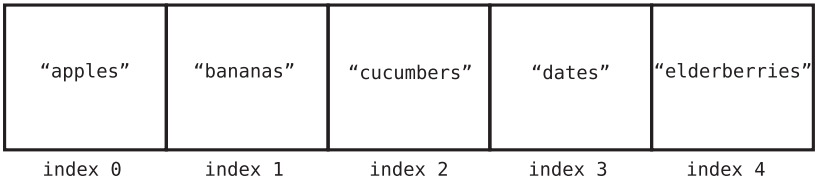
This array happens to contain five strings, each representing something that I might buy at the supermarket. (You've *got* to try elderberries.)

Arrays come with their own technical jargon.

The *size* of an array is how many data elements the array holds. Our grocery list array has a size of 5, since it contains five values.

The *index* of an array is the number that identifies where a piece of data lives inside the array.

In most programming languages, we begin counting the index at 0. So, for our example array, "apples" is at index 0, and "elderberries" is at index 4, like this:



Data Structure Operations

To understand the performance of any data structure—such as the array—we need to analyze the common ways our code might interact with that data structure.

Many data structures are used in four basic ways, which we refer to as *operations*. These operations are:

- *Read:* Reading refers to looking something up at a particular spot within the data structure. With an array, this means looking up a value at a particular index. For example, looking up which grocery item is located at index 2 would be *reading* from the array.
- *Search:* Searching refers to looking for a particular value within a data structure. With an array, this means looking to see if a particular value exists within the array, and if so, at which index. For example, looking up the index of "dates" in our grocery list would be *searching* the array.
- *Insert:* Insertion refers to adding a new value to our data structure. With an array, this means adding a new value to an additional slot within the array. If we were to add "figs" to our shopping list, we'd be *inserting* a new value into the array.
- *Delete:* Deletion refers to removing a value from our data structure. With an array, this means removing one of the values from the array. For example, if we removed "bananas" from our grocery list, this value would be *deleted* from the array.

In this chapter, we'll analyze how fast each of these operations are when applied to an array.

Measuring Speed

So, how do we measure the speed of an operation?

If you take away just one thing from this book, let it be this: when we measure how "fast" an operation takes, we do not refer to how fast the operation takes in terms of pure *time,* but instead in how many *steps* it takes.

We've actually seen this earlier in the context of printing the even numbers from 2 to 100. The second version of that function was faster because it took half as many steps as the first version did.

Why do we measure code's speed in terms of steps?

We do this because we can never say definitively that any operation takes, say, five seconds. While a piece of code may take five seconds on a particular computer, that same piece of code may take longer on an older piece of hardware. For that matter, that same code might run much faster on the supercomputers of tomorrow. Measuring the speed of an operation in terms of time is undependable, since the time will always change depending on the hardware it is run on.

However, we *can* measure the speed of an operation in terms of how many computational *steps* it takes. If Operation A takes 5 steps, and Operation B takes 500 steps, we can assume that Operation A will always be faster than Operation B on *all* pieces of hardware. Measuring the number of steps is, therefore, the key to analyzing the speed of an operation.

Measuring the speed of an operation is also known as measuring its *time complexity.* Throughout this book, I'll use the terms *speed, time complexity, efficiency, performance,* and *runtime* interchangeably. They all refer to the number of steps a given operation takes.

Let's jump into the four operations of an array and determine how many steps each one takes.

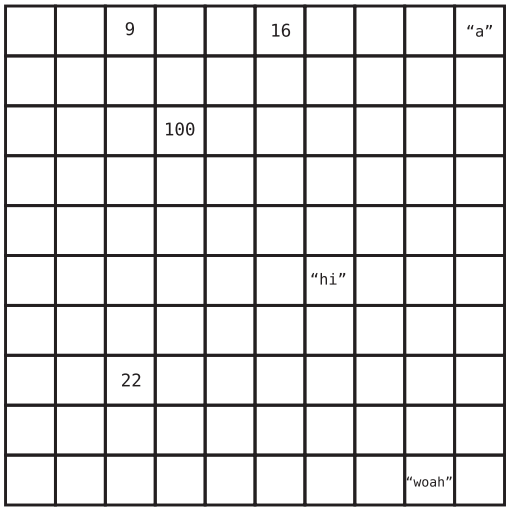
Reading

The first operation we'll look at is *reading,* which looks up what value is contained at a particular index inside the array.

A computer can read from an array in just one step. This is because the computer has the ability to jump to any particular index in the array and peer inside. In our example of `["apples", "bananas", "cucumbers", "dates", "elderberries"]`, if we looked up index 2, the computer would jump right to index 2 and report that it contains the value `"cucumbers"`.

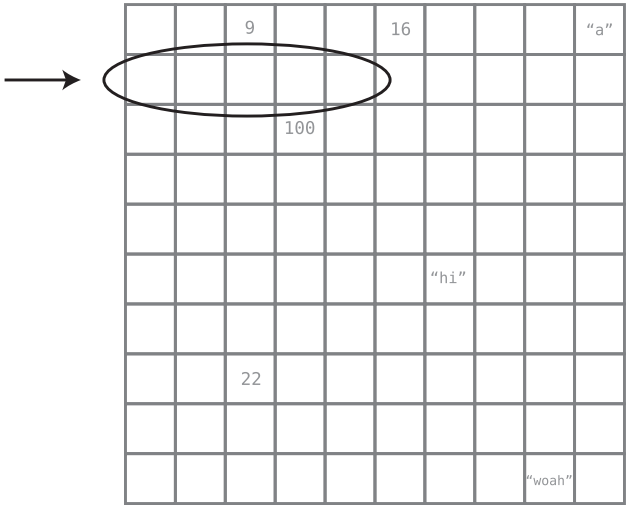
How is the computer able to look up an array's index in just one step? Let's see how.

A computer's memory can be viewed as a giant collection of cells. In the following diagram, you can see a grid of cells in which some are empty and some contain bits of data:



While this visual is a simplification of how computer memory works under the hood, it represents the essential idea.

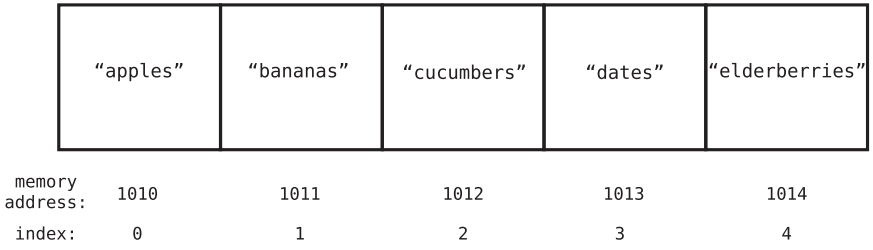
When a program declares an array, it allocates a contiguous set of empty cells for use in the program. So, if you were creating an array meant to hold five elements, your computer would find a group of five empty cells in a row and designate it to serve as your array:



Now, every cell in a computer's memory has a specific address. It's sort of like a street address (for example, 123 Main St.), except that it's represented with a number. Each cell's memory address is one number greater than the previous cell's address. Here's a visual that shows each cell's memory address:

1000	1001	1002	1003	1004	1005	1006	1007	1008	1009
1010	1011	1012	1013	1014	1015	1016	1017	1018	1019
1020	1021	1022	1023	1024	1025	1026	1027	1028	1029
1030	1031	1032	1033	1034	1035	1036	1037	1038	1039
1040	1041	1042	1043	1044	1045	1046	1047	1048	1049
1050	1051	1052	1053	1054	1055	1056	1057	1058	1059
1060	1061	1062	1063	1064	1065	1066	1067	1068	1069
1070	1071	1072	1073	1074	1075	1076	1077	1078	1079
1080	1081	1082	1083	1084	1085	1086	1087	1088	1089
1090	1091	1092	1093	1094	1095	1096	1097	1098	1099

In the next diagram, you can see our shopping list array with its indexes and memory addresses:



When the computer reads a value at a particular index of an array, it can jump straight to that index because of the combination of the following facts about computers:

1. A computer can jump to any *memory address* in one step. For example, if you asked a computer to inspect whatever's at memory address 1063, it can access that without having to perform any search process. As an analogy, if I ask you to raise your right pinky finger, you wouldn't have to search all your fingers to find which one is your right pinky. You'd be able to identify it immediately.

2. Whenever a computer allocates an array, it also makes note at which memory address the array *begins*. So, if we asked the computer to find the first element of the array, it would be able to instantly jump to the appropriate memory address to find it.

Now, these facts explain how the computer can find the *first* value of an array in a single step. However, a computer can also find the value at *any* index by performing simple addition. If we asked the computer to find the value at index 3, the computer would simply take the memory address at index 0 and add 3. (Memory addresses are sequential, after all.)

Let's apply this to our grocery list array. Our example array begins at memory address 1010. So, if we told the computer to read the value at index 3, the computer would go through the following thought process:

1. The array begins with index 0, which is at memory address 1010.
2. Index 3 will be exactly three slots past index 0.
3. By logical extension, index 3 would be located at memory address 1013, since 1010 + 3 is 1013.

Once the computer knows that index 3 is at memory address 1013, it can jump right there and see that it contains the value `"dates"`.

Reading from an array is, therefore, an efficient operation, since the computer can read any index by jumping to any memory address in one step. Although I described the computer's thought process by breaking it down into three parts, we are currently focusing on the main step of the computer jumping to a memory address. (In later chapters, we'll explore how to know which steps are the ones worth focusing on.)

Naturally, an operation that takes just one step is the fastest type of operation. Besides being a foundational data structure, arrays are also a very powerful data structure because we can read from them with such speed.

Now, what if instead of asking the computer what value is contained at index 3, we flipped the question around and asked at what index `"dates"` can be found? That is the search operation, and we'll explore that next.

Searching

As I stated previously, *searching* an array means looking to see whether a particular value exists within an array and if so, at which index it's located.

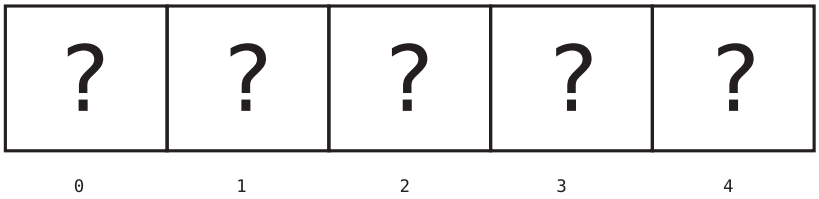
In a sense, it's the inverse of reading. Reading means providing the computer an *index* and asking it to return the value contained there. Searching, on the

other hand, means providing the computer a *value* and asking it to return the index of that value's location.

While these two operations sound similar, there's a world of difference between them when it comes to efficiency. Reading from an index is fast, since a computer can jump immediately to any index and discover the value contained there. Searching, though, is tedious, since the computer has no way to jump to a particular value.

This is an important fact about computers: a computer has immediate access to all of its memory addresses, but it has no idea offhand what *values* are contained at each memory address.

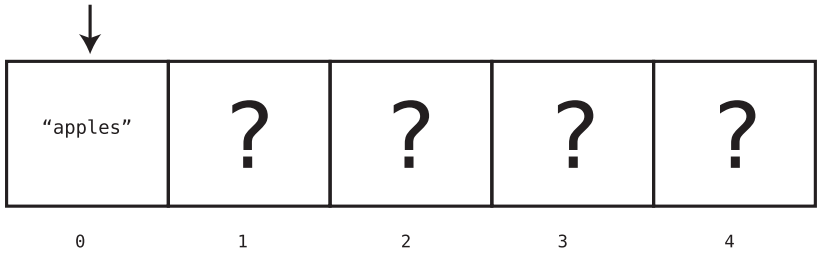
Let's take our earlier array of fruits and veggies, for example. The computer can't immediately see the actual contents of each cell. To the computer, the array looks something like this:



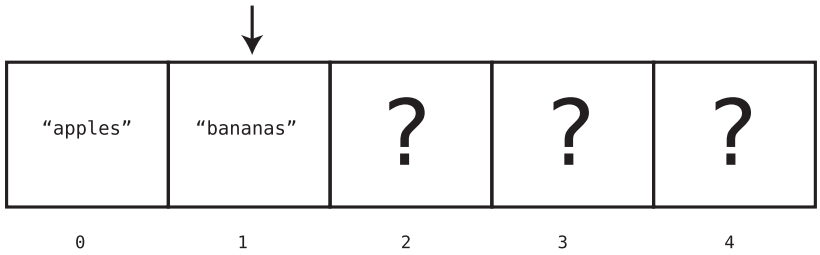
To search for a fruit within the array, the computer has no choice but to inspect each cell one at a time.

The following diagrams demonstrate the process the computer would use to search for `"dates"` within our array.

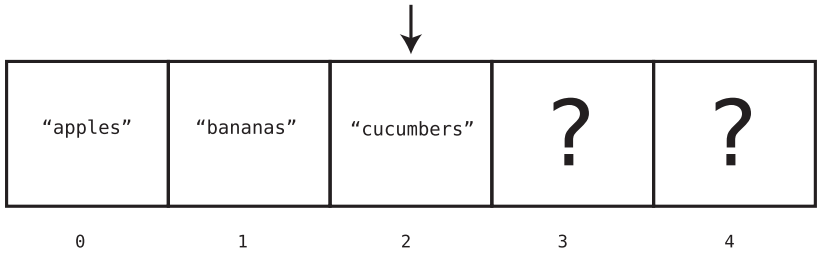
First, the computer checks index 0:



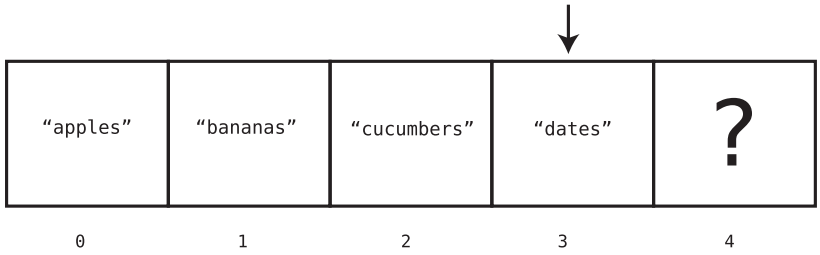
Since the value at index 0 is `"apples"`, and not the `"dates"` we're looking for, the computer moves on to the next index as shown in the diagram on page 10.



Since index 1 doesn't contain the "dates" we're looking for either, the computer moves on to index 2:



Once again, we're out of luck, so the computer moves to the next cell:



Aha! We've found the elusive "dates", and now know that the "dates" are found at index 3. At this point, the computer does not need to move on to the next cell of the array, since it already found what we're looking for.

In this example, because the computer had to check four different cells until it found the value we were searching for, we'd say that this particular operation took a total of four steps.

In *Why Algorithms Matter*, you'll learn about another way to search an array, but this basic search operation—in which the computer checks each cell one at a time—is known as *linear search*.

Now, what is the *maximum* number of steps a computer would need to perform to conduct a linear search on an array?

If the value we're seeking happens to be in the final cell in the array (like "elderberries"), then the computer would end up searching through *every* cell of the

array until it finally finds the value it's looking for. Also, if the value we're looking for doesn't occur in the array at all, the computer likewise would have to search every cell so that it can be sure the value doesn't exist within the array.

So, it turns out that for an array of five cells, the maximum number of steps linear search would take is five. For an array of 500 cells, the maximum number of steps linear search would take is 500.

Another way of saying this is that for N cells in an array, linear search would take a maximum of N steps. In this context, N is just a variable that can be replaced by any number.

In any case, it's clear that searching is less efficient than reading, since searching can take many steps, while reading always takes just one step no matter the size of the array.

Next, we'll analyze the operation of insertion.

Insertion

The efficiency of inserting a new piece of data into an array depends on *where* within the array you're inserting it.

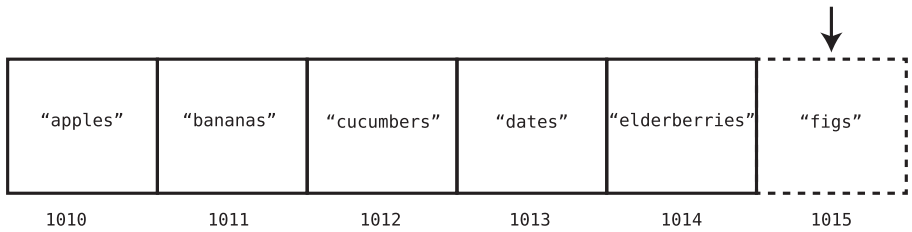
Let's say we want to add "figs" to the end of our shopping list. Such an insertion takes just one step.

This is true due to another fact about computers: when allocating an array, the computer always keeps track of the array's size.

When we couple this with the fact that the computer also knows at which memory address the array begins, computing the memory address of the last item of the array is a cinch: if the array begins at memory address 1010 and is of size 5, that means its final memory address is 1014. So, to insert an item beyond that would mean adding it to the *next* memory address, which is 1015.

Once the computer calculates which memory address to insert the new value into, it can do so in one step.

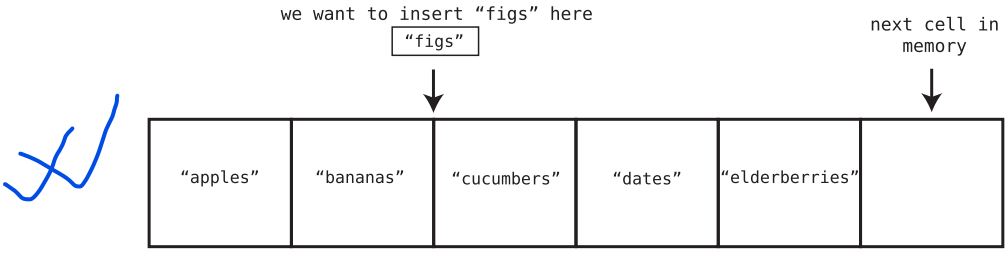
This is what inserting "figs" at the end of the array looks like:



But there's one hitch. Because the computer initially allocated only five cells in memory for the array, and now we're adding a sixth element, the computer may have to allocate additional cells toward this array. In many programming languages, this is done under the hood automatically, but each language handles this differently, so I won't get into the details of it.

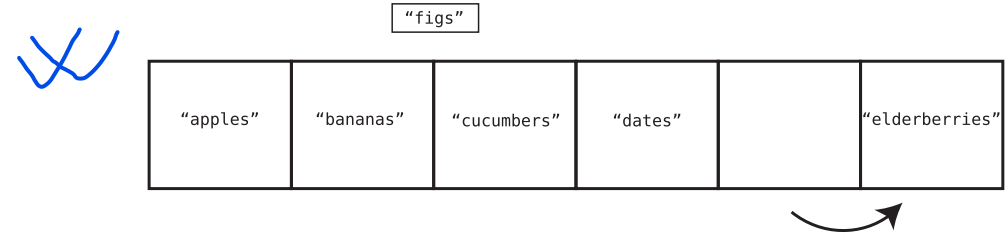
We've dealt with insertions at the end of an array, but inserting a new piece of data at the *beginning* or in the *middle* of an array is a different story. In these cases, we need to *shift* pieces of data to make room for what we're inserting, leading to additional steps.

For example, let's say we want to add "figs" to index 2 within the array. Take a look at the following diagram:

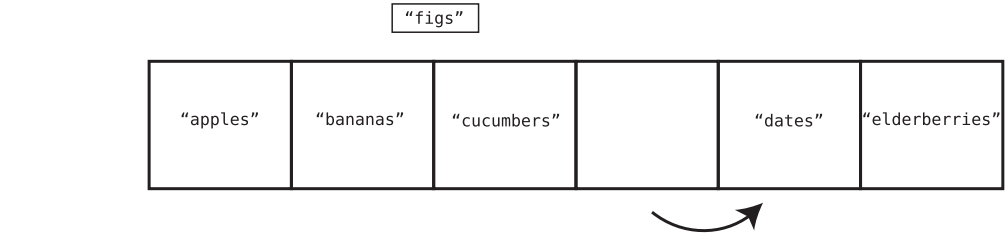


To do this, we need to move "cucumbers", "dates", and "elderberries" to the right to make room for "figs". This takes multiple steps, since we need to first move "elderberries" one cell to the right to make room to move "dates". We then need to move "dates" to make room for "cucumbers". Let's walk through this process.

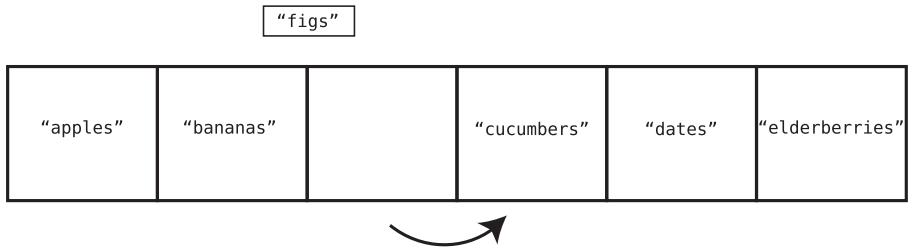
Step 1: We move "elderberries" to the right:



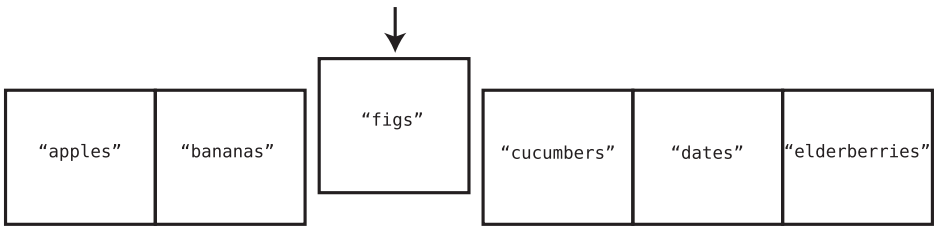
Step 2: We now move "dates" to the right:



Step 3: We now move "cucumbers" to the right:



Step 4: Finally, we can insert "figs" into index 2:



Notice that in the preceding example, insertion took four steps. Three of the steps involved shifting data to the right, while one step involved the actual insertion of the new value.

The worst-case scenario for insertion into an array—that is, the scenario in which insertion takes the most steps—is when we insert data at the *beginning* of the array. This is because when inserting at the beginning of the array, we have to move *all* the other values one cell to the right.

We can say that insertion in a worst-case scenario can take *N + 1 steps* for an array containing N elements. This is because we need to shift all N elements over, and then finally execute the actual insertion step.

Now that we've covered insertion, we're up to the array's final operation: deletion.

Deletion

Deletion from an array is the process of eliminating the value at a particular index.

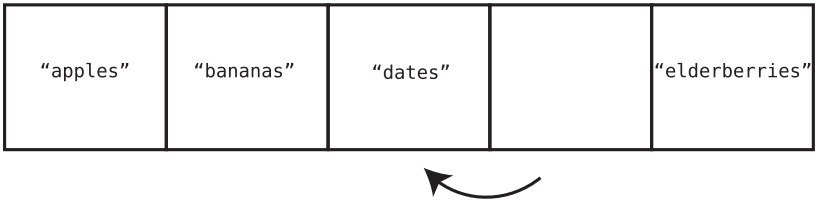
Let's return to our original example array and delete the value at index 2. In our example, this value is "cucumbers".

Step 1: We delete "cucumbers" from the array:

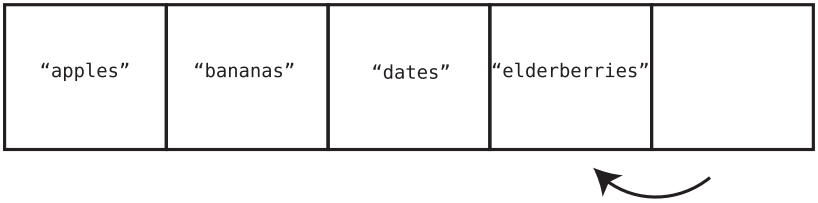


While the actual deletion of "cucumbers" technically took just one step, we now have a problem: we have an empty cell sitting smack in the middle of our array. An array is not effective when there are gaps in the middle of it, so to resolve this issue, we need to shift "dates" and "elderberries" to the left. This means our deletion process requires additional steps.

Step 2: We shift "dates" to the left:



Step 3: We shift "elderberries" to the left:



It turns out that for this deletion, the entire operation took three steps. The first step involved the actual deletion, and the other two steps involved data shifts to close the gap.

Like insertion, the worst-case scenario of deleting an element is deleting the very first element of the array. This is because index 0 would become empty, and we'd have to shift *all* the remaining elements to the left to fill the gap.

For an array of five elements, we'd spend one step deleting the first element, and four steps shifting the four remaining elements. For an array of 500 elements, we'd spend one step deleting the first element, and 499 steps shifting

the remaining data. We can say then, that for an array containing N elements, the maximum number of steps that deletion would take is N steps.

Congratulations! We've analyzed the time complexity of our first data structure. Now that you've learned how to analyze a data structure's efficiency, you can now discover how different data structures have different efficiencies. This is crucial, because choosing the correct data structure for your code can have serious ramifications on your software's performance.

The next data structure—the *set*—seems so similar to the array at first glance. However, you'll see that the operations performed on arrays and sets have different efficiencies.

Sets: How a Single Rule Can Affect Efficiency

Let's explore another data structure: the *set*. A set is a data structure that does not allow duplicate values to be contained within it.

There are different types of sets, but for this discussion, I'll talk about an *array-based set*. This set is just like an array—it is a simple list of values. The only difference between this set and a classic array is that the set never allows duplicate values to be inserted into it.

For example, if you had the set `["a", "b", "c"]` and tried to add another `"b"`, the computer just wouldn't allow it, since a `"b"` already exists within the set.

Sets are useful when you need to ensure that you don't have duplicate data.

For instance, if you're creating an online phone book, you don't want the same phone number appearing twice. In fact, I'm currently suffering from this with my local phone book: my home phone number is not just listed for me, but also it is erroneously listed as the phone number for some family named Zirkind. (Yes, this is a true story.) Let me tell you—it's quite annoying to receive phone calls and voicemail from people looking for the Zirkinds. For that matter, I'm sure the Zirkinds are also wondering why no one ever calls them. And when I call the Zirkinds to let them know about the error, my wife picks up the phone because I've called my own number. (Okay, that last part never happened.) If only the program that produced the phone book had used a set…

In any case, an array-based set is an array with one additional constraint of barring duplicates. While not allowing duplicates is a useful feature, this simple constraint also causes the set to have a *different efficiency* for one of the four primary operations.

Let's analyze the reading, searching, insertion, and deletion operations in the context of an array-based set.

Reading from a set is exactly the same as reading from an array—it takes just one step for the computer to look up what's contained within a particular index. As I described earlier, this is because the computer can jump to any index within the set since it can easily calculate and jump to its memory address.

Searching a set also turns out to be no different than searching an array—it takes up to N steps to search for a value within a set. And deletion is also identical between a set and an array—it takes up to N steps to delete a value and move data to the left to close the gap.

Insertion, however, is where arrays and sets diverge. Let's first explore inserting a value at the *end* of a set, which was a best-case scenario for an array. We saw that with an array, the computer can insert a value at its end in a single step.

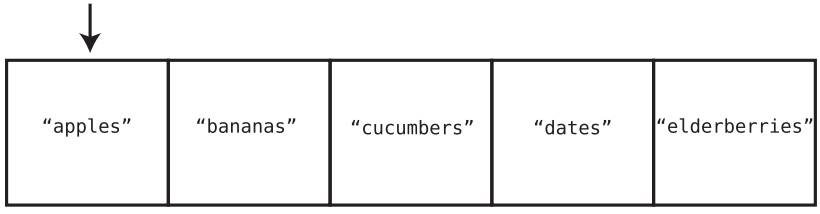
With a set, however, the computer first needs to determine that this value doesn't already exist in this set—because that's what sets do: they prevent duplicate data from being inserted into them.

Now, how will the computer ensure that the new data isn't already contained in the set? Remember, a computer doesn't know offhand what values are contained within the cells of an array or set. Because of this, the computer will first need to *search* the set to see whether the value we want to insert is already there. Only if the set does not yet contain our new value will the computer allow the insertion to take place.

So, every insertion into a set *first requires a search.*

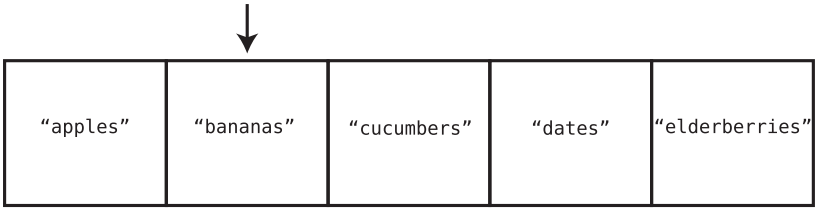
Let's see this in action with an example. Imagine our grocery list from earlier was stored as a set—which would be a decent choice since we don't want to buy the same thing twice, after all. If our current set is `["apples", "bananas", "cucumbers", "dates", "elderberries"]`, and we want to insert `"figs"` into the set, the computer must execute the following steps, beginning with a search for `"figs"`.

Step 1: Search index 0 for `"figs"`:

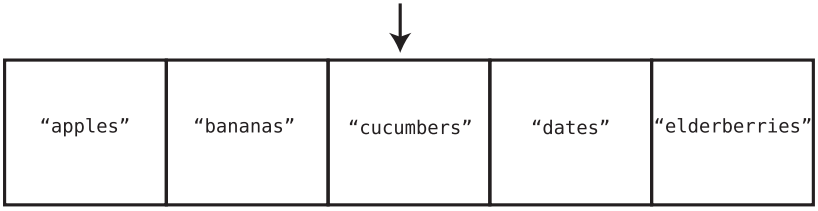


It's not there, but it might be somewhere else in the set. We need to make sure "figs" does not exist anywhere before we can insert it.

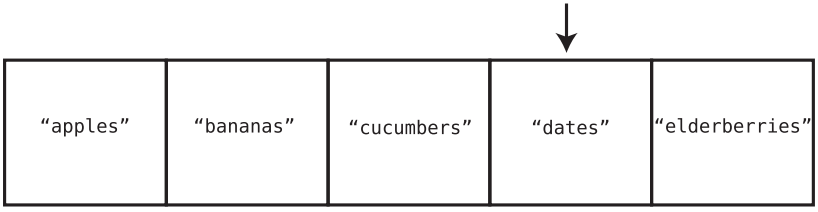
Step 2: Search index 1:



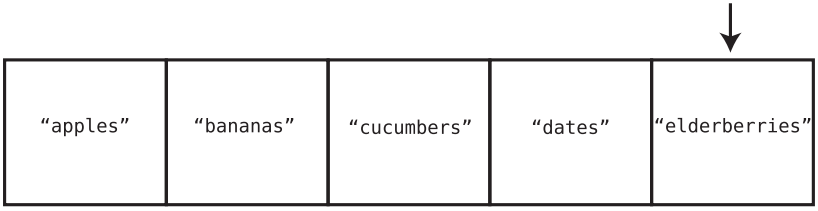
Step 3: Search index 2:



Step 4: Search index 3:

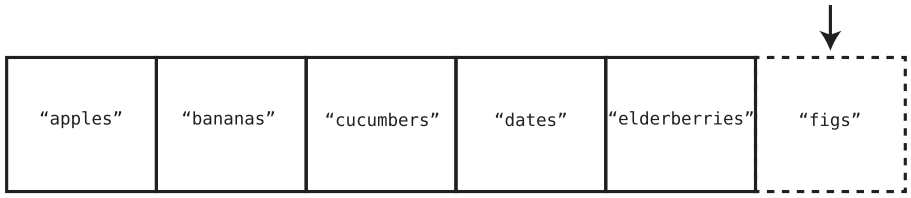


Step 5: Search index 4:



Now that we've searched the entire set, we know with certainty that it doesn't already contain "figs". At this point, it's safe to complete the insertion. And that brings us to our final step:

Step 6: Insert "figs" at the end of the set:



Inserting a value at the end of a set is the best-case scenario, but we still had to perform six steps for a set originally containing five elements. That is, we had to search all five elements before performing the final insertion step.

Said another way: insertion into the end of a set will take up to N + 1 steps for N elements. This is because there are N steps of search to ensure that the value doesn't already exist within the set, and then one step for the actual insertion. Contrast this with the regular array, in which such an insertion takes a grand total of one step.

In the worst-case scenario, where we're inserting a value at the *beginning* of a set, the computer needs to search N cells to ensure that the set doesn't already contain that value, another N steps to shift all the data to the right, and another final step to insert the new value. That's a total of 2N + 1 steps. Contrast this to insertion into the beginning of a regular array, which only takes N + 1 steps.

Now, does this mean you should avoid sets just because insertion is slower for sets than regular arrays? Absolutely not. Sets are important when you need to ensure that there is no duplicate data. (Hopefully, one day my phone book will be fixed.) But when you don't have such a need, an array may be preferable, since insertions for arrays are more efficient than insertions for sets. You must analyze the needs of your own application and decide which data structure is a better fit.

Wrapping Up

Analyzing the number of steps an operation takes is the heart of understanding the performance of data structures. Choosing the right data structure for your program can spell the difference between bearing a heavy load versus collapsing under it. In this chapter, you've learned to use this analysis to weigh whether an array or a set might be the appropriate choice for a given application.

Now that you've begun to learn how to think about the time complexity of data structures, we can use the same analysis to compare competing algorithms (even within the *same* data structure) to ensure the ultimate speed and performance of our code. And that's exactly what the next chapter is about.

Exercises

The following exercises provide you with the opportunity to practice with arrays. The solutions to these exercises are found in the section, Chapter 1, on page 439.

1. For an array containing 100 elements, provide the number of steps the following operations would take:
   a. Reading
   b. Searching for a value not contained within the array
   c. Insertion at the beginning of the array
   d. Insertion at the end of the array
   e. Deletion at the beginning of the array
   f. Deletion at the end of the array
2. For an array-based set containing 100 elements, provide the number of steps the following operations would take:
   a. Reading
   b. Searching for a value not contained within the array
   c. Insertion of a new value at the beginning of the set
   d. Insertion of a new value at the end of the set
   e. Deletion at the beginning of the set
   f. Deletion at the end of the set
3. Normally the search operation in an array looks for the first instance of a given value. But sometimes we may want to look for *every* instance of a given value. For example, say we want to count how many times the value "apple" is found inside an array. How many steps would it take to find all the "apples"? Give your answer in terms of N.

CHAPTER 2

Why Algorithms Matter

In the previous chapter, we took a look at our first data structures and saw how choosing the right data structure can affect the performance of our code. Even two data structures that seem so similar, such as the array and the set, can have very different levels of efficiency.

In this chapter, we're going to discover that even if we decide on a particular data structure, another major factor can affect the efficiency of our code: the proper selection of which *algorithm* to use.

Although the word *algorithm* sounds like something complex, it really isn't. An algorithm is simply *a set of instructions for completing a specific task.*

Even a process as simple as preparing a bowl of cereal is technically an algorithm, as it involves following a defined set of steps to achieve the task at hand. The cereal-preparation algorithm follows these four steps (for me, at least):

1. Grab a bowl.
2. Pour cereal into the bowl.
3. Pour milk into the bowl.
4. Dip a spoon into the bowl.

By following these steps in this particular order, we can now enjoy our breakfast.

When applied to computing, an algorithm refers to the set of instructions given to a computer to achieve a particular task. When we write any code, then, we're creating algorithms for the computer to follow and execute.

We can also express algorithms using plain English to set out the details of the instructions we plan on providing the computer. Throughout this book, I'll use both plain English as well as code to show how various algorithms work.

Sometimes, it's possible to have two different algorithms that accomplish the same task. We saw an example of this at the beginning of *Why Data Structures Matter*, where we had two different approaches for printing out even numbers. In that case, one algorithm had twice as many steps as the other.

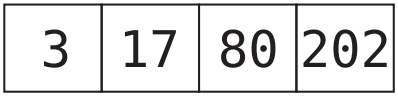
In this chapter, we'll encounter another two algorithms that solve the same problem. In this case, though, one algorithm will be faster than the other by *orders of magnitude.*

To explore these new algorithms, we'll need to take a look at a new data structure.

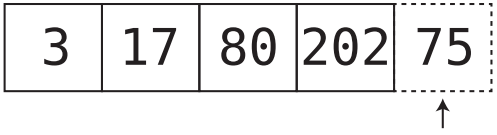
Ordered Arrays

The *ordered array* is almost identical to the "classic" array we saw in the previous chapter. The only difference is that ordered arrays require that the values are always kept—you guessed it—*in order.* That is, every time a value is added, it gets placed in the proper cell so that the values in the array remain sorted.

For example, let's take the array `[3, 17, 80, 202]`:

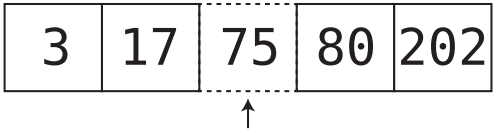


Assume we want to insert the value 75 into the array. If this array were a classic array, we could insert the 75 at the end, as follows:



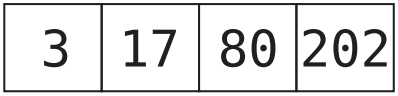
As we saw in the previous chapter, the computer can accomplish this in a single step.

On the other hand, if this were an *ordered array,* we'd have no choice but to insert the 75 in the proper spot so that the values remain in ascending order:

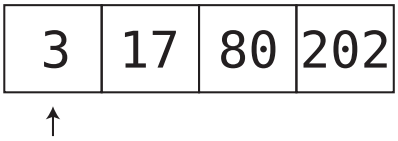


Now, this is easier said than done. The computer cannot simply drop the 75 into the right slot in a single step because it first has to *find* the right place to insert the 75, and then shift the other values to make room for it. Let's break down this process step by step.

Let's start again with our original ordered array:



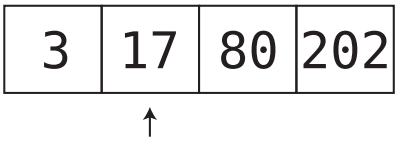
Step 1: We check the value at index 0 to determine whether the value we want to insert—the 75—should go to its left or to its right:



Because 75 is greater than 3, we know that the 75 will be inserted somewhere to its right. However, we don't know yet exactly which cell it should be inserted into, so we need to check the next cell.

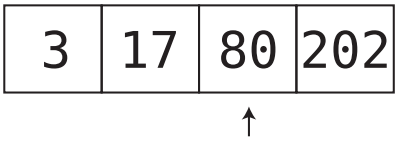
We'll call this type of step a *comparison*, where we compare the value we're inserting to a number already present in the ordered array.

Step 2: We inspect the value at the next cell:



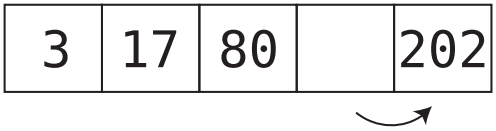
75 is greater than 17, so we need to move on.

Step 3: We check the value at the next cell:

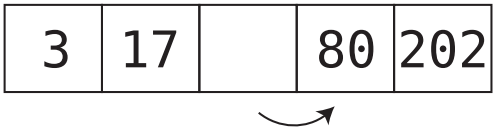


We've encountered the value 80, which is *greater* than the 75 we wish to insert. Since we've reached the first value that is greater than 75, we can conclude that the 75 must be placed immediately to the left of this 80 to maintain the order of this ordered array. To do this, we need to shift data to make room for the 75.

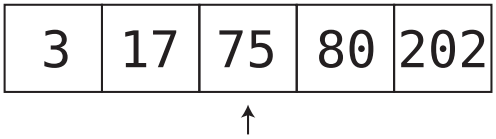
Step 4: Move the final value to the right:



Step 5: Move the next-to-last value to the right:



Step 6: We can finally insert the 75 into its correct spot:



It emerges that when inserting into an ordered array, we need to always conduct a search before the actual insertion to determine the correct spot for the insertion. This is one difference in performance between a classic array and an ordered array.

We can see in this example that there were initially four elements, and that insertion took six steps. In terms of N, we'd say that for N elements in an ordered array, the insertion took N + 2 steps in total.

Interestingly, the number of steps for insertion remains similar no matter where in the ordered array our new value ends up. If our value ends up toward the beginning of the ordered array, we have fewer comparisons and more shifts. If our value ends up toward the end, we get more comparisons but fewer shifts. The fewest steps occur when the new value winds up at the very end, since no shifts are necessary. In this case, we take N steps to compare the new value with all N existing values, plus one step for the insertion itself, yielding a total of N + 1 steps.

While insertion is less efficient for an ordered array than for a classic array, the ordered array has a secret superpower when it comes to searching.

Searching an Ordered Array

In the previous chapter, I described the process for searching for a particular value within a classic array: we check each cell one at a time—from left to

right—until we find the value we're looking for. I noted that this process is referred to as linear search.

Let's see how linear search differs between a classic and an ordered array.

Say that we have a regular array of [17, 3, 75, 202, 80]. If we were to search for the value 22—which happens to be nonexistent in this array—we would need to search each and every element because the 22 could potentially be anywhere in the array. The only time we could stop our search before we reach the array's end is if we happen to find the value we're looking for before we reach the end.

With an ordered array, however, we can stop a search early even if the value isn't contained within the array. Let's say we're searching for a 22 within an ordered array of [3, 17, 75, 80, 202]. We can stop the search as soon as we reach the 75, since it's impossible for the 22 to be anywhere to the right of it.

Here's a Ruby implementation of linear search on an ordered array:

```
def linear_search(array, search_value)

  # We iterate through every element in the array:
  array.each_with_index do |element, index|

    # If we find the value we're looking for, we return its index:
    if element == search_value
      return index

    # If we reach an element that is greater than the value
    # we're looking for, we can exit the loop early:
    elsif element > search_value
      break
    end
  end

  # We return nil if we do not find the value within the array:
  return nil

end
```

This method accepts two arguments: `array` is the ordered array we're searching, and `search_value` is the value we're searching for.

Here's how to use this function to find the `22` in our example array:

```
p linear_search([3, 17, 75, 80, 202], 22)
```

As you can see, this `linear_search` method iterates over every element of the array, looking for the `search_value`. The search stops as soon as the `element` it's iterating over is greater than the `search_value`, since we know that the `search_value` will not be found further within the array.

In this light, linear search can take fewer steps in an ordered array than in a classic array in certain situations. That being said, if we're searching for a value that happens to be the final value or not within the array at all, we will still end up searching each and every cell.

At first glance, then, standard arrays and ordered arrays don't have tremendous differences in efficiency, or at least not in worst-case scenarios. For both kinds of arrays, if they contain N elements, linear search can take up to N steps.

But we're about to unleash an algorithm that is so powerful that it will leave linear search in the dust.

We've been assuming until now that the only way to search for a value within an ordered array is linear search. The truth, however, is that linear search is only *one possible algorithm* for searching for a value. But it is not the *only* algorithm we can use.

The big advantage of an ordered array over a classic array is that an ordered array allows for an alternative searching algorithm. This algorithm is known as *binary search*, and it is a much, *much* faster algorithm than linear search.

Binary Search

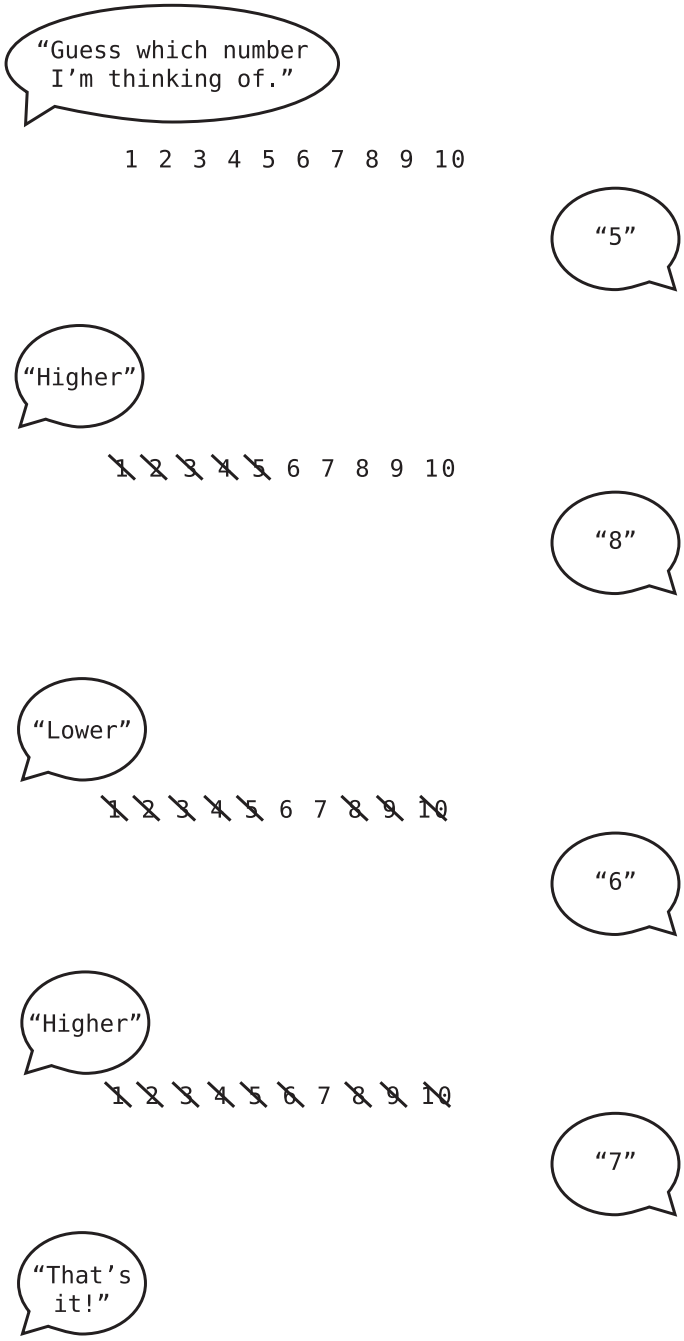
You've probably played this guessing game as a child: I'm thinking of a number between 1 and 100. Keep guessing which number I'm thinking of, and I'll let you know whether you need to guess higher or lower.

You may know intuitively how to play this game. You wouldn't begin by guessing number 1. Instead, you'd probably start with 50, which is smack in the middle. Why? Because by selecting 50, no matter whether I tell you to guess higher or lower, you've automatically eliminated half the possible numbers!

If you guess 50 and I tell you to guess higher, you'd then pick 75, to eliminate half of the *remaining* numbers. If after guessing 75, I told you to guess lower, you'd pick 62 or 63. You'd keep on choosing the halfway mark in order to keep eliminating half of the remaining numbers.

Let's visualize this process where we're told to guess a number between 1 and 10 as shown in the image on page 27.

This, in a nutshell, is binary search.

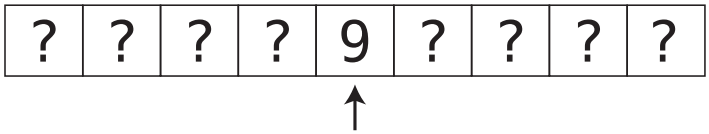

“Guess which number I’m thinking of.”
1 2 3 4 5 6 7 8 9 10
“5”
“Higher”
1 2 3 4 5 6 7 8 9 10
“8”
“Lower”
1 2 3 4 5 6 7 8 9 10
“6”
“Higher”
1 2 3 4 5 6 7 8 9 10
“7”
“That’s it!”

Let's see how binary search is applied to an ordered array. Say we have an ordered array containing nine elements. The computer doesn't know offhand what value each cell contains, so we will portray the array like this:



Say we'd like to search for the value 7 inside this ordered array. Here's how binary search would work:

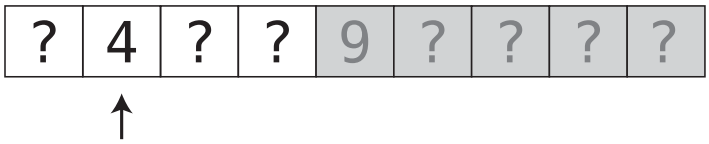
Step 1: We begin our search from the central cell. We can immediately jump to this cell, since we can calculate its index by taking the array's length and dividing it by 2. We check the value at this cell:



Because the value uncovered is a 9, we can conclude that the 7 is somewhere to its left. We've just successfully eliminated half of the array's cells—that is, all the cells to the right of the 9 (and the 9 itself):



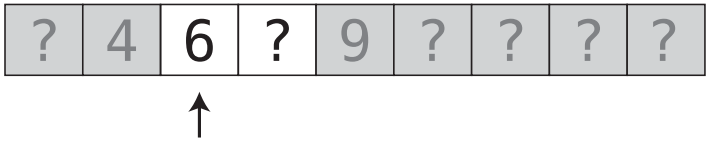
Step 2: Among the cells to the left of the 9, we inspect the middlemost value. There are two middlemost values, so we arbitrarily choose the left one:



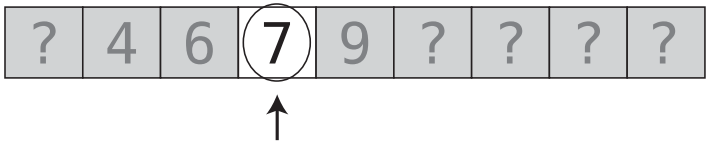
It's a 4, so the 7 must be somewhere to its right. We can eliminate the 4 and the cell to its left:



Step 3: There are two more cells where the 7 can be. We arbitrarily choose the left one as shown in the image on page 29.



Step 4: We inspect the final remaining cell. (If it's not there, that means there is no 7 within this ordered array.)



We found the 7 in four steps. While this is the same number of steps linear search would have taken in this example, we'll take a look at the power of binary search shortly.

Note that binary search is only possible within an ordered array. With a classic array, values can be in any order, and we'd never know whether to look to the left or right of any given value. This is one of the advantages of ordered arrays: we have the option of binary search.

Code Implementation: Binary Search

Here's an implementation of binary search in Ruby:

```
def binary_search(array, search_value)

  # First, we establish the lower and upper bounds of where the value
  # we're searching for can be. To start, the lower bound is the first
  # value in the array, while the upper bound is the last value:

  lower_bound = 0
  upper_bound = array.length - 1

  # We begin a loop in which we keep inspecting the middlemost value
  # between the upper and lower bounds:

  while lower_bound <= upper_bound do

    # We find the midpoint between the upper and lower bounds:
    # (We don't have to worry about the result being a non-integer
    # since in Ruby, the result of division of integers will always
    # be rounded down to the nearest integer.)

    midpoint = (upper_bound + lower_bound) / 2

    # We inspect the value at the midpoint:

    value_at_midpoint = array[midpoint]
```

```
    # If the value at the midpoint is the one we're looking for, we're done.
    # If not, we change the lower or upper bound based on whether we need
    # to guess higher or lower:

    if search_value == value_at_midpoint
      return midpoint
    elsif search_value < value_at_midpoint
      upper_bound = midpoint - 1
    elsif search_value > value_at_midpoint
      lower_bound = midpoint + 1
    end
  end

  # If we've narrowed the bounds until they've reached each other, that
  # means that the value we're searching for is not contained within
  # this array:

  return nil
end
```

Let's break this down. As with the linear_search method, binary_search accepts the array and the search_value as arguments.

Here's an example of how to call this method:

```
p binary_search([3, 17, 75, 80, 202], 22)
```

The method first establishes the range of indexes in which the search_value might be found. We do this with:

```
lower_bound = 0
upper_bound = array.length - 1
```

Because, when starting our search, the search_value might be found anywhere within the entire array, we establish the lower_bound as the first index and the upper_bound as the last index.

The essence of the search takes place within the while loop:

```
while lower_bound <= upper_bound do
```

This loop runs while we still have a range of elements in which the search_value may lie. As we'll see shortly, our algorithm will keep narrowing this range as we go. The clause lower_bound <= upper_bound will be reached when there's no more range left, and we can conclude that the search_value is not present in the array.

Within the loop, our code inspects the value at the midpoint of the range. This is accomplished with:

```
midpoint = (upper_bound + lower_bound) / 2
value_at_midpoint = array[midpoint]
```

The value_at_midpoint is the item found at the center of the range.

Now, if the value_at_midpoint is the search_value we're looking for, we've struck gold, and can return the index in which the search_value is found:

```
if search_value == value_at_midpoint
  return midpoint
```

If the search_value is less than the value_at_midpoint, it means the search_value must be found somewhere earlier in the array. We can then narrow the range of our search by making the upper_bound the index to the left of the midpoint, since the search_value cannot possibly be found anywhere further than that:

```
elsif search_value < value_at_midpoint
  upper_bound = midpoint - 1
```

Conversely, if the search_value is greater than the value_at_midpoint, it means the search_value can only be found somewhere to the right of the midpoint, so we raise the lower_bound appropriately:

```
elsif search_value > value_at_midpoint
  lower_bound = midpoint + 1
```

We return nil once the range has been narrowed down to 0 elements, and we know with certainty that the search_value does not exist within the array.

Binary Search vs. Linear Search

With ordered arrays of a small size, the algorithm of binary search doesn't have much of an advantage over linear search. But let's see what happens with larger arrays.

With an array containing 100 values, here are the maximum number of steps each type of search would take:

- Linear search: 100 steps
- Binary search: 7 steps

With linear search, if the value we're searching for is in the final cell or is greater than the value in the final cell, we have to inspect each and every element. For an array of size 100, this would take 100 steps.

When we use binary search, however, each guess we make eliminates half of the possible cells we'd have to search. In our very first guess, we get to eliminate a whopping 50 cells.

Let's look at this another way, and we'll see a pattern emerge.

With an array of size 3, binary search would take a maximum of two steps.

If we double the number of cells in the array (and add one more to keep the number odd for simplicity's sake), there are seven cells. For such an array, the maximum number of steps to find something using binary search is three.

If we double it again (and add one) so that the ordered array contains 15 elements, the maximum number of steps for binary search is four.

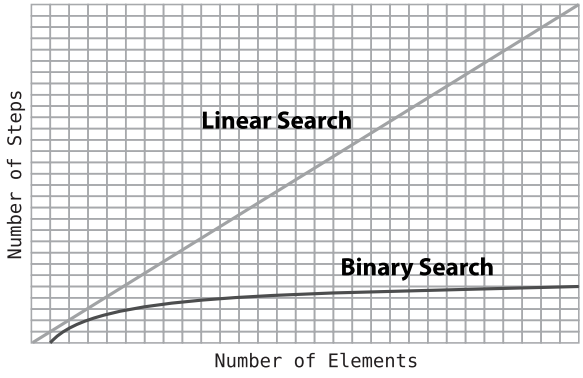
The pattern that emerges is that for each time we double the size of the ordered array, the number of steps needed for binary search increases by one. This makes sense, as each lookup eliminates half of the elements from the search.

This pattern is unusually efficient: each time we double the data, the binary search algorithm adds *just one more step*.

Contrast this with linear search. If you had 3 elements, you'd need up to three steps. For 7 elements, you'd need a maximum of seven steps. For 100 values, you'd need up to 100 steps. With linear search, then, *there are as many steps as there are items.* So for linear search, each time we double the size of the array, we *double* the number of steps of our search. For binary search, though, each time we double the size of the array, we only need to add *one more step.*

Let's see how this plays out for larger arrays. With an array of 10,000 elements, linear search can take up to 10,000 steps, while binary search takes up to a maximum of just 13 steps. For an array of size one million, linear search would take up to one million steps, while binary search would take up to just *20 steps.*

We can visualize the difference in performance between linear and binary search with this graph:



We'll be analyzing a bunch of graphs that look like this, so let's take a moment to digest what is going on. The x-axis represents the number of elements

inside the array. That is, as we move from left to right, we are dealing with an increasing amount of data.

The y-axis represents how many steps the algorithm takes. As we move up the graph, we are looking at a greater number of steps.

If you look at the line representing linear search, you'll see that as an array has more elements, linear search takes a proportionally increasing number of steps. Essentially, for each additional element in the array, linear search takes one additional step. This produces a straight diagonal line.

With binary search, on the other hand, you'll see that as the data increases, the algorithm's steps only increase marginally. This makes perfect sense with what we know: you have to double the amount of data just to add one additional step to binary search.

Keep in mind that ordered arrays aren't faster in every respect. As you've seen, insertion in ordered arrays is slower than in standard arrays. But here's the trade-off: by using an ordered array, you have somewhat slower insertion, but much faster search. Again, you must always analyze your application to see which is a better fit. Will your software be doing many insertions? Will searching be a significant feature of the app you are building?

Pop Quiz

I find the following pop quiz question really forces one to grasp the efficiency of binary search. Cover the answer and see if you get it right.

The question: We said that for an ordered array with 100 elements, binary search takes seven steps. How many steps would binary search take on an ordered array containing *200* elements?

The answer: 8 steps.

The intuitive answer I often hear is 14 steps, but this is incorrect. The whole beauty of binary search is that each inspection eliminates half of the remaining elements. Therefore, each time we *double* the amount of data, we add only one step. After all, this doubling of data gets totally eliminated with the first inspection!

It's worth noting that now that we've added binary search to our toolkit, insertion within an ordered array can become faster as well. Insertion requires a search before the actual insertion, but we can now upgrade that search from a linear search to a binary search. However, insertion within an ordered array still remains slower than within a regular array, as the regular array's insertion doesn't require a search at all.

Wrapping Up

Often there is more than one way to achieve a particular computing goal, and the algorithm you choose can seriously affect the speed of your code.

It's also important to realize that there usually isn't a single data structure or algorithm that is perfect for every situation. For example, just because ordered arrays allow for binary search doesn't mean you should always use ordered arrays. In situations where you don't anticipate the need to search the data much, only adding data, standard arrays may be a better choice because their insertion is faster.

As we've seen, the way to analyze competing algorithms is to count the number of steps each one takes. In the next chapter, we're going to look at a formalized way of expressing the time complexity of competing data structures and algorithms. Having this common language will give us clearer information that will allow us to make better decisions about which algorithms we choose.

Exercises

The following exercises provide you with the opportunity to practice with binary search. The solutions to these exercises are found in the section, Chapter 2, on page 440.

1. How many steps would it take to perform a linear search for the number 8 in the ordered array, `[2, 4, 6, 8, 10, 12, 13]`?
2. How many steps would binary search take for the previous example?
3. What is the maximum number of steps it would take to perform a binary search on an array of size 100,000?

CHAPTER 3

O Yes! Big O Notation

We've seen in the preceding chapters that the primary factor in determining an algorithm's efficiency is the number of steps it takes.

However, we can't simply label one algorithm a "22-step algorithm" and another a "400-step algorithm." This is because the number of steps an algorithm takes cannot be pinned down to a single number. Let's take linear search, for example. The number of steps linear search takes varies, as it takes as many steps as there are elements in the array. If the array contains 22 elements, linear search takes 22 steps. If the array contains 400 elements, however, linear search takes 400 steps.

The more effective way, then, to quantify the efficiency of linear search is to say that linear search takes *N steps* for *N elements in the array.* That is, if an array has N elements, linear search takes N steps. Now, this is a pretty wordy way of expressing this concept.

To help ease communication regarding time complexity, computer scientists have borrowed a concept from the world of mathematics to describe a concise and consistent language around the efficiency of data structures and algorithms. Known as *Big O Notation,* this formalized expression of these concepts allows us to easily categorize the efficiency of a given algorithm and convey it to others.

Once you understand Big O Notation, you'll have the tools to analyze each algorithm going forward in a consistent and concise way—and it's the way the pros do it.

While Big O Notation comes from the math world, I'm going to leave out all the mathematical jargon and explain it as it relates to computer science. Additionally, I'm going to begin by explaining Big O Notation in simple terms and then continue to refine it as we proceed through this and the next three

chapters. It's not a difficult concept, but it'll be made even easier if I explain it in chunks over multiple chapters.

Big O: How Many Steps Relative to N Elements?

Big O achieves consistency by focusing on the number of steps an algorithm takes, but in a specific way. Let's start off by applying Big O to the algorithm of linear search.

In a worst-case scenario, linear search will take as many steps as there are elements in the array. As we've previously phrased it: for N elements in the array, linear search can take up to N steps. The appropriate way to express this in Big O Notation is:

O(N)

Some pronounce this as "Big Oh of N." Others call it "Order of N." My personal preference, however, is "Oh of N."

Here's what the notation means. It expresses the answer to what we'll call the "key question." The key question is: *if there are N data elements, how many steps will the algorithm take?* Go ahead and read that sentence again. Then, emblazon it on your forehead, as this is the definition of Big O Notation that we'll be using throughout the rest of this book.

The *answer* to the key question lies within the *parentheses* of our Big O expression. O(N) says that the answer to the key question is that *the algorithm will take N steps.*

Let's quickly review the thought process for expressing time complexity with Big O Notation, again using the example of linear search. First, we ask the key question: if there are N data elements in an array, how many steps will linear search take? Because the answer to this question is that linear search will take N steps, we express this as O(N). For the record, an algorithm that is O(N) is also known as having *linear time.*

Let's contrast this with how Big O would express the efficiency of *reading* from a standard array. As you learned in *Why Data Structures Matter*, reading from an array takes just one step, no matter how large the array is. To figure out how to express this in Big O terms, we are going to again ask the key question: if there are N data elements, how many steps will reading from an array take? The answer is that reading takes just one step. So, we express this as O(1), which I pronounce "Oh of 1."

O(1) is interesting, since although our key question revolves around N ("If there are N data elements, how many steps will the algorithm take?"), the answer has nothing to do with N. And that's actually the whole point. That is, *no matter how many elements* an array has, reading from the array *always* takes one step.

And this is why O(1) is considered the "fastest" kind of algorithm. Even as the data increases, an O(1) algorithm doesn't take any additional steps. The algorithm always takes a constant number of steps no matter what N is. In fact, an O(1) algorithm can also be referred to as having *constant time.*

So, Where's the Math?

As I mentioned earlier in this book, I'm taking an easy-to-understand approach to the topic of Big O. That's not the only way to do it; if you were to take a traditional college course on algorithms, you'd probably be introduced to Big O from a mathematical perspective. Big O is originally a concept from mathematics, and therefore, it's often described in mathematical terms. For example, one way of describing Big O is that it describes the upper bound of the growth rate of a function, or that if a function g(x) grows no faster than a function f(x), then g is said to be a member of O(f). Depending on your mathematics background, that either makes sense or doesn't help very much. I've written this book so that you don't need as much math to understand the concept.

If you want to dig further into the math behind Big O, check out *Introduction to Algorithms* by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein (MIT Press, 2009) for a full mathematical explanation. Justin Abrahms also provides a pretty good definition in his article: https://justin.abrah.ms/computer-science/understanding-big-o-formal-definition.html.

The Soul of Big O

Now that we've encountered O(N) and O(1), we begin to see that Big O Notation does more than simply describe the number of steps an algorithm takes, such as with a hard number like 22 or 400. Rather, it's an answer to that key question on your forehead: if there are N data elements, how many steps will the algorithm take?

While that key question is indeed the strict definition of Big O, there's actually more to Big O than meets the eye.

Let's say we have an algorithm that always takes three steps no matter how much data there is. That is, for N elements, the algorithm always takes three steps. How would you express that in terms of Big O?

Based on everything you've learned up to this point, you'd probably say that it's O(3).

However, it's actually O(1). And that's because of the next layer of understanding Big O, which I will reveal now.

While Big O is an expression of the number of an algorithm's steps relative to N data elements, that alone misses the deeper *why* behind Big O, what I dub the "soul of Big O."

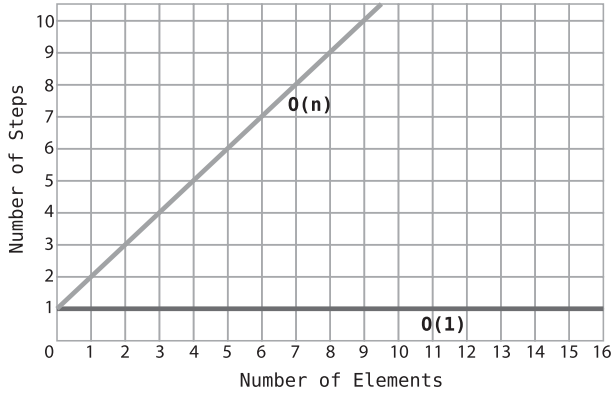
The soul of Big O is what Big O is truly concerned about: how will an algorithm's performance *change as the data increases?*

This is the soul of Big O. Big O doesn't want to simply tell you how many steps an algorithm takes. It wants to tell you the story of how the number of steps increase as the data *changes.*

Viewed with this lens, we don't care very much whether an algorithm is O(1) or O(3). Because both algorithms are the type that aren't affected by increased data, as their number of steps remain constant, they're essentially the same kind of algorithm. They're both algorithms whose steps remain constant irrespective of the data, and we don't care to make a distinction between the two.

An algorithm that is O(N), on the other hand, is a different type of algorithm. It's an algorithm whose performance *is* affected as we increase the data. More specifically, it's the kind of algorithm whose steps increase in direct proportion to the data as the data increases. This is the story O(N) tells. It tells you about the proportional relationship between the data and the algorithm's efficiency. It describes exactly how the number of steps increase as the data increases.

Look at how these two types of algorithms are plotted on a graph:



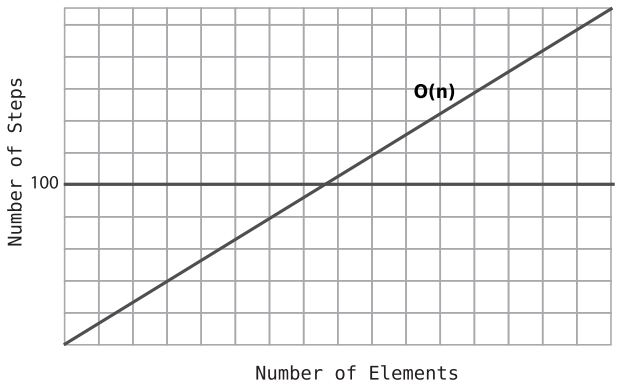
Notice that O(N) makes a perfect diagonal line. This is because for every additional piece of data, the algorithm takes one additional step. Accordingly, the more data, the more steps the algorithm will take.

Contrast this with O(1), which is a perfect horizontal line. No matter how much data there is, the number of steps remain constant.

Deeper into the Soul of Big O

To see why the "soul of Big O" is so important, let's go one level deeper. Say we had an algorithm of constant time that always took 100 steps no matter how much data there was. Would you consider that to be more or less performant than an algorithm that is O(N)?

Take a look at the following graph:



As the graph depicts, for a data set that is fewer than 100 elements, an O(N) algorithm takes fewer steps than the O(1) 100-step algorithm. At exactly 100 elements, the lines cross, meaning the two algorithms take the same number of steps, namely 100. But here's the key point: for *all arrays greater than 100,* the O(N) algorithm takes more steps.

Because there will always be *some* amount of data at which the tides turn, and O(N) takes more steps from that point until infinity, O(N) is considered to be, on the whole, less efficient than O(1) no matter how many steps the O(1) algorithm actually takes.

The same is true even for an O(1) algorithm that always takes one million steps. As the data increases, there will inevitably reach a point at which O(N) becomes less efficient than the O(1) algorithm, and will remain so up toward an infinite amount of data.

Same Algorithm, Different Scenarios

As you learned in the previous chapters, linear search isn't *always* O(N). It's true that if the item we're looking for is in the final cell of the array, it will take N steps to find it. But when the item we're searching for is found in the *first* cell of the array, linear search will find the item in just one step. So, this case of linear search would be described as O(1). If we were to describe the efficiency of linear search in its totality, we'd say that linear search is O(1) in a *best-case* scenario, and O(N) in a *worst-case* scenario.

While Big O effectively describes both the best- and worst-case scenarios of a given algorithm, Big O Notation generally refers to the *worst-case scenario* unless specified otherwise. This is why most references will describe linear search as being O(N) even though it *can* be O(1) in a best-case scenario.

This is because a "pessimistic" approach can be a useful tool: knowing exactly how inefficient an algorithm can get in a worst-case scenario prepares us for the worst and may have a strong impact on our choices.

An Algorithm of the Third Kind

In the previous chapter, you learned that binary search on an ordered array is much faster than linear search on the same array. Let's now look at how to describe binary search in terms of Big O Notation.

We can't describe binary search as being O(1), because the number of steps increases as the data increases. It also doesn't fit into the category of O(N), since the number of steps is much fewer than the N data elements. As we have seen, binary search takes only seven steps for an array containing 100 elements.

Binary search, then, seems to fall somewhere *in between* O(1) and O(N). So what is it?

In Big O terms, we describe binary search as having a time complexity of:

O(log N)

I pronounce this as "Oh of log N." This type of algorithm is also known as having a time complexity of *log time.*

Simply put, O(log N) is the Big O way of describing an algorithm that *increases one step each time the data is doubled.* As you learned in the previous chapter, binary search does just that. You'll see momentarily *why* this is expressed as O(log N), but let's first summarize what you've learned so far.

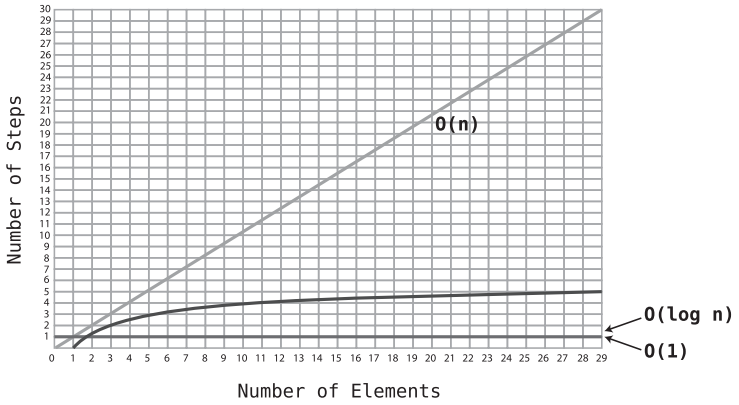
Of the three types of algorithms you've learned about so far, they can be sorted from most efficient to least efficient as follows:

O(1)

O(log N)

O(N)

Let's look at a graph that compares the three types:



Note how O(log N) curves ever so slightly upward, making it less efficient than O(1), but much more efficient than O(N).

To understand why this algorithm is called "O(log N)," you need to first understand what *logarithms* are. If you're already familiar with this mathematical concept, feel free to skip the next section.

Logarithms

Let's examine why algorithms such as binary search are described as O(log N). What is a log, anyway?

Log is shorthand for *logarithm*. The first thing to note is that logarithms have nothing to do with algorithms, even though the two words look and sound so similar.

Logarithms are the inverse of *exponents*. Here's a quick refresher on what exponents are:

2^3 is the equivalent of:

2 * 2 * 2

which just happens to be 8.

Now, $\log_2 8$ is the converse. It means: how many times do you have to multiply 2 by itself to get a result of 8?

Because you have to multiply 2 by itself 3 times to get 8, $\log_2 8 = 3$.

Here's another example:

2^6 translates to:

$2 * 2 * 2 * 2 * 2 * 2 = 64$

Because we had to multiply 2 by itself six times to get 64, therefore:

$\log_2 64 = 6$.

While the preceding explanation is the "textbook" definition of logarithms, I like to use an alternative way of describing the same concept because many people find that they can wrap their heads around it more easily, especially when it comes to Big O Notation.

Another way of explaining $\log_2 8$ is: if we kept *dividing* 8 by 2 until we ended up with 1, how many 2s would we have in our equation?

$8 / 2 / 2 / 2 = 1$

In other words, how many times do we need to halve 8 until we end up with 1? In this example, it takes us three times. Therefore,

$\log_2 8 = 3$.

Similarly, we could explain $\log_2 64$ as: how many times do we need to halve 64 until we end up with 1?

$64 / 2 / 2 / 2 / 2 / 2 / 2 = 1$

Since there are six 2s, $\log_2 64 = 6$.

Now that you understand what logarithms are, the meaning behind O(log N) will become clear.

O(log N) Explained

Let's bring this all back to Big O Notation. In computer science, whenever we say O(log N), it's actually shorthand for saying $O(\log_2 N)$. We just omit that small 2 for convenience.

Recall that Big O Notation resolves the key question: if there are N data elements, how many steps will the algorithm take?

O(log N) means that for N data elements, the algorithm would take $log_2 N$ *steps*. If there are 8 elements, the algorithm would take three steps, since $\log_2 8 = 3$.

Said another way, if we keep dividing the 8 elements in half, it would take us three steps until we end up with 1 element.

This is *exactly* what happens with binary search. As we search for a particular item, we keep dividing the array's cells in half until we narrow it down to the correct number.

Said simply: *O(log N) means the algorithm takes as many steps as it takes to keep halving the data elements until we remain with 1.*

The following table demonstrates a striking difference between the efficiencies of O(N) and O(log N):

N Elements	O(N)	O(log N)
8	8	3
16	16	4
32	32	5
64	64	6
128	128	7
256	256	8
512	512	9
1024	1024	10

While the O(N) algorithm takes as many steps as there are data elements, the O(log N) algorithm takes just one additional step each time the data is doubled.

In future chapters, you will encounter algorithms that fall under categories of Big O Notation other than the three you've learned about so far. But in the meantime, let's apply these concepts to some examples of everyday code.

Practical Examples

Here's some typical Python code that prints all the items from a list:

```
things = ['apples', 'baboons', 'cribs', 'dulcimers']

for thing in things:
    print("Here's a thing: %s" % thing)
```

How would we describe the efficiency of this algorithm in Big O Notation?

The first thing to realize is that this is an example of an algorithm. While it may not be fancy, any code that does anything at all is technically an algorithm—it's a particular process for solving a problem. In this case, the problem is that we want to print all the items from a list. The algorithm we use to solve this problem is a `for` loop containing a `print` statement.

To break this down, we need to analyze how many steps this algorithm takes. In this case, the main part of the algorithm—the `for` loop—takes four steps. In this example, there are four things in the list, and we print each one out a single time.

However, the number of steps isn't constant. If the list contained 10 elements, the `for` loop would take 10 steps. Since this `for` loop takes as many steps as there are elements, we'd say that this algorithm has an efficiency of O(N).

The next example is a simple Python-based algorithm for determining whether a number is prime:

```
def is_prime(number):
    for i in range(2, number):
        if number % i == 0:
            return False
    return True
```

The preceding code accepts a `number` as an argument and begins a `for` loop in which we divide the `number` by every integer from 2 up to that `number` and see if there's a remainder. If there's no remainder, we know that the number is not prime and we immediately return `False`. If we make it all the way up to the number and always find a remainder, then we know that the number is prime and we return `True`.

In this case, the key question is slightly different than in the previous examples. In the previous examples, our key question asked how many steps the algorithm would take if there were N data elements in an array. Here, we're not dealing with an array, but we *are* dealing with a `number` that we pass into this function. Depending on the `number` we pass in, this will affect how many times the function's loop runs.

In this case, then, our key question will be: when passing in the `number` N, how many steps will the algorithm take?

If we pass the number 7 into `is_prime`, the `for` loop runs about seven times. (It technically runs five times, since it starts at 2 and ends right before the actual `number`.) For the number 101, the loop runs about 101 times. Because the number of steps increases in lockstep with the number passed into the function, this is a classic example of O(N).

Again, the key question here dealt with a different kind of N, since our primary piece of data was a number rather than an array. We'll get more practice in identifying our Ns as we progress through the future chapters.

Wrapping Up

With Big O Notation, we have a consistent system that allows us to compare any two algorithms. With it, we will be able to examine real-life scenarios and choose between competing data structures and algorithms to make our code faster and able to handle heavier loads.

In the next chapter, we'll encounter a real-life example in which we use Big O Notation to speed up our code significantly.

Exercises

The following exercises provide you with the opportunity to practice with Big O Notation. The solutions to these exercises are found in the section, Chapter 3, on page 440.

1. Use Big O Notation to describe the time complexity of the following function that determines whether a given year is a leap year:

```
function isLeapYear(year) {
  return (year % 100 === 0) ? (year % 400 === 0) : (year % 4 === 0);
}
```

2. Use Big O Notation to describe the time complexity of the following function that sums up all the numbers from a given array:

```
function arraySum(array) {
  let sum = 0;

  for(let i = 0; i < array.length; i++) {
    sum += array[i];
  }

  return sum;
}
```

3. The following function is based on the age-old analogy used to describe the power of compounding interest:

   Imagine you have a chessboard and put a single grain of rice on one square. On the second square, you put 2 grains of rice, since that is double the amount of rice on the previous square. On the third square, you put 4 grains. On the fourth square, you put 8 grains, and on the fifth square, you put 16 grains, and so on.

The following function calculates which square you'll need to place a certain number of rice grains. For example, for 16 grains, the function will return 5, since you will place the 16 grains on the fifth square.

Use Big O Notation to describe the time complexity of this function, which is below:

```
function chessboardSpace(numberOfGrains) {
  let chessboardSpaces = 1;
  let placedGrains = 1;

  while (placedGrains < numberOfGrains) {
    placedGrains *= 2;
    chessboardSpaces += 1;
  }

  return chessboardSpaces;
}
```

4. The following function accepts an array of strings and returns a new array that only contains the strings that start with the character `"a"`. Use Big O Notation to describe the time complexity of the function:

```
function selectAStrings(array) {
  let newArray = [];

  for(let i = 0; i < array.length; i++) {
    if (array[i].startsWith("a")) {
      newArray.push(array[i]);
    }
  }

  return newArray;
}
```

5. The following function calculates the median from an *ordered* array. Describe its time complexity in terms of Big O Notation:

```
function median(array) {
  const middle = Math.floor(array.length / 2);

  // If array has even amount of numbers:
  if (array.length % 2 === 0) {
    return (array[middle - 1] + array[middle]) / 2;
  } else { // If array has odd amount of numbers:
    return array[middle];
  }
}
```

CHAPTER 4

Speeding Up Your Code with Big O

Big O Notation is a great tool for expressing the efficiency of an algorithm. We've already been able to use it to quantify the difference between binary search vs. linear search, as binary search is O(log N)—a much faster algorithm than linear search, which is O(N).

With Big O, you also have the opportunity to compare your algorithm to *general algorithms out there in the world,* and you can say to yourself, "Is this a fast or slow algorithm as far as algorithms generally go?"

If you find that Big O labels your algorithm as a "slow" one, you can now take a step back and try to figure out if there's a way to optimize it by trying to get it to fall under a faster category of Big O. This may not always be possible, of course, but it's certainly worth thinking about.

In this chapter, we'll write some code to solve a practical problem and then measure our algorithm using Big O. We'll then see if we might be able to modify the algorithm in order to give it a nice efficiency bump. (Spoiler: we will.)

Bubble Sort

Before jumping into our practical problem, though, we need to first look at a new category of algorithmic efficiency in the world of Big O. To demonstrate it, we'll get to use one of the classic algorithms of computer-science lore.

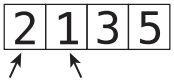
Sorting algorithms have been the subject of extensive research in computer science, and tens of such algorithms have been developed over the years. They all solve the following problem:

Given an array of unsorted values, how can we sort them so that they end up in ascending order?

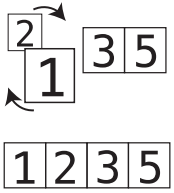
In this and the following chapters, we're going to encounter a number of these sorting algorithms. Some of the first ones you'll learn about are known as "simple sorts," in that they are easy to understand, but are not as efficient as some of the faster sorting algorithms out there.

Bubble Sort is a basic sorting algorithm and follows these steps:

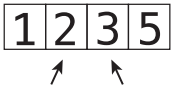
1. Point to two consecutive values in the array. (Initially, we start by pointing to the array's first two values.) Compare the first item with the second one:



2. If the two items are out of order (in other words, the left value is greater than the right value), swap them (if they already happen to be in the correct order, do nothing for this step):



3. Move the "pointers" one cell to the right:



4. Repeat Steps 1 through 3 until we reach the end of the array, or if we reach the values that have already been sorted. (This will make more sense in the walk-through that follows.) At this point, we have completed our first *pass-through* of the array. That is, we "passed through" the array by pointing to each of its values until we reached the end.

5. We then move the two pointers back to the first two values of the array, and execute another pass-through of the array by running Steps 1 through 4 again. We keep on executing these pass-throughs until we have a pass-through in which we did not perform any swaps. When this happens, it means our array is fully sorted and our work is done.

Bubble Sort in Action

Let's walk through a complete example of Bubble Sort.

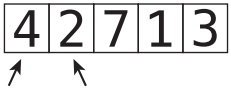
Assume we want to sort the array, `[4, 2, 7, 1, 3]`. It's currently out of order, and we want to produce an array that contains the same values in ascending order.

Let's begin the first pass-through:

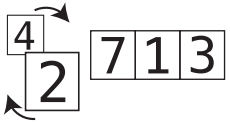
This is our starting array:



Step 1: First, we compare the 4 and the 2:

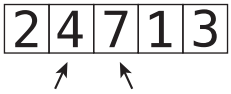


Step 2: They're out of order, so we swap them:



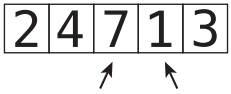


Step 3: Next, we compare the 4 and the 7:

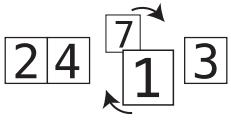


They're in the correct order, so we don't need to perform a swap.

Step 4: We now compare the 7 and the 1:

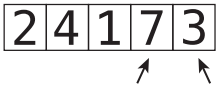


Step 5: They're out of order, so we swap them:

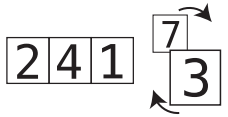


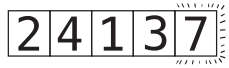


Step 6: We compare the 7 and the 3:



Step 7: They're out of order, so we swap them:





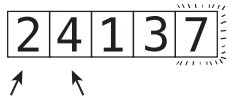
We now know for a fact that the 7 is in its correct position within the array, because we kept moving it along to the right until it reached its proper place. The previous diagram has little lines surrounding the 7 to indicate that the 7 is officially in its correct position.

This is actually the reason why this algorithm is called *Bubble* Sort: in each pass-through, the highest unsorted value "bubbles" up to its correct position.

Because we made at least one swap during this pass-through, we need to conduct another pass-through.

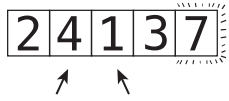
We begin the second pass-through:

Step 8: We compare the 2 and the 4:

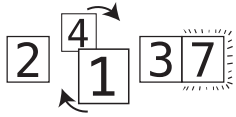


They're in the correct order, so we can move on.

Step 9: We compare the 4 and the 1:

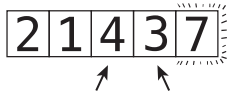


Step 10: They're out of order, so we swap them:

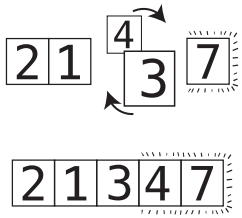




Step 11: We compare the 4 and the 3:



Step 12: They're out of order, so we swap them:



We don't have to compare the 4 and the 7 because we know that the 7 is already in its correct position from the previous pass-through. And now we also know that the 4 has bubbled up to its correct position as well. This concludes our second pass-through.

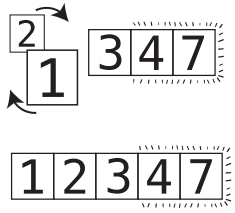
Because we made at least one swap during this pass-through, we need to conduct another pass-through.

We begin the third pass-through:

Step 13: We compare the 2 and the 1:



Step 14: They're out of order, so we swap them:



Step 15: We compare the 2 and the 3:



They're in the correct order, so we don't need to swap them.

We now know that the 3 has bubbled up to its correct spot:



Since we made at least one swap during this pass-through, we need to perform another one.

And so begins the fourth pass-through:

Step 16: We compare the 1 and the 2:



Because they're in order, we don't need to swap. We can end this pass-through, since all the remaining values are already correctly sorted.

Now that we've made a pass-through that didn't require any swaps, we know that our array is completely sorted:



Code Implementation: Bubble Sort

Here's an implementation of Bubble Sort in Python:

```
def bubble_sort(list):
    unsorted_until_index = len(list) - 1
    sorted = False

    while not sorted:
        sorted = True
        for i in range(unsorted_until_index):
            if list[i] > list[i+1]:
                list[i], list[i+1] = list[i+1], list[i]
                sorted = False
        unsorted_until_index -= 1

    return list
```

To use this function, we can pass an unsorted array to it, like so:

```
print(bubble_sort([65, 55, 45, 35, 25, 15, 10]))
```

This function will then return the sorted array.

Let's break the function down line by line to see how it works. I'll explain each line by first providing the explanation, followed by the line of code itself.

The first thing we do is create a variable called `unsorted_until_index`. This keeps track of the rightmost index of the array that has *not* yet been sorted. When we first start the algorithm, the array is completely unsorted, so we initialize this variable to be the final index in the array:

```
unsorted_until_index = len(list) - 1
```

We also create a variable called sorted that will keep track of whether the array is fully sorted. Of course, when our code first runs, it isn't, so we set it to False:

```
sorted = False
```

We begin a while loop that continues to run as long as the array is not sorted. Each round of this loop represents a pass-through of the array:

```
while not sorted:
```

Next, we preliminarily establish sorted to be True:

```
sorted = True
```

The approach here is that in each pass-through, we'll assume the array is sorted until we encounter a swap, in which case we'll change the variable back to False. If we get through an entire pass-through without having to make any swaps, sorted will remain True, and we'll know that the array is completely sorted.

Within the while loop, we begin a for loop in which we point to each pair of values in the array. We use the variable i as our first pointer, and it starts from the beginning of the array and goes until the index that has not yet been sorted:

```
for i in range(unsorted_until_index):
```

Within this loop, we compare each pair of adjacent values and swap those values if they're out of order. We also change sorted to False if we have to make a swap:

```
if list[i] > list[i+1]:
    list[i], list[i+1] = list[i+1], list[i]
    sorted = False
```

At the end of each pass-through, we know that the value we bubbled up all the way to the right is now in its correct position. Because of this, we decrement the unsorted_until_index by 1, since the index it was already pointing to is now sorted:

```
unsorted_until_index -= 1
```

The while loop ends once sorted is True, meaning the array is completely sorted. Once this is the case, we return the sorted array:

```
return list
```

The Efficiency of Bubble Sort

The Bubble Sort algorithm contains two significant kinds of steps:

- *Comparisons:* two numbers are compared with one another to determine which is greater.
- *Swaps:* two numbers are swapped with one another in order to sort them.

Let's start by determining how many *comparisons* take place in Bubble Sort.

Our example array has five elements. Looking back, you can see that in our first pass-through, we had to make four comparisons between sets of two numbers.

In our second pass-through, we only had to make three comparisons. This is because we didn't have to compare the final two numbers, since we knew that the final number was in the correct spot due to the first pass-through.

In our third pass-through, we made two comparisons, and in our fourth pass-through, we made just one comparison.

So, that's:

4 + 3 + 2 + 1 = 10 comparisons.

To put it this in a way that would hold true for arrays of all sizes, we'd say that for N elements, we make

(N - 1) + (N - 2) + (N - 3) … + 1 comparisons.

Now that we've analyzed the number of comparisons that take place in Bubble Sort, let's analyze the *swaps.*

In a worst-case scenario, where the array is sorted in descending order (the exact opposite of what we want), we'd actually need a swap for each comparison. So, we'd have 10 comparisons and 10 swaps in such a scenario for a grand total of 20 steps.

Let's look at the big picture. With an array containing five values in reverse order, we make 4 + 3 + 2 + 1 = 10 comparisons. Along with the 10 comparisons, we also have 10 swaps, totaling 20 steps.

For such an array with 10 values, we get 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 45 comparisons, and another 45 swaps. That's a total of 90 steps.

With an array containing *20* values, we'd have:

19 + 18 + 17 + 16 + 15 + 14 + 13 + 12 + 11 + 10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = *190* comparisons, and approximately 190 swaps, for a total of 380 steps.

Notice the inefficiency here. As the number of elements increases, the number of steps grows *exponentially*. We can see this clearly in the following table:

N Data Elements	Max # of Steps
5	20
10	90
20	380
40	1560
80	6320

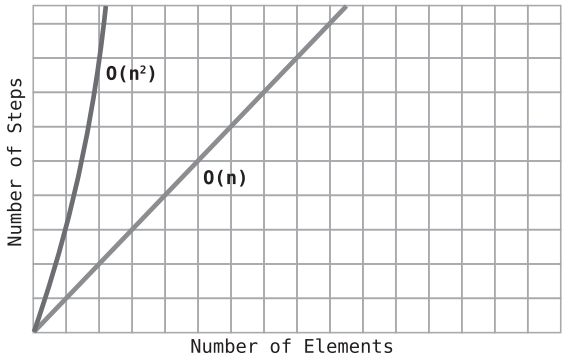
If you look at the growth of steps as N increases, you'll see that it's growing by approximately N^2. Take a look at the following table:

N Data Elements	# of Bubble Sort Steps	N^2
5	20	25
10	90	100
20	380	400
40	1560	1600
80	6320	6400

Let's express the time complexity of Bubble Sort with Big O Notation. Remember, Big O always answers the key question: if there are N data elements, how many steps will the algorithm take?

Because for N values, Bubble Sort takes N^2 steps, in Big O, we say that Bubble Sort has an efficiency of $O(N^2)$.

$O(N^2)$ is considered to be a relatively inefficient algorithm, since as the data increases, the steps increase dramatically. Look at this graph, which compares $O(N^2)$ against the faster O(N):



Note how $O(N^2)$ curves sharply upward in terms of number of steps as the data grows. Compare this with O(N), which plots along a simple, diagonal line.

One last note: $O(N^2)$ is also referred to as *quadratic time*.

A Quadratic Problem

Here's a practical example of where we can replace a slow $O(N^2)$ algorithm with a speedy O(N) one.

Let's say you're working on a JavaScript application that analyzes the ratings people give to products, where users leave ratings from 1 to 10. Specifically, you're writing a function that checks whether an array of ratings contains any duplicate numbers. This will be used in more complex calculations in other parts of the software.

For example, the array `[1, 5, 3, 9, 1, 4]` has two instances of the number `1`, so we'd return `true` to indicate that the array has a case of duplicate numbers.

One of the first approaches that may come to mind is the use of nested loops, as follows:

```
function hasDuplicateValue(array) {
    for(let i = 0; i < array.length; i++) {
        for(let j = 0; j < array.length; j++) {
            if(i !== j && array[i] === array[j]) {
                return true;
            }
        }
    }
    return false;
}
```

In this function, we iterate through each value of the array using the variable `i`. As we focus on each value in `i`, we then run a *second* loop that looks through all the values in the array—using `j`—and checks if the values at positions `i` and `j` are the same. If they are, it means we've encountered duplicate values and we return `true`. If we get through all of the looping and we haven't encountered any duplicates, we return `false`, since we know that there are no duplicates in the array.

While this certainly works, is it efficient? Now that we know a bit about Big O Notation, let's take a step back and see what Big O would say about this function.

Remember that Big O expresses how many steps the algorithm takes relative to N data values. To apply this to our situation, we'd ask ourselves: for N

values in the array provided to our hasDuplicateValue function, how many steps would our algorithm take in a worst-case scenario?

To answer the preceding question, we need to analyze what steps our function takes, as well as what the worst-case scenario would be.

The preceding function has one type of step, namely *comparisons*. It repeatedly compares array[i] and array[j] to see if they are equal and therefore represent a duplicate pair. In a worst-case scenario, the array contains no duplicates, which would force our code to complete all of the loops and exhaust every possible comparison before returning false.

Based on this, we can conclude that for N values in the array, our function would perform N^2 comparisons. This is because we perform an outer loop that must iterate N times to get through the entire array, and for *each iteration,* we must iterate *another N times* with our inner loop. That's N steps * N steps, which is N^2 steps, leaving us with an algorithm of $O(N^2)$.

We can actually prove that our function takes N^2 steps by adding some code to our function that tracks the algorithm's number of steps:

```
function hasDuplicateValue(array) {
    let steps = 0; // count of steps
    for(let i = 0; i < array.length; i++) {
        for(let j = 0; j < array.length; j++) {
            steps++; // increment number of steps
            if(i !== j && array[i] === array[j]) {
                return true;
            }
        }
    }
    console.log(steps); // print number of steps if no duplicates
    return false;
}
```

This added code will print the number of steps taken when there are no duplicates. If we run hasDuplicateValue([1, 4, 5, 2, 9]), for example, we'll see an output of 25 in the JavaScript console, indicating that there were 25 comparisons for the 5 elements in the array. If we test this for other values, we'll see that the output is always the size of the array squared. This is classic $O(N^2)$.

Very often (but not always), when an algorithm nests one loop inside another, the algorithm is $O(N^2)$. So, whenever you see a nested loop, $O(N^2)$ alarm bells should go off in your head.

Now, the fact that our function is $O(N^2)$ should give us pause. This is because $O(N^2)$ is considered a relatively slow algorithm. Whenever you encounter a

slow algorithm, it's worth spending some time to consider whether there are any faster alternatives. There may *not* be any better alternatives, but let's first make sure.

A Linear Solution

Following is another implementation of the hasDuplicateValue function that doesn't rely on nested loops. It's a bit clever, so let's first look at how it works and then we'll see if it's any more efficient than our first implementation.

```
function hasDuplicateValue(array) {
    let existingNumbers = [];
    for(let i = 0; i < array.length; i++) {
        if(existingNumbers[array[i]] === 1) {
            return true;
        } else {
            existingNumbers[array[i]] = 1;
        }
    }
    return false;
}
```

Here's what this function does. It creates an array called existingNumbers, which starts out empty.

Then, we use a loop to check each number in the array. As it encounters each number, it places an arbitrary value (we've chosen to use a 1) in the existingNumbers array at the *index* of the number we're encountering.

For example, let's say our array is [3, 5, 8]. When we encounter the 3, we place a 1 at index 3 of existingNumbers. So, the existingNumbers array will now be the rough equivalent of this:

```
[undefined, undefined, undefined, 1]
```

There's now a 1 at index 3 of existingNumbers, to indicate and remember for the future that we've already encountered a 3 in our given array.

When our loop then encounters the 5 from the given array, it adds a 1 to index 5 of existingNumbers:

```
[undefined, undefined, undefined, 1, undefined, 1]
```

Finally, when we reach the 8, existingNumbers will now look like this:

```
[undefined, undefined, undefined, 1, undefined, 1, undefined, undefined, 1]
```

Essentially, we're using the indexes of existingNumbers to remember which numbers from the array we've seen so far.

Now, here's the real trick. Before the code stores a 1 in the appropriate index, it *first checks to see whether that index already has a 1 as its value.* If it does, this means we've already encountered that number, meaning we found a duplicate. If this is the case, we simply return true and cut the function short. If we get to the end of the loop without having returned true, it means there are no duplicates and we return false.

To determine the efficiency of this new algorithm in terms of Big O, we once again need to determine the number of steps the algorithm takes in a worst-case scenario.

Here, the significant type of step is looking at each number and checking whether the value of its index in existingNumbers is a 1:

```
if(existingNumbers[array[i]] === 1)
```

(In addition to the comparisons, we also make *insertions* into the existingNumbers array, but we're considering that kind of step trivial in this analysis. More on this in the next chapter.)

In terms of the worst-case scenario, such a scenario would occur when the array contains no duplicates, in which case our function must complete the entire loop.

This new algorithm appears to make N comparisons for N data elements. This is because there's only one loop, and it simply iterates for as many numbers as there are in the array. We can test out this theory by tracking the steps in the JavaScript console:

```
function hasDuplicateValue(array) {
    let steps = 0;
    let existingNumbers = [];
    for(let i = 0; i < array.length; i++) {
        steps++;
        if(existingNumbers[array[i]] === 1) {
            return true;
        } else {
            existingNumbers[array[i]] = 1;
        }
    }
    console.log(steps);
    return false;
}
```

If we run hasDuplicateValue([1, 4, 5, 2, 9]) now, we'll see that the output in the JavaScript console is 5, which is the same as the size of our array. We'd find this to be true across arrays of all sizes. This algorithm, then, is O(N).

We know that O(N) is much faster than $O(N^2)$, so by using this second approach, we've optimized our hasDuplicateValue function significantly. This is a *huge* speed boost.

(There is actually one disadvantage with this new implementation, namely that this approach will consume more memory than the first approach. Don't worry about this for now; we'll discuss this at length in *Dealing with Space Constraints*.)

Wrapping Up

It's clear that having a solid understanding of Big O Notation can enable you to identify slow code and select the faster of two competing algorithms.

However, there are situations in which Big O Notation will have us believe that two algorithms have the same speed, while one is actually faster. In the next chapter, you're going to learn how to evaluate the efficiencies of various algorithms even when Big O isn't nuanced enough to do so.

Exercises

The following exercises provide you with the opportunity to practice with speeding up your code. The solutions to these exercises are found in the section, Chapter 4, on page 441.

1. Replace the question marks in the following table to describe how many steps occur for a given number of data elements across various types of Big O:

N Elements	O(N)	O(log N)	$O(N^2)$
100	100	?	?
2000	?	?	?

2. If we have an $O(N^2)$ algorithm that processes an array and find that it takes 256 steps, what is the size of the array?

3. Use Big O Notation to describe the time complexity of the following function. It finds the greatest product of any pair of two numbers within a given array:

```
def greatestProduct(array):
  greatestProductSoFar = array[0] * array[1]

  for i, iVal in enumerate(array):
    for j, jVal in enumerate(array):
      if i != j and iVal * jVal > greatestProductSoFar:
        greatestProductSoFar = iVal * jVal

  return greatestProductSoFar
```

4. The following function finds the greatest single number within an array, but has an efficiency of $O(N^2)$. Rewrite the function so that it becomes a speedy O(N):

```
def greatestNumber(array):
  for i in array:
    # Assume for now that i is the greatest:
    isIValTheGreatest = True

    for j in array:
      # If we find another value that is greater than i,
      # i is not the greatest:
      if j > i:
        isIValTheGreatest = False

    # If, by the time we checked all the other numbers, i
    # is still the greatest, it means that i is the greatest number:
    if isIValTheGreatest:
      return i
```

CHAPTER 5

Optimizing Code with and Without Big O

We've seen that Big O Notation is a great tool for comparing algorithms and determining which algorithm should be used for a given situation. However, it's certainly not the *only* tool. In fact, there may be times when two competing algorithms are described in the same way using Big O, yet one algorithm is faster than the other.

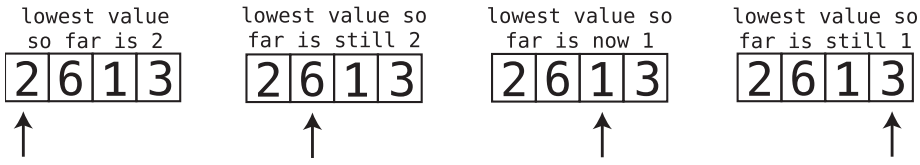
In this chapter, you're going to learn how to discern between two algorithms that *seem* to have the same efficiency, and how to select the faster of the two.

Selection Sort

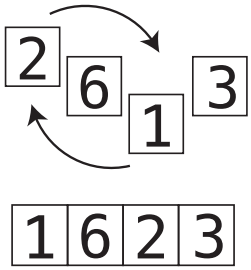
In the previous chapter, we explored a sorting algorithm known as Bubble Sort, which had an efficiency of $O(N^2)$. We're now going to dig into another sorting algorithm called *Selection Sort,* and see how it measures up to Bubble Sort.

The steps of Selection Sort are as follows:

1. We check each cell of the array from left to right to determine which value is least. As we move from cell to cell, we keep track of the lowest value we've encountered so far. (We'll do this by storing its index in a variable.) If we encounter a cell that contains a value that is even lower than the one in our variable, we replace it so that the variable now points to the new index. See the following diagram:



2. Once we've determined which index contains the lowest value, we swap its value with the value we began the pass-through with. This would be index 0 in the first pass-through, index 1 in the second pass-through, and so on. The diagram here illustrates making the swap of the first pass-through.



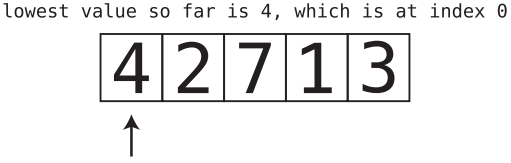
3. Each pass-through consists of Steps 1 and 2. We repeat the pass-throughs until we reach a pass-through that would start at the end of the array. By this point, the array will have been fully sorted.

Selection Sort in Action

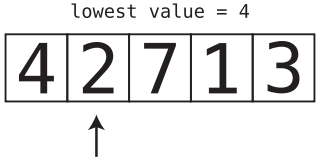
Let's walk through the steps of Selection Sort using the example array, `[4, 2, 7, 1, 3]`.

We begin our first pass-through:

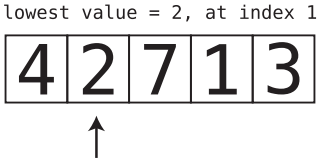
We set things up by inspecting the value at index 0. By definition, it's the lowest value in the array we've encountered so far (as it's the *only* value we've encountered so far), so we keep track of its index in a variable:



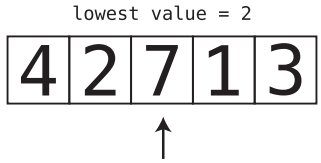
Step 1: We compare the 2 with the lowest value so far (which happens to be 4):



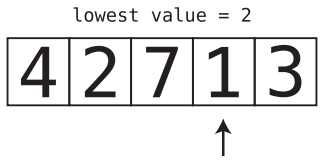
The 2 is even less than the 4, so it becomes the lowest value so far:



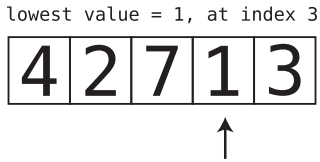
Step 2: We compare the next value—the 7—with the lowest value so far. The 7 is greater than the 2, so 2 remains our lowest value:



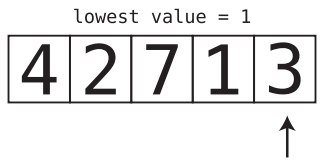
Step 3: We compare the 1 with the lowest value so far:



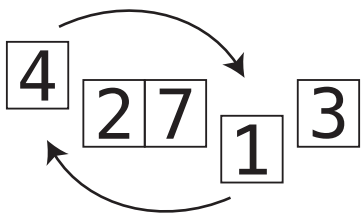
Because the 1 is even less than the 2, 1 becomes our new lowest value:



Step 4: We compare the 3 to the lowest value so far, which is the 1. We've reached the end of the array, and we've determined that 1 is the lowest value out of the entire array:



Step 5: Because 1 is the lowest value, we swap it with whatever value is at index 0—the index we began this pass-through with:

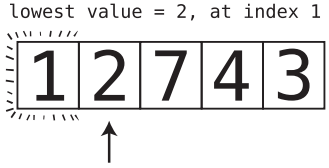


Since we've moved the lowest value to the beginning of the array, that means the lowest value is now in its correct spot:

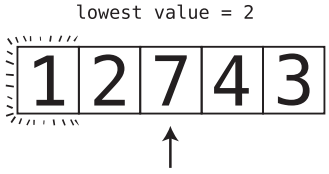


We are now ready to begin our second pass-through.

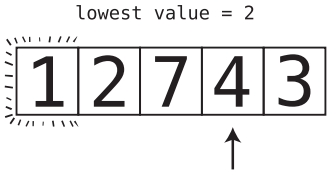
Setup: The first cell—index 0—is already sorted, so this pass-through begins at the next cell, which is index 1. The value at index 1 is the number 2, and it is the lowest value we've encountered in this pass-through so far:



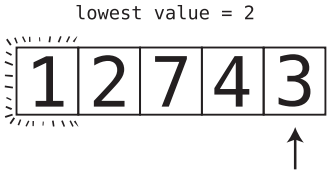
Step 6: We compare the 7 with the lowest value so far. The 2 is less than the 7, so 2 remains our lowest value:



Step 7: We compare the 4 with the lowest value so far. The 2 is less than the 4, so 2 remains our lowest value:



Step 8: We compare the 3 with the lowest value so far. The 2 is less than the 3, so 2 remains our lowest value:

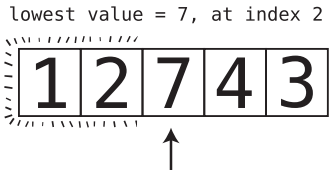


We've reached the end of the array. Since the lowest value from this pass-through was already in its correct spot, we don't need to perform a swap. This ends our second pass-through, leaving us with:

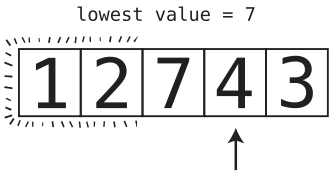


We now begin the third pass-through.

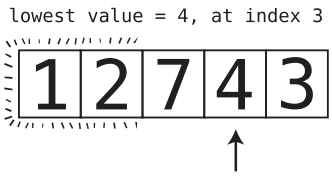
Setup: We begin at index 2, which contains the value 7. The 7 is the lowest value we've encountered so far in this pass-through:



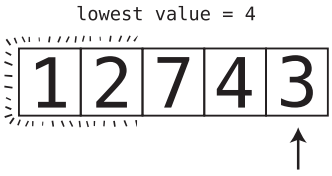
Step 9: We compare the 4 with the 7:



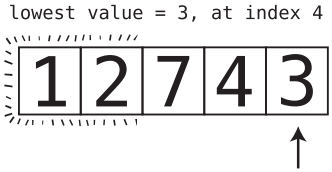
We note that 4 is our new lowest value:



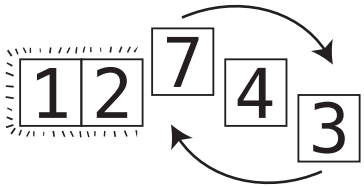
Step 10: We encounter the 3, which is even lower than the 4:



The 3 becomes our new lowest value:



Step 11: We've reached the end of the array, so we swap the 3 with the value we started our pass-through with, which is the 7:

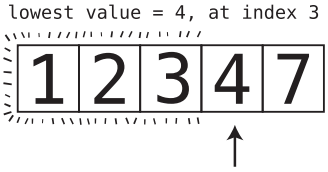


We now know that the 3 is in the correct place within the array:

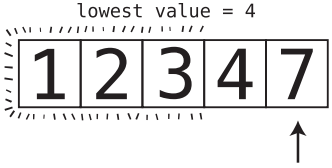


While you and I can both see that the entire array is correctly sorted at this point, the *computer* does not know this yet, so it must begin a fourth pass-through.

Setup: We begin the pass-through with index 3. The 4 is the lowest value so far:



Step 12: We compare the 7 with the 4:



The 4 remains the lowest value we've encountered in this pass-through so far, so we don't need to swap it, since it's already in the correct place.

Because all the cells besides the last one are correctly sorted, that must mean the last cell is also in the correct order, and our entire array is properly sorted:



Code Implementation: Selection Sort

Here's a JavaScript implementation of Selection Sort:

```
function selectionSort(array) {
  for(let i = 0; i < array.length - 1; i++) {
    let lowestNumberIndex = i;
    for(let j = i + 1; j < array.length; j++) {
      if(array[j] < array[lowestNumberIndex]) {
        lowestNumberIndex = j;
      }
    }

    if(lowestNumberIndex != i) {
      let temp = array[i];
      array[i] = array[lowestNumberIndex];
      array[lowestNumberIndex] = temp;
    }
  }
  return array;
}
```

Let's break this down line by line.

We begin a loop that represents each pass-through. It uses the variable `i` to point to each value of the `array`, and goes up through the second-to-last value:

```
for(let i = 0; i < array.length - 1; i++) {
```

It doesn't need to run for the last value itself, since the array will be fully sorted by that point.

Next, begin keeping track of the *index* containing the lowest value we encounter so far:

```
let lowestNumberIndex = i;
```

This `lowestNumberIndex` will be 0 at the beginning of the first pass-through, 1 at the beginning of the second, and so on.

The reason we specifically keep track of the index is because we'll need access to both the lowest value and its index in the rest of our code, and we can use the index to reference both. (We can check the lowest value by calling `array[lowestNumberIndex]`).

Within each pass-through, we check the remaining values of the array to see if there might be a lower value than the current lowest value:

```
for(let j = i + 1; j < array.length; j++) {
```

Indeed, if we find a lower value, we store this new value's index in lowestNumberIndex:

```
if(array[j] < array[lowestNumberIndex]) {
  lowestNumberIndex = j;
}
```

By the end of the inner loop, we'll have found the index of the lowest number from this pass-through.

If the lowest value from this pass-through is already in its correct place (which would happen when the lowest value is the first value we encounter in the pass-through), we don't need to do anything. But if the lowest value is *not* in its correct place, we need to perform a swap. Specifically, we swap the lowest value with the value at i, which was the index we started the pass-through with:

```
if(lowestNumberIndex != i) {
  let temp = array[i];
  array[i] = array[lowestNumberIndex];
  array[lowestNumberIndex] = temp;
}
```

Finally, we return the sorted array:

```
return array;
```

The Efficiency of Selection Sort

Selection Sort contains two types of steps: comparisons and swaps. That is, we compare each value with the lowest number we've encountered in each pass-through, and we swap the lowest number into its correct position.

Looking back at our example array that contains five elements, we had to make a total of 10 comparisons. Let's break it down in the following table:

Pass-Through #	# of Comparisons
1	4 comparisons
2	3 comparisons
3	2 comparisons
4	1 comparison

That's a grand total of 4 + 3 + 2 + 1 = 10 comparisons.

To put it in a way that works for arrays of all sizes, we'd say that for N elements, we make

(N - 1) + (N - 2) + (N - 3) … + 1 comparisons.

As for *swaps*, though, we only need to make a maximum of one swap per pass-through. This is because in each pass-through, we make either one or zero swaps, depending on whether the lowest number of that pass-through is already in the correct position. Contrast this with Bubble Sort, where in a worst-case scenario—an array in descending order—we have to make a swap for *each and every* comparison.

Here's a side-by-side comparison of Bubble Sort and Selection Sort:

N Elements	Max # of Steps in Bubble Sort	Max # of Steps in Selection Sort
5	20	14 (10 comparisons + 4 swaps)
10	90	54 (45 comparisons + 9 swaps)
20	380	199 (180 comparisons + 19 swaps)
40	1560	819 (780 comparisons + 39 swaps)
80	6320	3239 (3160 comparisons + 79 swaps)

From this comparison, it's clear Selection Sort takes about half the number of steps Bubble Sort does, indicating that Selection Sort is twice as fast.

Ignoring Constants

But here's the funny thing: in the world of Big O Notation, Selection Sort and Bubble Sort are described in *exactly the same way.*

Again, Big O Notation answers the key question: if there are N data elements, how many steps will the algorithm take? Because Selection Sort takes roughly half of N^2 steps, it would seem reasonable that we'd describe the efficiency of Selection Sort as being $O(N^2 / 2)$. That is, for N data elements, there are $N^2 / 2$ steps. The following table bears this out:

N Elements	$N^2 / 2$	Max # of Steps in Selection Sort
5	$5^2 / 2 = 12.5$	14
10	$10^2 / 2 = 50$	54
20	$20^2 / 2 = 200$	199
40	$40^2 / 2 = 800$	819
80	$80^2 / 2 = 3200$	3239

In reality, however, Selection Sort is described in Big O as $O(N^2)$, just like Bubble Sort. This is because of a major rule of Big O that I'm now introducing for the first time:

Big O Notation ignores constants.

This is simply a mathematical way of saying that Big O Notation never includes regular numbers that aren't an exponent. We simply drop these regular numbers from the expression.

In our case, then, even though the algorithm takes $N^2 / 2$ steps, we drop the "/ 2" because it's a regular number, and express the efficiency as $O(N^2)$.

Here's a few more examples:

For an algorithm that takes N / 2 steps, we'd call it O(N).

An algorithm that takes $N^2 + 10$ steps would be expressed as $O(N^2)$, since we drop the 10, which is a regular number.

With an algorithm that takes 2N steps (meaning N * 2), we drop the regular number and call it O(N).

Even O(100N), which is *100 times slower than O(N),* is also referred to as O(N).

Offhand, it would seem that this rule would render Big O Notation entirely useless, as you can have two algorithms that are described in the same exact way with Big O, and yet one can be *100 times faster* than the other. And that's exactly what we're seeing here with Selection Sort and Bubble Sort. Both are described in Big O as $O(N^2)$, but Selection Sort is actually twice as fast as Bubble Sort.

So, what gives?

Big O Categories

This leads us to the next concept within Big O: Big O Notation only concerns itself with *general categories* of algorithm speeds.

As an analogy, let's talk about physical buildings. There are, of course, many different types of buildings. There are one-floor single-family homes, and two-floor single-family homes, and three-floor single-family homes. There are high-rise apartment buildings with varying numbers of floors. And there are skyscrapers with various heights and shapes.

If we were to compare two buildings, one of which is a single-family home and one of which is a skyscraper, it becomes almost moot to mention how many floors each one has. Because the two buildings are so incredibly different

in their sizes and functions, we don't need to say, "This one is a two-story home, while the other is a one-hundred-floor skyscraper." We may as well just call one a house and the other a skyscraper. Calling them by their general categories is enough to signify their vast differences.

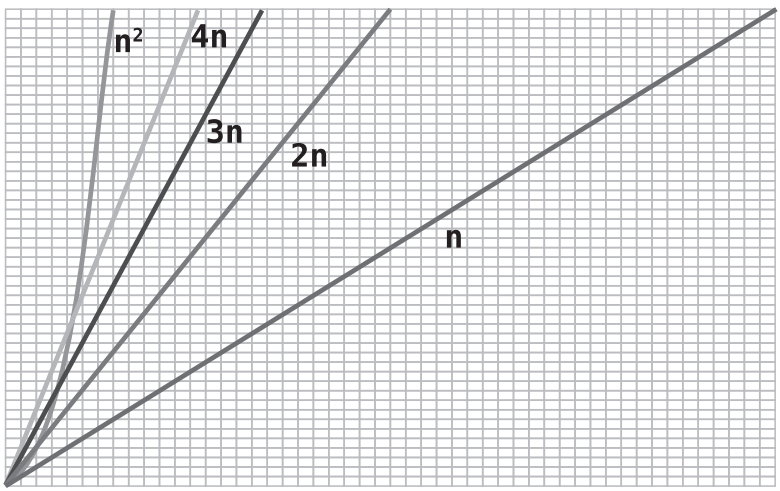
The same applies to algorithm efficiencies. If we compare, say, an O(N) algorithm with an $O(N^2)$ algorithm, the two efficiencies are so different that it doesn't really matter whether the O(N) algorithm is actually O(2N), or O(N / 2) or even O(100N).

Now, here's why O(N) and $O(N^2)$ are considered two separate categories, while O(N) and O(100N) are part of the same category.

Remember The Soul of Big O, on page 37. Big O Notation doesn't care merely about the number of steps an algorithm takes. It cares about the long-term trajectory of the algorithm's steps as the data increases. O(N) tells a story of straight growth—that the steps increase in a straight line according to some proportion of the data. This is true even when the steps are 100N. $O(N^2)$ tells a different story—one of exponential growth.

Exponential growth is a completely different category compared to any form of O(N). This point is really driven home when we consider that $O(N^2)$ will, at some point in data growth, become slower than O(N) multiplied by *any* factor.

In the following graph, you can see how $O(N^2)$ becomes slower than various factors of N:



Therefore, when comparing two efficiencies that belong to two different categories of Big O, it's enough to identify them by their general category. Talking about O(2N) when compared to $O(N^2)$ is like talking about a two-story house compared to a skyscraper. We may as well just say that O(2N) is part of the general category of O(N).

All the types of Big O we've encountered, whether it's O(1), O(log N), O(N), $O(N^2)$, or the types we'll encounter later in this book, are general categories of Big O that are widely different from each other. Multiplying or dividing the number of steps by a regular number doesn't make them change to another category.

However, when two algorithms fall under the *same* classification of Big O, it doesn't necessarily mean that both algorithms have the same speed. After all, Bubble Sort is twice as slow as Selection Sort even though both are $O(N^2)$. So, while Big O is perfect for contrasting algorithms that fall under different classifications of Big O, when two algorithms fall under the *same* classification, further analysis is required to determine which algorithm is faster.

A Practical Example

Let's return to the very first code example from Chapter 1, with minor changes:

```
def print_numbers_version_one(upperLimit):
  number = 2

  while number <= upperLimit:
    # If number is even, print it:
    if number % 2 == 0:
      print(number)

    number += 1

def print_numbers_version_two(upperLimit):
  number = 2

  while number <= upperLimit:
    print(number)

    # Increase number by 2, which, by definition,
    # is the next even number:
    number += 2
```

Here we have two algorithms for accomplishing the same task, namely printing all even numbers starting from 2 to some upperLimit. (In Chapter 1, the upper limit was fixed at 100, while here, we let the user pass in a number as the upperLimit.)

I noted in Chapter 1 that the first version takes twice as many steps as the second version, but now let's see how this plays out in terms of Big O.

Again, Big O expresses the answer to the key question: if there are N data elements, how many steps will the algorithm take? In this case, though, N isn't the size of an array, but simply the number we pass into the function to serve as the `upperLimit`.

The first version takes about N steps. That is, if the `upperLimit` is 100, the function takes about 100 steps. (It really takes 99 steps, since it starts the count at 2.) So, we can safely say that the first algorithm has a time complexity of O(N).

The second version takes N / 2 steps. When the `upperLimit` is 100, the function takes just 50 steps. While it would be tempting to call this O(N / 2), you've now learned that we drop the constants and reduce the expression to O(N) as well.

Now, the second version is twice as fast as the first one and would naturally be the better choice. This is another great example of where two algorithms can be expressed the same way using Big O Notation, but further analysis is needed to figure out which algorithm is faster.

Significant Steps

Let's apply one more level of analysis to the previous example. If we look again at the first version, `print_numbers_version_one`, we said that it takes N steps. This is because the loop runs N times, with N being the `upperLimit`.

But does the function really take just N steps?

If we really break things down, we can see that *multiple* steps occur in each round of the loop.

First, we have the comparison step (`if number % 2 == 0`), which checks whether the `number` is divisible by 2. This comparison happens in each round of the loop.

Second, we have the print step (`print(number)`), which happens just for the even numbers. This, then, occurs in every *other* round of the loop.

And third, we have `number += 1`, which runs in each round of the loop.

In the previous chapters, I alluded to the fact that you'd learn how to determine which steps are significant enough to be counted when expressing the Big O of an algorithm. In our case, then, which of these steps are considered significant? Do we care about the comparisons, the printing, or the incrementing of `number`?

The answer is that *all* steps are significant. It's just that when we express the steps in Big O terms, we drop the constants and thereby simplify the expression.

Let's apply this here. If we count all the steps, we have N comparisons, N incrementings, and N / 2 printings. This adds up to 2.5N steps. However, because we eliminate the constant of 2.5, we express this as O(N). So, which step was significant? They all were, but by dropping the constant, we effectively focus more on the number of times the loop runs, rather than the exact details of what happens within the loop.

Wrapping Up

We now have some powerful analysis tools at our disposal. We can use Big O to broadly determine the efficiency of an algorithm, and we can also compare two algorithms that fall within one classification of Big O.

However, another important factor must be taken into account when comparing the efficiencies of two algorithms. Until now, we've focused on how fast an algorithm is in a worst-case scenario. Now, worst-case scenarios, by definition, don't happen all the time. On average, most scenarios that occur are…well…average-case scenarios. In the next chapter, you'll learn how to take all scenarios into account.

Exercises

The following exercises provide you with the opportunity to practice analyzing algorithms. The solutions to these exercises are found in the section, Chapter 5, on page 442.

1. Use Big O Notation to describe the time complexity of an algorithm that takes 4N + 16 steps.
2. Use Big O Notation to describe the time complexity of an algorithm that takes $2N^2$.
3. Use Big O Notation to describe the time complexity of the following function, which returns the sum of all numbers of an array after the numbers have been doubled:

```
def double_then_sum(array)
  doubled_array = []

  array.each do |number|
    doubled_array << number *= 2
  end

  sum = 0

  doubled_array.each do |number|
    sum += number
  end

  return sum
end
```

4. Use Big O Notation to describe the time complexity of the following function, which accepts an array of strings and prints each string in multiple cases:

```
def multiple_cases(array)
  array.each do |string|
    puts string.upcase
    puts string.downcase
    puts string.capitalize
  end
end
```

5. The next function iterates over an array of numbers, and for each number whose *index* is even, it prints the sum of that number plus every number in the array. What is this function's efficiency in terms of Big O Notation?

```
def every_other(array)
  array.each_with_index do |number, index|
    if index.even?
      array.each do |other_number|
        puts number + other_number
      end
    end
  end
end
```

CHAPTER 6

Optimizing for Optimistic Scenarios

Until this point, we've focused primarily on how many steps an algorithm would take in a worst-case scenario. The rationale behind this is simple: if you're prepared for the worst, things will turn out okay.

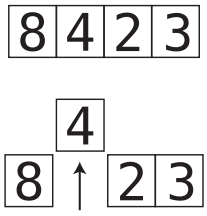
However, you'll discover in this chapter that the worst-case scenario isn't the *only* situation worth considering. Being able to consider *all* scenarios is an important skill that can help you choose the appropriate algorithm for every situation.

Insertion Sort

We've previously encountered two different sorting algorithms: Bubble Sort and Selection Sort. Both have efficiencies of $O(N^2)$, but Selection Sort is actually twice as fast. Now you'll learn about a third sorting algorithm called *Insertion Sort* that will reveal the power of analyzing scenarios beyond the worst case.

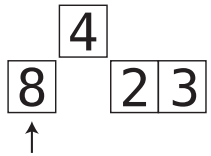
Insertion Sort consists of the following steps:

1. In the first pass-through, we temporarily remove the value at index 1 (the second cell) and store it in a temporary variable. This will leave a gap at that index, since it contains no value:

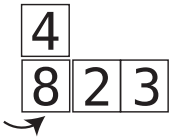


In subsequent pass-throughs, we remove the values at the subsequent indexes.

2. We then begin a shifting phase, where we take each value to the left of the gap and compare it to the value in the temporary variable:

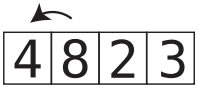


If the value to the left of the gap is greater than the temporary variable, we shift that value to the right:



As we shift values to the right, inherently the gap moves leftward. As soon as we encounter a value that is lower than the temporarily removed value, or we reach the left end of the array, this shifting phase is over.

3. We then insert the temporarily removed value into the current gap:

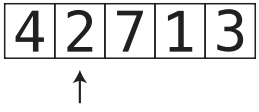


4. Steps 1 through 3 represent a single pass-through. We repeat these pass-throughs until the pass-through begins at the final index of the array. By then, the array will have been fully sorted.

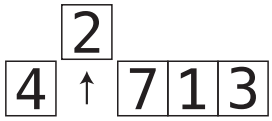
Insertion Sort in Action

Let's apply Insertion Sort to the array, `[4, 2, 7, 1, 3]`.

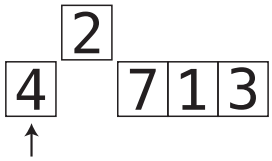
We begin the first pass-through by inspecting the value at index 1. This happens to contain the value 2:



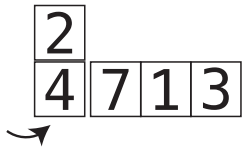
Step 1: We temporarily remove the 2 and keep it inside a variable called temp_value. We represent this value by shifting it above the rest of the array:



Step 2: We compare the 4 to the temp_value, which is 2:

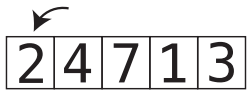


Step 3: Because 4 is greater than 2, we shift the 4 to the right:



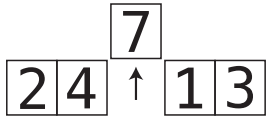
There's nothing left to shift, as the gap is now at the left end of the array.

Step 4: We insert the temp_value into the gap, completing our first pass-through:

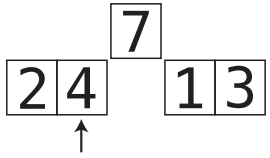


Next, we begin the second pass-through:

Step 5: In our second pass-through, we temporarily remove the value at index 2. We'll store this in temp_value. In this case, the temp_value is 7:

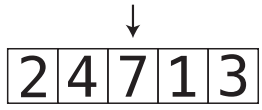


Step 6: We compare the 4 to the temp_value:



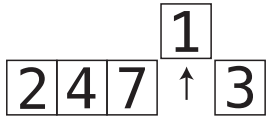
The 4 is lower, so we won't shift it. Since we reached a value that is less than the temp_value, this shifting phase is over.

Step 7: We insert the temp_value back into the gap, ending the second pass-through:

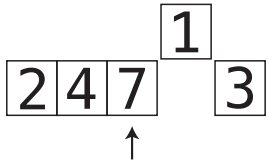


We now begin the third pass-through:

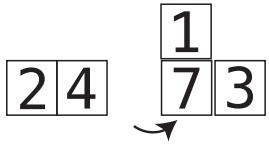
Step 8: We temporarily remove the 1 and store it in temp_value:



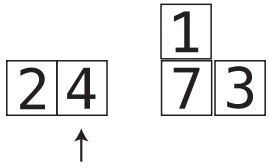
Step 9: We compare the 7 to the temp_value:



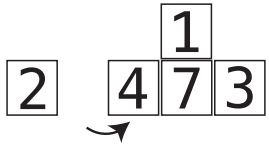
Step 10: The 7 is greater than 1, so we shift the 7 to the right:



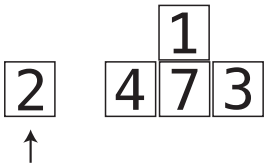
Step 11: We compare the 4 to the temp_value:



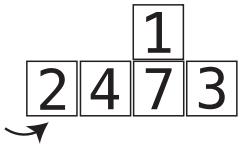
Step 12: The 4 is greater than 1, so we shift it as well:



Step 13: We compare the 2 to the temp_value:



Step 14: The 2 is greater, so we shift it:

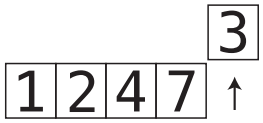


Step 15: The gap has reached the left end of the array, so we insert the temp_value into the gap, concluding this pass-through:

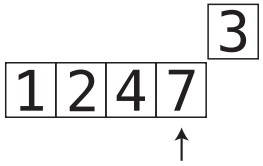


Now, we begin the fourth pass-through:

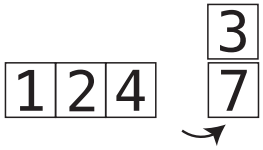
Step 16: We temporarily remove the value from index 4, making it our temp_value. This is the value 3:



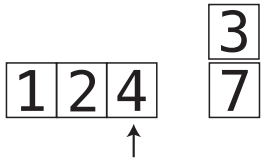
Step 17: We compare the 7 to the temp_value:



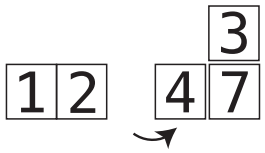
Step 18: The 7 is greater, so we shift the 7 to the right:



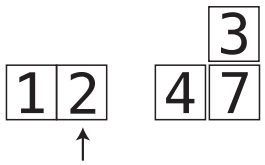
Step 19: We compare the 4 to the temp_value:



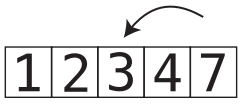
Step 20: The 4 is greater than the 3, so we shift the 4:



Step 21: We compare the 2 to the temp_value. The 2 is less than 3, so our shifting phase is complete:



Step 22: We insert the temp_value back into the gap:



Our array is now fully sorted:



Code Implementation: Insertion Sort

Here is a Python implementation of Insertion Sort:

```
def insertion_sort(array):
  for index in range(1, len(array)):

    temp_value = array[index]
    position = index - 1

    while position >= 0:
      if array[position] > temp_value:
        array[position + 1] = array[position]
        position = position - 1
```

```
      else:
        break

    array[position + 1] = temp_value

  return array
```

Let's walk through this code step by step.

First, we start a loop beginning at index 1 that runs through the entire array. Each round of this loop represents a pass-through:

```
for index in range(1, len(array)):
```

Within each pass-through, we save the value we're "removing" in a variable called temp_value:

```
temp_value = array[index]
```

Next, we create a variable called position, which will start immediately to the left of the index of the temp_value. This position will represent each value we compare against the temp_value:

```
position = index - 1
```

As we move through the pass-through, this position will keep moving leftward as we compare each value to the temp_value.

We then begin an inner while loop, which runs as long as position is greater or equal to 0:

```
while position >= 0:
```

We then perform our comparison. That is, we check whether the value at position is greater than the temp_value:

```
if array[position] > temp_value:
```

If it is, we shift that left value to the right:

```
array[position + 1] = array[position]
```

We then decrement position by 1 to compare the next left value against the temp_value in the next round of the while loop:

```
position = position - 1
```

If at any point we encounter a value at position that is greater or equal to the temp_value, we can get ready to end our pass-through, since it's time to move the temp_value into the gap:

```
else:
  break
```

The final step of each pass-through is moving the temp_value into the gap:

```
array[position + 1] = temp_value
```

After all pass-throughs have been completed, we return the sorted array:

```
return array
```

The Efficiency of Insertion Sort

Four types of steps occur in Insertion Sort: removals, comparisons, shifts, and insertions. To analyze the efficiency of Insertion Sort, we need to tally up each of these steps.

First, let's dig into comparisons. A comparison takes place each time we compare a value to the left of the gap with the temp_value. In a worst-case scenario, where the array is sorted in reverse order, we have to compare every number to the left of temp_value with temp_value in each pass-through. This is because each value to the left of temp_value will always be greater than temp_value, so the pass-through will only end when the gap reaches the left end of the array.

During the first pass-through, in which temp_value is the value at index 1, a maximum of one comparison is made, since there is only one value to the left of the temp_value. On the second pass-through, a maximum of two comparisons are made, and so on. On the final pass-through, we need to compare the temp_value with every single value in the array besides temp_value itself. In other words, if there are N elements in the array, a maximum of N - 1 comparisons are made in the final pass-through.

We can, therefore, formulate the total number of comparisons as:

1 + 2 + 3 + … + (N - 1) comparisons.

In our example array that contains five elements, that's a maximum of:

1 + 2 + 3 + 4 = 10 comparisons.

For an array containing 10 elements, there would be:

1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45 comparisons.

For an array containing 20 elements, there would be a total of 190 comparisons, and so on.

When examining this pattern, it emerges that for an array containing N elements, there are approximately $N^2 / 2$ comparisons. ($10^2 / 2$ is 50, and $20^2 / 2$ is 200. We'll look at this pattern more closely in the next chapter.)

Let's continue analyzing the other types of steps.

Shifts occur each time we move a value one cell to the right. When an array is sorted in reverse order, there will be as many shifts as there are comparisons, since every comparison will force us to shift a value to the right.

Let's add up comparisons and shifts for a worst-case scenario:

N^2 / 2 comparisons

+ N^2 / 2 shifts

N^2 steps

Removing and inserting the `temp_value` from the array happens once per pass-through. Since there are always N - 1 pass-throughs, we can conclude that there are N - 1 removals and N - 1 insertions.

So, now we've got:

N^2 comparisons and shifts combined

N - 1 removals

+ N - 1 insertions

N^2 + 2N - 2 steps

You've already learned one major rule of Big O: that Big O ignores constants. With this rule in mind, we'd—at first glance—simplify this to $O(N^2 + N)$.

However, there is another major rule of Big O that I'll reveal now:

Big O Notation only takes into account the highest order of N when we have multiple orders added together.

That is, if we have an algorithm that takes $N^4 + N^3 + N^2 + N$ steps, we only consider N^4 to be significant—and just call it $O(N^4)$. Why is this?

Look at the following table:

N	N^2	N^3	N^4
2	4	8	16
5	25	125	625
10	100	1,000	10,000
100	10,000	1,000,000	100,000,000

As N increases, N^4 becomes so much more significant than any other order of N that the smaller orders are considered trivial. For example, when looking at the bottom row of the table, when we add $N^4 + N^3 + N^2 + N$, we get a total of 101,010,100. But we may as well round that down to 100,000,000, which is accomplished by ignoring those lower orders of N.

We can apply this same concept to Insertion Sort. Even though we've already simplified Insertion Sort down to $N^2 + N$ steps, we simplify the expression further by throwing out the lower order, reducing it to $O(N^2)$.

It emerges that in a worst-case scenario, Insertion Sort has the same time complexity as Bubble Sort and Selection Sort. They're all $O(N^2)$.

I noted in the previous chapter that although Bubble Sort and Selection Sort are both $O(N^2)$, Selection Sort is faster because Selection Sort has $N^2 / 2$ steps compared with Bubble Sort's N^2 steps. At first glance, then, we'd say that Insertion Sort is as slow as Bubble Sort, since it too takes about N^2 steps.

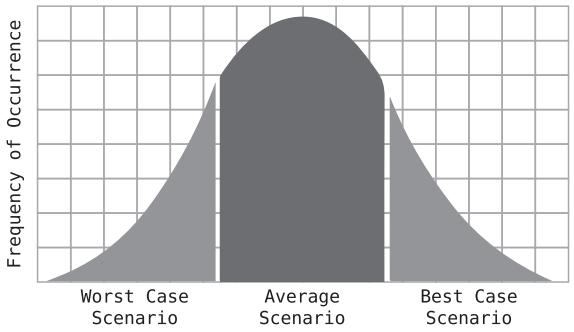
If I stop the book here, you'd walk away thinking that Selection Sort is the best choice out of the three, since it's twice as fast as either Bubble Sort or Insertion Sort. But it's actually not that simple.

The Average Case

Indeed, in a worst-case scenario, Selection Sort *is* faster than Insertion Sort. However, it is critical we also take into account the *average-case scenario.*

Why?

By definition, the cases that occur most frequently are average scenarios. Take a look at this simple bell curve:



Best- and worst-case scenarios happen relatively infrequently. In the real world, average scenarios are what occur most of the time.

Take a randomly sorted array, for example. What are the odds that the values will occur in perfect ascending or descending order? It's much more likely that the values will be all over the place.

Let's examine Insertion Sort, then, in the context of all scenarios.

We've looked at how Insertion Sort performs in a worst-case scenario—where the array is sorted in descending order. In the worst case, we saw that in each pass-through, we compare and shift every value we encounter. (We calculated this to be a total of N^2 comparisons and shifts.)

In the best-case scenario, where the data is already sorted in ascending order, we end up making just one comparison per pass-through and not a single shift, since each value is already in its correct place.

Where data is randomly sorted, however, we'll have pass-throughs in which we compare and shift all of the data, some of the data, or possibly none of the data. If you look at the preceding walk-through example in Insertion Sort in Action, on page 80, you'll notice that in the first and third pass-throughs, we compare and shift all the data we encounter. In the fourth pass-through, we compare and shift just some of it, and in the second pass-through, we make just one comparison and shift no data at all.

(This variance occurs because some pass-throughs compare all the data to the left of the `temp_value`, while other pass-throughs end early, due to encountering a value that is less than the `temp_value`.)

So, in the worst-case scenario, we compare and shift *all* the data, and in the best-case scenario, we shift *none* of the data (and just make one comparison per pass-through). For the average scenario, we can say that in the aggregate, we probably compare and shift about *half* the data. Thus, if Insertion Sort takes N^2 steps for the worst-case scenario, we'd say that it takes about $N^2 / 2$ steps for the average scenario. (In terms of Big O, however, both scenarios are $O(N^2)$.)

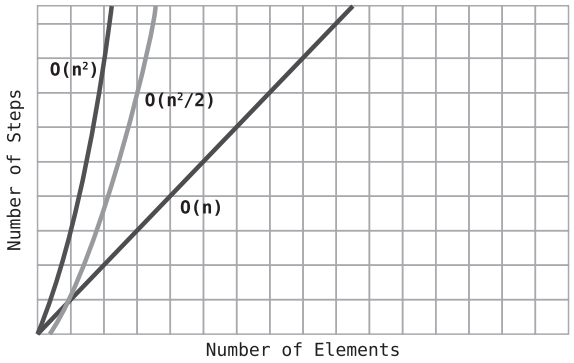
Let's dive into some specific examples.

The array, `[1, 2, 3, 4]` is already presorted, which is the best case. The worst case for the same data would be `[4, 3, 2, 1]`, and an example of an average case might be `[1, 3, 4, 2]`.

In the worst case (`[4, 3, 2, 1]`), there are six comparisons and six shifts, for a total of 12 steps. In an average case of `[1, 3, 4, 2]`, there are four comparisons and two shifts, for a total of six steps. In the best case (`[1, 2, 3, 4]`), there are three comparisons and zero shifts.

We can now see that the performance of Insertion Sort *varies greatly* based on the scenario. In the worst-case scenario, Insertion Sort takes N^2 steps. In an average scenario, it takes N^2 / 2 steps. And in the best-case scenario, it takes about N steps.

You can see these three types of performance in the following graph:



Contrast this with Selection Sort. Selection Sort takes N^2 / 2 steps in *all* cases, from worst to average to best-case scenarios. This is because Selection Sort doesn't have any mechanism for ending a pass-through early at any point. Each pass-through compares every value to the right of the chosen index no matter what.

Here's a table that compares Selection Sort and Insertion Sort:

	Best Case	Average Case	Worst Case
Selection Sort	N^2 / 2	N^2 / 2	N^2 / 2
Insertion Sort	N	N^2 / 2	N^2

So, which is better: Selection Sort or Insertion Sort? The answer is: well, it depends. In an average case—where an array is randomly sorted—they perform similarly. If you have reason to assume you'll be dealing with data that is *mostly* sorted, Insertion Sort will be a better choice. If you have reason to assume you'll be dealing with data that is mostly sorted in reverse order, Selection Sort will be faster. If you have no idea what the data will be like, that's essentially an average case, and both will be equal.

A Practical Example

Suppose you are writing a JavaScript application, and somewhere in your code you find that you need to get the intersection between two arrays. The intersection is a list of all the values that occur in *both* of the arrays. For example, if you have the arrays, `[3, 1, 4, 2]` and `[4, 5, 3, 6]`, the intersection would be a third array, `[3, 4]`, since both of those values are common to the two arrays.

Here's one possible implementation:

```
function intersection(firstArray, secondArray){
    let result = [];

    for (let i = 0; i < firstArray.length; i++) {
        for (let j = 0; j < secondArray.length; j++) {
            if (firstArray[i] == secondArray[j]) {
                result.push(firstArray[i]);
            }
        }
    }
    return result;
}
```

Here, we are running nested loops. In the outer loop, we iterate over each value in the first array. As we point to each value in the first array, we then run an inner loop that checks each value of the second array to see if it can find a match with the value being pointed to in the first array.

There are two types of steps taking place in this algorithm: comparisons and insertions. We compare every value of the two arrays against each other, and we insert matching values into the array `result`. Let's start by seeing how many comparisons there are.

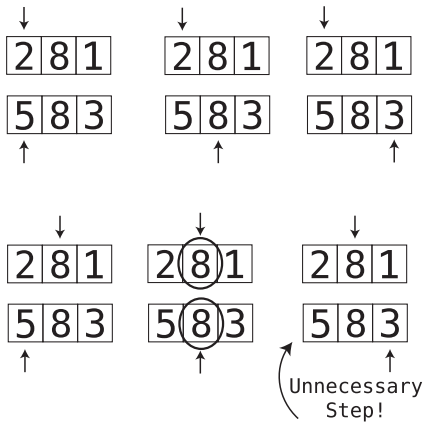
If the two arrays are of equal size, and say that N is the size of either array, the number of comparisons performed are N^2. This is because we compare each element of the first array to each element of the second array. Thus, if we have two arrays that each contain five elements, we'd end up making 25 comparisons. So, this intersection algorithm has an efficiency of $O(N^2)$.

The insertions, at most, would take N steps (if the two arrays happened to be identical). This is a lower order compared to N^2, so we'd still consider the algorithm to be $O(N^2)$. If the arrays are different sizes—say N and M—we'd say that the efficiency of this function is O(N * M). (More on this in *Big O in Everyday Code*.)

Is there any way we can improve this algorithm?

This is where it's important to consider scenarios beyond the worst case. In the current implementation of the intersection function, we make N^2 comparisons in *all* scenarios, no matter whether the arrays are identical or the arrays do not share a single common value.

However, where the two arrays share common values, we really shouldn't have to check *every* value of the second array against a value of the first array. Let's see why:



In this example, as soon as we find a common value (the 8), there's really no reason to complete the second loop. What are we checking for at this point? We've already determined that the second array contains the same 8 that the first array does, and can add it to result. We're performing an unnecessary step.

To fix this, we can add a single word to our implementation:

```
function intersection(firstArray, secondArray){
    let result = [];

    for (let i = 0; i < firstArray.length; i++) {
        for (let j = 0; j < secondArray.length; j++) {
            if (firstArray[i] == secondArray[j]) {
                result.push(firstArray[i]);
                break;
            }
        }
    }
    return result;
}
```

With the addition of the break, we can cut the inner loop short and save steps (and therefore time).

It's still true that in a worst-case scenario, where the two elements do not contain a single shared value, we have no choice but to perform N^2 comparisons. But now, in the best-case scenario, where the two arrays are identical, we only have to perform N comparisons. In an average case, where the two arrays are different but share *some* values, the performance will be somewhere between N and N^2.

This is a significant optimization to our intersection function, since our first implementation would make N^2 comparisons in all cases.

Wrapping Up

Being able to discern between best-, average-, and worst-case scenarios is a key skill in choosing the best algorithm for your needs, as well as taking existing algorithms and optimizing them further to make them significantly faster. Remember, while it's good to be prepared for the worst case, average cases are what happen most of the time.

Now that we've covered the important concepts related to Big O Notation, let's apply our knowledge to practical algorithms. In the next chapter, we're going to take a look at various everyday algorithms that may appear in real codebases, and identify the time complexity of each one in terms of Big O.

Exercises

The following exercises provide you with the opportunity to practice with optimizing for best- and worst-case scenarios. The solutions to these exercises are found in the section, Chapter 6, on page 442.

1. Use Big O Notation to describe the efficiency of an algorithm that takes $3N^2 + 2N + 1$ steps.

2. Use Big O Notation to describe the efficiency of an algorithm that takes $N + \log N$ steps.

3. The following function checks whether an array of numbers contains a pair of two numbers that add up to 10.

```
function twoSum(array) {
  for (let i = 0; i < array.length; i++) {
    for (let j = 0; j < array.length; j++) {
      if (i !== j && array[i] + array[j] === 10) {
        return true;
      }
    }
  }
  return false;
```

```
}
```

What are the best-, average-, and worst-case scenarios? Then, express the worst-case scenario in terms of Big O Notation.

4. The following function returns whether or not a capital "X" is present within a string.

```
function containsX(string) {

  foundX = false;

  for(let i = 0; i < string.length; i++) {
    if (string[i] === "X") {
      foundX = true;
    }
  }

  return foundX;

}
```

What is this function's time complexity in terms of Big O Notation?

Then, modify the code to improve the algorithm's efficiency for best- and average-case scenarios.

CHAPTER 7

Big O in Everyday Code

In the previous chapters, you learned how to use Big O notation to express the time complexity of code. As you've seen, there are quite a few details that go into Big O analysis. In this chapter, we'll use everything you've learned so far to analyze the efficiency of practical code samples that might be found in real-world codebases.

Determining the efficiency of our code is the first step in optimizing it. After all, if we don't know how fast our code is, how would we know if our modifications would make it faster?

Additionally, once we know how our code is categorized in terms of Big O Notation, we can make a judgment call as to whether it may need optimization in the first place. For example, an algorithm that is $O(N^2)$ is generally considered to be a "slow" algorithm. So, if we've determined that our algorithm falls into such a category, we should take pause and wonder if there are ways to optimize it.

Of course, $O(N^2)$ may be the best we can do for a given problem. However, knowing that our algorithm is considered slow can signal to us to dig deeper and analyze whether faster alternatives are available.

In the future chapters of this book, you're going to learn many techniques for optimizing our code for speed. But the first step of optimization is being able to determine how fast our code currently is.

So, let's begin.

Mean Average of Even Numbers

The following Ruby method accepts an array of numbers and returns the mean average of all its *even* numbers. How would we express its efficiency in terms of Big O?

```
def average_of_even_numbers(array)

  # The mean average of even numbers will be defined as the sum of
  # the even numbers divided by the count of even numbers, so we
  # keep track of both the sum and the count:

  sum = 0.0
  count_of_even_numbers = 0

  # We iterate through each number in the array, and if we encounter
  # an even number, we modify the sum and the count:

  array.each do |number|
    if number.even?
      sum += number
      count_of_even_numbers += 1
    end
  end

  # We return the mean average:

  return sum / count_of_even_numbers
end
```

Here's how to break the code down to determine its efficiency.

Remember that Big O is all about answering the key question: if there are N data elements, how many steps will the algorithm take? Therefore, the first thing we want to do is determine what the "N" data elements are.

In this case, the algorithm is processing the array of numbers passed into this method. These, then, would be the "N" data elements, with N being the size of the array.

Next, we have to determine how many steps the algorithm takes to process these N values.

We can see that the guts of the algorithm is the loop that iterates over each number inside the array, so we'll want to analyze that first. Since the loop iterates over each of the N elements, we know that the algorithm takes at least N steps.

Looking *inside* the loop, though, we can see that a varying number of steps occur within each round of the loop. For each and every number, we check whether the number is even. Then, if the number is even, we perform two more steps: we modify the `sum` variable, and we modify the `count_of_even_numbers` variable. So, we execute three more steps for even numbers than we do for odd numbers.

As you've learned, Big O focuses primarily on worst-case scenarios. In our case, the worst case is when all the numbers are even, in which case we

perform three steps during each round of the loop. Because of this, we can say that for N data elements, our algorithm takes 3N steps. That is, for each of the N numbers, our algorithm executes three steps.

Now, our method performs a few other steps outside of the loop as well. Before the loop, we initialize the two variables and set them to 0. Technically, these are two steps. After the loop, we perform another step: the division of `sum / count_of_even_numbers`. Technically, then, our algorithm takes three extra steps in addition to the 3N steps, so the total number of steps is 3N + 3.

However, you also learned that Big O notation ignores constant numbers, so instead of calling our algorithm O(3N + 3), we simply call it O(N).

Word Builder

The next example is an algorithm that collects every combination of two-character strings built from an array of single characters. For example, given the array: `["a", "b", "c", "d"]`, we'd return a new array containing the following string combinations:

```
[
  'ab', 'ac', 'ad', 'ba', 'bc', 'bd',
  'ca', 'cb', 'cd', 'da', 'db', 'dc'
]
```

Following is a JavaScript implementation of this algorithm. Let's see if we can figure out its Big O efficiency:

```
function wordBuilder(array) {
  let collection = [];

  for(let i = 0; i < array.length; i++) {
    for(let j = 0; j < array.length; j++) {
      if (i !== j) {
        collection.push(array[i] + array[j]);
      }
    }
  }

  return collection;
}
```

Here we're running one loop nested inside another. The outer loop iterates over each character in the array, keeping track of the index `i`. For each index `i`, we run an inner loop that iterates again over each character in the same array using the index `j`. Within this inner loop, we concatenate the characters at `i` and `j`, with the exception of when `i` and `j` are pointing to the same index.

To determine the efficiency of our algorithm, we once again need to determine what the N data elements are. In our case, as in the previous example, N is the number of items inside the array passed to the function.

The next step is to determine the number of steps our algorithm takes relative to the N data elements. In our case, the outer loop iterates over all N elements, and for each element, the inner loop iterates again over all N elements, which amounts to N steps multiplied by N steps. This is the classic case of $O(N^2)$, and is often what nested-loop algorithms turn out to be.

Now, what would happen if we modified our algorithm to compute each combination of *three-character* strings? That is, for our example array of `["a", "b", "c", "d"]`, our function would return the array:

```
[
  'abc', 'abd', 'acb',
  'acd', 'adb', 'adc',
  'bac', 'bad', 'bca',
  'bcd', 'bda', 'bdc',
  'cab', 'cad', 'cba',
  'cbd', 'cda', 'cdb',
  'dab', 'dac', 'dba',
  'dbc', 'dca', 'dcb'
]
```

Here's an implementation that uses three nested loops. What is its time complexity?

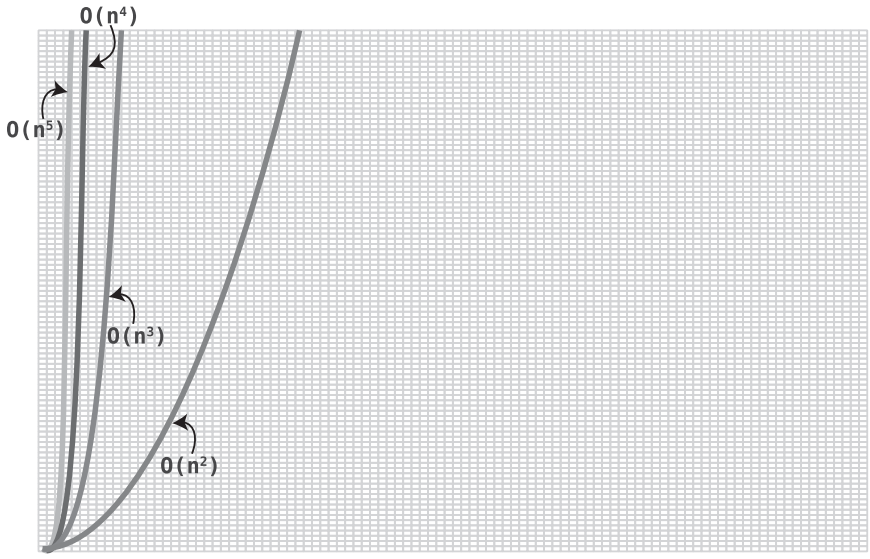
```
function wordBuilder(array) {
  let collection = [];

  for(let i = 0; i < array.length; i++) {
    for(let j = 0; j < array.length; j++) {
      for(let k = 0; k < array.length; k++) {
        if (i !== j && j !== k && i !== k) {
          collection.push(array[i] + array[j] + array[k]);
        }
      }
    }
  }

  return collection;
}
```

In this algorithm, for N data elements, we have N steps of the `i` loop multiplied by the N steps of the `j` loop multiplied by the N steps of the `k` loop. This is N * N * N, which is N^3 steps, which is described as $O(N^3)$.

If we had four or five nested loops, we'd have algorithms that are $O(N^4)$ and $O(N^5)$, respectively. Let's see how these all appear on a graph on page 99.



Optimizing code from a speed of $O(N^3)$ to $O(N^2)$ would be a big win, since the code becomes exponentially faster.

Array Sample

In the next example, we create a function that takes a small sample of an array. We expect to have very large arrays, so our sample is just the first, middlemost, and last value from the array.

Here is a Python implementation of this function. See if you can identify its efficiency in Big O:

```
def sample(array):
  first = array[0]
  middle = array[int(len(array) / 2)]
  last = array[-1]

  return [first, middle, last]
```

In this case again, the array passed into this function is the primary data, so we can say that N is the number of elements in this array.

However, our function ends up taking the same number of steps no matter what N is. Reading from the beginning, midpoint, and last indexes of an array each takes one step no matter the size of the array. Similarly, finding the array's length and dividing it by 2 also takes one step.

Since the number of steps is constant—that is, it remains the same no matter what N is—this algorithm is considered O(1).

Average Celsius Reading

Here's another example that involves mean averages. Let's say we're building weather-forecasting software. To determine the temperature of a city, we take temperature readings from across many thermometers across the city, and we calculate the mean average of those temperatures.

We'd also like to display the temperatures in both Fahrenheit and Celsius, but our readings are initially only provided to us in Fahrenheit.

To get the average Celsius temperature, our algorithm does two things: first, it converts all the readings from Fahrenheit to Celsius. Then, it calculates the mean average of all the Celsius numbers.

Following is some Ruby code that accomplishes this. What is its Big O?

```
def average_celsius(fahrenheit_readings)

  # Collect Celsius numbers here:
  celsius_numbers = []

  # Convert each reading to Celsius and add to array:
  fahrenheit_readings.each do |fahrenheit_reading|
    celsius_conversion = (fahrenheit_reading - 32) / 1.8
    celsius_numbers.push(celsius_conversion)
  end

  # Get sum of all Celsius numbers:
  sum = 0.0

  celsius_numbers.each do |celsius_number|
    sum += celsius_number
  end

  # Return mean average:
  return sum / celsius_numbers.length
end
```

First, we can say that N is the number of `fahrenheit_readings` passed into our method.

Inside the method, we run two loops. The first converts the readings to Celsius, and the second sums all the Celsius numbers. Since we have two loops that each iterate over all N elements, we have N + N, which is 2N (plus a few constant steps). Because Big O Notation drops the constants, this gets reduced to O(N).

Don't get thrown off by the fact that in the earlier Word Builder example, two loops led to an efficiency of $O(N^2)$. There, the loops were *nested*, which led to N steps *multiplied* by N steps. In our case, however, we simply have two loops, one after the other. This is N steps *plus* N steps (2N), which is a mere O(N).

Clothing Labels

Suppose we're writing software for a clothing manufacturer. Our code accepts an array of newly produced clothing items (stored as strings), and creates text for every possible label we'll need.

Specifically, our labels should contain the item name plus its size, ranging from 1 to 5. For example, if we have the array, ["Purple Shirt", "Green Shirt"], we want to produce label text for those shirts like this:

```
[
"Purple Shirt Size: 1",
"Purple Shirt Size: 2",
"Purple Shirt Size: 3",
"Purple Shirt Size: 4",
"Purple Shirt Size: 5",
"Green Shirt Size: 1",
"Green Shirt Size: 2",
"Green Shirt Size: 3",
"Green Shirt Size: 4",
"Green Shirt Size: 5"
]
```

Here is Python code that will create this text for an entire array of clothing items:

```
def mark_inventory(clothing_items):
  clothing_options = []

  for item in clothing_items:
    # For sizes 1 through 5 (Python ranges go UP TO second
    # number, but do not include it):
    for size in range(1, 6):
      clothing_options.append(item + " Size: " + str(size))

  return clothing_options
```

Let's determine this algorithm's efficiency. The clothing_items are the primary data being processed, so N is the number of clothing_items.

This code contains nested loops, so it's tempting to declare this algorithm to be $O(N^2)$. However, we need to analyze this case a little more carefully. While code containing nested loops often is $O(N^2)$, in this case, it's not.

Nested loops that result in $O(N^2)$ occur when each loop revolves around N. In our case, however, while our outer loop runs N times, our inner loop runs a constant five times. That is, this inner loop will always run five times no matter what N is.

So, while our outer loop runs N times, the inner loop runs five times for each of the N strings. While this means our algorithm runs 5N times, this is reduced to O(N), since Big O notation ignores constants.

Count the Ones

Here's another algorithm where the Big O is different from what it seems at first glance. This function accepts an *array of arrays*, where the inner arrays contain 1's and 0's. The function then returns how many 1's there are.

So, for this example input:

```
[
  [0, 1, 1, 1, 0],
  [0, 1, 0, 1, 0, 1],
  [1, 0]
]
```

our function will return 7, since there are seven 1's. Here's the function in Python:

```
def count_ones(outer_array):
  count = 0

  for inner_array in outer_array:
    for number in inner_array:
      if number == 1:
        count += 1

  return count
```

What's the Big O of this algorithm?

Again, it's easy to notice the nested loops and jump to the conclusion that it's $O(N^2)$. However, the two loops are iterating over two completely different things.

The outer loop is iterating over the inner arrays, and the inner loop is iterating over the actual numbers. At the end of the day, our inner loop only runs for as many numbers as there are *in total.*

Because of this, we can say that N represents how many numbers there are. And since our algorithm simply processes each number, the function's time complexity is O(N).

Palindrome Checker

A *palindrome* is a word or phrase that reads the same both forward and backward. Some examples include "racecar," "kayak," and "deified."

Here's a JavaScript function that determines whether a string is a palindrome:

```
function isPalindrome(string) {

  // Start the leftIndex at index 0:
  let leftIndex = 0;
  // Start rightIndex at last index of array:
  let rightIndex = string.length - 1;

  // Iterate until leftIndex reaches the middle of the array:
  while (leftIndex < string.length / 2) {

    // If the character on the left doesn't equal the character
    // on the right, the string is not a palindrome:
    if (string[leftIndex] !== string[rightIndex]) {
      return false;
    }

    // Move leftIndex one to the right:
    leftIndex++;
    // Move rightIndex one to the left:
    rightIndex--;
  }

  // If we got through the entire loop without finding any
  // mismatches, the string must be a palindrome:
  return true;
}
```

Let's determine the Big O of this algorithm.

In this case, N is the size of the `string` passed to this function.

The guts of the algorithm takes place within the `while` loop. Now, this loop is somewhat interesting because it only runs until it reaches the midpoint of the string. That would mean that the loop runs N / 2 steps.

However, Big O ignores constants. Because of this, we drop the division by 2, and our algorithm is O(N).

Get All the Products

Our next example is an algorithm that accepts an array of numbers and returns the product of every combination of two numbers.

For example, if we passed in the array, `[1, 2, 3, 4, 5]`, the function returns:

```
[2, 3, 4, 5, 6, 8, 10, 12, 15, 20]
```

This is because we first multiply the 1 by the 2, 3, 4, and 5. Then we multiply the 2 by the 3, 4, and 5. Next, we multiply the 3 by the 4 and the 5. And finally, we multiply the 4 by the 5.

Note something interesting: when we multiply, say, the 2 by the other numbers, we only have to multiply it by the numbers that are to the right of it. We don't have to go back and multiply 2 by the 1, because that was already covered back when we multiplied by the 1 by the 2. So, each number only needs to be multiplied by the remaining numbers to the right of it.

Here's a JavaScript implementation of this algorithm:

```
function twoNumberProducts(array) {
  let products = [];

  // Outer array:
  for(let i = 0; i < array.length - 1; i++) {

    // Inner array, in which j always begins one index
    // to the right of i:
    for(let j = i + 1; j < array.length; j++) {
      products.push(array[i] * array[j]);
    }
  }

  return products;
}
```

Let's break this down. N is the number of items in the array passed to this function.

We run the outer loop N times. (We actually run it N - 1 times, but we'll drop that constant.) The inner loop, though, is different. Since j always begins one index to the right of i, the inner loop's number of steps decrease each time that it's launched by the outer loop.

Let's see how many times the inner loop runs for our example array, which contains five elements:

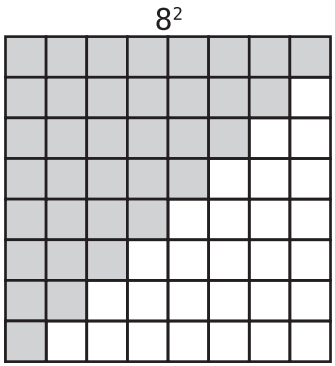
When i is 0, the inner loop runs while j is 1, 2, 3, and 4. When i is 1, the inner loop runs while j is 2, 3, and 4. When i is 2, the inner loop runs while j is 3, and 4. When i is 3, the inner loop runs while j is 4. When all is said and done, the inner loop runs:

4 + 3 + 2 + 1 times.

To put this in terms of N, we can say that the inner loop runs approximately:

N + (N - 1) + (N - 2) + (N - 3) ... + 1 times.

This formula always turns out to compute to about N^2 / 2. We can visualize this in the diagram on page 105. For the purposes of the diagram, we'll say that N is 8, so there are 8^2, or 64, squares.



If you work your way from the top row to the bottom, you'll see that the top row has all N squares shaded gray. The next row has N - 1 squares shaded gray, and the one after that has N - 2 gray squares. This pattern continues until the bottom row, which has just one shaded square.

You can also see at a glance that half of the squares are shaded. This demonstrates that the pattern of N + (N - 1) + (N - 2) + (N - 3)... + 1 is equivalent to $N^2 / 2$.

We've figured out, then, that the inner loop runs for $N^2 / 2$ steps. But because Big O ignores constants, we express this as $O(N^2)$.

Dealing with Multiple Datasets

Now, what happens if instead of computing the product of every two numbers from a single array, we instead compute the product of every number from one array by every number of a *second* array?

For example, if we had the array, `[1, 2, 3]` and the array, `[10, 100, 1000]`, we'd compute the products as:

```
[10, 100, 1000, 20, 200, 2000, 30, 300, 3000]
```

Our code would be similar to the previous snippet, with some slight modifications:

```
function twoNumberProducts(array1, array2) {
  let products = [];

  for(let i = 0; i < array1.length; i++) {
    for(let j = 0; j < array2.length; j++) {
      products.push(array1[i] * array2[j]);
    }
  }

  return products;
}
```

Let's analyze the time complexity of this function.

First, what is N? This is the first hurdle, as we now have *two* datasets, namely, the two arrays.

It's tempting to lump everything together and say that N is the total number of items of both arrays combined. However, this is problematic for the following reason:

Here's a tale of two scenarios. In Scenario 1, there are two arrays of size 5. In Scenario 2, there's one array of size 9, and another of size 1.

In both scenarios, we'd end up saying that N is 10, since 5 + 5 = 10 and 9 + 1 = 10. However, the efficiency of both scenarios is *very* different.

In Scenario 1, our code takes 25 (5 * 5) steps. Because N is 10, this is equivalent to $(N / 2)^2$ steps.

In Scenario 2, though, our code takes 9 (9 * 1) steps, which is close to about N steps. This is dramatically faster than Scenario 1!

So, we don't want to consider N to be the total number of integers from both arrays, since we'd never be able to pin down the efficiency in terms of Big O Notation, as it varies based on the different scenarios.

We're in a bit of a bind here. We have no choice but to express the time complexity as O(N * M), where N is the size of one array, and M is the size of the other.

This is a new concept: whenever we have two distinct datasets that have to interact with each other through multiplication, we have to identify both sources separately when we describe the efficiency in terms of Big O.

While this is the correct way of expressing this algorithm in terms of Big O Notation, it's a little less useful than other expressions of Big O. Comparing an O(N * M) algorithm to algorithms that only have an N (and not an M) is a little like comparing apples to oranges.

However, we do know that there's a specific range in which O(N * M) lies. That is, if N and M are the same, it's equivalent to $O(N^2)$. And if they're not the same, and we arbitrarily assign the smaller number to be M, even if M is as low as 1, we end up with O(N). In a sense then, O(N * M) can be construed as a range between O(N) and $O(N^2)$.

Is this great? No, but it's the best we can do.

Password Cracker

You're a hacker (an ethical one, of course) who's trying to figure out someone's password. You decide on a brute-force approach, and write some code that produces every possible string of a given length. Here's the code you whipped up:

```
def every_password(n)
  (("a" * n)..("z" * n)).each do |str|
    puts str
  end
end
```

The way this works is you pass in a number to the function, which becomes the variable `n`.

If `n` is 3, for example, the code `"a" * n` produces the string `"aaa"`. The code then sets a loop between all possible strings within the range of `"aaa"` and `"zzz"`. Running this code will print:

```
aaa
aab
aac
aad
aae

...

zzx
zzy
zzz
```

If `n` is 4, your code will print all possible strings of length 4:

```
aaaa
aaab
aaac
aaad
aaae

...

zzzx
zzzy
zzzz
```

If you try running this code even for a mere length of 5, you may be waiting quite some time for it to finish. This is a slow algorithm! But how do we express it in terms of Big O?

Let's break it down.

If we simply print each letter from the alphabet once, it would take 26 steps.

When we print every two-character combination, we end up with 26 characters multiplied by 26 characters.

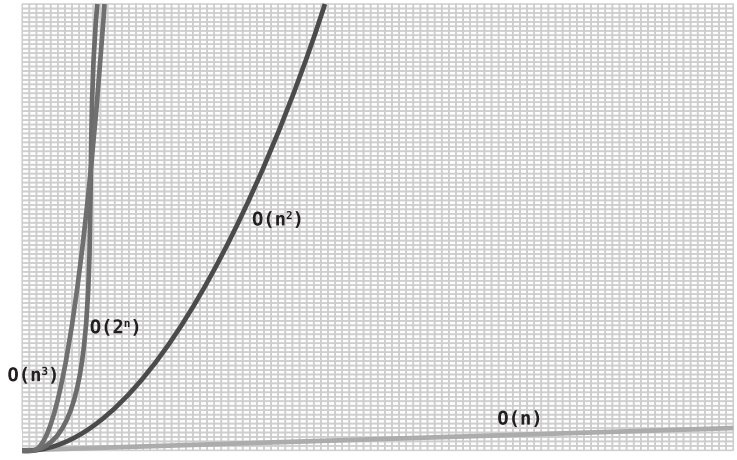
When printing every three-character combination, we end up with 26 * 26 * 26 combinations.

Do you see the pattern?

Length	Combinations
1	26
2	26^2
3	26^3
4	26^4

If we look at this in terms of N, it emerges that if N is the length of each string, *the number of combinations is* 26^N.

Therefore, in Big O Notation, we express this as $O(26^N)$. This is an utterly glacial algorithm! The truth is that even an algorithm that is a "mere" $O(2^N)$ is incredibly slow. Let's see how it looks on a graph compared to some of the other algorithms we've seen so far:



As you can see, $O(2^N)$ gets even slower than $O(N^3)$ at a point.

In a certain sense, $O(2^N)$ is the opposite of O(log N). With an algorithm of O(log N) (like binary search), each time the data is doubled, the algorithm takes one additional step. With an algorithm of $O(2^N)$, each time we add *one* element of data, the algorithm *doubles* in steps!

In our password cracker, each time we increase N by one, the number of steps get multiplied by *26*. This takes an incredible amount of time, which is why brute force is such an inefficient way to crack a password.

Wrapping Up

Congratulations! You're now a Big O pro. You can analyze all sorts of algorithms and categorize their time complexities. Armed with this knowledge, you'll be able to methodically optimize your code for speed.

Speaking of which, in the next chapter, we'll discover a new data structure that is the one of the most useful and common tools for speeding up algorithms. And I'm talking about some serious speed.

Exercises

The following exercises provide you with the opportunity to practice with algorithms in practical situations. The solutions to these exercises are found in the section, Chapter 7, on page 443.

1. Use Big O Notation to describe the time complexity of the following function. The function returns true if the array is a "100-Sum Array," and false if it is not.

   A "100-Sum Array" meets the following criteria:

   - Its first and last numbers add up to 100.
   - Its second and second-to-last numbers add up to 100.
   - Its third and third-to-last numbers add up to 100, and so on.

   Here is the function:

```
def one_hundred_sum?(array)
  left_index = 0
  right_index = array.length - 1

  while left_index < array.length / 2
    if array[left_index] + array[right_index] != 100
      return false
    end

    left_index += 1
    right_index -= 1
  end

  return true
end
```

2. Use Big O Notation to describe the time complexity of the following function. It merges two sorted arrays together to create a new sorted array containing all the values from both arrays:

```
def merge(array_1, array_2)
  new_array = []
  array_1_pointer = 0
  array_2_pointer = 0

  # Run the loop until we've reached end of both arrays:
  while array_1_pointer < array_1.length ||
      array_2_pointer < array_2.length

    # If we already reached the end of the first array,
    # add item from second array:
    if !array_1[array_1_pointer]
      new_array << array_2[array_2_pointer]
      array_2_pointer += 1
    # If we already reached the end of the second array,
    # add item from first array:
    elsif !array_2[array_2_pointer]
      new_array << array_1[array_1_pointer]
      array_1_pointer += 1
    # If the current number in first array is less than current
    # number in second array, add from first array:
    elsif array_1[array_1_pointer] < array_2[array_2_pointer]
      new_array << array_1[array_1_pointer]
      array_1_pointer += 1
    # If the current number in second array is less than or equal
    # to current number in first array, add from second array:
    else
      new_array << array_2[array_2_pointer]
      array_2_pointer += 1
    end
  end

  return new_array
end
```

3. Use Big O Notation to describe the time complexity of the following function. This function solves a famous problem known as "finding a needle in the haystack."

   Both the needle and haystack are strings. For example, if the needle is `"def"` and the haystack is `"abcdefghi"`, the needle is contained somewhere in the haystack, as `"def"` is a substring of `"abcdefghi"`. However, if the needle is `"dd"`, it cannot be found in the haystack of `"abcdefghi"`.

   This function returns true or false, depending on whether the needle can be found in the haystack:

```
def find_needle(needle, haystack)
  needle_index = 0
  haystack_index = 0

  while haystack_index < haystack.length
    if needle[needle_index] == haystack[haystack_index]
      found_needle = true

      while needle_index < needle.length
        if needle[needle_index] != haystack[haystack_index + needle_index]
          found_needle = false
          break
        end
        needle_index += 1
      end

      return true if found_needle
      needle_index = 0
    end

    haystack_index += 1
  end

  return false
end
```

4. Use Big O Notation to describe the time complexity of the following function. This function finds the greatest product of three numbers from a given array:

```
def largest_product(array)
  largest_product_so_far = array[0] * array[1] * array[2]
  i = 0

  while i < array.length
    j = i + 1
    while j < array.length
      k = j + 1
      while k < array.length
        if array[i] * array[j] * array[k] > largest_product_so_far
          largest_product_so_far = array[i] * array[j] * array[k]
        end
        k += 1
      end
      j += 1
    end
    i += 1
  end

  return largest_product_so_far

end
```

5. I once saw a joke aimed at HR people: "Want to immediately eliminate the unluckiest people from your hiring process? Just take half of the resumes on your desk and throw them in the trash."

   If we were to write software that kept reducing a pile of resumes until we had one left, it might take the approach of alternating between throwing out the top half and the bottom half. That is, it will first eliminate the top half of the pile, and then proceed to eliminate the bottom half of what remains. It keeps alternating between eliminating the top and bottom until one lucky resume remains, and that's who we'll hire!

   Describe the efficiency of this function in terms of Big O:

```
def pick_resume(resumes)
  eliminate = "top"

  while resumes.length > 1
    if eliminate == "top"
      resumes = resumes[resumes.length / 2, resumes.length - 1]
      eliminate = "bottom"
    elsif eliminate == "bottom"
      resumes = resumes[0, resumes.length / 2]
      eliminate = "top"
    end
  end

  return resumes[0]
end
```

CHAPTER 8

Blazing Fast Lookup with Hash Tables

Imagine you're writing a program that allows customers to order fast food from a restaurant, and you're implementing a menu of foods with their respective prices. You could, technically, use an array:

```
menu = [ ["french fries", 0.75], ["hamburger", 2.5],
["hot dog", 1.5], ["soda", 0.6] ]
```

This array contains several subarrays, and each subarray contains two elements. The first element is a string representing the food on the menu, and the second element represents the price of that food.

As you learned in *Why Algorithms Matter*, if this array were unordered, searching for the price of a given food would take O(N) steps since the computer would have to perform a linear search. If it's an *ordered* array, the computer could do a binary search, which would take O(log N).

While O(log N) isn't bad, we can do better. In fact, we can do *much* better. By the end of this chapter, you'll learn how to use a special data structure called a *hash table*, which can be used to look up data in just O(1) time. By knowing how hash tables work under the hood and the right places to use them, you can leverage their tremendous lookup speeds in many situations.

Hash Tables

Most programming languages include a data structure called a *hash table*, and it has an amazing superpower: fast reading. Note that hash tables are called by different names in various programming languages. Other names include hashes, maps, hash maps, dictionaries, and associative arrays.

Here's an example of the menu as implemented with a hash table using Ruby:

```
menu = { "french fries" => 0.75, "hamburger" => 2.5,
"hot dog" => 1.5, "soda" => 0.6 }
```

A hash table is a list of paired values. The first item in each pair is called the *key,* and the second item is called the *value.* In a hash table, the key and value have some significant association with one another. In this example, the string, "french fries" is the key, and 0.75 is the value. They are paired together to indicate that french fries cost 75 cents.

In Ruby, you can look up a key's value using this syntax:

```
menu["french fries"]
```

This would return the value 0.75.

Looking up a value in a hash table has an efficiency of O(1) on average, as it usually takes *just one step.* Let's see why.

Hashing with Hash Functions

Do you remember those secret codes you used as a kid to create and decipher messages?

For example, here's a simple way to map letters to numbers:

A = 1
B = 2
C = 3
D = 4
E = 5

and so on.

According to this code,

ACE converts to 135,
CAB converts to 312,
DAB converts to 412,

and

BAD converts to 214.

This process of taking characters and converting them to numbers is known as *hashing.* And the code that is used to convert those letters into particular numbers is called a *hash function.*

There are many other hash functions besides this one. Another example of a hash function may be to take each letter's corresponding number and return the *sum* of all the numbers. If we did that, BAD would become the number 7 following a two-step process:

Step 1: First, BAD converts to 214.

Step 2: We then take each of these digits and get their sum:

2 + 1 + 4 = 7

Another example of a hash function may be to return the *product* of all the letters' corresponding numbers. This would convert the word BAD into the number 8:

Step 1: First, BAD converts to 214.

Step 2: We then take the product of these digits:

2 * 1 * 4 = 8

In our examples for the remainder of this chapter, we're going to stick with this last version of the hash function. Real-world hash functions are more complex than this, but this "multiplication" hash function will keep our examples clear and simple.

The truth is that a hash function needs to meet only one criterion to be valid: a hash function must convert the same string to the *same number* every single time it's applied. If the hash function can return inconsistent results for a given string, it's not valid.

Examples of invalid hash functions include functions that use random numbers or the current time as part of their calculation. With these functions, BAD might convert to 12 one time, and 106 another time.

With our "multiplication" hash function, however, BAD will *always* convert to 8. That's because B is always 2, A is always 1, and D is always 4. And 2 * 1 * 4 is *always* 8. There's no way around this.

Note that with this hash function, DAB will *also* convert into 8 just as BAD will. This will actually cause some issues that I'll address later.

Armed with the concept of hash functions, we can now understand how a hash table actually works.

Building a Thesaurus for Fun and Profit, but Mainly Profit

On nights and weekends, you're single-handedly working on a stealth startup that will take over the world. It's…a thesaurus app. But this isn't any *old* thesaurus app—this is Quickasaurus. And you know that it will totally disrupt the billion-dollar thesaurus market. When a user looks up a word in Quick-asaurus, it returns just *one* synonym, instead of *every* possible synonym as old-fashioned thesaurus apps do.

Since every word has an associated synonym, this is a great use case for a hash table. After all, a hash table is a list of paired items. Let's get started.

We can represent our thesaurus using a hash table:

```
thesaurus = {}
```

Under the hood, a hash table stores its data in a bunch of cells in a row, similar to an array. Each cell has a corresponding number. For example:



(We left off index 0 since nothing would be stored there given our "multiplication" hash function.)

Let's add our first entry into the hash table:

```
thesaurus["bad"] = "evil"
```

In code, our hash table now looks like this:

```
{"bad" => "evil"}
```

Let's explore how the hash table stores this data.

First, the computer applies the hash function to the key. Again, we'll be using the "multiplication" hash function described previously. So, this would compute as:

BAD = 2 * 1 * 4 = 8

Since our key ("bad") hashes into 8, the computer places the value ("evil") into cell 8:



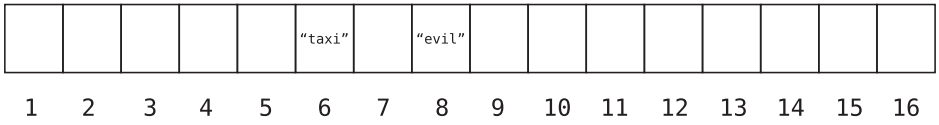
Now, let's add another key-value pair:

```
thesaurus["cab"] = "taxi"
```

Again, the computer hashes the key:

CAB = 3 * 1 * 2 = 6

Since the resulting value is 6, the computer stores the value ("taxi") inside cell 6.

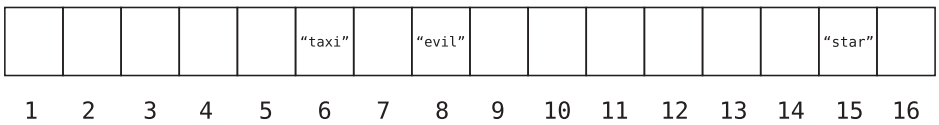


Let's add one more key-value pair:

```
thesaurus["ace"] = "star"
```

To sum up what's happening here: for every key-value pair, each *value* is stored at the *index* of the *key*, after the key has been hashed.

ACE hashes into 15, since ACE = 1 * 3 * 5 = 15, so "star" gets placed into cell 15:



In code, our hash table currently looks like this:

```
{"bad" => "evil", "cab" => "taxi", "ace" => "star"}
```

Hash Table Lookups

When we look up items from a hash table, we use a key to find its associated value. Let's see how this works with our Quickasaurus example hash table.

Suppose we want to look up the value associated with the key "bad". In our code, we'd say:

```
thesaurus["bad"]
```

To find the value associated with "bad", the computer executes two simple steps:

1. The computer hashes the key we're looking up: BAD = 2 * 1 * 4 = 8.
2. Since the result is 8, the computer looks inside cell 8 and returns the value stored there. In this case, that is the string, "evil".

Let's take a step back and look at the big picture here. In a hash table, the placement of each value is determined by its key. That is, by hashing the key itself, we compute the index number where the key's associated value should be placed.

Because the key determines the placement of the value, we use this principle to make lookups a cinch. When we have any key and want to find its value, the key itself tells us where the value will be found. Just as we hashed the key to insert the value in the appropriate cell, we can hash the key again to find where we previously put that value.

It now becomes clear why looking up a value in a hash table is typically O(1): it's a process that takes a constant amount of time. The computer hashes the key, turns it into a number, and jumps to the index with that number to retrieve the value stored there.

We can now understand why a hash table would yield faster lookups for our restaurant menu than an array. With an array, when we look up the price of a menu item, we would have to search through each cell until we find it. For an unordered array, this would take up to O(N), and for an ordered array, this would take up to O(log N). Using a hash table, however, we can now use the actual menu items as keys, allowing us to do a hash table lookup of O(1). And *that's* the beauty of a hash table.

One-Directional Lookups

It's important to point out that the ability to find any value within the hash table in a single step only works if we know the value's key. If we tried to find a particular value without knowing its key, we'd still have to resort to searching each and every key-value pair within the hash table, which is O(N).

Similarly, we can only do O(1) lookups when using a *key* to find the *value.* If, on the other hand, we want to use a *value* to find its associated *key,* we cannot take advantage of the hash table's fast lookup ability.

This is because the whole premise of the hash table is that the key determines the value's location. But this premise only works in one direction: we use the key to find the value. The value does not determine the key's location, so we have no way to easily find any key without combing through all of them.

Come to think of it, where are the keys stored? In the previous diagrams, we only saw how the values are stored in the hash table.

While this detail may vary from language to language, some languages store the keys next to the values themselves. This is useful in case of collisions, which I'll discuss in the next section.

In any case, there's another aspect of the hash table's one-directional nature that's worth noting. Each key can exist only once in the hash table, but there can be multiple instances of a value.

If we think about the menu example from the beginning of this chapter, we can't have the hamburger listed twice (nor would we want to, as it only has one price). However, we *could* have multiple foods that cost $2.50.

In many languages, if we try to store a key-value pair where the key already exists, it simply overwrites the old value while keeping the same key.

Dealing with Collisions

Hash tables are awesome, but are not without complications.

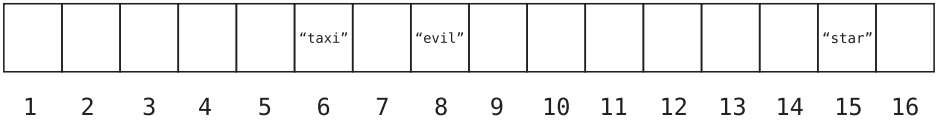
Continuing our thesaurus example: what happens if we want to add the following entry into our thesaurus?

```
thesaurus["dab"] = "pat"
```

First, the computer would hash the key:

DAB = 4 * 1 * 2 = 8

Then, it would try to add "pat" to our hash table's cell 8:

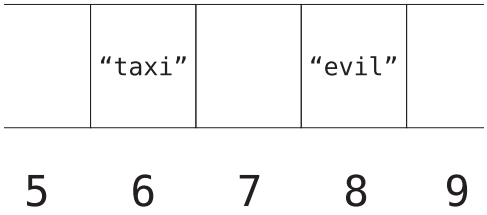


Uh-oh. Cell 8 is already filled with "evil"—literally!

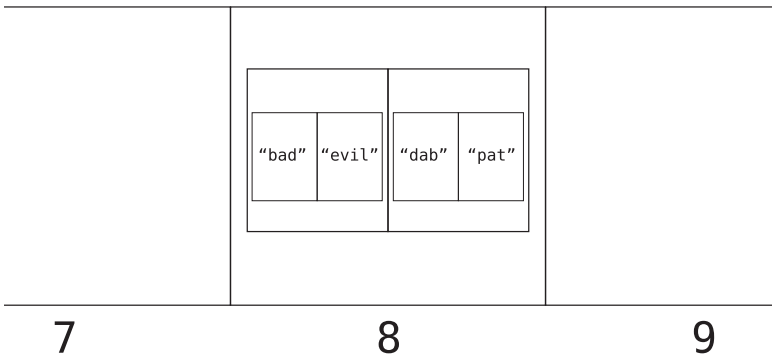
Trying to add data to a cell that is already filled is known as a *collision*. Fortunately, there are ways around it.

One classic approach for handling collisions is known as *separate chaining*. When a collision occurs, instead of placing a *single* value in the cell, it places in it a reference to an array.

Let's look more carefully at a subsection of our hash table's underlying data storage:



In our example, the computer wants to add "pat" to cell 8, but it already contains "evil". So, it replaces the contents of cell 8 with an array as shown on page 120.



This array contains subarrays where the first value is the word, and the second value is its synonym.

Let's walk through how a hash table lookup works in this case. If we look up:

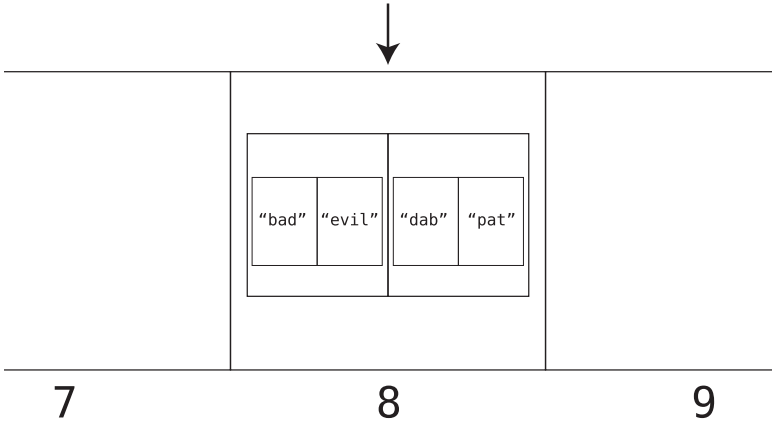
```
thesaurus["dab"]
```

the computer takes the following steps:

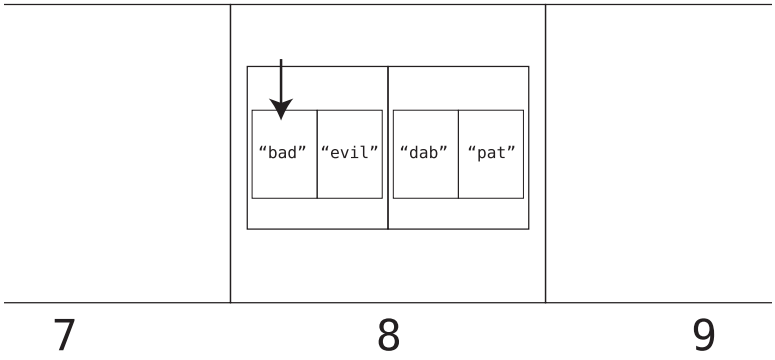
1. It hashes the key: DAB = 4 * 1 * 2 = 8.
2. It looks up cell 8. The computer takes note that cell 8 contains an array of arrays rather than a single value.
3. It searches through the array linearly, looking at index 0 of each subarray until it finds our key (`"dab"`). It then returns the value at index 1 of the correct subarray.

Let's walk through these steps visually.

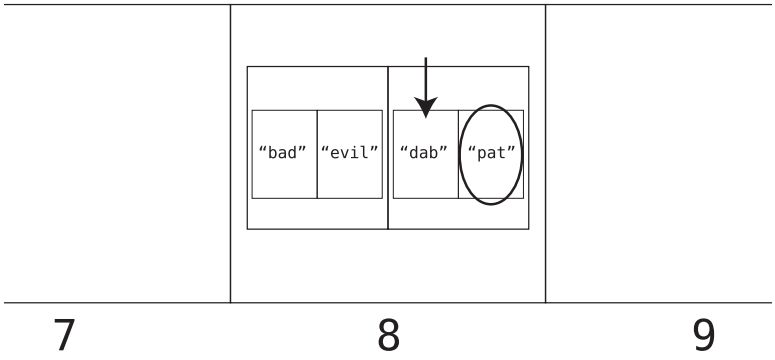
We hash DAB into 8, so the computer inspects that cell:



Since cell 8 contains an array of subarrays, we begin a linear search through each subarray, starting at the first one. We inspect index 0 of the first subarray:



It does not contain the key we are looking for (`"dab"`), so we move on to index 0 of the next subarray:



We found `"dab"`, which would indicate that the value at index 1 of that subarray (`"pat"`) is the value we're looking for.

In a scenario where the computer hits upon a cell that references an array, its search can take some extra steps, as it needs to conduct a linear search within an array of multiple values. If somehow all of our data ended up within a single cell of our hash table, our hash table would be no better than an array. So, it actually turns out that the worst-case performance for a hash table lookup is O(N).

Because of this, it is critical that a hash table is designed in such a way that it will have few collisions, and therefore, typically perform lookups in O(1) time rather than O(N) time.

Luckily, most programming languages implement hash tables and handle these details for us. However, by understanding how it all works under the hood, we can appreciate how hash tables eke out O(1) performance.

Let's see how hash tables can be set up to avoid frequent collisions.

Making an Efficient Hash Table

Ultimately, a hash table's efficiency depends on three factors:

- How much data we're storing in the hash table
- How many cells are available in the hash table
- Which hash function we're using

It makes sense why the first two factors are important. If you have a lot of data and only a few cells, there will be many collisions and the hash table will lose its efficiency. Let's explore, however, why the hash function itself is important for efficiency.

Let's say we're using a hash function that always produces a value that falls in the range from 1 to 9. An example of this is a hash function that converts letters into their corresponding numbers and keeps adding the resulting digits together until it ends up with a single digit.

For example:

PUT = 16 + 21 + 20 = 57

Because 57 contains more than one digit, the hash function breaks up the 57 into 5 + 7:

5 + 7 = 12

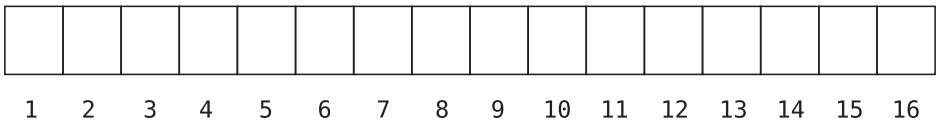
12 also contains more than one digit, so it breaks up the 12 into 1 + 2:

1 + 2 = 3

In the end, PUT hashes into 3.

This hash function by its very nature will *always* return a number 1 through 9.

Let's return to our example hash table:



With this hash function, the computer would never even use cells 10 through 16 even though they exist. All data would be stuffed into cells 1 through 9.

A good hash function, therefore, is one that distributes its data across *all* available cells. The more we can spread out our data, the fewer collisions we will have.

The Great Balancing Act

You learned that a hash table's efficiency goes up as its number of collisions goes down. In theory, then, the best way to avoid collisions would be to have a hash table with a large number of cells. Imagine we want to store just five items in our hash table. A hash table with 1,000 cells would seem to be wonderful for our case, since odds are there would be no collisions.

However, while avoiding collisions is important, we have to balance that with avoiding memory hogging as well.

Although a hash table with 1,000 cells for our five pieces of data is great for avoiding collisions, we'd be using up 1,000 cells just to store just five pieces of data, and that's a poor use of memory.

This is the balancing act that a hash table must perform. A good hash table *strikes a balance of avoiding collisions while not consuming lots of memory.*

To accomplish this, computer scientists have developed the following rule of thumb: for every 7 data elements stored in a hash table, it should have 10 cells.

So, if you're planning on storing 14 elements, you'd want to have 20 available cells, and so on.

This ratio of data to cells is called the *load factor.* Using this terminology, we'd say that the ideal load factor is 0.7 (7 elements / 10 cells).

If you initially stored 7 pieces of data in a hash table, the computer might allocate a hash table with 10 cells. When you begin to add more data, though, the computer will expand the hash table by adding more cells and changing the hash function so that the new data will be distributed evenly across the new cells.

Again, most of the internals of a hash table are managed by the computer language you're using. It decides how big the hash table needs to be, what hash function to use, and when it's time to expand the hash table. You have the right to assume that your programming language has implemented its hash table to allow for peak performance.

Now that we've seen how hashes work, it's clear that they have a superior lookup efficiency of O(1). We're going to use this knowledge shortly to optimize our code for speed.

But first, let's first take a quick tour of the many different use cases for hash tables when it comes to simple data organization.

Hash Tables for Organization

Because hash tables keep data in pairs, they are useful in many scenarios for organizing data.

Some data exists naturally in paired form. The fast-food menu and thesaurus scenarios from this chapter are classic examples of this. The menu contains each food item paired with its price. The thesaurus contains each word paired with its synonym. In fact, in Python, hash tables are known as *dictionaries* since a dictionary is a common form of paired data: it's a list of words with their respective definitions.

Other examples of naturally paired data can include tallies, such as political candidates and the number of votes each received:

```
{"Candidate A" => 1402021, "Candidate B" => 2321443, "Candidate C" => 432}
```

An inventory tracking system, which keeps track of how much of each item is in supply, is another tally example:

```
{"Yellow Shirt" => 1203, "Blue Jeans" => 598, "Green Felt Hat" => 65}
```

Hash tables are such a natural fit for paired data that we can even use them to simplify conditional logic in certain instances.

Say we encounter a function that returns the meaning of common HTTP status code numbers:

```
def status_code_meaning(number)
  if number == 200
    return "OK"
  elsif number == 301
    return "Moved Permanently"
  elsif number == 401
    return "Unauthorized"
  elsif number == 404
    return "Not Found"
  elsif number == 500
    return "Internal Server Error"
  end
end
```

If we think about this code, we'll realize that the conditional logic revolves around paired data, namely, the status code numbers and their respective meanings.

By using a hash table, we can completely eliminate the conditional logic:

```
STATUS_CODES = {200 => "OK", 301 => "Moved Permanently",
                401 => "Unauthorized", 404 => "Not Found",
                500 => "Internal Server Error"}

def status_code_meaning(number)
  return STATUS_CODES[number]
end
```

Another common use for hash tables is to represent objects that have various attributes. For example, here's a representation of a dog:

```
{"Name" => "Fido", "Breed" => "Pug", "Age" => 3, "Gender" => "Male"}
```

As you can see, attributes are a kind of paired data, since the attribute name becomes the key, and the actual attribute becomes the value.

We can create an entire list of dogs if we place multiple hash tables inside an array:

```
[
  {"Name" => "Fido", "Breed" => "Pug", "Age" => 3, "Gender" => "Male"},
  {"Name" => "Lady", "Breed" => "Poodle", "Age" => 6, "Gender" => "Female"},
  {"Name" => "Spot", "Breed" => "Dalmatian", "Age" => 2, "Gender" => "Male"}
]
```

Hash Tables for Speed

While hash tables are a perfect fit for paired data, they can also be used to make your code faster—even if your data doesn't exist as pairs. And this is where things really get exciting.

Here's a simple array:

```
array = [61, 30, 91, 11, 54, 38, 72]
```

If you want to search for a number in this array, how many steps would it take?

Because the array is unordered, you would have to perform a linear search, which would take N steps. You learned this all the way back at the beginning of the book.

However, what would happen if we ran some code that would convert these numbers into a hash table that looked like this?

```
hash_table = {61 => true, 30 => true, 91 => true,
11 => true, 54 => true, 38 => true, 72 => true}
```

Here, we've stored each number as a key and assigned the boolean true as the associated value for each number.

Now, if I asked you to search this hash table for a certain number as a key, how many steps would it take?

Well, with the simple code:

```
hash_table[72]
```

I could look up the number 72 in a single step.

That is, by doing a hash table lookup using 72 as the key, I can determine in one step whether the 72 is present in the hash table. The reasoning is straightforward: if 72 is a key in the hash table, I'd get back true, since the 7 has true as its value. On the other hand, if the 72 is *not* a key in the hash table, I'd get back nil. (Various languages return different values when a key is not present in a hash table. Ruby returns nil.)

Since doing a hash table lookup takes just one step, I can therefore find any number in the hash table (as a key) in one step.

Can you see the magic?

By converting an array into a hash table in this way, we can go from O(N) searches to O(1) searches.

Here's what's interesting about using a hash table in this way. Even though hash tables are often used for naturally paired data, our data here is *not* paired. We just care about a list of single numbers.

While we did assign a value to each key, it doesn't really matter what the value is. We used true as the value for each key, but any arbitrary value (that is "truthy") would achieve the same results.

The trick here is that by placing each number in the hash table as a key, we can later look up each of those keys in one step. If our lookup returns any value, it means the key itself must be in the hash table. If we get back nil, then the key must not be in the hash table.

I refer to using a hash table in this way as "using it as an index." (It's my own term.) An index at the back of a book tells you whether the topic can be found in the book, instead of you having to flip through all the pages to find it. Here as well, we created the hash table to serve as a kind of index; in our case, it's an index that tells us whether a specific item is contained within the original array.

Let's use this technique to boost the speed of a very practical algorithm.

Array Subset

Let's say we need to determine whether one array is a subset of another array. Take these two arrays, for example:

```
["a", "b", "c", "d", "e", "f"]
["b", "d", "f"]
```

The second array, ["b", "d", "f"] is a subset of the first array, ["a", "b", "c", "d", "e", "f"] because every value of the second array is contained within the first array.

However, if our arrays were:

```
["a", "b", "c", "d", "e", "f"]
["b", "d", "f", "h"]
```

the second array is *not* a subset of the first array, because the second array contains the value "h", which does not exist within the first array.

How would we write a function that compares two arrays and lets us know if one is a subset of the other?

One way we can do this is by using nested loops. Essentially, we'd iterate through every element of the smaller array, and for each element in the smaller array, we'd then begin a second loop that iterates through each element of the larger array. If we ever find an element in the smaller array that isn't contained within the larger array, our function will return false. If the code gets past the loops, it means it never encountered a value in the smaller array that wasn't contained within the larger array, so it returns true.

Here's a JavaScript implementation of this approach:

```
function isSubset(array1, array2) {

  let largerArray;
  let smallerArray;

  // Determine which array is smaller:
  if(array1.length > array2.length) {
    largerArray = array1;
    smallerArray = array2;
  } else {
    largerArray = array2;
    smallerArray = array1;
  }

  // Iterate through smaller array:
  for(let i = 0; i < smallerArray.length; i++) {

    // Assume temporarily that the current value from
    // smaller array is not found in larger array:
    let foundMatch = false;
```

```
    // For each value in smaller array, iterate through
    // larger array:
    for(let j = 0; j < largerArray.length; j++) {

      // If the two values are equal, it means the current
      // value in smaller array is present in the larger array:
      if(smallerArray[i] === largerArray[j]) {
        foundMatch = true;
        break;
      }
    }

    // If the current value in smaller array doesn't exist
    // in larger array, return false:
    if(foundMatch === false) { return false; }
  }

  // If we get to the end of the loops, it means that all
  // values from smaller array are present in larger array:
  return true;
}
```

When we analyze the efficiency of this algorithm, we find that it's O(N * M), since it runs for the number of items in the first array multiplied by the number of items in the second array.

Now, let's harness the power of a hash table to dramatically improve the efficiency of our algorithm. Let's ditch our original approach and start again from scratch.

In our new approach, after we've determined which array is larger and which is smaller, we're going to run a single loop through the larger array, and store each value inside of a hash table:

```
let hashTable = {};

for(const value of largerArray) {
  hashTable[value] = true;
}
```

In this code snippet, we create an empty hash table inside the `hashTable` variable. Then, we iterate through each value in the `largerArray`, and add the item from the array to the hash table. We add the item itself as a key, and `true` as the value.

For the earlier example, `["a", "b", "c", "d", "e", "f"]`, once we've run it through this loop, we end up with a hash table that looks like this:

```
{"a": true, "b": true, "c": true, "d": true, "e": true, "f": true}
```

This becomes our "index" that will allow us to conduct O(1) lookups of these items later on.

Now, here's the brilliant part. Once the first loop is complete and we now have this hash table to work with, we can then begin a second (non-nested) loop that iterates through the *smaller* array:

```
for(const value of smallerArray) {
  if(!hashTable[value]) { return false; }
}
```

This loop looks at each item in the smallerArray and checks to see whether it exists as a key inside the hashTable. Remember, the hashTable stores all the items from largerArray as its keys. So, if we find an item in hashTable, it means the item is also in largerArray. And if we don't find an item in hashTable, it means it's also not inside the largerArray.

So, for each item in smallerArray, we check whether it's a key in hashTable. If it's not, that means the item isn't contained within the largerArray, and the smallerArray is therefore not a subset of the larger array and we return false. (However, if we get past this loop, it means the smaller array *is* a subset of the larger one.)

Let's put this altogether in one complete function:

```
function isSubset(array1, array2) {
  let largerArray;
  let smallerArray;
  let hashTable = {};

  // Determine which array is smaller:
  if(array1.length > array2.length) {
    largerArray = array1;
    smallerArray = array2;
  } else {
    largerArray = array2;
    smallerArray = array1;
  }

  // Store all items from largerArray inside hashTable:
  for(const value of largerArray) {
    hashTable[value] = true;
  }

  // Iterate through each item in smallerArray and return false
  // if we encounter an item not inside hashTable:
  for(const value of smallerArray) {
    if(!hashTable[value]) { return false; }
  }
```

```
  // If we got this far in our code without returning false,
  // it means that all the items in the smallerArray
  // must be contained within largerArray:
  return true;
}
```

Now, how many steps did this algorithm take? We iterated through each item of the *larger* array once in order to build the hash table.

And we iterated through each item of the *smaller* array taking just one step per item to look up the item in the hash table. Remember, a hash table lookup takes just one step.

If we say that N is the total number of items of both arrays combined, our algorithm is O(N), since we touched each item just once. That is, we spent one step on each item from the larger array followed by one step on each item from the smaller array.

That's a *huge* win over our first algorithm, which was O(N * M).

This technique of using a hash table as an "index" comes up frequently in algorithms that require multiple searches within an array. That is, if your algorithm will need to keep searching for values inside an array, each search would itself take up to N steps. By creating a hash table "index" of the array, we reduce each search to only one step.

As I pointed out, what makes this technique particularly interesting is that when using a hash table as an "index," we aren't even dealing with naturally paired data. Instead, we just want to know whether the key itself is in the hash table. When we use the key to perform a lookup in the hash table and receive any value (no matter how arbitrary it is), it means the key must be present in the hash table.

Wrapping Up

Hash tables are indispensable when it comes to building efficient software. With their O(1) reads and insertions, it's a difficult data structure to beat.

Until now, our analysis of various data structures revolved around their efficiency and speed. But did you know that some data structures provide advantages other than speed? In the next lesson, we're going to explore two data structures that can help improve code elegance and maintainability.

Exercises

The following exercises provide you with the opportunity to practice with hash tables. The solutions to these exercises are found in the section, Chapter 8, on page 444.

1. Write a function that returns the intersection of two arrays. The intersection is a third array that contains all values contained within the first two arrays. For example, the intersection of `[1, 2, 3, 4, 5]` and `[0, 2, 4, 6, 8]` is `[2, 4]`. Your function should have a complexity of O(N). (If your programming language has a built-in way of doing this, don't use it. The idea is to build the algorithm yourself.)

2. Write a function that accepts an array of strings and returns the first duplicate value it finds. For example, if the array is `["a", "b", "c", "d", "c", "e", "f"]`, the function should return `"c"`, since it's duplicated within the array. (You can assume that there's one pair of duplicates within the array.) Make sure the function has an efficiency of O(N).

3. Write a function that accepts a string that contains all the letters of the alphabet except one and returns the missing letter. For example, the string, `"the quick brown box jumps over a lazy dog"` contains all the letters of the alphabet except the letter, `"f"`. The function should have a time complexity of O(N).

4. Write a function that returns the first *non*-duplicated character in a string. For example, the string, `"minimum"` has two characters that only exist once—the `"n"` and the `"u"`, so your function should return the `"n"`, since it occurs first. The function should have an efficiency of O(N).

CHAPTER 9

Crafting Elegant Code with Stacks and Queues

Until now, our discussion around data structures has focused primarily on how they affect the *performance* of various operations. However, having a variety of data structures in your programming arsenal also allows you to create code that is simpler and easier to read.

In this chapter, you're going to discover two new data structures: stacks and queues. The truth is that these two structures are not entirely new. They're simply arrays with restrictions. Yet, these restrictions are exactly what make them so elegant.

More specifically, stacks and queues are elegant tools for handling temporary data. From operating system architecture to printing jobs to traversing data, stacks and queues serve as temporary containers that can be used to form beautiful algorithms.

Think of temporary data like the food orders in a diner. What each customer orders is important until the meal is made and delivered; then you throw the order slip away. You don't need to keep that information around. Temporary data is information that doesn't have any meaning after it's processed, so you can throw it away once you're done with it.

Stacks and queues handle this kind of temporary data, but have a special focus on the *order* in which the data is handled, as you'll now learn.

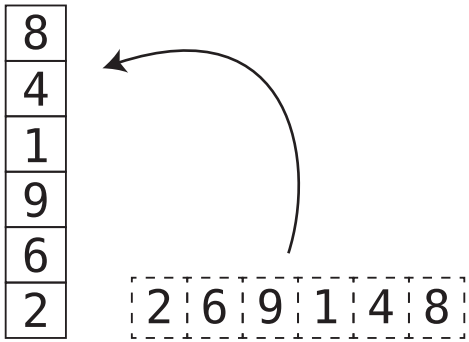
Stacks

A *stack* stores data in the same way arrays do—it's simply a list of elements. The one catch is that stacks have the following three constraints:

- Data can be inserted only at the end of a stack.
- Data can be deleted only from the end of a stack.
- Only the last element of a stack can be read.

You can think of a stack as an actual stack of dishes; you can't look at the face of any dish other than the one at the top. Similarly, you can't add any dish except to the top of the stack, nor can you remove any dish besides the one at the top. (At least, you shouldn't.) In fact, most computer science literature refers to the end of the stack as its *top,* and the beginning of the stack as its *bottom.*

Our diagrams will reflect this terminology by viewing stacks as vertical arrays, like so:



As you can see, the first item in the array becomes the bottom of the stack, while the last item becomes the stack's top.

While the restrictions of a stack seem—well—restrictive, we'll see shortly how they are to our benefit.

To see a stack in action, let's start with an empty stack.

Inserting a new value into a stack is also called *pushing onto the stack.* Think of it as adding a dish onto the top of the dish stack.

Let's push a 5 onto the stack:

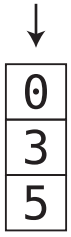


Again, there's nothing fancy going on here. All we're really doing is just inserting a data element into the end of an array.

Now, let's push a 3 onto the stack:



Next, let's push a 0 onto the stack:

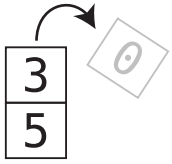


Note that we're always adding data to the top (that is, the end) of the stack. If we wanted to insert the 0 into the bottom or middle of the stack, we couldn't, because that is the nature of a stack: data can only be added to the top.

Removing elements from the top of the stack is called *popping from the stack.* Because of a stack's restrictions, we can only pop data from the top.

Let's pop some elements from our example stack.

First, we pop the 0:



Our stack now contains only two elements: the 5 and the 3.

Next, we pop the 3:



Our stack now only contains the 5:



A handy acronym used to describe stack operations is *LIFO*, which stands for "Last In, First Out." All this means is that the *last* item *pushed* onto a stack is always the *first* item *popped* from it. It's sort of like students who slack off—they're always the last to arrive to class, but the first to leave.

Abstract Data Types

Most programming languages don't actually come with the stack as a built-in data type or class. Instead, it's up to you to implement it yourself. This is a stark contrast with arrays, which are available in most languages.

To create a stack, then, you generally have to use one of the built-in data structures to actually hold the data. Here is one way to implement a stack using Ruby, which uses an array under the hood:

```
class Stack
  def initialize
    @data = []
  end

  def push(element)
    @data << element
  end

  def pop
    @data.pop
  end

  def read
    @data.last
  end
end
```

As you can see, our stack implementation stores the data in an array called `@data`.

Whenever a stack is initiated, we automatically build an empty array with `@data = []`. Our stack also contains methods that `push` new elements to the `@data` array, `pop` elements from `@data` array, and `read` elements from the `@data` array.

However, by building the `Stack` class around the array, we have built an interface that forces the user to interact with the array in limited ways. While one can normally read from any index of an array, when using the array through the stack interface, one can only read the last item. The same goes for inserting and deleting data.

The stack data structure, then, isn't the same kind of data structure that an array is. The array is built into most programming languages and interacts directly with the computer's memory. The stack, on the other hand, is really

a set of rules and processes around how we should interact with an array so that we can achieve a particular result.

In fact, a stack doesn't even care about *what* data structure is under the hood. All it cares about is that there's a list of data elements that act in a LIFO way. Whether we accomplish this with an array or some other type of built-in data structure doesn't actually matter. Because of this, the stack is an example of what is known as an *abstract data type*—it's a kind of data structure that is a set of theoretical rules that revolve around some other built-in data structure.

The set that we encountered in *Why Data Structures Matter* is another example of an abstract data type. Some implementations of sets use arrays under the hood, while other implementations actually use hash tables. The set itself, though, is simply a theoretical concept: it's a list of non-duplicated data elements.

Many of the data structures we'll encounter in the remainder of this book are abstract data types—they are pieces of code that are written on top of other built-in data structures.

It should be noted that even a built-in data structure can be an abstract data type. Even if a programming language does implement its own `Stack` class, it doesn't change the fact that the stack data structure is still a concept that allows for various data structures to be used under the hood.

Stacks in Action

Although a stack is not typically used to store data on a long-term basis, it can be a great tool to handle temporary data as part of various algorithms. Here's an example:

Let's create the beginnings of a JavaScript linter—that is, a program that inspects a programmer's JavaScript code and ensures that each line is syntactically correct. It can be very complicated to create a linter, as there are many different aspects of syntax to inspect.

We'll focus on just one specific aspect of the linter—opening and closing braces. This includes parentheses, square brackets, and curly braces—all common causes of frustrating syntax errors.

To solve this problem, let's first analyze what type of syntax is incorrect when it comes to braces. If we break it down, we'll find that there are three situations of erroneous syntax.

The first is when there is an opening brace that doesn't have a corresponding closing brace, such as this:

```
(var x = 2;
```

We'll call this Syntax Error Type #1.

The second is when there is a closing brace that was never preceded by a corresponding opening brace:

```
var x = 2;)
```

We'll call that Syntax Error Type #2.

The third, which we'll refer to as Syntax Error Type #3, is when a closing brace is not the same *type* of brace as the immediately preceding opening brace, such as:

```
(var x = [1, 2, 3)];
```

In the preceding example, there is a matching set of parentheses, and a matching pair of square brackets, but the closing parenthesis is in the wrong place, as it doesn't match the immediately preceding opening brace, which is a square bracket.

How can we implement an algorithm that inspects a line of JavaScript code and ensures that there are no brace-related syntax errors? This is where a stack allows us to implement a beautiful linting algorithm, which works as follows:

We prepare an empty stack, and then we read each character from left to right following these rules:

1. If we find any character that isn't a type of brace (parenthesis, square bracket, or curly brace), we ignore it and move on.
2. If we find an *opening* brace, we push it onto the stack. Having it on the stack means we're waiting to close that particular brace.
3. If we find a *closing* brace, we pop the top element in the stack and inspect it. We then analyze:
    - If the item we popped (which is always an opening brace) does not match the current closing brace, it means we've encountered Syntax Error Type #3.
    - If we couldn't pop an element because the stack was empty, that means the current closing brace doesn't have a corresponding opening brace beforehand. This is Syntax Error Type #2.

- If the item we popped *is* a corresponding match for the current closing brace, it means we've successfully closed that opening brace, and we can continue parsing the line of JavaScript code.

4. If we make it to the end of the line and there's still something left on the stack, that means there's an opening brace without a corresponding closing brace, which is Syntax Error Type #1.

Let's see this in action using the following example:

```
(var x = {y: [1, 2, 3]})
```

After we prepare an empty stack, we begin reading each character from left to right.

Step 1: We begin with the first character, which happens to be an opening parenthesis:

```
↓
(var x = {y: [1, 2, 3]})
```

Step 2: Since it's a type of opening brace, we push it onto the stack:

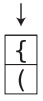


We then ignore all the characters, `var x = `, since they aren't brace characters.

Step 3: We encounter our next opening brace:

```
          ↓
(var x = {y: [1, 2, 3]})
```

Step 4: We push it onto the stack:

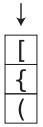


We then ignore the `y:`.

Step 5: We encounter the opening square bracket:

```
               ↓
(var x = {y: [1, 2, 3]})
```

Step 6: We add that to the stack as well:



We then ignore the 1, 2, 3.

Step 7: We encounter our first closing brace—a closing square bracket:

```
(var x = {y: [1, 2, 3]})
                    ↓
```

Step 8: We pop the element at the top of the stack, which happens to be an *opening* square bracket:



Since our closing square bracket is a corresponding match to this top element of the stack, it means we can continue with our algorithm without throwing any errors.

Step 9: We move on, encountering a closing curly brace:

```
(var x = {y: [1, 2, 3]})
                     ↓
```

Step 10: We pop the top item from stack:



It's an opening curly brace, so we've found a match with the current closing brace.

Step 11: We encounter a closing parenthesis:

```
(var x = {y: [1, 2, 3]})
                      ↓
```

Step 12: We pop the last element in the stack. It's a corresponding match, so there are no errors so far.

Since we've made it through the entire line of code, and our stack is empty, our linter can conclude that there are no syntactical errors on this line (relating to opening and closing braces).

Code Implementation: Stack-Based Code Linter

Here's a Ruby implementation of the preceding algorithm. Note that we are using our earlier Ruby implementation of the Stack class:

```
class Linter

  def initialize
    # We use a simple array to serve as our stack:
    @stack = Stack.new
  end
```

```
  def lint(text)
    # We start a loop which reads each character in our text:
    text.each_char do |char|

      # If the character is an opening brace:
      if is_opening_brace?(char)

        # We push it onto the stack:
        @stack.push(char)

      # If the character is a closing brace:
      elsif is_closing_brace?(char)

        # Pop from stack:
        popped_opening_brace = @stack.pop

        # If the stack was empty, so what we popped was nil,
        # it means that an opening brace is missing:
        if !popped_opening_brace
          return "#{char} doesn't have opening brace"
        end

        # If the popped opening brace doesn't match the
        # current closing brace, we produce an error:
        if is_not_a_match(popped_opening_brace, char)
          return "#{char} has mismatched opening brace"
        end
      end
    end

    # If we get to the end of line, and the stack isn't empty:
    if @stack.read

      # It means we have an opening brace without a
      # corresponding closing brace, so we produce an error:
      return "#{@stack.read} does not have closing brace"
    end

    # Return true if line has no errors:
    return true
  end

  private

  def is_opening_brace?(char)
    ["(", "[", "{"].include?(char)
  end

  def is_closing_brace?(char)
    [")", "]", "}"].include?(char)
  end

  def is_not_a_match(opening_brace, closing_brace)
    closing_brace != {"(" => ")", "[" => "]", "{" => "}"}[opening_brace]
  end
end
```

The lint method accepts a string containing JavaScript code and iterates over each character with:

```
text.each_char do |char|
```

If we encounter an opening brace, we push it onto the stack:

```
if is_opening_brace?(char)
  @stack.push(char)
```

Note that we're using a helper method defined as follows called is_opening_brace?, which checks whether the character is an opening brace:

```
["(", "[", "{"].include?(char)
```

When we encounter a closing brace, we pop the top of the stack and store it in a variable called popped_opening_brace:

```
popped_opening_brace = @stack.pop
```

Our stack only stores opening braces, so whatever we popped will be some sort of opening brace, assuming there was something on the stack to pop.

However, it's possible the stack was empty, in which case the result of our pop will be nil. If this is the case, it means we've encountered Syntax Error Type #2:

```
if !popped_opening_brace
  return "#{char} doesn't have opening brace"
end
```

For simplicity's sake, we return a basic string containing an error message when we encounter any errors during our linting process.

Assuming that we *did* pop an opening brace off the stack, we then check it to see whether it matches the current closing brace. If it doesn't match, this is Syntax Error Type #3:

```
if is_not_a_match(popped_opening_brace, char)
  return "#{char} has mismatched opening brace"
end
```

(The is_not_a_match helper method is defined later in our code.)

Finally, when our code finishes parsing the line, we check whether there are any opening braces left in the stack with @stack.read. If there are, it means we have an opening brace that was never closed, and we produce an error message. This is Syntax Error Type #1:

```
if @stack.read
  return "#{@stack.read} does not have closing brace"
end
```

Finally, we return true if the JavaScript contains no errors.

We can then use our Linter class as follows:

```
linter = Linter.new
puts linter.lint("( var x = { y: [1, 2, 3] } )")
```

In this example, the JavaScript code is correct, so we just get back true.

However, if we input a line that has an error, such as a missing opening parenthesis:

```
"var x = { y: [1, 2, 3] })"
```

we'll get the error message,) doesn't have opening brace.

In this example, we used a stack to implement our linter with a really neat algorithm. But if a stack actually uses an array under the hood, why bother with a stack? Couldn't we have accomplished the same task using an array?

The Importance of Constrained Data Structures

By definition, if a stack is just a constrained version of an array, that means an array can do anything a stack can do. If so, what advantage does a stack provide?

Constrained data structures like the stack (and the queue, which we'll encounter shortly) are important for several reasons.

First, when we work with a constrained data structure, we can prevent potential bugs. The linting algorithm, for example, only works if we exclusively remove items from the top of the stack. If a programmer inadvertently writes code that removes items from the middle of the array, the algorithm will break down. By using a stack, we're forced into only removing items from the top, as it's impossible to get the stack to remove any other item.

Second, data structures like stacks give us a new mental model for tackling problems. The stack, for example, gives us the whole idea of a Last-In-First-Out process. We can then apply this LIFO mindset to solve all sorts of problems, such as the linter just described.

And once we're familiar with the stack and its LIFO nature, the code we write using the stack becomes familiar and elegant to other developers who read our code. As soon as someone sees a stack being used within an algorithm, they immediately know that the algorithm is working with a LIFO-based process.

Stack Wrap-Up

Stacks are ideal for processing any data that should be handled last in, first out. The "undo" function in a word processor, for example, is a great use case for a stack. As the user types, we keep track of each keystroke by pushing the keystroke onto the stack. Then, when the user hits the "undo" key, we pop the most recent keystroke from the stack and eliminate it from the document. At this point, their next-to-most recent keystroke is now sitting at the top of the stack, ready to be undone if need be.

Queues

A *queue* is another data structure designed to process temporary data. It's like a stack in many ways, except that it processes data in a different order. Like a stack, a queue is also an abstract data type.

You can think of a queue as a line of people at the movie theater. The first one in the line is the first one to leave the line and enter the theater. With queues, the first item added to the queue is the first item to be removed. That's why computer scientists apply the acronym "FIFO" to queues: First In, First Out.

As with a line of people, a queue is usually depicted horizontally. It's also common to refer to the beginning of the queue as its *front,* and the end of the queue as its *back.*

Like stacks, queues are arrays with three restrictions (it's just a different set of restrictions):

- Data can be inserted only at the *end* of a queue. (This is identical behavior to the stack.)
- Data can be deleted only from the *front* of a queue. (This is the opposite behavior of the stack.)
- Only the element at the *front* of a queue can be read. (This, too, is the opposite of behavior of the stack.)

Let's see a queue in action, beginning with an empty queue.

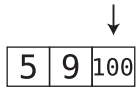
First, we insert a 5 (a common term for inserting into a queue is *enqueue,* but we'll use the terms insert and enqueue interchangeably):



Next, we insert a 9:

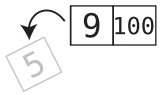


Next, we insert a 100:

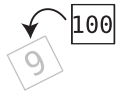


As of now, the queue has functioned just like a stack. However, removing data happens in the reverse, as we remove data from the *front* of the queue. (Removing an element from a queue is also known as *dequeuing*.)

If we want to remove data, we must start with the 5, since it's at the front of the queue:



Next, we remove the 9:



Our queue now only contains one element, the 100.

Queue Implementation

I mentioned that the queue is an abstract data type. Like many other abstract data types, it doesn't come implemented in many programming languages. Here is a Ruby implementation of a queue:

```
class Queue
  def initialize
    @data = []
  end

  def enqueue(element)
    @data << element
  end

  def dequeue
    # The Ruby shift method removes and returns the
    # first element of an array:
    @data.shift
  end

  def read
    @data.first
  end
end
```

Again, our `Queue` class wraps the array with an interface that restricts our interaction with the data, only allowing us to process the data in specific ways. The `enqueue` method allows us to insert data at the end of the array, while the `dequeue` removes the first item from the array. And the `read` method allows us to peek at just the very first element of the array.

Queues in Action

Queues are common in many applications, ranging from printing jobs to background workers in web applications.

Let's say we were programming a simple Ruby interface for a printer that can accept printing jobs from various computers across a network. We want to make sure we print each document in the order in which they were received.

This code uses our Ruby implementation of the `Queue` class from earlier:

```
class PrintManager

  def initialize
    @queue = Queue.new
  end

  def queue_print_job(document)
    @queue.enqueue(document)
  end

  def run
    # Each time this loop runs, we read the document
    # at the front of the queue:
    while @queue.read
      # We dequeue the document and print it:
      print(@queue.dequeue)
    end
  end

  private

  def print(document)
    # Code to run the actual printer goes here.
    # For demo purposes, we'll print to the terminal:
    puts document
  end

end
```

We can then utilize this class as follows:

```
print_manager = PrintManager.new
print_manager.queue_print_job("First Document")
print_manager.queue_print_job("Second Document")
print_manager.queue_print_job("Third Document")
print_manager.run
```

Each time we call queue_print_job, we add the "document" (represented by a string, in this example) to the queue:

```
def queue_print_job(document)
  @queue.enqueue(document)
end
```

When we call run, we print each document by processing them in the order in which they were received:

```
def run
  while @queue.read
    print(@queue.dequeue)
  end
end
```

Note how we dequeue each document as we print it.

When we run the previous code, the program will output the three documents in the same order in which they were received:

```
First Document
Second Document
Third Document
```

While this example is simplified and abstracts away some of the nitty-gritty details that a real live printing system may have to deal with, the fundamental use of a queue for such an application is very real and serves as the foundation for building such a system.

Queues are also the perfect tool for handling asynchronous requests—they ensure that the requests are processed in the order in which they were received. They are also commonly used to model real-world scenarios where events need to occur in a certain order, such as airplanes waiting for takeoff and patients waiting for their doctor.

Wrapping Up

As you've seen, stacks and queues are programmers' tools for elegantly handling all sorts of practical algorithms.

Now that you've learned about stacks and queues, you've unlocked a new achievement: you can learn about recursion, which depends upon a stack. Recursion also serves as the foundation for many of the more advanced and super-efficient algorithms that I'll cover in the rest of this book.

Exercises

The following exercises provide you with the opportunity to practice with stacks and queues. The solutions to these exercises are found in the section, Chapter 9, on page 446.

1. If you were writing software for a call center that places callers on hold and then assigns them to "the next available representative," would you use a stack or a queue?

2. If you pushed numbers onto a stack in the following order: 1, 2, 3, 4, 5, 6, and then popped two items, which number would you be able to read from the stack?

3. If you inserted numbers into a queue in the following order: 1, 2, 3, 4, 5, 6, and then dequeued two items, which number would you be able to read from the queue?

4. Write a function that uses a stack to reverse a string. (For example, `"abcde"` would become `"edcba"`.) You can work with our earlier implementation of the Stack class.

CHAPTER 10

Recursively Recurse with Recursion

Recursion is a key concept in computer science that will unlock the more advanced algorithms we're going to encounter in this book. When used correctly, recursion can be used to solve certain types of tricky problems in surprisingly simple ways. Sometimes, it even seems like magic.

But before we dive in, a pop quiz!

What happens when the `blah()` function defined here is called?

```
function blah() {
  blah();
}
```

As you probably guessed, it will call itself infinitely, since `blah()` calls itself, which in turn calls itself, and so on.

Recursion is the term for a function calling itself. Indeed, infinite recursion, as in the above example, is utterly useless. When harnessed correctly, though, recursion can be a powerful tool.

Recurse Instead of Loop

Let's say you work at NASA and need to program a countdown function for launching spacecraft. The particular function that you're asked to write should accept a number—such as 10—and display the numbers from 10 down to 0.

Take a moment and implement this function in the language of your choice. When you're done, read on.

Odds are that you wrote a simple loop, along the lines of this JavaScript implementation:

```
function countdown(number) {
  for(let i = number; i >= 0; i--) {
    console.log(i);
  }
}

countdown(10);
```

There's nothing wrong with this implementation, but it may have never occurred to you that you don't *have* to use a loop.

How?

Let's try recursion instead. Here's a first attempt at using recursion to implement our `countdown` function:

```
function countdown(number) {
  console.log(number);

  countdown(number - 1);
}
```

Let's walk through this code step by step.

Step 1: We call `countdown(10)`, so the argument variable `number` starts out as 10.

Step 2: We print `number` (which contains the value 10) to the console.

Step 3: Before the `countdown` function is complete, it calls `countdown(9)`, since `number - 1` is 9.

Step 4: `countdown(9)` begins running. In it, we print `number` (which is currently 9) to the console.

Step 5: Before `countdown(9)` is complete, it calls `countdown(8)`.

Step 6: `countdown(8)` begins running. We print 8 to the console.

Before we continue stepping through the code, note how we're using recursion to achieve our goal. We're not using any loop constructs, but by simply having the `countdown` function call itself, we are able to count down from 10 and print each number to the console.

In almost any case in which you can use a loop, you can also use recursion. Now, just because you *can* use recursion doesn't mean that you *should* use recursion. Recursion is a tool that allows for writing elegant code. In the preceding example, the recursive approach is not necessarily any more beautiful or efficient than using a classic `for` loop. However, we will soon encounter examples in which recursion shines. In the meantime, let's continue exploring how recursion works.

The Base Case

Let's continue our walk-through of the `countdown` function. We'll skip a few steps for brevity…

Step 21: We call `countdown(0)`.

Step 22: We print `number` (that is, 0) to the console.

Step 23: We call `countdown(-1)`.

Step 24: We print `number` (that is, -1) to the console.

Uh-oh. As you can see, our solution is not perfect, as we'll end up infinitely printing negative numbers.

To perfect our solution, we need a way to end our countdown at 0 and prevent the recursion from continuing on forever.

We can solve this problem by adding a conditional statement that ensures that if `number` is currently 0, we don't call `countdown()` again:

```
function countdown(number) {
  console.log(number);
  if(number === 0) {
    return;
  } else {
    countdown(number - 1);
  }
}
```

Now, when `number` is 0, our code will not call the `countdown()` function again, but instead just return, thereby preventing another call of `countdown()`.

In recursion terminology, the case in which our function will *not* recurse is known as the *base case.* So, 0 is the base case for our `countdown()` function. Again, every recursive function needs at least one base case to prevent it from calling itself indefinitely.

Reading Recursive Code

It takes time and practice to get used to recursion, and you will ultimately learn *two* sets of skills: *reading* recursive code, and *writing* recursive code. Reading recursive code is somewhat easier, so let's get some practice with that first.

We'll do this by looking at another example: calculating factorials.

A *factorial* is best illustrated with some examples.

The factorial of 3 is:

3 * 2 * 1 = 6

The factorial of 5 is:

5 * 4 * 3 * 2 * 1 = 120

And so on and so forth. Here's a recursive implementation that returns a number's factorial using Ruby:

```
def factorial(number)
  if number == 1
    return 1
  else
    return number * factorial(number - 1)
  end
end
```

This code can look somewhat confusing at first glance. To walk through the code to see what it does, here's the process I recommend:

1. Identify the base case.
2. Walk through the function for the base case.
3. Identify the "next-to-last" case. This is the case just before the base case, as I'll demonstrate momentarily.
4. Walk through the function for the "next-to-last" case.
5. Repeat this process by identifying the case before the one you just analyzed, and walking though the function for that case.

Let's apply this process to the preceding code. If we analyze the code, we'll quickly notice that there are two paths:

```
if number == 1
  return 1
else
  return number * factorial(number - 1)
end
```

We can see that the recursion happens here, as `factorial` calls itself:

```
else
  return number * factorial(number - 1)
end
```

So, it must be that the following code refers to the base case, since this is the case in which the function does *not* call itself:

```
if number == 1
  return 1
```

We can conclude, then, that number being 1 is the base case.

Next, let's walk through the factorial method assuming it's dealing with the base case, which would be factorial(1). Again, the relevant code from our method is:

```
if number == 1
  return 1
```

Well, that's pretty simple—it's the base case, so no recursion actually happens. If we call factorial(1), the method simply returns 1. Okay, so grab a napkin and write this fact down:

factorial (1) returns 1

Next, let's move up to the next case, which would be factorial(2). The relevant line of code from our method is:

```
else
  return number * factorial(number - 1)
end
```

So, calling factorial(2) will return 2 * factorial(1). To calculate 2 * factorial(1), we need to know what factorial(1) returns. If you check your napkin, you'll see that it returns 1. So, 2 * factorial(1) will return 2 * 1, which just happens to be 2.

Add this fact to your napkin:

factorial (2) returns 2

factorial (1) returns 1

Now, what happens if we call factorial(3)? Again, the relevant line of code is:

```
else
  return number * factorial(number - 1)
end
```

So, that would translate into return 3 * factorial(2). What does factorial(2) return? You don't have to figure that out all over again, since it's on your napkin! It returns 2. So, factorial(3) will return 6 (because 3 * 2 = 6). Go ahead and add this wonderful factoid to your napkin:

factorial (3) returns 6

factorial (2) returns 2

factorial (1) returns 1

Take a moment and figure out for yourself what factorial(4) will return.

As you can see, starting the analysis from the base case and building up is a great way to reason about recursive code.

Recursion in the Eyes of the Computer

To complete your understanding of recursion, we need to see how the computer itself processes a recursive function. It's one thing for humans to reason about recursion using the earlier "napkin" method. However, the computer has to do the tricky work of calling a function from within the function itself. So, let's break down the process of how the computer executes a recursive function.

Say that we call factorial(3). Since 3 isn't the base case, the computer reaches the line:

```
return number * factorial(number - 1)
```

which launches the function factorial(2).

But there's a catch. When the computer begins to run factorial(2), did the computer yet complete running factorial(3)?

This is what makes recursion tricky for the computer. Until the computer reaches the end keyword of factorial(3), it's not done with factorial(3). So, we enter into a weird situation. The computer did not yet complete executing factorial(3), yet it's starting to run factorial(2) while *still in the middle* of factorial(3).

And factorial(2) isn't the end of the story, because factorial(2) triggers factorial(1). So, it's a pretty crazy thing: while still in the middle of running factorial(3), the computer calls factorial(2). And while running factorial(2), the computer runs factorial(1). It turns out then, that factorial(1) runs in the middle of *both* factorial(2) and factorial(3).

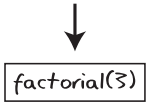
How does the computer keep track of all of this? It needs some way to remember that after it finishes factorial(1), it needs to go back and finish running factorial(2). And then it needs to remember to complete factorial(3) once it completed factorial(2).

The Call Stack

Luckily, you recently learned about stacks in *Crafting Elegant Code*. The computer uses a stack to keep track of which functions it's in the middle of calling. This stack is known, appropriately enough, as the *call stack*.

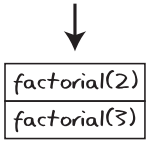
Here's how the call stack works in the context of our factorial example.

The computer begins by calling factorial(3). Before the method completes executing, however, factorial(2) gets called. In order to track that the computer is still in the middle of factorial(3), the computer pushes that information onto a call stack:



This indicates that the computer is in the middle of factorial(3). (Really, the computer also needs to save which line it's in the middle of, and some other things like variable values, but I'm keeping the diagrams simple.)

The computer then proceeds to execute factorial(2). Now, factorial(2), in turn, calls factorial(1). Before the computer dives into factorial(1), though, the computer needs to remember that it's still in the middle of factorial(2), so it pushes that onto the call stack as well:

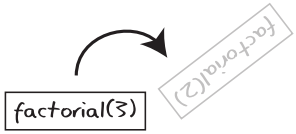


The computer then executes factorial(1). Since 1 is the base case, factorial(1) completes without calling the factorial method again.

After the computer completes factorial(1), it checks the call stack to see whether it's in the middle of any other functions. If there's something in the call stack, it means that the computer still has work to do, namely, wrap up some other functions it was in the middle of.

If you recall, stacks are restricted in that we can only pop its top element. This is ideal for recursion, since the top element will be *the most recently called function,* which is what the computer needs to wrap up next. It's a LIFO situation: the function that was called last (that is, most recently), is the function we need to complete first.

The next thing the computer does is pop the top element of the call stack, which currently is factorial(2):



The computer then completes its execution of factorial(2).

Now, the computer pops the next item from the stack. By this time, only factorial(3) is left on the stack, so the computer pops it and therefore completes running factorial(3).

At this point, the stack is empty, so the computer knows it's done executing all of the methods, and the recursion is complete.

Looking back at this example from a bird's-eye view, you'll see that the order in which the computer calculates the factorial of 3 is as follows:

1. factorial(3) is called first. Before it's done…
2. factorial(2) is called second. Before it's done…
3. factorial(1) is called third.
4. factorial(1) is *completed* first.
5. factorial(2) is completed based on the result of factorial(1).
6. Finally, factorial(3) is completed based on the result of factorial(2).

The factorial function is a calculation made based on recursion. The calculation is ultimately made by factorial(1) passing its result (which is 1) to factorial(2). Then, factorial(2) multiplies this 1 by 2, yielding 2, and passes this result to factorial(3). Finally, factorial(3) takes this result, and multiplies it by 3, computing the result of 6.

Some refer to this idea as *passing a value up through the call stack*. That is, each recursive function returns its computed value to its "parent" function. Eventually, the function that was initially called first computes the final value.

Stack Overflow

Let's take a look back at the infinite recursion example from the beginning of the chapter. Recall that blah() called itself ad infinitum. What do you think will happen to the call stack?

In the case of infinite recursion, the computer keeps pushing the same function again and again onto the call stack. The call stack grows and grows until, eventually, the computer reaches a point where there's simply no more room in its short-term memory to hold all this data. This causes an error known as *stack overflow*—the computer just shuts down the recursion and says, "I refuse to call the function again, because I'm running out of memory!"

Filesystem Traversal

Now that you've seen how recursion works, we can use it to solve certain problems that would otherwise be uncrackable.

One type of problem in which recursion is a natural fit is when we need to delve into multiple layers of a problem without knowing how many layers there are.

Take the example of traversing through a filesystem. Let's say you have a script that does something with all the contents inside a directory, such as printing all the subdirectory names. However, you don't want the script to only handle the immediate subdirectories—you want it to act on all the subdirectories *within* the subdirectories of the directory and all of *their* subdirectories, and so on.

Let's create a simple Ruby script that prints the names of all subdirectories within a given directory:

```
def find_directories(directory)
  # Inspect each file within the directory. Some of these "files" may
  # actually be subdirectories.
  Dir.foreach(directory) do |filename|

    # If the current file is itself a subdirectory:
    if File.directory?("#{directory}/#{filename}") &&
    filename != "." && filename != ".."

      # Print out the full path name:
      puts "#{directory}/#{filename}"
    end
  end
end
```

We can call this function by passing in a directory name. If we want to call it on the current directory, we could write:

```
find_directories(".")
```

In this script, we look through each file within the given directory. If the file is itself a subdirectory (and isn't a single or double period, which represent the current and previous directories, respectively), we print the subdirectory name.

While this works well, it only prints the names of the subdirectories *immediately* within the current directory. It does not print the names of the subdirectories *within* those subdirectories.

Let's update our script so that it can search one level deeper:

```
def find_directories(directory)
  # Loop through first-level directory:
  Dir.foreach(directory) do |filename|
    if File.directory?("#{directory}/#{filename}") &&
    filename != "." && filename != ".."
      puts "#{directory}/#{filename}"
```

```
      # Loop through second-level subdirectory:
      Dir.foreach("#{directory}/#{filename}") do |inner_filename|
        if File.directory?("#{directory}/#{filename}/#{inner_filename}") &&
        inner_filename != "." && inner_filename != ".."
          puts "#{directory}/#{filename}/#{inner_filename}"
        end
      end

    end
  end
end
```

Now, every time our script discovers a directory, it then conducts an identical loop through the subdirectories of *that* directory and prints the names of the subdirectories. But this script also has its limitations because it's only searching two levels deep. What if we want to search three, four, or five levels deep? We would need five levels of nested loops.

And what if we want to search as deep as our subdirectories go? That would seem to be impossible, as we don't even know how many levels there are.

And *this* is where recursion really shines. With recursion, we can write a script that goes arbitrarily deep—and is also simple!

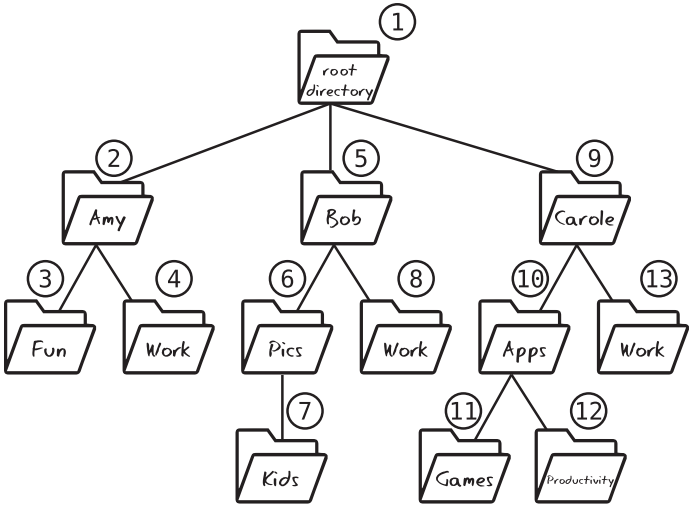
```
def find_directories(directory)
  Dir.foreach(directory) do |filename|
    if File.directory?("#{directory}/#{filename}") &&
    filename != "." && filename != ".."
      puts "#{directory}/#{filename}"

      # Recursively call this function on the subdirectory:
      find_directories("#{directory}/#{filename}")
    end
  end
end
```

As this script encounters files that are themselves subdirectories, it calls the find_directories method upon that very subdirectory. The script can therefore dig as deep as it needs to, leaving no subdirectory unturned.

To visualize how this algorithm acts on an example filesystem, examine the diagram on page 159, which specifies the order in which the script traverses the subdirectories.

We'll encounter this process again with a detailed visual walk-through in Depth-First Search, on page 339.



Wrapping Up

As you've seen with the filesystem example, recursion is often a great choice for an algorithm in which the algorithm needs to dig into an arbitrary number of levels deep into something.

You've now seen how recursion works and how incredibly useful it can be. You've also learned how to walk through and read recursive code. However, most people have a difficult time writing their own recursive functions when they're first starting out. In the next chapter, we'll explore techniques to help you learn to write recursively. Along the way, you'll also discover other important use cases where recursion can be an incredible tool.

Exercises

The following exercises provide you with the opportunity to practice with recursion. The solutions to these exercises are found in the section, Chapter 10, on page 447.

1. The following function prints every other number from a `low` number to a `high` number. For example, if `low` is 0 and `high` is 10, it would print:

```
0
2
4
6
8
10
```

Identify the base case in the function:

```
def print_every_other(low, high)
  return if low > high
  puts low
  print_every_other(low + 2, high)
end
```

2. My kid was playing with my computer and changed my factorial function so that it computes factorial based on (n - 2) instead of (n - 1). Predict what will happen when we run factorial(10) using this function:

```
def factorial(n)
  return 1 if n == 1
  return n * factorial(n - 2)
end
```

3. Following is a function in which we pass in two numbers called low and high. The function returns the sum of all the numbers from low to high. For example, if low is 1, and high is 10, the function will return the sum of all numbers from 1 to 10, which is 55. However, our code is missing the base case, and will run indefinitely! Fix the code by adding the correct base case:

```
def sum(low, high)
  return high + sum(low, high - 1)
end
```

4. Here is an array containing both numbers as well as other arrays, which in turn contain numbers and arrays:

```
array = [  1,
           2,
           3,
           [4, 5, 6],
           7,
           [8,
             [9, 10, 11,
               [12, 13, 14]
             ]
           ],
           [15, 16, 17, 18, 19,
             [20, 21, 22,
               [23, 24, 25,
                 [26, 27, 29]
               ], 30, 31
             ], 32
           ], 33
        ]
```

Write a recursive function that prints all the numbers (and *just* numbers).

CHAPTER 11

Learning to Write in Recursive

In the previous chapter, you learned what recursion is and how it works. However, I found on my own learning journey that even once I understood how recursion works, I still struggled to write my own recursive functions.

Through deliberate practice and taking note of various recursive patterns, I discovered some techniques that helped me to learn to "write in recursive" more easily, and I'd like to share them with you. Along the way, you'll discover additional areas where recursion shines.

Note that we will not be discussing the efficiency of recursion in this chapter. Recursion can actually have a terribly negative impact on the time complexity of an algorithm, but this is the subject of the next chapter. For now, we're just going to focus on developing the recursive mindset.

Recursive Category: Repeatedly Execute

Over the course of tackling various recursive problems, I began to find that there are various "categories" of problems. Once I learned an effective technique for a certain category, when I found another problem that belonged to the same category, I was able to apply the same technique to solve it.

The category that I found to be the easiest was one in which the goal of the algorithm was to repeatedly execute a task.

The NASA spacecraft countdown algorithm from the previous chapter is a great example of this. The code prints a number such as 10, and then 9, then 8, all the way down to 0. While the number the function prints is different each time, we boil down the code's essence to the fact that it is repeatedly executing a task—namely, the printing of a number.

Here was our JavaScript implementation of this algorithm:

```
function countdown(number) {
  console.log(number);

  if(number === 0) { // number being 0 is the base case
    return;
  } else {
    countdown(number - 1);
  }
}
```

I found that for problems of this category, the last line of code in the function was a simple, single call to the function again. In the previous snippet, this takes the form of `countdown(number - 1)`. This line does one thing: it makes the next recursive call.

The directory-printing algorithm from the previous chapter is another example of this. This function repeatedly executes the task of printing directory names.

Our Ruby code looked like this:

```
def find_directories(directory)
  Dir.foreach(directory) do |filename|
    if File.directory?("#{directory}/#{filename}") &&
    filename != "." && filename != ".."
      puts "#{directory}/#{filename}"

      # Recursively call this function on the subdirectory:
      find_directories("#{directory}/#{filename}")
    end
  end
end
```

Here as well, the last line of code is `find_directories("#{directory}/#{filename}")`, which is a simple call to the recursive function, triggering it again.

Recursive Trick: Passing Extra Parameters

Let's try our hand at another problem of the "repeatedly execute" category. We're going to write an algorithm that takes an array of numbers and doubles each of the numbers within the array. Note that we're not going to produce a new array; instead, we're going to modify the array in place.

This algorithm, too, is one in which we repeatedly execute a task. Specifically, we repeatedly double a number. We start with the first number and double it. We move onto the second number and double it—and so on.

In-Place Modifications

Let's take a look at the concept of *in-place* modifications, just in case it's not clear.

Generally speaking, there are two basic approaches to manipulating data. Let's take the example of doubling the values in an array. If I had the array, `[1, 2, 3, 4, 5]`, and wanted to "double" it to produce the array, `[2, 4, 6, 8, 10]`, I could do one of two things.

The first option is to create a new array that contains the "doubled" data, but leave the original array alone. Take the following code:

```
a = [1, 2, 3, 4, 5]
b = double_array(a)
```

Since the `double_array` function is creating and returning a brand-new array, if we were to check the values of `a` and `b`, we'd get:

```
a # [1, 2, 3, 4, 5]
b # [2, 4, 6, 8, 10]
```

The original array `a` was unmodified, and `b` holds a brand-new array.

The second option is called *in-place* modification, meaning the function actually changes the *original array* passed into function.

With in-place modification, if we now check `a` and `b`, we'd find:

```
a # [2, 4, 6, 8, 10]
b # [2, 4, 6, 8, 10]
```

The in-place function actually modifies `a`, and `b` is actually just pointing to the very same array that `a` is.

Whether we choose to create a new array or modify the original one in place is up to us and depends on the context of our project. We're going to talk more about in-place algorithms in *Dealing with Space Constraints*.

Let's take a stab at writing this algorithm in Python, which we'll call `double_array()`. We know that our final line will be the recursive call, so let's include that line:

```
def double_array(array):
  double_array(array)
```

Next, we need to add the code that will actually double the number. But which number will we double? Let's try to double the first number:

```
def double_array(array):
  array[0] *= 2
  double_array(array)
```

Okay, so we've doubled the number at index 0, but how do we proceed to double the number at index 1?

Now, if we were to use a loop instead of recursion, we would have used a variable to keep track of the index and continuously increased it by 1, along the lines of:

```
def double_array(array):
  index = 0

  while (index < len(array)):
    array[index] *= 2
    index += 1
```

In our recursive version, though, the only argument to our function is the array. We need some way to keep track of and increment an index. How do we pull this off?

And now for our next trick…

Let's pass in extra parameters!

Let's modify the beginning of our function so that it accepts *two* arguments: the array itself, and an index to keep track of:

```
def double_array(array, index):
```

As things stand right now, when we call this function, we need to pass in the array and the starting index, which will be 0:

```
double_array([1, 2, 3, 4, 5], 0)
```

Once we have the index as a function argument, we now have a way of incrementing and tracking the index as we make each successive recursive call. Here's the code for this:

```
def double_array(array, index):
  array[index] *= 2
  double_array(array, index + 1)
```

In each successive call, we pass in the array again as the first argument, but we also pass along an incremented index. This allows us to keep track of an index just as we would in a classical loop.

Our code isn't perfect just yet, though. Our function will throw an error once the index goes past the end of the array and tries to multiply a nonexistent number. To solve this, we need our base case:

```
def double_array(array, index):
  # Base case: when the index goes past the end of the array
  if index >= len(array):
    return

  array[index] *= 2
  double_array(array, index + 1)
```

We can test this function out with the following code:

```
array = [1, 2, 3, 4]
double_array(array, 0)
print(array)
```

Our recursive function is now complete. However, if your programming language supports default arguments, we can make things even prettier.

Right now, we need to call the function as

```
double_array([1, 2, 3, 4, 5], 0)
```

Admittedly, passing in that 0 as a second parameter isn't beautiful—it's just so we can achieve our trick of maintaining an index. After all, we *always* want to start our index off at 0.

However, we can use default parameters to allow us to simply call the function the original way:

```
double_array([1, 2, 3, 4, 5])
```

Here's our updated code to make this work:

```
def double_array(array, index=0):
  # Base case: when the index goes past the end of the array
  if (index >= len(array)):
    return

  array[index] *= 2
  double_array(array, index + 1)
```

All we updated here was setting a default argument of `index=0`. This way, the first time we call the function, we don't have to pass in the index parameter. However, we still get to use the index parameter for all successive calls.

The "trick" of using extra function parameters is a common technique in writing recursive functions, and a handy one.

Recursive Category: Calculations

In the previous section, we discussed the first category of recursive functions—those whose job it is to repeatedly execute a task. For the remainder of this chapter, I'll elaborate on a second general category: performing a calculation based on a subproblem.

There are many functions whose goal it is to perform a calculation. A function that returns the sum of two numbers, or a function that finds the greatest value within an array, are examples of these. These functions receive some sort of input, and return the result of calculations involving that input.

In *Recursively Recurse with Recursion*, we found that one area in which recursion shines is where we need to act on a problem that has an arbitrary number of levels of depth. A second area in which recursion shines is where *it is able to make a calculation based on a subproblem of the problem at hand.*

Before I define what a *subproblem* is, let's refer back to the factorial problem of the previous chapter. As you learned, the factorial of 6 is:

6 * 5 * 4 * 3 * 2 * 1

To write a function that calculates the factorial of a number, we could use a classic loop that starts with the 1 and builds up from there. That is, we'd multiply the 2 by the 1, and then multiply 3 by the result, and then 4, and so on until we reach 6.

Such a function may look like this in Ruby:

```
def factorial(n)
  product = 1

  (1..n).each do |num|
    product *= num
  end

  return product
end
```

However, we could approach the problem differently, and that is to calculate the factorial based on its *subproblem.*

A *subproblem* is a version of the very same problem applied to a smaller input. Let's apply this to our case.

If you think about it, factorial(6) will be 6 multiplied by whatever the result of factorial(5) is.

Since factorial(6) is:

6 * 5 * 4 * 3 * 2 * 1

and factorial(5) is:

5 * 4 * 3 * 2 * 1

we can conclude that factorial(6) is equivalent to:

6 * factorial(5).

That is, once we have the result of factorial(5), we can simply multiply that result by 6 to get the answer to factorial(6).

Since factorial(5) is the smaller problem that can be used to compute the result for the bigger problem, we call factorial(5) a *subproblem* of factorial(6).

Here's the implementation of this from the last chapter:

```
def factorial(number)
  if number == 1
    return 1
  else
    return number * factorial(number - 1)
  end
end
```

Again, the key line here is return number * factorial(number - 1), in which we compute the result as number multiplied by our subproblem, which is factorial(number - 1).

Two Approaches to Calculations

We've seen that when writing a function that makes a calculation, there are two potential approaches: we can try to build the solution from the "bottom up," or we can attack the problem going "top down" by making the calculation based on the problem's subproblem. Indeed, computer science literature refers to the terms *bottom up* and *top down* in regard to recursion strategies.

The truth is that both approaches can be achieved through recursion. While we previously saw the bottom-up approach using a classic loop, we can also use recursion to implement the bottom-up strategy.

To do this, we need to use our trick of passing extra parameters, as follows:

```
def factorial(n, i=1, product=1)
  return product if i > n
  return factorial(n, i + 1, product * i)
end
```

In this implementation, we have three parameters. n, as before, is the number whose factorial we're computing. i is a simple variable that starts at 1 and increments by one in each successive call until it reaches n. Finally, product is the variable in which we store the calculation as we keep multiplying each successive number. We keep passing the product to the successive call so we can keep track of it as we go.

While we can use recursion in this way to achieve the bottom-up approach, it's not particularly elegant and does not add much value over using a classic loop.

When going bottom up, we're employing the same strategy for making the calculation whether we're using a loop or recursion. The computational approach is the same.

But to go top down, we *need* recursion. And because recursion is the only way to achieve a top-down strategy, it's one of the key factors that makes recursion a powerful tool.

Top-Down Recursion: A New Way of Thinking

This brings us to the central point of this chapter: recursion shines when implementing a top-down approach because going top down *offers a new mental strategy for tackling a problem.* That is, a recursive top-down approach allows one to think about a problem in a completely different way.

Specifically, when we go top down, we get to mentally "kick the problem down the road." We can free our mind from some of the nitty-gritty details we normally have to think about when going bottom up.

To see what I mean, let's take another look at the key line from our top-down factorial implementation:

```
return number * factorial(number - 1)
```

This line of code makes its calculation based on factorial(number - 1). When we write this line of code, do we have to understand how the factorial function it's calling works? Technically, we don't. Whenever we write code that calls another function, we assume that the function will return the correct value without necessarily understanding how its internals work.

Here as well, when we calculate our answer based on calling the factorial function, we don't need to understand how the factorial function works; we can just expect it to return the correct result. Of course, the weird part is that *we're the ones writing the factorial function!* This line of code exists *within the*

factorial function itself. But that's what is so great about top-down thinking: in a way, we can solve the problem without even knowing how to solve the problem.

When we write "in recursive" to implement a top-down strategy, we get to relax our brains a little bit. We can even choose to ignore the details of how the calculation actually works. We get to say, "Let's just rely on the subproblem to deal with the details."

The Top-Down Thought Process

If you haven't done a lot of top-down recursion before, it takes time and practice to learn to think in this way. However, I found that when tackling a top-down problem, it helps to think the following three thoughts:

1. Imagine the function you're writing has already been implemented by someone else.
2. Identify the subproblem of the problem.
3. See what happens when you call the function on the subproblem and go from there.

While these steps sound vague at the moment, they'll become more clear through the following examples.

Array Sum

Say we have to write a function called `sum` that sums up all the numbers in a given array. For example, if we pass the array, `[1, 2, 3, 4, 5]` into the function, it'll return 15, which is the sum of those numbers.

The first thing we'll do is imagine that the `sum` function has already been implemented. Admittedly, this takes a certain suspension of disbelief, since we know that we're in the middle of writing this function as we speak! But let's try to let go and pretend that the `sum` function already works.

Next, let's identify the subproblem. This can be more of an art than a science, but practice will help you get better at this. In our case, we can say that the subproblem is the array, `[2, 3, 4, 5]`—that is, all the numbers from the array save the first one.

Finally, let's see what happens when we apply the `sum` function to our subproblem. If the `sum` function "already works," and the subproblem is `[2, 3, 4, 5]`, what happens when we call `sum([2, 3, 4, 5])`? Well, we get the sum of 2 + 3 + 4 + 5, which is 14.

To solve our problem of finding the sum of [1, 2, 3, 4, 5] then, we can just add the first number, 1, to the result of sum([2, 3, 4, 5]).

In pseudocode, we'd write something like this:

```
return array[0] + sum(the remainder of the array)
```

In Ruby, we can write this as:

```
return array[0] + sum(array[1, array.length - 1])
```

(In many languages, the syntax array[x, y] returns an array from index x through index y.)

Now, believe it or not, we're done! Save for the base case, which we'll get to in a moment, our sum function can be written as:

```
def sum(array)
  return array[0] + sum(array[1, array.length - 1])
end
```

Note that we didn't think about how we're going to add all the numbers together. All we did was imagine that someone else wrote the sum function for us, which we applied to the subproblem. We kicked the problem down the road, but in doing so, we solved the entire problem.

The last thing we need to do is handle the base case. That is, if each subproblem recursively calls its own subproblem, we'll eventually reach the subproblem of sum([5]). This function will eventually try to add the 5 to the remainder of the array, but there *are* no other elements in the array.

To deal with this, we can add the base case:

```
def sum(array)
  # Base case: only one element in the array:
  return array[0] if array.length == 1

  return array[0] + sum(array[1, array.length - 1])
end
```

And we're done.

String Reversal

Let's try another example. We're going to write a reverse function that reverses a string. So, if the function accepts the argument "abcde", it'll return "edcba".

First, let's identify the subproblem. Again, this takes practice, but very often the first thing to try is the next-to-smallest version of the problem at hand.

So, for the string "abcde", let's assume the subproblem is "bcde". This subproblem is the same as the original string minus its first character.

Next, let's imagine that someone did us the great favor of implementing the reverse function for us. How nice of them!

Now, if the reverse function is available for our use and our subproblem is "bcde", that means we can already call reverse("bcde"), which would return "edcb".

Once we can do that, dealing with the "a" is a piece of cake. We just need to throw it onto the end of the string.

So, we can write:

```
def reverse(string)
  return reverse(string[1, string.length - 1]) + string[0]
end
```

Our computation is simply the result of calling reverse on the subproblem, and then adding the first character to the end.

Once again, save for the base case, we are done. I know, it's crazy magical.

The base case occurs when the string has one character, so we can add the following line of code to handle it:

```
def reverse(string)
  # Base case: string with just one character
  return string[0] if string.length == 1

  return reverse(string[1, string.length - 1]) + string[0]
end
```

And we're done.

Counting X

We're on a roll, so let's try another example. Let's write a function called count_x that returns the number of "x's" in a given string. If our function is passed the string, "axbxcxd", it'll return 3, since there are three instances of the character "x."

Let's first identify the subproblem. As in the previous example, we'll say the subproblem is the original string minus its first character. So, for "axbxcxd", the subproblem is "xbxcxd".

Next, let's imagine count_x has already been implemented. If we call count_x on our subproblem, by calling count_x("xbxcxd"), we get 3. To that, we just need to

add 1 if our first character is also an "x." (If our first character is *not* an "x," we don't need to add anything to the result of our subproblem.)

So, we can write:

```
def count_x(string)
  if string[0] == "x"
    return 1 + count_x(string[1, string.length - 1])
  else
    return count_x(string[1, string.length - 1])
  end
end
```

This conditional statement is straightforward. If the first character is an "x," we add 1 to the result of the subproblem. Otherwise, we return the result of our subproblem as is.

Here too, we're basically done. All we need to do is deal with the base case.

We can say that the base case here is when a string has only one character. But this leads to some awkward code, since we really have two base cases, as the single character may be an "x," but may also not be an "x":

```
def count_x(string)

  # Base case:
  if string.length == 1
    if string[0] == "x"
      return 1
    else
      return 0
    end
  end

  if string[0] == "x"
    return 1 + count_x(string[1, string.length - 1])
  else
    return count_x(string[1, string.length - 1])
  end
end
```

Thankfully, we can simplify this using another simple trick. In many languages, calling string[1, 0] returns an empty string.

With this in mind, we can actually write our code more simply:

```
def count_x(string)

  # Base case: an empty string
  return 0 if string.length == 0
```

```
  if string[0] == "x"
    return 1 + count_x(string[1, string.length - 1])
  else
    return count_x(string[1, string.length - 1])
  end
end
```

In this new version, we say that the base case occurs when the string is empty (`string.length == 0`). We return 0 because there will never be an "x" in an empty string.

When the string has just one character, the function will add 1 or 0 to the result of the next function call. This next function call is `count_x(string[1, 0])`, since `string.length - 1` is 0. Because `string[1, 0]` is an empty string, this final call is the base case and returns 0.

And now we're done.

For the record, calling `array[1, 0]` also returns an empty array in many languages, so we could have used the same trick in the previous two examples as well.

The Staircase Problem

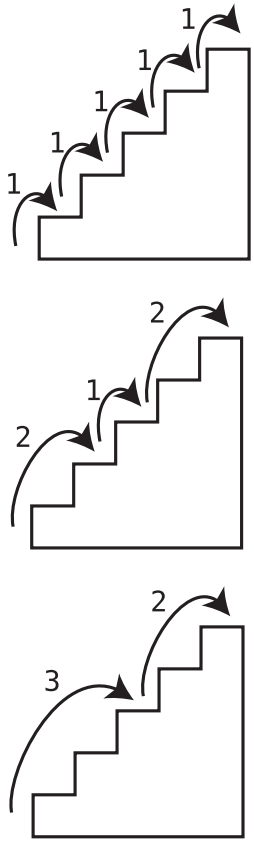
We've now learned to use a new mental strategy for solving certain computational problems using top-down recursion. However, you may still be skeptical and ask, "Why do we need this new mental strategy anyway? I've been able to solve these problems with loops just fine until now."

Indeed, you may not need a new mental strategy for simpler computations. However, when it comes to more complex functions, you may find that the recursive mindset makes the writing of code much easier. It certainly does for me!

Here's one of my favorite examples. There's a famous question—known as the "Staircase Problem"—that goes like this:

Let's say we have a staircase of N steps, and a person has the ability to climb one, two, or three steps at a time. How many different possible "paths" can someone take to reach the top? Write a function that will calculate this for N steps.

The image on page 174 displays three possible paths of jumping up a five-step staircase.



These are just three of many possible paths.

Let's first explore this problem with a bottom-up approach. That is, we'll work our way up from the simplest cases to the more complex ones.

Obviously, if there's only one step, there's only one possible path.

With two steps, there are two paths. The person can climb one step twice, or the person can jump up two steps at once. I'll write this as:

```
1, 1
2
```

With a staircase of three steps, someone could take one of four possible paths:

```
1, 1, 1
1, 2
2, 1
3
```

With four steps, there are seven options:

```
1, 1, 1, 1
1, 1, 2
1, 2, 1
1, 3
2, 1, 1
2, 2
3, 1
```

Go ahead and try to draw up all the combinations for a five-step staircase. It's not that easy! And this is just five steps. Imagine how many combinations there are for 11 steps.

Now, let's get to the question at hand: how would we *write the code* to count all the paths?

Without the recursive mindset, it can be difficult to wrap one's mind around the algorithm for making this calculation. However, with the top-down way of thinking, the problem can become surprisingly easy.

For an 11-step staircase, the first subproblem that comes to mind is a 10-step staircase. Let's go with that for now. If we knew how many possible paths there are to climb a 10-step staircase, can we use that as a base for calculating the paths for an 11-step staircase?

For starters, we do know that climbing an 11-step staircase will take *at least* as many steps as climbing a 10-step staircase. That is, we have all the paths to get to stair number 10, and from there, one can climb one more step to get to the top.

However, this can't be the complete solution, since we know that someone can also jump to the top from stair numbers 9 and 8 as well.

If we think about it further, we'll realize that if you're taking any path that includes going from stair 10 to stair 11, you're not taking any of the paths that include jumping from stair 9 to stair 11. Conversely, if you jump from stair 9 to stair 11, you're not taking any of the paths that include stepping on stair 10.

So, we know that the number of paths to the top will include at least the number of paths to stair 10 plus the number of paths to stair 9.

And since it's possible to also jump from stair 8 to stair 11, as one can jump three steps at a time, we need to include the count of those paths as well.

We've determined, then, that the number of steps to the top is at least the sum of all the paths to stairs 10, 9, and 8.

However, in thinking about it even further, it's evident there aren't any other possible paths to the top beyond these. After all, one cannot jump from stair 7 to stair 11. So, we can conclude that for N steps, the number of paths is:

```
number_of_paths(n - 1) + number_of_paths(n - 2) + number_of_paths(n - 3)
```

Other than the base case, this will be the code for our function!

```
def number_of_paths(n)
  number_of_paths(n - 1) + number_of_paths(n - 2) + number_of_paths(n - 3)
end
```

It's seems too good to be true that this is almost all the code we need. But it *is* true. All that's left to deal with is the base case.

Staircase Problem Base Case

Determining the base case for this problem is slightly tricky. That's because when this function gets to an `n` of 3, 2, or 1, the function will call itself on `n` of 0 or below. For example, `number_of_paths(2)` calls itself for `number_of_paths(1)`, `number_of_paths(0)`, and `number_of_paths(-1)`.

One way we can deal with this is by "hardcoding" all the bottom cases:

```
def number_of_paths(n)
  return 0 if n <= 0
  return 1 if n == 1
  return 2 if n == 2
  return 4 if n == 3
  return number_of_paths(n - 1) + number_of_paths(n - 2) +
    number_of_paths(n - 3)
end
```

Another way to devise the base cases here is to cleverly "rig" the system by using strange but effective base cases that just happen to compute the right numbers. Let me show you what I mean.

We know that we definitely want the result of `number_of_paths(1)` to be 1, so we'll start with the base case of:

```
return 1 if n == 1
```

Now, we know that we want `number_of_paths(2)` to return 2, but we don't *have* to create that base case explicitly. Instead, we can take advantage of the fact that `number_of_paths(2)` will compute as `number_of_paths(1) + number_of_paths(0) + number_of_paths(-1)`. Since `number_of_paths(1)` returns 1, if we made `number_of_paths(0)` also return 1, and `number_of_paths(-1)` return 0, we'd end up with a sum of 2, which is what we want.

So, we can add the following base cases:

```
return 0 if n < 0
return 1 if n == 1 || n == 0
```

Let's move onto number_of_paths(3), which will return the sum of number_of_paths(2) + number_of_paths(1) + number_of_paths(0). We know that we want the result to be 4, so let's see if the math works out. We already rigged number_of_paths(2) to return 2. number_of_paths(1) will return 1, and number_of_paths(0) will also return 1, so we end up getting the sum of 4, which is just what we need.

Our complete function can also be written as:

```
def number_of_paths(n)
  return 0 if n < 0
  return 1 if n == 1 || n == 0
  return number_of_paths(n - 1) + number_of_paths(n - 2) +
    number_of_paths(n - 3)
end
```

While this is less intuitive than our previous version, we cover all the base cases with just two lines of code.

As you can see, the top-down recursive approach made solving this problem much easier than it might have been otherwise.

Anagram Generation

To top off our conversation, let's tackle our most complex recursive problem yet. We're going to use everything we've got in our recursion toolbox to make this work.

We're going to write a function that returns an array of all anagrams of a given string. An anagram is a reordering of all the characters within a string. For example, the anagrams of "abc" are:

```
["abc",
"acb",
"bac",
"bca",
"cab",
"cba"]
```

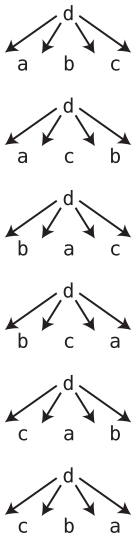
Now, let's say we were to collect all the anagrams of the string "abcd". Let's apply our top-down mindset to this problem.

Presumably, we could say that the subproblem of "abcd" is "abc". The question then, is: if we had a working anagrams function that returned all the anagrams

of "abc", how can we use them to produce all the anagrams of "abcd"? Think about this for a bit and see if you can come up with any approaches.

Here's the approach that occurred to me. (There are others, though.)

If we had all six anagrams of "abc", we can come up with every permutation of "abcd" if we stick the "d" in every possible spot within each anagram of "abc":



Here is a Ruby implementation of this algorithm. You'll note that it's certainly more involved than the previous examples in this chapter:

```
def anagrams_of(string)
  # Base case: if the string is only one character,
  # return an array containing just a single-character string:
  return [string[0]] if string.length == 1

  # Create an array to hold all the anagrams:
  collection = []

  # Find all anagrams of the substring from the second character until
  # the end. For example, if the string is "abcd", the substring is "bcd",
  # so we'll find all anagrams of "bcd":
  substring_anagrams = anagrams_of(string[1, string.length - 1])

  # Iterate over each substring
  substring_anagrams.each do |substring_anagram|

    # Iterate over each index of the substring, from 0 until
    # one index past the end of the string:
    (0..substring_anagram.length).each do |index|

      # Create a copy of the substring anagram:
      copy = String.new(substring_anagram)
```

```
      # Insert the first character of our string into the
      # substring anagram copy. Where it will go depends
      # on the index we're up to within this loop.
      # Then, take this new string and add it to our collection of anagrams:
      collection << copy.insert(index, string[0])
    end
  end

  # Return the entire collection of anagrams:
  return collection
end
```

This code is not trivial, so let's break it down. For now, we'll skip over the base case.

We start by creating an empty array in which we'll collect the entire collection of anagrams:

```
collection = []
```

This is the same array we'll return at the end of our function.

Next, we grab the array of all anagrams from the substring of our string. This substring is the subproblem string, namely, from the second character until the end. For example, if the string is "hello", the substring is "ello":

```
substring_anagrams = anagrams_of(string[1, string.length - 1])
```

Note how we use the top-down mentality to assume that the anagrams_of function already works on the substring.

We then iterate over each of the substring_anagrams:

```
substring_anagrams.each do |substring_anagram|
```

Before moving on, it's worth noting at this point that we are using a combination of loops and recursion together. Using recursion doesn't mean that you *have* to eliminate loops from your code altogether! We're using each tool in the way that most naturally helps us solve the problem at hand.

For each substring anagram, we iterate over all of its indexes, make a copy of the substring anagram, and insert the first character of our string (the only character not contained within the substring) into that index. Each time we do this, we have created a new anagram, which we then add to our collection:

```
(0..substring_anagram.length).each do |index|
  copy = String.new(substring_anagram)
  collection << copy.insert(index, string[0])
end
```

When we're done, we return the collection of anagrams.

The base case is where the substring contains only one character, in which case there's only one anagram—the character itself!

The Efficiency of Anagram Generation

As an aside, let's stop for a moment to analyze the efficiency of our anagram-generating algorithm, since we'll discover something interesting. In fact, the time complexity of generating anagrams is a new category of Big O that we haven't encountered before.

If we think about how many anagrams we generate, we'll notice an interesting pattern.

For a string containing three characters, we create permutations that start with each of the three characters. Each permutation then picks its middle character from one of the two remaining characters, and its last character from the last character that's left. This is 3 * 2 * 1, which is six permutations.

Looking at this for other string lengths, we get:

```
4 characters: 4 * 3 * 2 * 1         anagrams
5 characters: 5 * 4 * 3 * 2 * 1     anagrams
6 characters: 6 * 5 * 4 * 3 * 2 * 1 anagrams
```

Do you recognize this pattern? It's a factorial!

That is, if the string has six characters, the number of anagrams is whatever the factorial of 6 is. This is `6 * 5 * 4 * 3 * 2 * 1`, which computes to 720.

The mathematical symbol for factorial is the exclamation point. So, factorial 6 is expressed as 6!, and the factorial of 10 is expressed as 10!.

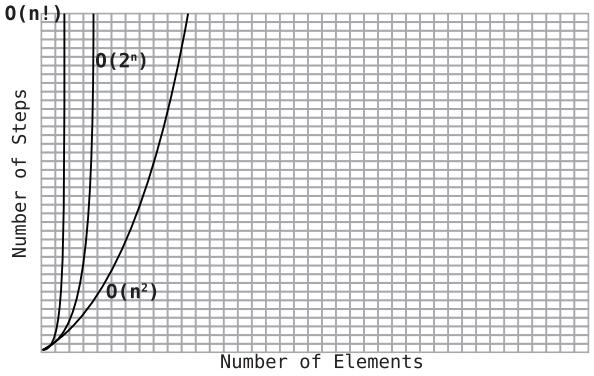
Remember that Big O expresses the answer to the key question: if there are N data elements, how many steps will the algorithm take? In our case, N would be the length of the string.

For a string of length N, we produce N! anagrams. In Big O Notation then, this is expressed as *O(N!)*. This is also known as *factorial time.*

O(N!) is the slowest category of Big O we'll encounter in this book. Let's see how it looks compared to other "slow" Big O categories in the graph on page 181.

Although O(N!) is extremely slow, we don't have a better option here, since our task is to generate *all* the anagrams, and there simply are N! anagrams for an N-character word.

In any case, recursion played a pivotal role in this algorithm, which is an important example of how recursion can be used to solve a complex problem.



Wrapping Up

Learning to write functions that use recursion certainly takes practice. But you're now armed with tricks and techniques that will make the learning process easier for you.

We're not done with our journey through recursion just yet, though. While recursion is a great tool for solving a variety of problems, it can actually slow your code down *a lot* if you're not careful. In the next chapter, you'll learn how to wield recursion while still keeping your code nice and speedy.

Exercises

The following exercises provide you with the opportunity to practice with recursion. The solutions to these exercises are found in the section, Chapter 11, on page 448.

1. Use recursion to write a function that accepts an array of strings and returns the total number of characters across all the strings. For example, if the input array is `["ab", "c", "def", "ghij"]`, the output should be `10` since there are 10 characters in total.

2. Use recursion to write a function that accepts an array of numbers and returns a new array containing just the even numbers.

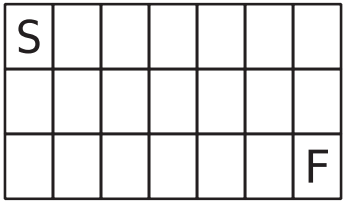
3. There is a numerical sequence known as "Triangular Numbers." The pattern begins as 1, 3, 6, 10, 15, 21, and continues onward with the Nth number in the pattern, which is N plus the previous number. For example, the 7th number in the sequence is 28, since it's 7 (which is N) plus 21 (the previous number in the sequence). Write a function that accepts a

number for N and returns the correct number from the series. That is, if the function was passed the number 7, the function would return 28.

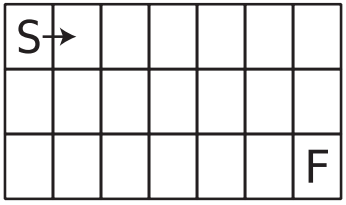
4. Use recursion to write a function that accepts a string and returns the first index that contains the character "x." For example, the string, "abcdefghijklmnopqrstuvwxyz" has an "x" at index 23. To keep things simple, assume the string *definitely* has at least one "x."

5. This problem is known as the "Unique Paths" problem: Let's say you have a grid of rows and columns. Write a function that accepts a number of rows and a number of columns, and calculates the number of possible "shortest" paths from the upper-leftmost square to the lower-rightmost square.

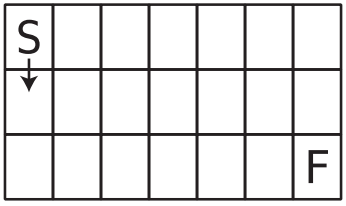
   For example, here's what the grid looks like with three rows and seven columns. You want to get from the "S" (Start) to the "F" (Finish).



   By "shortest" path, I mean that at every step, you're moving either one step to the right:



   or one step downward:



   Again, your function should calculate the *number* of shortest paths.

CHAPTER 12

Dynamic Programming

In the previous chapter, you learned how to write recursively and how to use recursion to solve some rather complex problems.

However, while recursion can certainly solve *some* problems, it can also create *new* ones if not used properly. In fact, recursion is often the culprit behind some of the slowest categories of Big O, such as $O(2^N)$.

The good news, though, is that many of these problems can be avoided. In this chapter, you'll learn how to identify some of the most common speed traps found in recursive code and how to express such algorithms in terms of Big O. More important, you'll learn the techniques to fix these problems.

Here's some more good news: the techniques found in this chapter are actually pretty simple. Let's take a look at how to use these easy but effective methods for turning our recursive nightmares into recursive bliss.

Unnecessary Recursive Calls

Here's a recursive function that finds the greatest number from an array:

```
def max(array)

  # Base case - if the array has only one element, it is
  # by definition the greatest number:

  return array[0] if array.length == 1

  # Compare the first element with the greatest element
  # from the remainder of the array. If the first element
  # is greater, return it as the greatest number:
  if array[0] > max(array[1, array.length - 1])
    return array[0]
```

```
  # Otherwise, return the greatest number from the remainder of the array:
  else
    return max(array[1, array.length - 1])
  end
end
```

The essence of each recursive call is the comparison of a single number (`array[0]`) to the maximum number from the remainder of the array. (To calculate the maximum number from the remainder of the array, we call the very `max` function we're in, which is what makes the function recursive.)

We achieve the comparison with a conditional statement. The first half of the conditional statement is:

```
if array[0] > max(array[1, array.length - 1])
    return array[0]
```

This snippet says that if the single number (`array[0]`) is greater than what has already been determined to be the maximum number of the rest of the array (`max(array[1, array.length - 1])`), then by definition, `array[0]` must be the greatest number, so we return it.

The second half of the conditional statement is:

```
else
  return max(array[1, array.length - 1])
```

This second snippet says that if `array[0]` is *not* greater than the greatest number from the rest of the array, then the greatest number from the rest of the array must be the greatest number overall, and we return it.

While this code works, it contains a hidden inefficiency. If you look carefully, you'll note that our code contains the phrase, `max(array[1, array.length - 1])` twice, once in each half of the conditional statement.

The problem with this is that each time we mention `max(array[1, array.length - 1])`, we trigger an entire avalanche of recursive calls.

Let's break this down for an example array of `[1, 2, 3, 4]`.

We know that we're going to start by comparing the 1 with the maximum number of the remaining array, `[2, 3, 4]`. That, in turn, will compare the 2 against the max of the remaining `[3, 4]`, which in turn will compare the 3 against the `[4]`. This, too, triggers one more recursive call on the `[4]` itself, which is the base case.

However, to really see how our code plays out, we're going to start by analyzing the "bottom" call and working our way up the call chain.

Let's begin.

Max Recursive Walk-Through

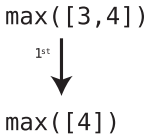
When we call max([4]), the function simply returns the number 4. Again, this is because our base case is when the array only contains one element, as dictated by the following line of code:

```
return array[0] if array.length == 1
```

This is pretty straightforward—it's a single function call:

```
max([4])
```

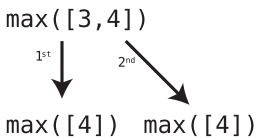
Moving up the call chain, let's see what happens when we call max([3, 4]). In the first half of the conditional statement (if array[0] > max(array[1, array.length - 1])), we compare the 3 to max([4]). But calling max([4]) is itself a recursive call. The following diagram depicts max([3, 4]) calling max([4]):



Note that next to the arrow, we put the label "1st" to indicate that this recursive call was triggered by the *first* half of the conditional statement within max([3, 4]).

After this step has been completed, our code can now compare the 3 with the result of max([4]). Since the 3 is not greater than that result (4), we trigger the second half of the conditional. (This is the code, return max(array[1, array.length - 1]).) In this case, we return max([4]).

But when our code returns max([4]), it triggers the actual function call of max([4]). This is now the second time we've triggered the max([4]) call:



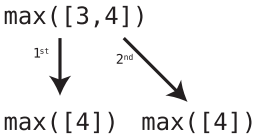
As you can see, the function, max([3, 4]) ends up calling max([4]) twice. Of course, we'd rather try to avoid doing this if we don't have to. If we've already computed

the result of max([4]) once, why should we call the same function again just to get the same result?

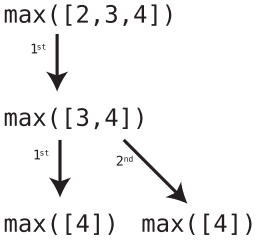
However, this problem gets a lot worse when we move just one level up the call chain.

Here's what happens when we call max([2, 3, 4]).

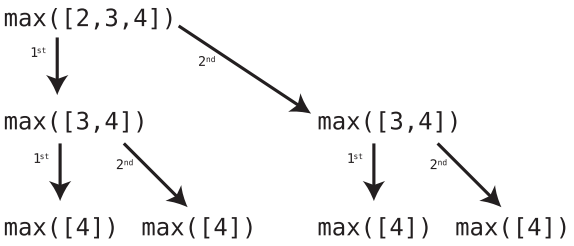
During the first half of the conditional, we compare the 2 against max([3, 4]), which we've already determined looks like this:



So, max([2, 3, 4]) calling max([3, 4]) then, would look like this:

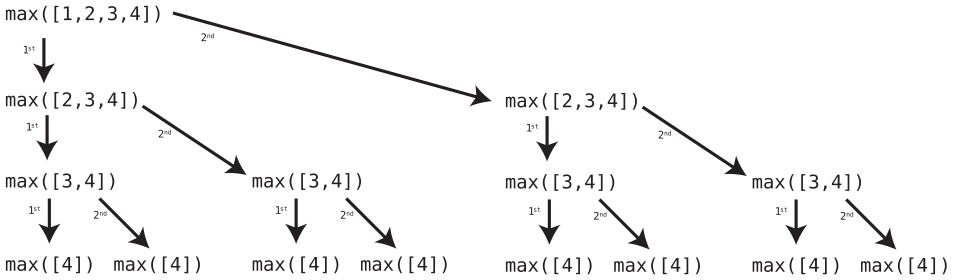


But here's the kicker. This is just for the *first* half of the conditional of max([2, 3, 4]). For the second half of the conditional, we end up calling max([3, 4]) *again:*



Yikes!

Let's dare to move to the top of the call chain, calling max([1, 2, 3, 4]). When all is said and done, after we call max for both halves of the conditional, we get what is shown in the diagram on page 187.



So, when we call `max([1, 2, 3, 4])`, we actually end up triggering the `max` function 15 times.

We can see this visually by adding the statement, `puts "RECURSION"` to the beginning of our function:

```
def max(array)
  puts "RECURSION"

  # remaining code omitted for brevity
```

When we then run our code, we'll see the word, RECURSION, printed to our terminal 15 times.

Now, we do need *some* of those calls, but not all of them. We do need to calculate `max([4])`, for example, but one such function call is enough to get the computed result. But here, we call that function *eight* times.

The Little Fix for Big O

Thankfully, there's an easy way to eliminate all these extra recursive calls. We'll call `max` only once within our code, and *save* the result to a variable:

```
def max(array)

  return array[0] if array.length == 1

  # Calculate the max of the remainder of the array
  # and store it inside a variable:

  max_of_remainder = max(array[1, array.length - 1])

  # Comparison of first number against this variable:

  if array[0] > max_of_remainder
    return array[0]
  else
    return max_of_remainder
  end
end
```

By implementing this simple modification, we end up calling `max` a mere four times. Try it out yourself by adding the `puts "RECURSION"` line and running the code.

The trick here is that we're making each necessary function call once, and saving the result in a variable so that we don't have to ever call that function again.

The difference in efficiency between our initial function and our ever-so-slightly modified function is quite stark.

The Efficiency of Recursion

In our second, improved version of the `max` function, the function recursively calls itself as many times as there are values in the array. We'd call this O(N).

Up until this point, the cases of O(N) we've seen involved loops, with a loop running N times. However, we can apply the same principles of Big O to recursion as well.

As you'll recall, Big O answers the key question: if there are N data elements, how many steps will the algorithm take?

Since the improved `max` function runs N times for N values in the array, it has a time complexity of O(N). Even if the function itself contains multiple steps, such as five, its time complexity would be O(5N), which is reduced to O(N).

In the first version, though, the function called itself *twice* during each run (save for the base case). Let's see how this plays out for different array sizes.

The following table shows how many times `max` gets called on arrays of various sizes:

N Elements	Number of Calls
1	1
2	3
3	7
4	15
5	31

Can you see the pattern? When we increase the data by one, we roughly *double* the number of steps the algorithm takes. As you learned by our discussion of the Password Cracker, on page 107, this is the pattern of $O(2^N)$. We already know that this is an extremely slow algorithm.

The improved version of the max function, however, only calls max for as many elements as there are inside the array. This means that our second max function has an efficiency of O(N).

This is a powerful lesson: avoiding extra recursive calls is key to keeping recursion fast. What at first glance was a very small change to our code—the mere storing of a computation in a variable—ended up changing the speed of our function from $O(2^N)$ to O(N).

Overlapping Subproblems

The Fibonacci sequence is a mathematical sequence of numbers that goes like this until infinity:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55...

This sequence begins with the numbers 0 and 1, and each subsequent number is the sum of the previous two numbers of the sequence. For example, the number 55 was computed because it is the sum of the previous two numbers, which are 21 and 34.

The following Python function returns the Nth number in the Fibonacci sequence. For example, if we pass the number 10 to the function, it will return 55, as 55 is the tenth number in the series. (The 0 is considered the 0th number in the series.)

```
def fib(n):
  # The base cases are the first two numbers in the series:
  if n == 0 or n == 1:
    return n

  # Return the sum of the previous two Fibonacci numbers:
  return fib(n - 2) + fib(n - 1)
```

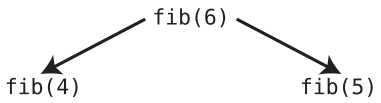
The key line in this function is:

```
return fib(n - 2) + fib(n - 1)
```

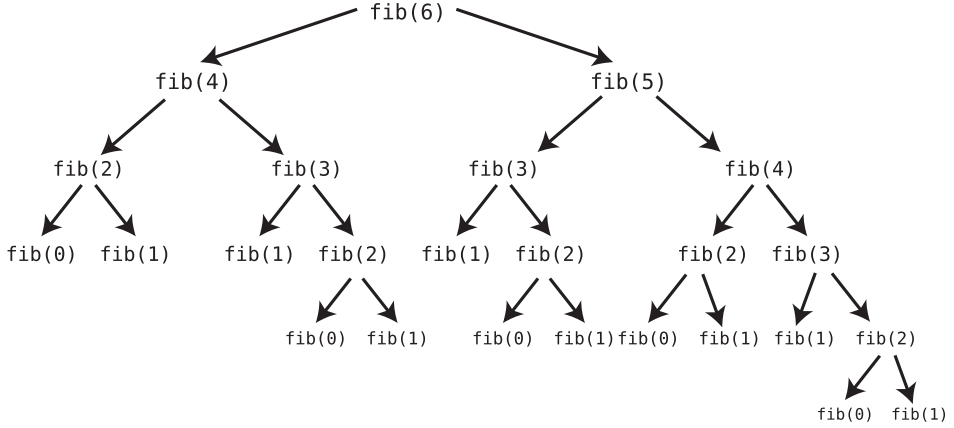
which sums the previous two numbers in the Fibonacci series. It's a beautiful recursive function.

However, alarm bells should be going off in your head right now because our function calls itself *twice*.

Let's take the computation of the sixth Fibonacci number, for example. The function fib(6) makes a call to both fib(4) and fib(5), as shown in the diagram on page 190.



As we've seen, a function calling itself twice can easily lead us down the road to $O(2^N)$. Indeed, here are all the recursive calls made by `fib(6)`:



You've got to admit, $O(2^N)$ can look pretty scary.

While one simple change worked to optimize the first example in this chapter, optimizing our Fibonacci sequence isn't as simple.

And that's because there isn't just one single piece of data we can save in a variable. We *need* to calculate both `fib(n - 2)` and `fib(n - 1)` (as each Fibonacci number is the sum of those two numbers), and storing the result of one won't alone give us the result for the other.

This is a case of what computer scientists call *overlapping subproblems.* Let's unpack that term.

When a problem is solved by solving smaller versions of the same problem, the smaller problem is called a *subproblem.* This concept isn't new—we've been dealing with it frequently throughout our discussion of recursion. In the case of the Fibonacci sequence, we compute each number by first computing the smaller numbers in the sequence. The computation of these smaller numbers are the subproblems.

What makes these subproblems *overlapping,* though, is the fact that `fib(n - 2)` and `fib(n - 1)` end up calling many of the same functions as each other. That is, `fib(n - 1)` ends up making some of the very same calculations previously made by `fib(n - 2)`. For example, you can see in the previous diagram that both `fib(4)` and `fib(5)` call `fib(3)` (as well as many other duplicated calls).

Well, it seems that we're at a dead end. Our Fibonacci example requires us to make many overlapping function calls, and our algorithm oozes along at a pace of $O(2^N)$. There's just nothing we can do.

Or is there?

Dynamic Programming through Memoization

Luckily, we *do* have options, and that is through something called dynamic programming. *Dynamic programming* is the process of optimizing recursive problems that have overlapping subproblems.

(Don't pay too much attention to the word "dynamic." There's some debate about how the term came about, and there's nothing obviously dynamic about the techniques I'm about to demonstrate.)

Optimizing an algorithm with dynamic programming is typically accomplished with one of two techniques.

The first technique is something called *memoization.* And no, that's not a typo. Pronounced meh-moe-ih-ZAY-shun, memoization is a simple, but brilliant technique for reducing recursive calls in cases of overlapping subproblems.

Essentially, memoization reduces recursive calls by *remembering* previously computed functions. (In this respect, memoization really is like its similar-sounding word memorization.)

In our Fibonacci example, the first time `fib(3)` is called, the function does its computation and returns the number 2. However, before moving on, the function stores this result inside a hash table. The hash table will look something like this:

```
{3: 2}
```

This indicates that the result of `fib(3)` is the number 2.

Similarly, our code will memoize the results of all new computations it encounters. After encountering `fib(4)`, `fib(5)`, and `fib(6)`, for example, our hash table will look like this:

```
{
  3: 2,
  4: 3,
  5: 5,
  6: 8
}
```

Now that we have this hash table, we can use it to prevent future recursive calls. Here's the way this works:

Without memoization, `fib(4)` would normally call `fib(3)` and `fib(2)`, which in turn make their own recursive calls. Now that we have this hash table, we can approach things differently. Instead of `fib(4)` just blithely calling `fib(3)`, for example, it first checks the hash table to see if the result of `fib(3)` has already been computed. Only if the `3` key is *not* in the hash table does the function proceed to call `fib(3)`.

Memoization goes for the jugular of overlapping subproblems. The whole issue with overlapping subproblems is that we end up computing the same recursive calls over and over again. With memoization, though, each time we make a new calculation, we store it in the hash table for future use. This way, we only make a calculation if it hadn't ever been made before.

Okay, this all sounds good, but there's one glaring problem. How does each recursive function get access to this hash table?

The answer is: we pass the hash table as a second parameter to the function.

Because the hash table is a specific object in memory, we're able to pass it from one recursive call to the next, even though we're modifying it as we go. This is true even as we unwind the call stack. Even though the hash table may have been empty when the original call was made, that same hash table can be full of data by the time the original call has finished executing.

Implementing Memoization

To pass the hash table along, we modify our function to accept *two* arguments, with the hash table as the second. We call this hash table, `memo`, as in memoization:

```
def fib(n, memo):
```

When calling the function the first time, we pass in both the number and an empty hash table:

```
fib(6, {})
```

Each time `fib` calls itself, it also passes along the hash table, which gets filled up along the way.

Here's the rest of the function:

```
def fib(n, memo):
  if n == 0 or n == 1:
    return n
```

```
  # Check the hash table (called memo) to see whether fib(n)
  # was already computed or not:
  if not memo.get(n):

    # If n is NOT in memo, compute fib(n) with recursion
    # and then store the result in the hash table:
    memo[n] = fib(n - 2, memo) + fib(n - 1, memo)

  # By now, fib(n)'s result is certainly in memo. (Perhaps
  # it was there before, or perhaps we just stored it there
  # in the previous line of code. But it's certainly there now.)
  # So let's return it:
  return memo[n]
```

Let's analyze this line by line.

Again, our function now accepts two parameters, namely `n` and the `memo` hash table:

```
def fib(n, memo):
```

We could also have `memo` set to a default value, so we don't have to explicitly pass in an empty hash table the first time we call it:

```
def fib(n, memo={}):
```

In any case, the base cases of 0 and 1 remain the same and are unaffected by memoization.

Before making any recursive calls, our code first checks to see whether `fib(n)` has already been calculated for the given `n`:

```
if not memo.get(n):
```

(If the calculation for `n` is already in the hash table, we simply return the result with `return memo[n]`.)

Only if the calculation for `n` has not yet been made do we proceed with the calculation:

```
memo[n] = fib(n - 2, memo) + fib(n - 1, memo)
```

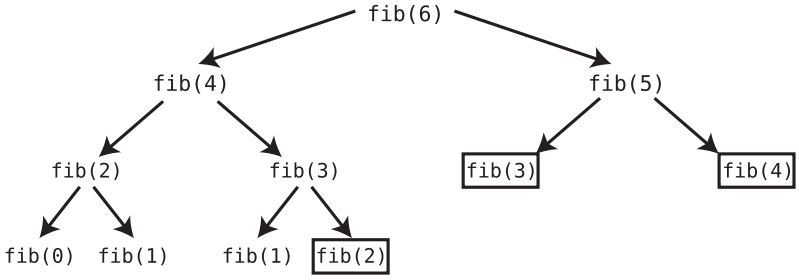
Here, we store the result of the calculation in the `memo` hash table so we never have to calculate it again.

Also note how we pass `memo` along as an argument to the `fib` function each time we call it. This is the key to sharing the `memo` hash table across all the calls to the `fib` function.

As you can see, the guts of the algorithm remain the same. We are still using recursion to solve our problem, as the computation of `fib` is still essentially

fib(n - 2) + fib(n - 1). However, if the number we're computing is new, we store the result in a hash table, and if the number we're computing was already computed once before, we simply grab the result from the hash table instead of computing it again.

When we map out the recursive calls in our memoized version, we get this:



In this diagram, each call that is surrounded by a box is one in which the result was retrieved from the hash table.

So, what is the Big O of our function now? Let's look at how many recursive calls we make for different types of N:

N Elements	Number of Calls
1	1
2	3
3	5
4	7
5	9
6	11

We can see that for N, we make (2N) - 1 calls. Since in Big O we drop the constants, this is an O(N) algorithm.

This is an incredible improvement over $O(2^N)$. Go memoization!

Dynamic Programming through Going Bottom-Up

I mentioned earlier that dynamic programming can be achieved through one of two techniques. We looked at one technique, memoization, which is quite nifty.

The second technique, known as *going bottom-up,* is a lot less fancy and may not even seem like a technique at all. All going bottom-up means is to ditch recursion and use some other approach (like a loop) to solve the same problem.

The reason that going bottom-up is considered part of dynamic programming is because dynamic programming means taking a problem *that could be solved recursively* and ensure that it doesn't make duplicate calls for overlapping subproblems. Using iteration (that is, loops) instead of recursion is, technically, a way to achieve this.

Going bottom-up becomes more of a "technique" when the problem is more naturally solved with recursion. Generating Fibonacci numbers is one example where recursion is a neat, elegant solution. Having to solve the same problem with iteration may take more brainpower, as an iterative approach may be less intuitive. (Imagine solving the Staircase Problem from the previous chapter with a loop. Ugh.)

Let's see how we might implement a bottom-up approach for our Fibonacci function.

Bottom-Up Fibonacci

In the following bottom-up approach, we start with the first two Fibonacci numbers: 0 and 1. We then use good ol' iteration to build up the sequence:

```
def fib(n):

  if n == 0:
    return 0

  # a and b start with the first two numbers in the
  # series, respectively:
  a = 0
  b = 1

  # Loop from 1 until n:
  for i in range(1, n):

    # a and b each move up to the next numbers in the series.
    # Namely, b becomes the sum of b + a, and a becomes what b used to be.
    # We utilize a temporary variable to make these changes:
    temp = a
    a = b
    b = temp + a

  return b
```

Here, we start variables `a` and `b` as 0 and 1, respectively, as those are the first two numbers of the Fibonacci sequence.

We then start a loop so we can calculate each number of the sequence until we reach `n`:

```
for i in range(1, n):
```

To calculate the next number in the sequence, we need to add the two previous numbers. We assign `temp` to the second most recent value, and `a` to the most recent value:

```
temp = a
a = b
```

The new number in the sequence, which we'll reassign to `b`, is the sum of those previous two numbers:

```
b = temp + a
```

Because our code is a simple loop from 1 to N, our code takes N steps. Like the memoization approach, it's O(N).

Memoization vs. Bottom-Up

You've now seen the two primary techniques of dynamic programming: memoization and going bottom-up. Is one technique better than the other?

Usually, it depends on the problem and why you're using recursion in the first place. If recursion presents an elegant and intuitive solution to a given problem, you may want to stick with it and use memoization to deal with any overlapping subproblems. However, if the iterative approach is equally intuitive, you may want to go with that.

It's important to point out that even with memoization, recursion does carry some extra overhead versus iteration. Specifically, with any recursion, the computer needs to keep track of all the calls in a call stack, which consumes memory. The memoization itself also requires the use of a hash table, which will take up additional space on your computer as well. (More on this in *Dealing with Space Constraints*.)

Generally speaking, going bottom-up is often the better choice unless the recursive solution is more intuitive. Where recursion is more intuitive, you can keep the recursion and keep it fast by using memoization.

Wrapping Up

Now that you're able to write efficient recursive code, you've also unlocked a superpower. You're about to encounter some really efficient—yet advanced—algorithms, and many of them rely on the principles of recursion.

Exercises

The following exercises provide you with the opportunity to practice with dynamic programming. The solutions to these exercises are found in the section, Chapter 12, on page 450.

1. The following function accepts an array of numbers and returns the sum, as long as a particular number doesn't bring the sum above 100. If adding a particular number will make the sum higher than 100, that number is ignored. However, this function makes unnecessary recursive calls. Fix the code to eliminate the unnecessary recursion:

```
def add_until_100(array)
  return 0 if array.length == 0
  if array[0] + add_until_100(array[1, array.length - 1]) > 100
    return add_until_100(array[1, array.length - 1])
  else
    return array[0] + add_until_100(array[1, array.length - 1])
  end
end
```

2. The following function uses recursion to calculate the Nth number from a mathematical sequence known as the "Golomb sequence." It's terribly inefficient, though! Use memoization to optimize it. (You don't have to actually understand how the Golomb sequence works to do this exercise.)

```
def golomb(n)
  return 1 if n == 1
  return 1 + golomb(n - golomb(golomb(n - 1)));
end
```

3. Here is a solution to the "Unique Paths" problem from an exercise in the previous chapter. Use memoization to improve its efficiency:

```
def unique_paths(rows, columns)
  return 1 if rows == 1 || columns == 1
  return unique_paths(rows - 1, columns) + unique_paths(rows, columns - 1)
end
```

CHAPTER 13

Recursive Algorithms for Speed

We've seen that understanding recursion unlocks all sorts of new algorithms, such as traversing a filesystem or producing anagrams. In this chapter, you're going to learn that recursion is also the key to algorithms that can make our code run much, much faster.

In previous chapters, we've encountered a number of sorting algorithms, including Bubble Sort, Selection Sort, and Insertion Sort. In real life, however, none of these methods are actually used to sort arrays. Most computer languages have built-in sorting functions for arrays that save us the time and effort of implementing our own. And in many of these languages, the sorting algorithm that is employed under the hood is *Quicksort.*

The reason we're going to dig into Quicksort (even though it's already built into many computer languages) is because by studying how it works, you can learn how to use recursion to greatly speed up an algorithm, and you can do the same for other practical algorithms of the real world.

Quicksort is an extremely fast sorting algorithm that is particularly efficient for average scenarios. While in worst-case scenarios (that is, inversely sorted arrays), it performs similarly to Insertion Sort and Selection Sort, it is much faster for average scenarios—which are what occur most of the time.

Quicksort relies on a concept called *partitioning,* so we'll jump into that first.

Partitioning

To *partition* an array is to take a random value from the array—which is then called the *pivot*—and make sure that every number that is less than the pivot ends up to the left of the pivot, and that every number greater than the pivot ends up to the right of the pivot. We accomplish partitioning through a simple algorithm that will be described in the following example.

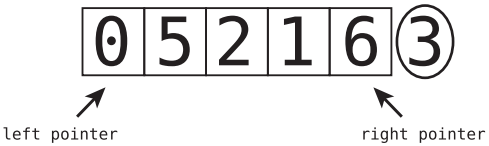
Let's say we have the following array:



For consistency's sake, we'll always select the rightmost value to be our pivot (although we can technically choose other values). In this case, the number 3 is our pivot. We indicate this by circling it:



We then assign "pointers"—one to the left-most value of the array, and one to the rightmost value of the array, excluding the pivot itself:



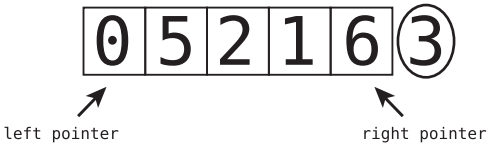
We're now ready to begin the actual partition, which follows these steps. Don't worry—the steps will become clearer when we walk through our example shortly.

1. The left pointer continuously moves one cell to the right until it reaches a value that is greater than or equal to the pivot, and then stops.
2. Then, the right pointer continuously moves one cell to the left until it reaches a value that is less than or equal to the pivot, and then stops. The right pointer will also stop if it reaches the beginning of the array.
3. Once the right pointer has stopped, we reach a crossroads. If the left pointer has reached (or gone beyond) the right pointer, we move on to Step 4. Otherwise, we swap the values that the left and right pointers are pointing to, and then go back to repeat Steps 1, 2, and 3 again.
4. Finally, we swap the pivot with the value that the left pointer is currently pointing to.

When we're done with a partition, we are now assured that all values to the left of the pivot are less than the pivot, and all values to the right of the pivot are greater than it. And that means the pivot itself is now in its correct place within the array, although the other values are not yet necessarily completely sorted.

Let's apply this to our example:

Step 1: Compare the left pointer (now pointing to 0) to our pivot (the value 3):



Since 0 is less than the pivot, the left pointer moves on in the next step.

Step 2: The left pointer moves on:



We compare the left pointer (the 5) to our pivot. Is the 5 lower than the pivot? It's not, so the left pointer stops, and we activate the right pointer in our next step.

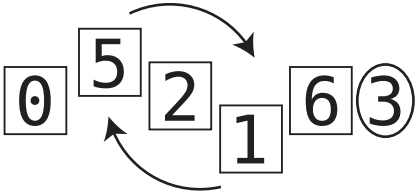
Step 3: Compare the right pointer (6) to our pivot. Is the value greater than the pivot? It is, so our pointer will move on in the next step.

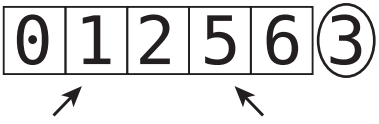
Step 4: The right pointer moves on:



We compare the right pointer (1) to our pivot. Is the value greater than the pivot? It's not, so our right pointer stops.

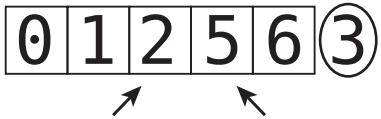
Step 5: Since both pointers have stopped, we swap the values of the two pointers:





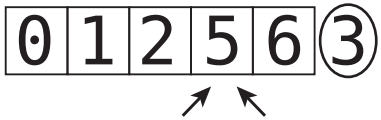
We then activate our left pointer again in the next step.

Step 6: The left pointer moves on:



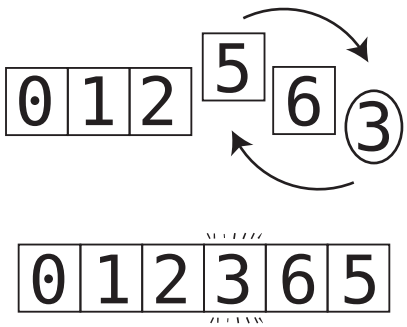
We compare the left pointer (2) to our pivot. Is the value less than the pivot? It is, so the left pointer moves on.

Step 7: The left pointer moves on to the next cell. Note that at this point, both the left and right pointers are pointing to the same value:



We compare the left pointer to our pivot. Because our left pointer is pointing to a value that is greater than our pivot, it stops. At this point, since our left pointer has reached our right pointer, we are done with moving pointers.

Step 8: For our final step of the partition, we swap the value that the left pointer is pointing to with the pivot:



Although our array isn't completely sorted, we have successfully completed a partition. That is, since our pivot was the number 3, all numbers that are less than 3 are to its left, while all numbers greater than 3 are to its right. This also means, by definition, that the 3 is *now in its correct place within the array.*

Code Implementation: Partitioning

Following is an implementation of a `SortableArray` class in Ruby that includes a `partition!` method that partitions the array as we've described:

```
class SortableArray

  attr_reader :array

  def initialize(array)
    @array = array
  end

  def partition!(left_pointer, right_pointer)

    # We always choose the right-most element as the pivot.
    # We keep the index of the pivot for later use:
    pivot_index = right_pointer

    # We grab the pivot value itself:
    pivot = @array[pivot_index]

    # We start the right pointer immediately to the left of the pivot
    right_pointer -= 1

    while true

      # Move the left pointer to the right as long as it
      # points to value that is less than the pivot:
      while @array[left_pointer] < pivot do
        left_pointer += 1
      end

      # Move the right pointer to the left as long as it
      # points to a value that is greater than the pivot:
      while @array[right_pointer] > pivot do
        right_pointer -= 1
      end

      # We've now reached the point where we've stopped
      # moving both the left and right pointers.

      # We check whether the left pointer has reached
      # or gone beyond the right pointer. If it has,
      # we break out of the loop so we can swap the pivot later
      # on in our code:
      if left_pointer >= right_pointer
        break

      # If the left pointer is still to the left of the right
      # pointer, we swap the values of the left and right pointers:
      else
        @array[left_pointer], @array[right_pointer] =
          @array[right_pointer], @array[left_pointer]

        # We move the left pointer over to the right, gearing up
        # for the next round of left and right pointer movements:
        left_pointer += 1
      end

    end
```

```
    # As the final step of the partition, we swap the value
    # of the left pointer with the pivot:
    @array[left_pointer], @array[pivot_index] =
      @array[pivot_index], @array[left_pointer]

    # We return the left_pointer for the sake of the quicksort method
    # which will appear later in this chapter:
    return left_pointer
  end
end
```

Let's break this code down a bit.

The partition! method accepts the starting points of the left and right pointers as parameters:

```
def partition!(left_pointer, right_pointer)
```

When this method is first called on an array, these pointers will point to the left and right ends of the array, respectively. However, we'll see that Quicksort will call this method on subsections of the array as well. Because of this, we can't always assume the left and right pointers are always the two extremities of the array, so they need to become method arguments. This point will be clearer when I explain the complete Quicksort algorithm.

Next, we select our pivot, which is always the rightmost element in the range we're dealing with:

```
pivot_index = right_pointer
pivot = @array[pivot_index]
```

Once our pivot has been identified, we move the right_pointer to the item immediately left of the pivot:

```
right_pointer -= 1
```

We then begin a loop that will run until the left_pointer and right_pointer meet. Within this loop, we use another loop to keep moving the left_pointer to the right until it reaches an item that is greater or equal to the pivot:

```
while @array[left_pointer] < pivot do
  left_pointer += 1
end
```

Similarly, we move the right_pointer to the left until it hits an item that is less or equal to the pivot:

```
while @array[right_pointer] > pivot do
  right_pointer -= 1
end
```

Once the left_pointer and right_pointer have stopped moving, we check whether the two pointers have met:

```
if left_pointer >= right_pointer
  break
```

If they have, we exit the loop and get ready to swap the pivot, which we'll get to momentarily. However, if the two pointers have not yet met, we swap the values at the two pointers:

```
@array[left_pointer], @array[right_pointer] =
  @array[right_pointer], @array[left_pointer]
```

Finally, once the two pointers have met, we swap the pivot with the value at the left_pointer:

```
@array[left_pointer], @array[pivot_index] =
  @array[pivot_index], @array[left_pointer]
```

The method returns the left_pointer, as this will be needed by the Quicksort algorithm (which I'll explain shortly).

Quicksort

The Quicksort algorithm is a combination of partitions and recursion. It works as follows:

1. Partition the array. The pivot is now in its proper place.
2. Treat the subarrays to the left and right of the pivot as their own arrays, and recursively repeat Steps 1 and 2. That means we'll partition each subarray and end up with even smaller sub-subarrays to the left and right of each subarray's pivot. We then partition those sub-subarrays, and so on and so forth.
3. When we have a subarray that has zero or one elements, that is our base case and we do nothing.

Let's return to our example. We began with the array of [0, 5, 2, 1, 6, 3] and ran a single partition on the entire array. Since Quicksort begins with such a partition, that means we're already partly through the Quicksort process. We left off with:



As you can see, the value 3 was the original pivot. Now that the pivot is in the correct place, we need to sort whatever is to the left and right of the pivot. Note that in our example, it just so happens that the numbers to the left of the pivot are already sorted, but the computer doesn't know that yet.

The next step after the partition is to treat everything to the left of the pivot as its own array and partition it.

We'll obscure the rest of the array for now, as we're not focusing on it at the moment:



Now, of this `[0, 1, 2]` subarray, we'll make the rightmost element the pivot. So, that would be the number 2:



We'll establish our left and right pointers:



And now we're ready to partition this subarray. Let's continue from after Step 8, where we left off previously.

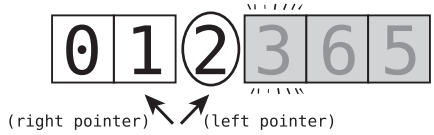
Step 9: We compare the left pointer (0) to the pivot (2). Since the 0 is less than the pivot, we continue moving the left pointer.

Step 10: We move the left pointer one cell to the right, and it now happens to be pointing to the same value the right pointer is pointing to:



We compare the left pointer to the pivot. Since the value 1 is less than the pivot, we move on.

Step 11: We move the left pointer one cell to the right, which just happens to be the pivot:



At this point, the left pointer is pointing to a value that is equal to the pivot (since it *is* the pivot!), and so the left pointer stops.

Note how the left pointer managed to sneak past the right pointer. That's okay, though. The algorithm is designed to work even with such an occurrence.

Step 12: Now, we activate the right pointer. However, because the right pointer's value (1) is less than the pivot, it stays still.

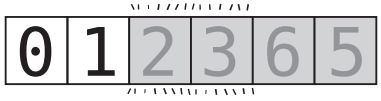
Since our left pointer has passed our right pointer, we're done moving pointers altogether in this partition.

Step 13: Next, we swap the pivot with the left pointer's value. Now, it just so happens that the left pointer is pointing to the pivot itself, so we swap the pivot with itself, which results in no change at all. At this point, the partition is complete and the pivot (2) is now in its correct spot:



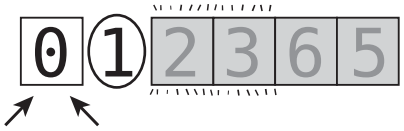
We now have a subarray of `[0, 1]` to the left of the pivot (the 2) and no subarray to its right. The next step is to recursively partition the subarray to the pivot's left, which again, is `[0, 1]`. We don't have to deal with any subarray to the right of the pivot since no such subarray exists.

Because all we'll focus on in the next step is the subarray `[0, 1]`, we'll block out the rest of the array so it looks like this:



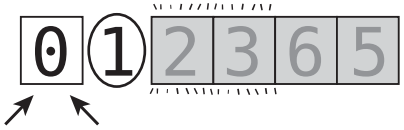
To partition the subarray, `[0, 1]`, we'll make the rightmost element (the 1) the pivot. Where will we put the left and right pointers? Well, the left pointer will

point to the 0, but the right pointer will also point to the 0, since we always start the right pointer at one cell to the left of the pivot. This gives us this:



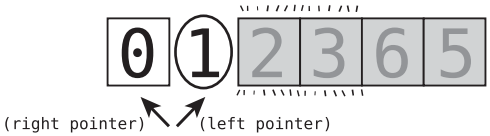
We're now ready to begin the partition.

Step 14: We compare the left pointer (0) with the pivot (1):



It's less than the pivot, so we move on.

Step 15: We shift the left pointer one cell to the right. It now points to the pivot:

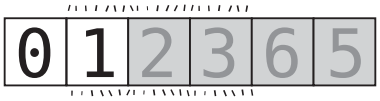


Since the left pointer's value (1) is not lower than the pivot (since it *is* the pivot), the left pointer stops moving.

Step 16: We compare the right pointer with the pivot. Since it's pointing to a value that is less than the pivot, we don't move it anymore. And since the left pointer has passed the right pointer, we're done moving pointers for this partition.

Step 17: We now swap the left pointer with the pivot. Again, in this case, the left pointer is actually pointing to the pivot itself, so the swap doesn't actually change anything. The pivot is now in its proper place, and we're done with this partition.

That leaves us with this:

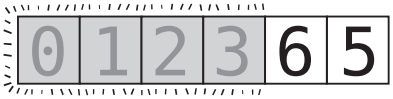


Next up, we need to partition the subarray to the left of the most recent pivot. In this case, that subarray is `[0]`—an array of just one element. An array of zero or one elements is our base case, so we don't do anything. The element is just considered to be in its proper place automatically. So, now we've got:

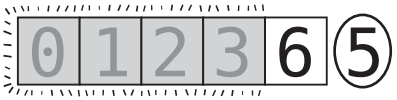


We started out by treating 3 as our pivot, and recursively partitioned the subarray to its left (`[0, 1, 2]`). As promised, we now need to come back to recursively partitioning the subarray to the right of the 3, which is `[6, 5]`.

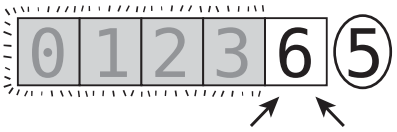
We'll obscure the `[0, 1, 2, 3]`, since we've already sorted those, and now we're only focusing on the `[6, 5]`:



In the next partition, we'll treat the rightmost element (the 5) as the pivot. That gives us:



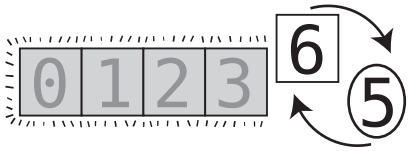
When setting up our next partition, our left and right pointers both end up pointing to the 6:



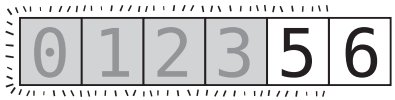
Step 18: We compare the left pointer (6) with the pivot (5). Since 6 is greater than the pivot, the left pointer doesn't move further.

Step 19: The right pointer is pointing to the 6 as well, so we would theoretically move on to the next cell to the left. However, there are no more cells to the left of the 6, so the right pointer stops moving. Since the left pointer has reached the right pointer, we're done moving pointers altogether for this partition. That means we're ready for the final step.

Step 20: We swap the pivot with the value of the left pointer:



Our pivot (5) is now in its correct spot, leaving us with:



Next up, we technically need to recursively partition the subarray to the left and right of the [5, 6] subarray. Since there is no subarray to its left, that means we only need to partition the subarray to the right. Since the subarray to the right of the 5 is a single element of [6], that's our base case and we do nothing—the 6 is automatically considered to be in its proper place:



And we're done!

Code Implementation: Quicksort

Following is a quicksort! method we can add to the previous SortableArray class that would successfully complete the Quicksort:

```
def quicksort!(left_index, right_index)
  # Base case: the subarray has 0 or 1 elements:
  if right_index - left_index <= 0
    return
  end

  # Partition the range of elements and grab the index of the pivot:
  pivot_index = partition!(left_index, right_index)

  # Recursively call this quicksort! method on whatever
  # is to the left of the pivot:
  quicksort!(left_index, pivot_index - 1)

  # Recursively call this quicksort! method on whatever
  # is to the right of the pivot:
  quicksort!(pivot_index + 1, right_index)
end
```

The code here is surprisingly concise, but let's look at each line. For now, we'll skip over the base case.

We begin by partitioning the range of elements between the left_index and right_index:

```
pivot_index = partition!(left_index, right_index)
```

The first time we run quicksort!, we partition the entire array. In subsequent calls, though, this line of code partitions whatever range of elements lie between the left_index and right_index, which may be a subsection of the original array.

Note that we assign the return value of partition! to a variable called pivot_index. If you'll recall, this value was the left_pointer, which pointed to the pivot by the time the partition! method was complete.

We then recursively call quicksort! on the subarrays to the left and right of the pivot:

```
quicksort!(left_index, pivot_index - 1)
quicksort!(pivot_index + 1, right_index)
```

The recursion ends when we reach the base case, which is when the subarray at hand contains no more than one element:

```
if right_index - left_index <= 0
  return
end
```

We can take our Quicksort implementation for a test drive using the following code:

```
array = [0, 5, 2, 1, 6, 3]
sortable_array = SortableArray.new(array)
sortable_array.quicksort!(0, array.length - 1)
p sortable_array.array
```

The Efficiency of Quicksort

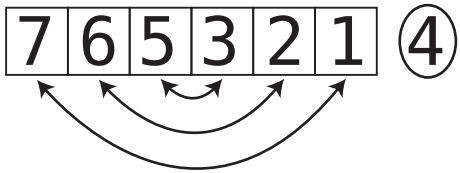
To figure out the efficiency of Quicksort, let's first determine the efficiency of a *single* partition.

When we break down the steps of a partition, we'll note that a partition involves two primary types of steps:

- *Comparisons:* We compare each of the values at hand to the pivot.
- *Swaps:* When appropriate, we swap the values being pointed to by the left and right pointers.

Each partition has at least N comparisons—that is, we compare each element of the array with the pivot. This is true because a partition always has the left and right pointers move through each cell until the left and right pointers reach each other.

The number of swaps, however, will depend upon how the data is sorted. A single partition can have, at most, N / 2 swaps, as even if we'd swap values at every opportunity, each swap takes care of two values. As you can see in the following diagram, we partition six elements in just three swaps:



Now, in most cases, we're not making a swap every step of the way. For *randomly* sorted data, we generally swap about half of the values. On average, then, we're making about N / 4 swaps.

So, on average, we make about N comparisons and N / 4 swaps. We can say, then, that there are about 1.25N steps for N data elements. In Big O Notation, we ignore constants, so we'd say that a partition runs in O(N) time.

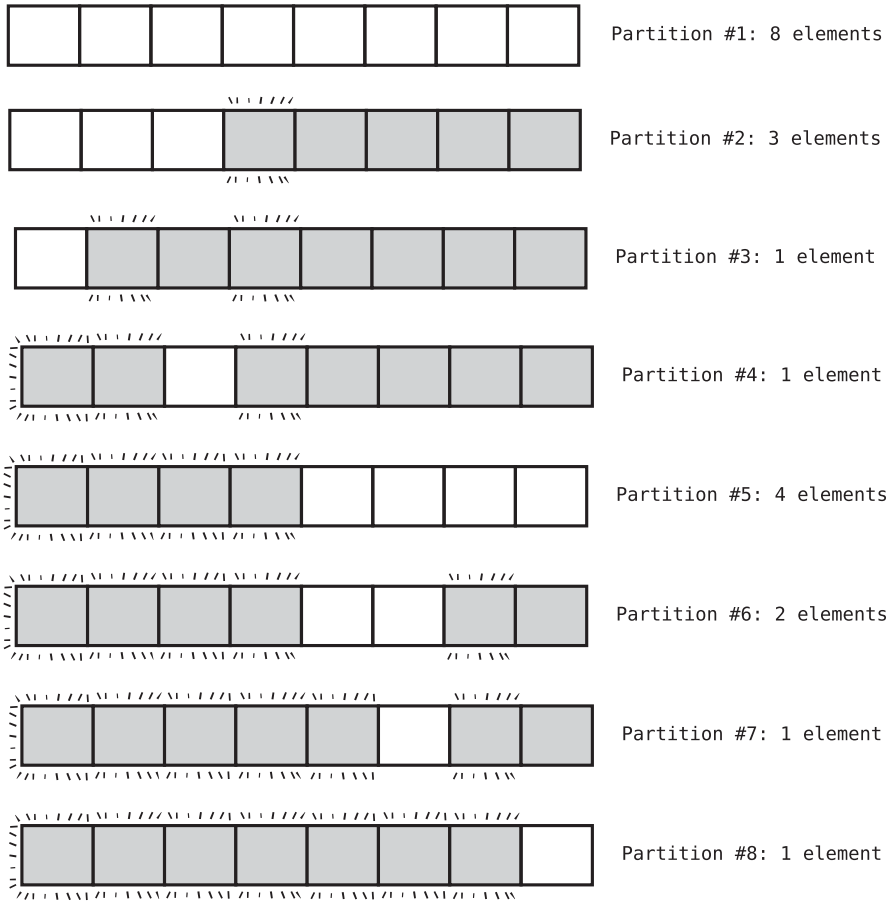
Now, that's the efficiency of a *single* partition. But Quicksort involves *many* partitions, so we need to conduct further analysis to determine the efficiency of Quicksort.

Quicksort from a Bird's-Eye View

To visualize this more easily, here's a diagram on page 213 depicting a typical Quicksort on an array of eight elements from a bird's-eye view. In particular, the diagram shows how many elements each partition acts upon. We've left out the actual numbers from the array since the exact values don't matter. Note that in the diagram, the active subarray is the group of cells that is not grayed out.

We can see that we have eight partitions, but each partition takes place on subarrays of various sizes. We perform a partition on the original array of eight elements, but also perform partitions on subarrays of sizes 4, 3, and 2, and another four partitions on arrays of size 1.

Since Quicksort is essentially comprised of this series of partitions, and each partition takes about N steps for N elements of each subarray, if we add the



sizes of all the subarrays together, we'll get the total number of steps Quicksort takes:

```
  8 elements
  3 elements
  1 element
  1 element
  4 elements
  2 elements
  1 element
+ 1 element
————————
Total = About 21 steps
```

We see that where the original array has eight elements, Quicksort takes about 21 steps. This assumes a best- or average-case scenario, where the pivot ends up roughly in the middle of the subarray after each partition.

For an array of 16 elements, Quicksort takes about 64 steps, and for an array of 32 elements, Quicksort takes about 160 steps. Take a look at this table:

N	Quicksort Steps (approx.)
4	8
8	24
16	64
32	160

(While in our example earlier, the number of Quicksort steps for an array of size 8 was 21, I put 24 in this table. The exact number can vary from case to case, and 24 is also a reasonable approximation. I specifically made it 24 to make the following explanation a little clearer.)

The Big O of Quicksort

How do we categorize Quicksort in terms of Big O Notation?

If we look at the pattern shown earlier, we'll note that the number of Quicksort steps for N elements in the array is about *N multiplied by log N*, as shown in the following table:

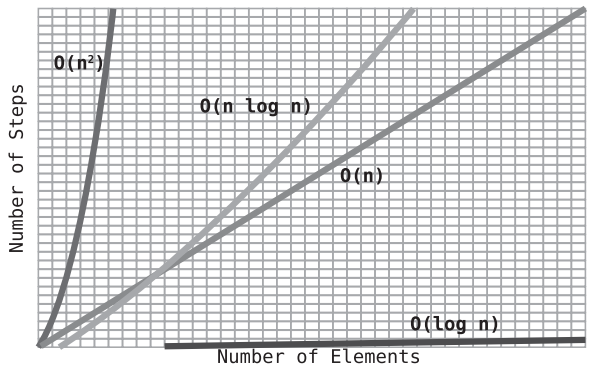
N	log N	N * log N	Quicksort Steps (approx.)
4	2	8	8
8	3	24	24
16	4	64	64
32	5	160	160

In fact, this is exactly how to express the efficiency of Quicksort. It's an algorithm of *O(N log N)*. We've discovered a new category of Big O!

The graph on page 215 shows how O(N log N) looks beside other categories of Big O.

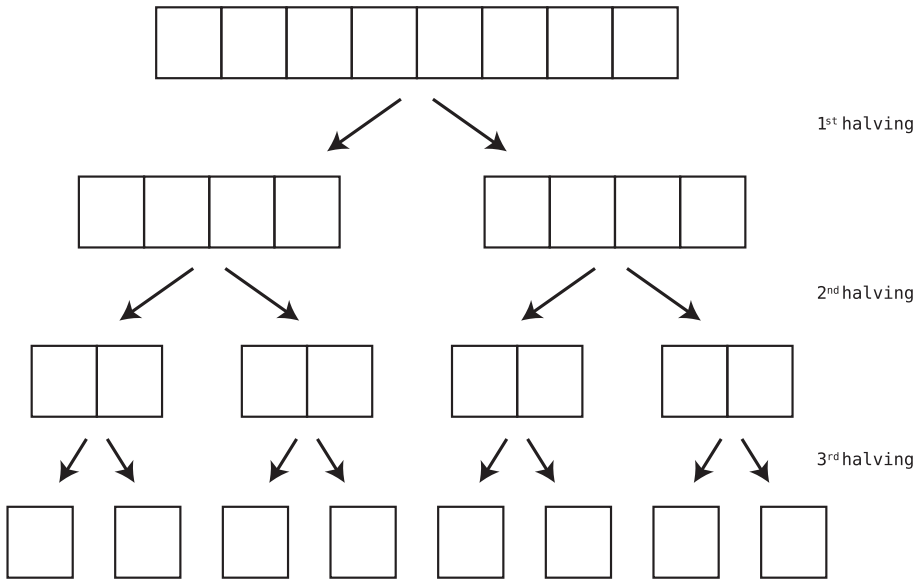
Now, it's not a coincidence that the number of steps in Quicksort just happens to align with N * log N. If we think about Quicksort more broadly, we can see *why* it is this way:

Each time we partition the array, we end up breaking it down into two subarrays. Assuming the pivot ends up somewhere in the middle of the array—which



is what happens in the average case—these two subarrays are of roughly equal sizes.

How many times can we break an array into halves until we've broken it completely down to the point of where each subarray is of size 1? For an array of size N, this will take us log N times. Take a look at the following diagram:



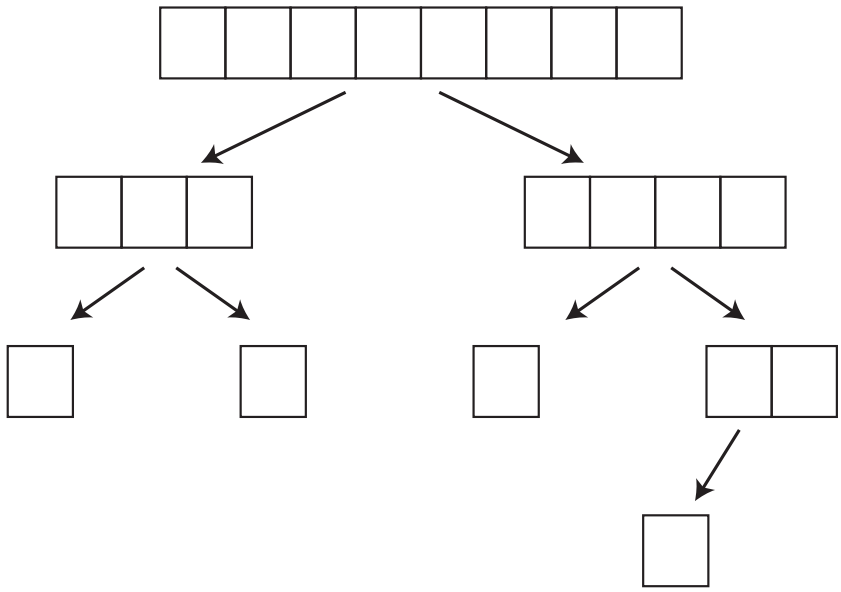
As you can see, for an array of size 8, it takes us three "halvings" until we've reduced the array into eight individual elements. This is log N, and fits with our definition of log N as being the number of times it takes to halve something until we reach 1.

So, this is why Quicksort takes N * log N steps. We have log N halvings, and for each halving, we perform a partition on all the subarrays whose elements add up to N. (They add up to N because all the subarrays are simply pieces of the original array of N elements.)

This is illustrated in the previous diagram. At the top of the diagram, for example, we partition the original array of eight elements, creating two subarrays of size 4. We then partition both subarrays of size 4, which means that we are again partitioning eight elements.

Bear in mind that O(N * log N) is just an approximation. In reality, we first perform an extra O(N) partition on the original array as well. Additionally, an array doesn't cleanly break into two even halves, since the pivot is not part of the "halving."

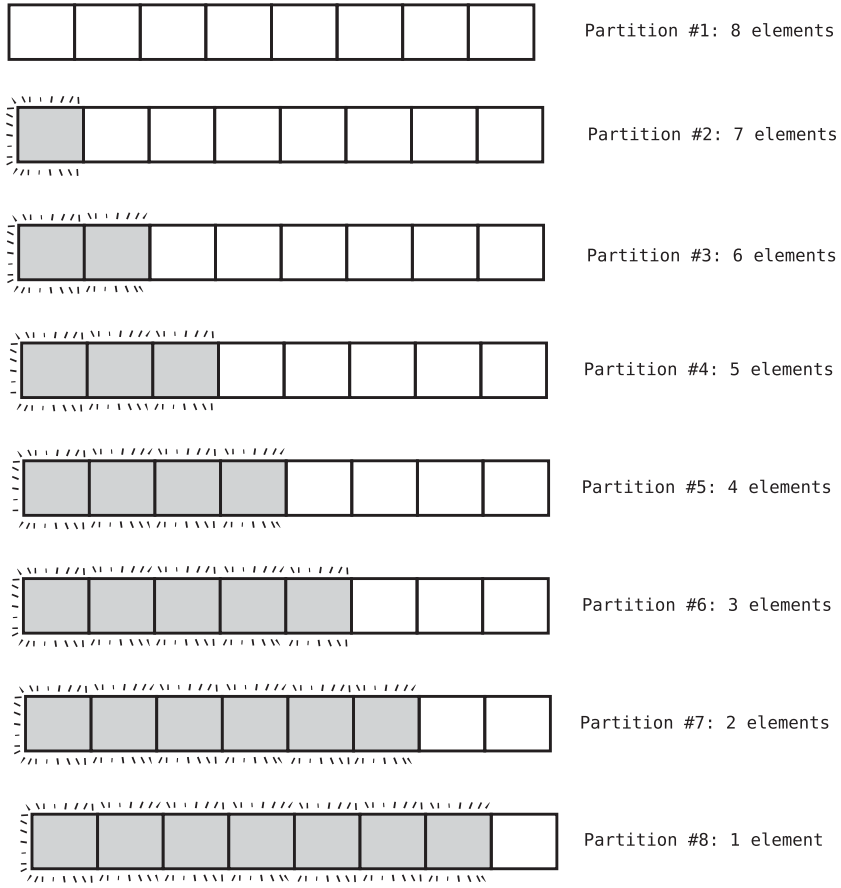
Here's what a more realistic example really looks like, where we ignore the pivot after each partition:



Quicksort in the Worst-Case Scenario

For many other algorithms we've encountered, the best case was one where the array was already sorted. When it comes to Quicksort, however, the best-case scenario is one in which the pivot always ends up smack in the middle of the subarray after the partition. Interestingly, this generally occurs when the values in the array are mixed up pretty well.

The worst-case scenario for Quicksort is one in which the pivot always ends up on one side of the subarray instead of in the middle. This can happen where the array is in perfect ascending or descending order. The visualization for this process is shown here:



In this diagram you can see that the pivot always ends up on the left end of each subarray.

While in this case each partition still involves only one swap, we lose out because of the increased number of comparisons. In the first example, when the pivot always ended up toward the middle, each partition after the first one was conducted on relatively small subarrays (the largest subarray had a size of 4). In this example, however, the first five partitions take place on subarrays of size 4 or greater. And each of these partitions have as many comparisons as there are elements in the subarray.

So, in this worst-case scenario, we have partitions of 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 elements, which yields a total of 36 comparisons.

To put this a little more formulaically, we'd say that for N elements, there are N + (N - 1) + (N - 2) + (N - 3) ... + 1 steps. We saw in our discussion of Get All the Products, on page 103 that this computes to N^2 / 2 steps, which for the purposes of Big O is $O(N^2)$.

So, in a worst-case scenario, Quicksort has an efficiency of $O(N^2)$.

Quicksort vs. Insertion Sort

Now that we've got Quicksort down, let's compare it with one of the simpler sorting algorithms, such as Insertion Sort:

	Best Case	Average Case	Worst Case
Insertion Sort	O(N)	$O(N^2)$	$O(N^2)$
Quicksort	O(N log N)	O(N log N)	$O(N^2)$

We can see they have identical worst-case scenarios, and that Insertion Sort is actually faster than Quicksort in a best-case scenario. However, the reason Quicksort is superior to Insertion Sort is because of the average scenario—which, again, is what happens most of the time. For average cases, Insertion Sort takes a whopping $O(N^2)$, while Quicksort is much faster at O(N log N).

Because of Quicksort's superiority in average circumstances, many programming languages use Quicksort under the hood of their built-in sorting functions. Because of this, it's unlikely you'll be implementing Quicksort yourself. However, there is a very similar algorithm that can come in handy for practical cases—and it's called Quickselect.

Quickselect

Let's say you have an array in random order, and you do not need to sort it, but you do want to know the tenth-lowest value in the array, or the fifth-highest. This can be useful if we had a lot of test grades and want to know what the 25th percentile was, or if we want to find the median grade.

One way to solve this would be to sort the entire array and then jump to the appropriate index.

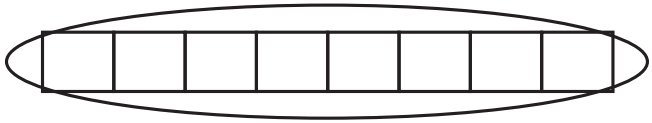
However, even were we to use a fast sorting algorithm like Quicksort, this algorithm would take at least O(N log N) for average cases. And while that isn't bad, we can do even better with a brilliant little algorithm known as

Quickselect. Like Quicksort, Quickselect relies on partitioning, and can be thought of as a hybrid of Quicksort and binary search.

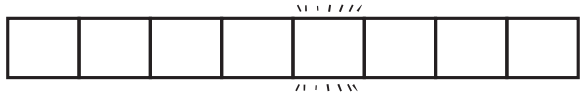
As you've seen earlier in this chapter, after a partition, the pivot value ends up in the appropriate spot in the array. Quickselect takes advantage of this in the following way:

Let's say we have an array of eight values, and we want to find the second-to-lowest value within the array.

First, we partition the entire array:



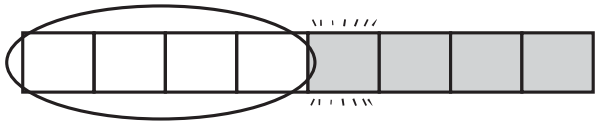
After the partition, the pivot will hopefully end up somewhere toward the middle of the array:



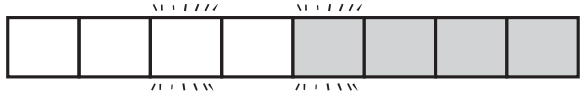
This pivot is now in its correct spot, and since it's in the fifth cell, we now know which value is the fifth-lowest value within the array.

Now, we're looking for the second-lowest value, not the fifth-lowest. But we *do* know that the second-lowest value is *somewhere to the left* of the pivot. We can now ignore everything to the right of the pivot and focus on the left subarray. It is in this respect that Quickselect is similar to binary search: we keep dividing the array in half and focus on the half in which we know the value we're seeking for will be found.

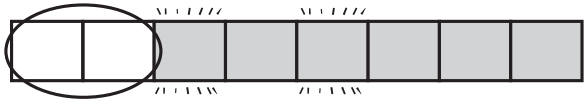
Next, we partition the subarray to the left of the pivot:



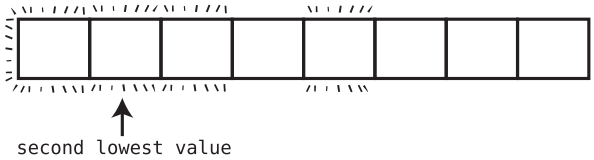
Let's say the new pivot of this subarray ends up the third cell:



We now know that the value in the third cell is in its correct spot, meaning it's the third-to-lowest value in the array. By definition, then, the second-to-lowest value will be somewhere to its left. We can now partition the subarray to the left of the third cell:



After this next partition, the lowest and second-lowest values will end up in their correct spots within the array:



We can then grab the value from the second cell and know with confidence that it's the second-lowest value in the entire array. One of the beautiful things about Quickselect is that we can find the correct value *without having to sort the entire array.*

With Quicksort, each time we halved the array, we needed to re-partition every single element again (in their subarray form), giving us O(N log N). With Quickselect, on the other hand, each time we cut the array in half, we only had to partition the one half we cared about—the half in which we know our value is to be found.

The Efficiency of Quickselect

When analyzing the efficiency of Quickselect, we'll see that it's O(N) for average scenarios. Why is this?

In our earlier example of an array of eight elements, we executed three partitions: one on an array of eight elements, one on a subarray of four elements, and one on a subarray of two elements.

Recall that each partition takes about N steps for the subarray it's run upon. The total steps, then, for the three partitions is 8 + 4 + 2 = 14 steps. So, an array of eight elements yields roughly 14 steps.

For an array of 64 elements, we run about 64 + 32 + 16 + 8 + 4 + 2 = 126 steps. For 128 elements, we would need about 254 steps. And for 256 elements, we would end up with 510 steps.

We can see that we need about 2N steps for N elements in the array.

(Another way to formulate this is to say that for N elements, we would need N + (N/2) + (N/4) + (N/8) + … 2 steps. This always turns out to be roughly 2N steps.)

Since Big O ignores constants, we drop the 2 from the 2N and say that Quickselect has an efficiency of O(N).

Code Implementation: Quickselect

Following is an implementation of a quickselect! method that can be dropped into the SortableArray class described earlier. You'll note that it's very similar to the quicksort! method:

```
def quickselect!(kth_lowest_value, left_index, right_index)
  # If we reach a base case - that is, that the subarray has one cell,
  # we know we've found the value we're looking for:
  if right_index - left_index <= 0
    return @array[left_index]
  end

  # Partition the array and grab the index of the pivot:
  pivot_index = partition!(left_index, right_index)

  # If what we're looking for is to the left of the pivot:
  if kth_lowest_value < pivot_index
    # Recursively perform quickselect on the subarray to
    # the left of the pivot:
    quickselect!(kth_lowest_value, left_index, pivot_index - 1)
  # If what we're looking for is to the right of the pivot:
  elsif kth_lowest_value > pivot_index
    # Recursively perform quickselect on the subarray to
    # the right of the pivot:
    quickselect!(kth_lowest_value, pivot_index + 1, right_index)
  else # if kth_lowest_value == pivot_index
    # if after the partition, the pivot position is in the same spot
    # as the kth lowest value, we've found the value we're looking for
    return @array[pivot_index]
  end
end
```

If you want to find the second-to-lowest value of an unsorted array, you'd run the following code:

```
array = [0, 50, 20, 10, 60, 30]
sortable_array = SortableArray.new(array)
p sortable_array.quickselect!(1, 0, array.length - 1)
```

The first argument of the quickselect! method accepts the position you're looking for, starting at index 0. We've put in a 1 to represent the second-to-lowest value. The second and third values are the left and right indexes of the array, respectively.

Sorting as a Key to Other Algorithms

As of this writing, the fastest sorting algorithms we know of have speeds of O(N log N). While Quicksort is one of the most popular among them, there are many others as well. Mergesort is another well-known O(N log N) sorting algorithm, and I recommend you look it up, as it is a beautiful recursive algorithm.

The fact that the fastest sorting algorithms are O(N log N) is quite important, as this has implications for other algorithms as well. This is because there are algorithms that use sorting as a component of a larger process.

For example, if you'll recall from *Speeding Up Your Code with Big O*, we dealt with the problem of checking whether there are duplicate values within an array.

The first solution we looked at involved nested loops and had an efficiency of $O(N^2)$. While we found a solution that took O(N), I parenthetically hinted at a disadvantage with that approach having to do with extra memory consumption. (I'll eventually discuss this at length in *Dealing with Space Constraints*.) So, let's assume that the O(N) approach is out. Are there any other ways we can improve upon the quadratic $O(N^2)$ solution? Hint: the solution has something to do with sorting!

We can actually build a beautiful algorithm if we presort the array.

Let's say the original array was `[5, 9, 3, 2, 4, 5, 6]`. There are two instances of `5`, so we do have a case of duplicates.

Now, if we first sorted this array, it would become `[2, 3, 4, 5, 5, 6, 9]`.

Next, we can use a single loop to iterate over each number. As we inspect each number, we'd check whether it's identical to the *next* number. If it is, we've found a duplicate. If we get to the end of the loop without finding a duplicate, then we know that we have no duplicates.

The trick here is that by presorting the numbers, we end up bunching duplicate numbers together.

In our example, we'd start by looking at the first number, which is the `2`. We'd check whether it's identical to the next number. The next number is `3`, so they're not duplicates.

We'd then check the `3` against the following number, which happens to be a `4`, allowing us to move on. We'd check the `4` against the `5`, and we'd move on again.

At this point, we'd inspect the first `5` and check it against the following number, which is the second `5`. Aha! We found a pair of duplicate numbers, and we can return `true`.

Here's a JavaScript implementation of this:

```
function hasDuplicateValue(array) {
  // Presort the array:
  // (In JavaScript, the following usage of the sort function
  // is required to ensure that the numbers are in numerical order,
  // instead of "alphabetical" order.)
  array.sort((a, b) => (a < b) ? -1 : 1);

  // Iterate through the values in the array up until the last one:
  for(let i = 0; i < array.length - 1; i++) {

    // If the value is identical to the next value
    // in the array, we found a duplicate:
    if(array[i] == array[i + 1]) {
      return true;
    }
  }

  // If we get to the end of the array without having
  // returned true, it means there are no duplicates:
  return false;
}
```

Now, this is an algorithm that used sorting as one of its components. What is the Big O of this algorithm?

We start out by sorting the array. We can assume that JavaScript's sort() function has an efficiency of O(N log N). Next, we spend up to N steps iterating through the array. Our algorithm, then, takes (N log N) + N steps.

You learned that when we have multiple orders added together, Big O Notation keeps only the highest order of N, because the lower orders are trivial beside the higher orders. Here as well, N is trivial beside N log N, so we reduce the expression to O(N log N).

So, that's it! We've used sorting to develop an algorithm that is O(N log N), which is a significant improvement over the original $O(N^2)$ algorithm.

Plenty of algorithms employ sorting as part of a larger process. We now know that any time we do so, we have an algorithm that is at *least* O(N log N). Of course, the algorithm may be slower than this if it has other things going on, but we know that O(N log N) will always be the baseline.

Wrapping Up

The Quicksort and Quickselect algorithms are recursive algorithms that present beautiful and efficient solutions to thorny problems. They're great examples of how a non-obvious but well-thought-out algorithm can boost performance.

Now that we've seen some more advanced algorithms, we're now going to head in a new direction and explore a trove of additional data structures. Some of these data structures have operations that involve recursion, so we'll be fully prepared to tackle those now. Besides for being really interesting, we'll see that each data structure has a special power that can bring significant advantages to a variety of applications.

Exercises

The following exercises provide you with the opportunity to practice with fast sorting. The solutions to these exercises are found in the section, Chapter 13, on page 451.

1. Given an array of positive numbers, write a function that returns the greatest product of any three numbers. The approach of using three nested loops would clock in at $O(N^3)$, which is very slow. Use sorting to implement the function in a way that it computes at O(N log N) speed. (There are even faster implementations, but we're focusing on using sorting as a technique to make code faster.)

2. The following function finds the "missing number" from an array of integers. That is, the array is expected to have all integers from 0 up to the array's length, but one is missing. As examples, the array, `[5, 2, 4, 1, 0]` is missing the number 3, and the array, `[9, 3, 2, 5, 6, 7, 1, 0, 4]` is missing the number 8.

   Here's an implementation that is $O(N^2)$ (the `includes` method alone is already O(N), since the computer needs to search the entire array to find `n`):

   ```
   function findMissingNumber(array) {
     for(let i = 0; i < array.length; i++) {
       if(!array.includes(i)) {
         return i;
       }
     }

     // If all numbers are present:
     return null;
   }
   ```

   Use sorting to write a new implementation of this function that only takes O(N log N). (There are even faster implementations, but we're focusing on using sorting as a technique to make code faster.)

3. Write three different implementations of a function that finds the greatest number within an array. Write one function that is $O(N^2)$, one that is O(N log N), and one that is O(N).

CHAPTER 14

Node-Based Data Structures

For the next several chapters, we're going to explore a variety of data structures that all build upon a single concept—the *node.* As you'll see shortly, nodes are pieces of data that may be dispersed throughout the computer's memory. Node-based data structures offer new ways to organize and access data that provide a number of major performance advantages.

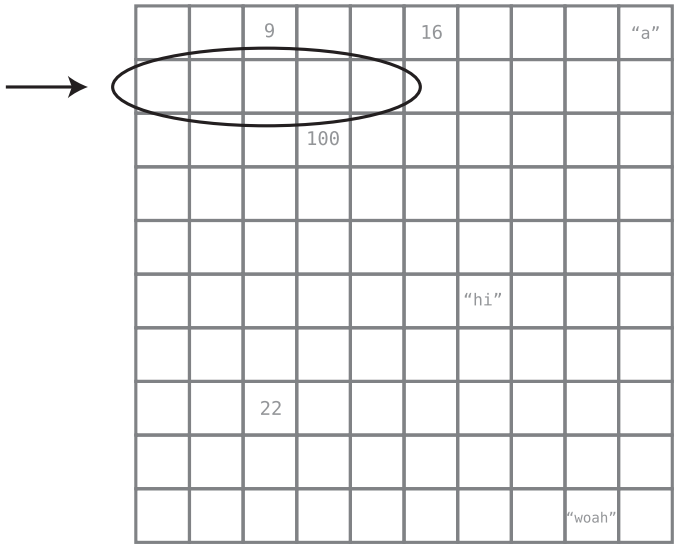
In this chapter, we will explore the linked list, which is the simplest node-based data structure and the foundation for future chapters. You'll also discover linked lists seem almost identical to arrays, but come with their own set of trade-offs in efficiency that can give us a performance boost for certain situations.

Linked Lists

Like an array, a *linked list* is a data structure that represents a list of items. While on the surface, arrays and linked lists look and act quite similarly, there are big differences under the hood.

As mentioned in *Why Data Structures Matter*, memory inside a computer can be visualized as a giant set of cells in which bits of data are stored. You learned that when creating an array, your code finds a contiguous group of empty cells in memory and designates them to store data for your application as shown on page 226.

You also saw that the computer has the ability to access any memory address in one step, and can use that power to also immediately access any index within the array. If you wrote code that said, "Look up the value at index 4," your computer could locate that cell in a single step. Again, this is because your program knows which memory address the array starts at—say, memory address 1000—and therefore knows that if it wants to look up index 4, it should simply jump to memory address 1004.

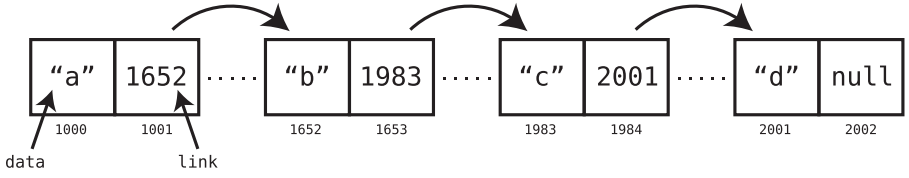


Linked lists, on the other hand, work quite differently. Instead of being a contiguous block of memory, the data from linked lists can be scattered across different cells throughout the computer's memory.

Connected data that is dispersed throughout memory are known as *nodes.* In a linked list, each node represents one item in the list. The big question, then, is: if the nodes are not next to each other in memory, how does the computer know which nodes are part of the same linked list?

This is the key to the linked list: each node also comes with a little extra information, namely, the memory address of the *next* node in the list.

This extra piece of data—this pointer to the next node's memory address—is known as a *link.* Here is a visual depiction of a linked list:



In this example, we have a linked list that contains four pieces of data: `"a"`, `"b"`, `"c"`, and `"d"`. However, it uses *eight* cells of memory to store this data, because each node consists of two memory cells. The first cell holds the actual data, while the second cell serves as a link that indicates where in

memory the next node begins. The final node's link contains `null` since the linked list ends there.

(A linked list's first node can also be referred to as its *head*, and its final node as its *tail*.)

If the computer knows at which memory address the linked list begins, it has all it needs to begin working with the list! Since each node contains a link to the next node, all the computer needs to do is follow each link to string together the entire list.

The fact that a linked list's data can be spread throughout the computer's memory is a potential advantage it has over the array. An array, by contrast, needs to find an entire block of contiguous cells to store its data, which can get increasingly difficult as the array size grows. Now, these details are managed by your programming language under the hood, so you may not have to worry about them. However, you'll see shortly that there are more tangible differences between linked lists and arrays that we can sink our teeth into.

Implementing a Linked List

Some programming languages, such as Java, come with linked lists built into the language. However, many languages do not, and it's fairly simple to implement them on our own.

Let's create our own linked list using Ruby. We'll use two classes to implement this: `Node` and `LinkedList`. Let's create the `Node` class first:

```
class Node

  attr_accessor :data, :next_node

  def initialize(data)
    @data = data
  end

end
```

The `Node` class has two attributes: `data` contains the node's primary value (for example, the string `"a"`), while `next_node` contains the link to the next node in the list. We can use this class as follows:

```
node_1 = Node.new("once")
node_2 = Node.new("upon")
node_3 = Node.new("a")
node_4 = Node.new("time")

node_1.next_node = node_2
node_2.next_node = node_3
node_3.next_node = node_4
```

With this code, we've created a list of four nodes that serve as a list containing the strings "once", "upon", "a", and "time".

Note that in our implementation, the next_node refers to another Node *instance* rather than an actual memory address number. The effect, however, is the same—the nodes are likely dispersed throughout the computer's memory, and yet we can use the nodes' links to string the list together.

Going forward, then, we're simply going to discuss each link as pointing to another node rather than to a specific memory address. Accordingly, we're going to use simplified diagrams to depict linked lists, such as this:



Each node in this diagram consists of two "cells." The first cell contains the node's data, and the second cell points to the next node.

This reflects our Ruby implementation of the Node class. In it, the data method returns the node's data, while the next_node method returns the next node in the list. *In this context, the next_node method serves as the node's link.*

While we've been able to create this linked list with the Node class alone, we need an easy way to tell our program where the linked list begins. To do this, we'll create a LinkedList class in addition to our previous Node class. Here's the LinkedList class in its basic form:

```
class LinkedList

  attr_accessor :first_node

  def initialize(first_node)
    @first_node = first_node
  end

end
```

At this point, all a LinkedList instance does is keep track of the first node of the list.

Previously we created a chain of nodes containing node_1, node_2, node_3, and node_4. We can now use our LinkedList class to reference this list by writing the following code:

```
list = LinkedList.new(node_1)
```

This list variable now acts as a handle on the linked list, as it is an instance of LinkedList that has access to the list's first node.

A really important point emerges: *when dealing with a linked list, we only have immediate access to its first node.* This is going to have serious ramifications as we'll see shortly.

At first glance, though, linked lists and arrays are pretty similar—they're both just lists of stuff. When we dig into the analysis, though, we'll see some pretty dramatic differences in these two data structures' performances! Let's jump into the four classic operations: reading, searching, insertion, and deletion.

Reading

As you know, a computer can read from an array in O(1) time. But now let's figure out the efficiency of reading from a linked list.

If you want to read, say, the value of the third item of a linked list, the computer cannot look it up in one step, because it wouldn't immediately know where to find it in the computer's memory. After all, each node of a linked list could be *anywhere* in memory! All our program knows immediately is the memory address of the *first* node of the linked list. However, it doesn't know offhand where any of the other nodes are.

To read from the third node, then, the computer must go through a process. First, it accesses the first node. It then follows the first node's link to the second node, and then the second node's link to the third node.

To get to any node, then, we always need to start with the first node (the only node we initially have access to), and follow the chain of nodes until we reach the node we want.

It turns out, then, that if we were to read from the last node in the list, it would take N steps for N nodes in the list. Linked lists having a worst-case read of O(N) is a major disadvantage when compared with arrays that can read any element in just O(1). But don't fret, as linked lists will have their moment to shine as we'll see soon.

Code Implementation: Linked List Reading

Let's go ahead and add a read method to our LinkedList class:

```
def read(index)
  # We begin at the first node of the list:
  current_node = first_node
  current_index = 0
```

```
  while current_index < index do
    # We keep following the links of each node until we get to the
    # index we're looking for:
    current_node = current_node.next_node
    current_index += 1

    # If we're past the end of the list, that means the
    # value cannot be found in the list, so return nil:
    return nil unless current_node
  end

  return current_node.data
end
```

If we want to read the fourth node from a list, for example, we'd call our method by passing in the node's index as follows:

```
list.read(3)
```

Let's walk through how this method works.

First, we create a variable called current_node that refers to the node we're currently accessing. Since we're going to start by accessing the first node, we say:

```
current_node = first_node
```

Recall that first_node is an attribute of the LinkedList class.

We also track the index of current_node so that we can know when we reach the desired index. We do this with:

```
current_index = 0
```

as the first node's index is 0.

We then launch a loop that runs while current_index is less than the index we're attempting to read:

```
while current_index < index do
```

In each pass-through of the loop, we access the next node in the list and make it the new current_node:

```
current_node = current_node.next_node
```

We also bump up the current_index by 1:

```
current_index += 1
```

At the end of each pass-through, we check whether we've reached the end of the list, and we return nil if the index we're trying to read isn't in our list:

```
return nil unless current_node
```

This works because the final node of the list will actually have a next_node that is nil, since the last node was never assigned a next_node of its own. This being the case, when we call current_node = current_node.next_node on the last node, current_node becomes nil.

Finally, if we do break out the loop, it's because we reached the desired index. We can then return the node's value with:

```
return current_node.data
```

Searching

As you know, searching means looking for a value within the list and returning its index. We've seen that linear search on an array has a speed of O(N), since the computer needs to inspect each value one at a time.

Linked lists also have a search speed of O(N). To search for a value, we need go through a similar process as we did with reading. That is, we begin with the first node and follow the links of each node to the next one. Along the way, we inspect each value until we find what we're looking for.

Code Implementation: Linked List Search

Here's how we can implement the search operation in Ruby. We'll call this method index_of, and pass in the value we're searching for:

```
def index_of(value)
  # We begin at the first node of the list:
  current_node = first_node
  current_index = 0

  begin
    # If we find the data we're looking for, we return it:
    if current_node.data == value
      return current_index
    end

    # Otherwise, we move on the next node:
    current_node = current_node.next_node
    current_index += 1
  end while current_node

  # If we get through the entire list without finding the
  # data, we return nil:
  return nil
end
```

We can then search for any value within the list using:

```
list.index_of("time")
```

As you can see, the mechanics of searching are similar to reading. The main difference is that the loop doesn't stop at a particular index, but runs until we either find the value or reach the end of the list.

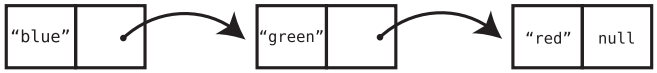
Insertion

Admittedly, linked lists have yet to impress us from a performance standpoint. They're no better than arrays at search, and much worse at reading. But not to worry—linked lists will have their moment. In fact, that moment is now.

Insertion is one operation in which linked lists have a distinct advantage over arrays *in certain situations.*

Recall that the worst-case scenario for insertion into an array is when the program inserts data into index 0, because it first has to shift the rest of the data one cell to the right, which ends up yielding an efficiency of O(N). With linked lists, however, insertion at the beginning of the list takes just one step—which is O(1). Let's see why.

Say we have the following linked list:



If we want to add "yellow" to the beginning of the list, all we have to do is create a new node and have its link point to the node containing "blue":



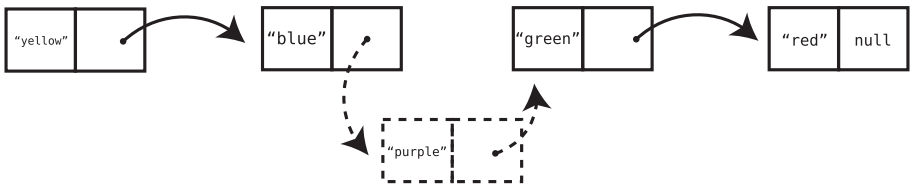
(In our code, we'd also need to update the LinkedList instance so that its first_node attribute now points to this "yellow" node.)

In contrast with an array, the linked list provides the flexibility of inserting data to the front of the list without requiring the shifting of any data. How sweet is that?

The truth is that, theoretically, inserting data *anywhere* within a linked list takes just one step, but there's one gotcha. Let's continue with our example. Here's our linked list now:

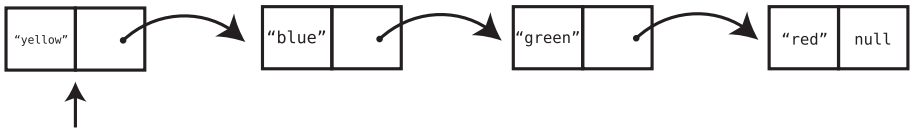


Say we now want to insert "purple" at index 2 (which would be between "blue" and "green"). The actual insertion takes just one step. That is, we can create the new purple node and simply change the blue node's link to point to the purple node as shown here:

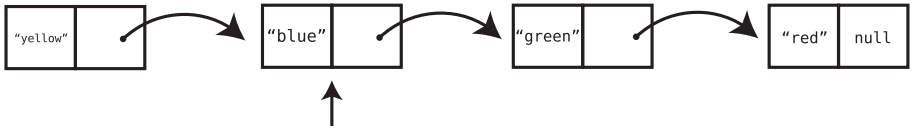


However, for the computer to do this, it first needs to *get to* the node at index 1 ("blue") so that it can modify its link to point to the newly created node. As we've seen, though, reading—which is accessing an item at a given index—from a linked list already takes O(N). Let's see this in action.

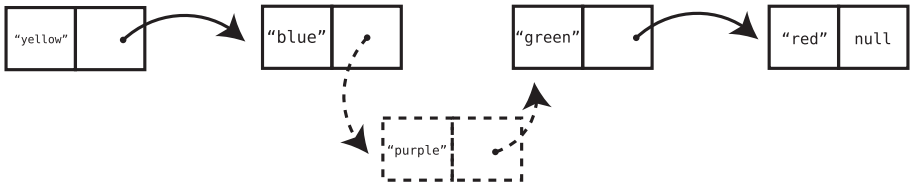
We know that we want to add a new node after index 1. So, the computer needs to get to index 1 of the list. To do this, we must start at the beginning of the list:



We then access the next node by following the first link:



Now that we've found index 1, we can finally add the new node:



In this case, adding "purple" took three steps. If we were to add it to the *end* of our list, it would take five steps: four steps to access index 3, and one step to insert the new node.

Practically speaking, then, inserting into a linked list is O(N), as the worst-case scenario of inserting at the end of the list will take N + 1 steps.

However, we've seen that in the best-case scenario of inserting at the *beginning* of the list is only O(1).

Interestingly, our analysis shows that the best- and worst-case scenarios for arrays and linked lists are the opposite of one another. The following table breaks this all down:

Scenario	Array	Linked List
Insert at beginning	Worst case	Best case
Insert at middle	Average case	Average case
Insert at end	Best case	Worst case

As you can see, arrays favor insertions at the end, while linked lists favor insertions at the beginning.

We've now found one thing that linked lists are great at—inserting things at the beginning of the list. Later in this chapter, we'll see a great practical example of where we can take advantage of this.

Code Implementation: Linked List Insertion

Let's add an insertion method to our LinkedList class. We'll call it insert_at_index:

```
def insert_at_index(index, value)
    # We create the new node with the provided value:
    new_node = Node.new(value)

    # If we are inserting at the beginning of the list:
    if index == 0
      # Have our new node link to what was the first node:
      new_node.next_node = first_node
      # Establish that our new node is now the list's first node:
      self.first_node = new_node
      return
    end

    # If we are inserting anywhere other than the beginning:

    current_node = first_node
    current_index = 0

    # First, we access the node immediately before where the
    # new node will go:
    while current_index < (index - 1) do
      current_node = current_node.next_node
      current_index += 1
    end
```

```
    # Have the new node link to the next node:
    new_node.next_node = current_node.next_node

    # Modify the link of the previous node to point to
    # our new node:
    current_node.next_node = new_node
  end
```

To use the method, we pass in both the new value as well as the index of where we want to insert it.

For example, to insert "purple" at index 3, we'd say:

```
list.insert_at_index(3, "purple")
```

Let's break this insert_at_index method down.

First, we create a new Node instance with the value provided to our method:

```
new_node = Node.new(value)
```

Next, we deal with the case where we're inserting into index 0, that is, at the beginning of our list. The algorithm for this case is different than if we insert elsewhere into the list, so we deal with this case separately.

To insert at the beginning of the list, we simply have our new_node link to the first node of the list and declare our new_node to be the first node going forward:

```
if index == 0
  new_node.next_node = first_node
  self.first_node = new_node
  return
end
```

The return keyword ends the method early, as there's nothing left to do.

The rest of the code deals with a case in which we're inserting anywhere other than at the beginning.

As with reading and searching, we start off by accessing the first node of the list:

```
current_node = first_node
current_index = 0
```

We then use a while loop to access the node *just before* the spot where we want to insert our new_node:

```
while current_index < (index - 1) do
  current_node = current_node.next_node
  current_index += 1
end
```

At this point, the current_node is the node that'll immediately precede our new_node.

Next, we set the link of our new_node to point the node after the current_node:

```
new_node.next_node = current_node.next_node
```

Finally, we change the link of the current_node (which, again, is to be the node that precedes our new_node) to point to our new_node:

```
current_node.next_node = new_node
```

And we're done!

Deletion

Linked lists shine when it comes to deletion as well, especially when deleting from the beginning of the list.

To delete a node from the beginning of a linked list, all we need to do is perform one step: we change the first_node of the linked list to now point to the second node.

Let's return to our example of the linked list containing the values "once", "upon", "a", and "time". If we want to delete the value "once", we could simply change the linked list to begin at "upon":

```
list.first_node = node_2
```

Contrast this with an array in which deleting the first element means shifting all remaining data one cell to the left, which takes O(N) time.

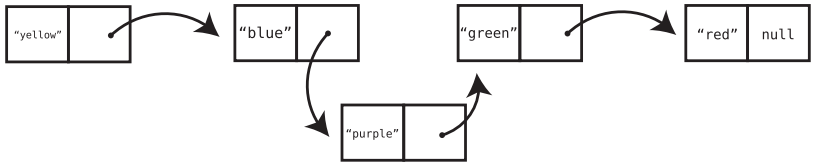
When it comes to deleting the *final* node of a linked list, the actual deletion takes one step—we just take the second-to-last node and make its link null. However, it takes N steps to even access the second-to-last node in the first place, since we need to start at the beginning of the list and follow the links until we reach it.

The following table contrasts the various scenarios of deletion for both arrays and linked lists. Note how it's identical to insertion:

Situation	Array	Linked List
Delete at beginning	Worst case	Best case
Delete at middle	Average case	Average case
Delete at end	Best case	Worst case

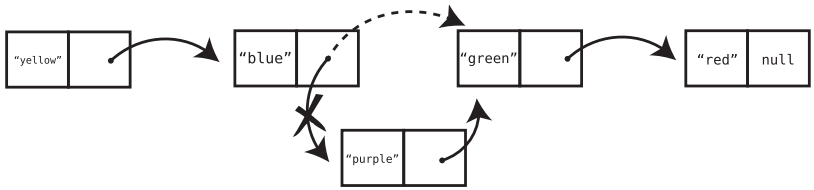
While deleting from the beginning or end of a linked list is straightforward, deleting from anywhere in the middle is slightly more involved.

Say we want to delete the value at index 2 ("purple") from our example linked list of colors as shown in the diagram on page 237.



To accomplish this, we need to first access the node immediately *preceding* the one we're deleting ("blue"). Then, we change its link to point to the node that is immediately *after* the node we're deleting ("green").

The following visualization demonstrates us changing the link of the "blue" node from "purple" to "green":



It's interesting to note that whenever we delete a node from our linked list, the node still exists in memory somewhere. We're just removing the node from our list by ensuring that no other node from the list links to it. This has the *effect* of deleting the node from our list, even if the node still exists in memory.

(Different programming languages handle these deleted nodes in various ways. Some will automatically detect that they're not being used and will "garbage collect" them, freeing up memory.)

Code Implementation: Linked List Deletion

Here's what the delete operation might look like in our LinkedList class. It's called delete_at_index, and we pass in the index we're going to delete:

```
def delete_at_index(index)
  # If we are deleting the first node:
  if index == 0
    # Simply set the first node to be what is currently the second node:
    self.first_node = first_node.next_node
    return
  end

  current_node = first_node
  current_index = 0

  # First, we find the node immediately before the one we
  # want to delete and call it current_node:
  while current_index < (index - 1) do
    current_node = current_node.next_node
    current_index += 1
  end
```

```
  # We find the node that comes after the one we're deleting:
  node_after_deleted_node = current_node.next_node.next_node

  # We change the link of the current_node to point to the
  # node_after_deleted_node, leaving the node we want
  # to delete out of the list:
  current_node.next_node = node_after_deleted_node
end
```

This method is pretty similar to the insert_at_index method we saw earlier. Let's highlight some of the novel points.

The method first deals with a case in which index is 0, meaning we intend to delete the first node of the list. The code for this is ridiculously simple:

```
self.first_node = first_node.next_node
```

All we do is change our list's first_node to be the second node and we're done!

The rest of the method handles deletions anywhere else from the list. To do this, we use a while loop to access the node immediately preceding the one we want to delete. This becomes our current_node.

We then grab the node that comes immediately *after* the node we're going to delete and store it in a variable called node_after_deleted_node:

```
node_after_deleted_node = current_node.next_node.next_node
```

Notice our little trick in accessing that node. It's simply the node that comes two nodes after the current_node!

Then, we modify the link of the current_node to point to the node_after_deleted_node:

```
current_node.next_node = node_after_deleted_node
```

Efficiency of Linked List Operations

After our analysis, it emerges that the comparison of linked lists and arrays breaks down as follows:

Operation	Array	Linked list
Reading	O(1)	O(N)
Search	O(N)	O(N)
Insertion	O(N) (O(1) at end)	O(N) (O(1) at beginning)
Deletion	O(N) (O(1) at end)	O(N) (O(1) at beginning)

In the grand scheme of things, linked lists seem to be lackluster when it comes to time complexity. They perform similarly to arrays for search,

insertion, and deletion, and are much slower when it comes to reading. If so, why would one ever want to use a linked list?

The key to unlocking the linked list's power is in the fact that the *actual insertion and deletion steps* are just O(1).

But isn't that only relevant when inserting or deleting at the beginning of the list? We saw that to insert or delete elsewhere, it takes up to N steps just to access the node we want to delete or insert after!

Well, it just so happens that there are scenarios in which we may already have accessed the right node for some other purpose. The next example is a case in point.

Linked Lists in Action

One case where linked lists shine is when we examine a single list and delete many elements from it. Let's say, for example, we're building an application that combs through existing lists of email addresses and removes any email address that has an invalid format.

No matter whether the list is an array or a linked list, we need to comb through the entire list one element at a time to inspect each email address. This, naturally, takes N steps. However, let's examine what happens when we actually delete each email address.

With an array, each time we delete an email address, we need another O(N) steps to shift the remaining data to the left to close the gap. All this shifting will happen before we can even inspect the next email address.

Let's assume that 1 in 10 email addresses are invalid. If we had a list of 1,000 email addresses, we'd have about 100 invalid ones. Our algorithm, then, would take 1,000 steps to read all 1,000 email addresses. On top of that, though, it might take up to an additional 100,000 steps for deletion, as for each of the 100 deleted addresses, we might shift up to 1,000 other elements.

With a linked list, however, as we comb through the list, each deletion takes just one step, as we can simply change a node's link to point to the appropriate node and move on. For our 1,000 emails, then, our algorithm would take just 1,100 steps, as there are 1,000 reading steps, and 100 deletion steps.

It turns out, then, that linked lists are an amazing data structure for moving through an entire list while making insertions or deletions, as we never have to worry about shifting other data as we make an insertion or deletion.

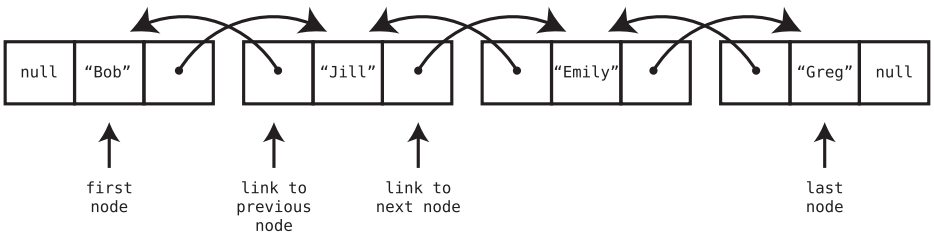
Doubly Linked Lists

Linked lists actually come in a number of different flavors. The linked list we've discussed until this point is the "classic" linked list, but with some slight modifications, we can grant linked lists additional superpowers.

One variant form of the linked list is the *doubly linked list.*

A doubly linked list is like a linked list except that each node has *two* links—one that points to the next node, and another that points to the *previous* node. In addition, the doubly linked list always keeps track of both the first and last nodes, instead of just the first node.

Here's what a doubly linked list looks like:



We can implement the core of a doubly linked list in Ruby like this:

```
class Node

  attr_accessor :data, :next_node, :previous_node

  def initialize(data)
    @data = data
  end

end

class DoublyLinkedList

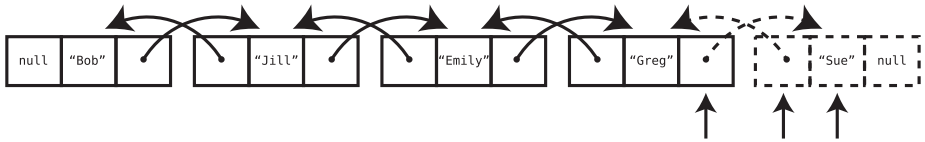
  attr_accessor :first_node, :last_node

  def initialize(first_node=nil, last_node=nil)
    @first_node = first_node
    @last_node = last_node
  end

end
```

Since a doubly linked list always knows where both its first and last nodes are, we can access each of them in a single step, or O(1). So, just as we can read, insert, or delete from the beginning of the list in O(1), we can do the same from the end of the list in O(1) as well.

Here's a depiction of inserting at the end of a doubly linked list:



As you can see, we create a new node ("Sue") and have its previous_node point to what used to be the last_node ("Greg") of the linked list. Then, we change the next_node of the last_node ("Greg") to point to this new node ("Sue"). Finally, we declare the new node ("Sue") to be the last_node of the linked list.

Code Implementation: Doubly Linked List Insertion

Here's the implementation of a new insert_at_end method that we can add to our DoublyLinkedList class:

```
def insert_at_end(value)
  new_node = Node.new(value)

  # If there are no elements yet in the linked list:
  if !first_node
    @first_node = new_node
    @last_node = new_node
  else # If the linked list already has at least one node:
    new_node.previous_node = @last_node
    @last_node.next_node = new_node
    @last_node = new_node
  end
end
```

Let's highlight the most important parts of this method.

First, we create the new node:

```
new_node = Node.new(value)
```

Then, we set the previous_node link of the new_node to point to what until this point was the last node:

```
new_node.previous_node = @last_node
```

Then, we change the last node's link (which was nil until this point) and have it point to our new_node:

```
@last_node.next_node = new_node
```

Last, we tell our instance of the DoublyLinkedList that its last node is our new_node:

```
@last_node = new_node
```

Moving Forward and Backward

With a "classic" linked list, we can only move *forward* through the list. That is, we can access the first node and follow the links to find all the other nodes of the list. But we're not able to move backward, as no node is aware of what the previous node is.

A doubly linked list allows for a lot more flexibility, as we can move both forward *and* backward through the list. In fact, we can even start with the last node and work our way backward to the first node.

Queues as Doubly Linked Lists

Because doubly linked lists have immediate access to both the front and end of the list, they can insert data on either side at O(1) as well as delete data on either side at O(1).

Because doubly linked lists can insert data at the end in O(1) time and delete data from the front in O(1) time, *they make the perfect underlying data structure for a queue.*

We looked at Queues, on page 144, and you'll recall that they are lists of items in which data can only be inserted at the end and removed from the beginning. You learned there that queues are an example of an abstract data type, and that we were able to use an array to implement them under the hood.

Now, since queues insert at the end and delete from the beginning, arrays are only so good as the underlying data structure. While arrays are O(1) for insertions at the end, they're O(N) for deleting from the beginning.

A doubly linked list, on the other hand, is O(1) for both inserting at the end *and* for deleting from the beginning. That's what makes it a perfect fit for serving as the queue's underlying data structure.

Code Implementation: Queue Built Upon a Doubly Linked List

Here's a complete example of a queue that is built upon a doubly linked list:

```
class Node

  attr_accessor :data, :next_node, :previous_node

  def initialize(data)
    @data = data
  end

end
```

```
class DoublyLinkedList

  attr_accessor :first_node, :last_node

  def initialize(first_node=nil, last_node=nil)
    @first_node = first_node
    @last_node = last_node
  end

  def insert_at_end(value)
    new_node = Node.new(value)

    # If there are no elements yet in the linked list:
    if !first_node
      @first_node = new_node
      @last_node = new_node
    else  # If the linked list already has at least one node:
      new_node.previous_node = @last_node
      @last_node.next_node = new_node
      @last_node = new_node
    end
  end

  def remove_from_front
    removed_node = @first_node
    @first_node = @first_node.next_node
    return removed_node
  end

end

class Queue
  attr_accessor :queue

  def initialize
    @data = DoublyLinkedList.new
  end

  def enqueue(element)
    @data.insert_at_end(element)
  end

  def dequeue
    removed_node = @data.remove_from_front
    return removed_node.data
  end

  def read
    return nil unless @data.first_node
    return @data.first_node.data
  end
end
```

To make this work, we added a remove_from_front method to the DoublyLinkedList class that looks like this:

```
def remove_from_front
  removed_node = @first_node
  @first_node = @first_node.next_node
  return removed_node
end
```

As you can see, we effectively delete the first node by changing the list's @first_node to be what is currently the second node. We then return the deleted node.

The Queue class implements its methods on top of our DoublyLinkedList. The enqueue method relies on the insert_at_end method of our DoublyLinkedList:

```
def enqueue(element)
  @data.insert_at_end(element)
end
```

Similarly, the dequeue method takes advantage of the linked list's ability to delete from the front of the list:

```
def dequeue
  removed_node = @data.remove_from_front
  return removed_node.data
end
```

By implementing our queue with a doubly linked list, we can now both insert and delete from the queue at a speedy O(1). And that's doubly awesome.

Wrapping Up

As we've seen, the subtle differences between arrays and linked lists unlock new ways to make our code faster than ever.

By looking at linked lists, you've also learned the concept of nodes. However, the linked list is only the simplest of node-based data structures. In the next chapters, you'll learn about node-based structures that are both more complex and more interesting—and will reveal new worlds about how nodes can yield tremendous power and efficiency.

Exercises

The following exercises provide you with the opportunity to practice with linked lists. The solutions to these exercises are found in the section, Chapter 14, on page 453.

1. Add a method to the classic LinkedList class that prints all the elements of the list.

2. Add a method to the DoublyLinkedList class that prints all the elements of the list in *reverse* order.

3. Add a method to the classic LinkedList class that returns the last element from the list. Assume you don't know how many elements are in the list.

4. Here's a tricky one. Add a method to the classic LinkedList class that reverses the list. That is, if the original list is A -> B -> C, all of the list's links should change so that C -> B -> A.

5. Here's a brilliant little linked list puzzle for you. Let's say you have access to a node from somewhere in the middle of a classic linked list, but not the linked list itself. That is, you have a variable that points to an instance of Node, but you don't have access to the LinkedList instance. In this situation, if you follow this node's link, you can find all the items from this middle node until the end, but you have no way to find the nodes that precede this node in the list.

   Write code that will effectively delete this node from the list. The entire remaining list should remain complete, with only this node removed.

CHAPTER 15

Speeding Up All the Things with Binary Search Trees

Sometimes, we may want to arrange our data in a specific order. For example, we may want an alphabetized list of names, or a list of products in order of lowest price to highest.

While we can use a sorting algorithm such as Quicksort to arrange our data into perfect ascending order, there's a cost to this. As we've seen, even the fastest algorithms take O(N log N) time. So, if we are going to want our data sorted *often*, it would be sensible to always keep our data in sorted order in the first place so that we never need to resort it.

An ordered array is a simple but effective tool for keeping data in order. It's also fast for certain operations, as it has O(1) reads and O(log N) search (when using binary search).

However, there is a drawback to ordered arrays.

When it comes to insertions and deletions, ordered arrays are relatively slow. Whenever a value is inserted into an ordered array, we first shift all greater values one cell to the right. And when a value is deleted from an ordered array, we shift all greater values one cell to the left. This takes N steps in a worst-case scenario (inserting into or deleting from the first cell of the array), and N / 2 steps on average. Either way, it's O(N), and O(N) is relatively slow for a simple insertion or deletion.

Now, if we were looking for a data structure that delivers all-around amazing speed, a hash table is a great choice. They are O(1) for search, insertion, and deletion. However, they do not maintain order, and order is what we need for our alphabetized-list application.

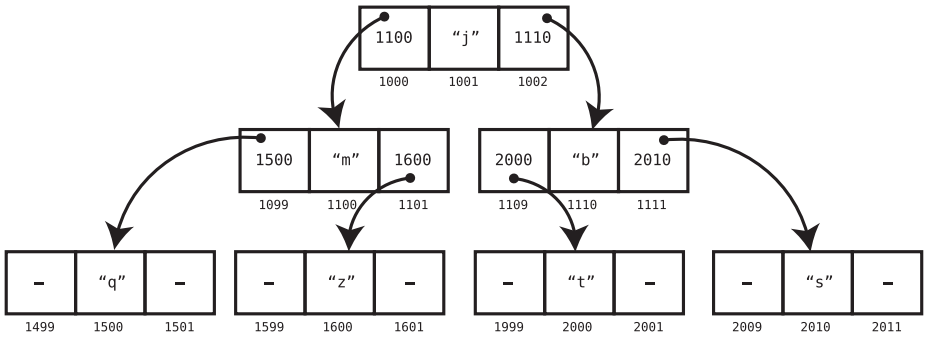
So, what do we do if we want a data structure that maintains order yet *also* has fast search, insertion, and deletion? Neither an ordered array nor a hash table is ideal.

Enter the binary search tree.

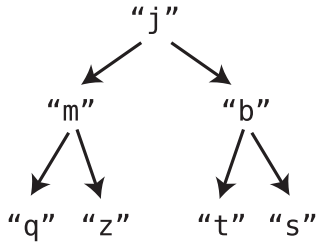
Trees

You were introduced to node-based data structures in the previous chapter with linked lists. In a classic linked list, each node contains a link that connects the node to a single other node. A *tree* is also a node-based data structure, but within a tree, each node can have links to *multiple* nodes.

Here is a visualization of a simple tree:



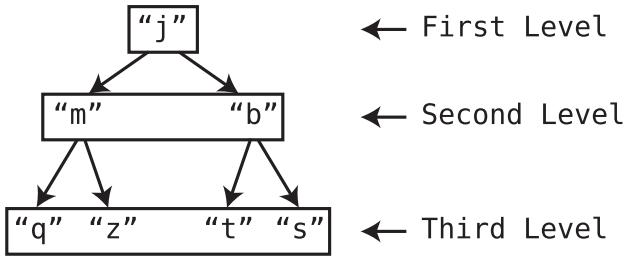
In this example, each node has links that lead to two other nodes. For the sake of simplicity, we can represent this tree visually without showing all the memory addresses:



Trees come with their own unique nomenclature:

- The uppermost node (in our example, the "j") is called the *root*. Yes, in our picture the root is at the *top* of the tree; it's how trees are typically depicted.

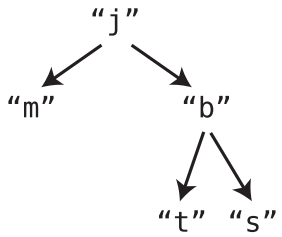
- In our example, we'd say that the "j" is a *parent* to "m" and "b." Conversely, "m" and "b" are *children* of "j." Similarly, the "m" is a parent of "q" and "z," and "q" and "z" are children of "m."
- As in a family tree, a node can have *descendants* and *ancestors.* A node's descendants are *all* the nodes that stem from a node, while a node's ancestors are *all* the nodes that it stems from. In our example, the "j" is the ancestor of all the other nodes in the tree, and all the other nodes are, therefore, the descendants of the "j."
- Trees are said to have *levels.* Each level is a row within the tree. Our example tree has three levels:



- One property of a tree is how *balanced* it is. A tree is balanced when its nodes' subtrees have the same number of nodes in it.

For instance, the above tree is said to be perfectly balanced. If you look at each node, its two subtrees have the same number of nodes. The root node ("j") has two subtrees, which each contain three nodes. You'll see that the same is true for every node in the tree as well. For example, the "m" node also has two subtrees where the two subtrees each contain one node.

The following tree, on the other hand, is *imbalanced:*



As you can see, the root's right subtree contains more nodes than its left subtree, causing an imbalance.

Binary Search Trees

There are many different kinds of tree-based data structures, but in this chapter, we'll focus on a particular tree known as a *binary search tree.*

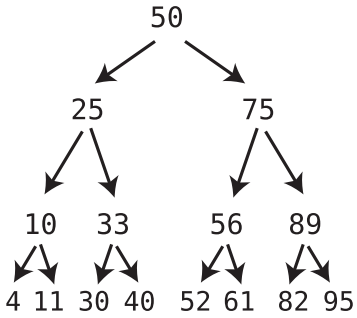
Note that there are two adjectives here: *binary* and *search.*

A *binary* tree is a tree in which each node has zero, one, or two children.

A binary *search* tree is a binary tree that also abides by the following rules:

- Each node can have at most one "left" child and one "right" child.
- A node's "left" descendants can only contain values that are less than the node itself. Likewise, a node's "right" descendants can only contain values that are greater than the node itself.

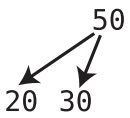
Here's an example of a binary search tree, in which the values are numbers:



Note that each node has one child with a lesser value than itself, which is depicted using a left arrow, and one child with a greater value than itself, which is depicted using a right arrow.

Additionally, notice that all of the 50's left descendants are less than it. At the same time, all of the 50's right descendants are greater than it. The same pattern goes for each and every node.

While the following example is a binary tree, it is not a binary *search* tree:



It's a binary tree because each node has zero, one, or two children. But it's not a binary *search* tree because the root node has two "left" children. That is, it has two children than are less than it. For a binary search tree to be valid, it can have at most one left (lesser) child and one right (greater) child.

The implementation of a tree node in Python might look something like this:

```
class TreeNode:
    def __init__(self,val,left=None,right=None):
        self.value = val
        self.leftChild = left
        self.rightChild = right
```

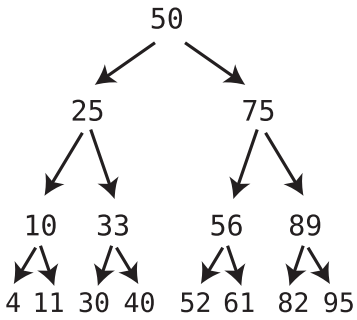
We can then build a simple tree like this:

```
node1 = TreeNode(25)
node2 = TreeNode(75)
root = TreeNode(50, node1, node2)
```

Because of the unique structure of a binary search tree, we can search for any value within it very quickly, as we'll now see.

Searching

Here, again, is a binary search tree:

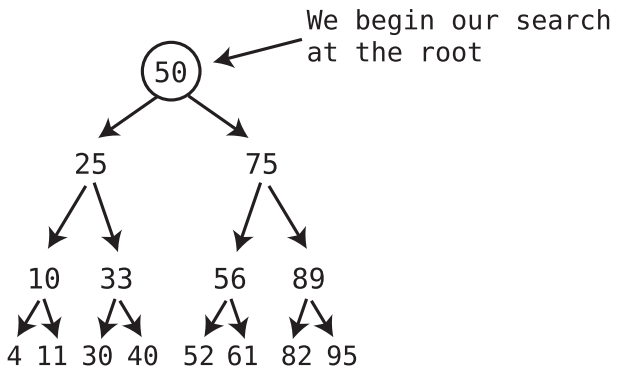


The algorithm for searching within a binary search tree is as follows:

1. Designate a node to be the "current node." (At the beginning of the algorithm, the root node is the first "current node.")
2. Inspect the value at the current node.
3. If we've found the value we're looking for, great!
4. If the value we're looking for is less than the current node, search for it in its left subtree.
5. If the value we're looking for is greater than the current node, search for it in its right subtree.
6. Repeat Steps 1 through 5 until we find the value we're searching for, or until we hit the bottom of the tree, in which case our value must not be in the tree.

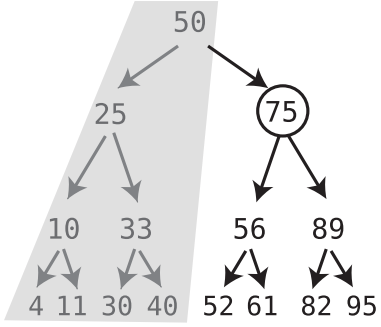
Say we want to search for the 61. Let's see how many steps it would take by walking through this visually.

When searching through a tree, we must always begin at the root:

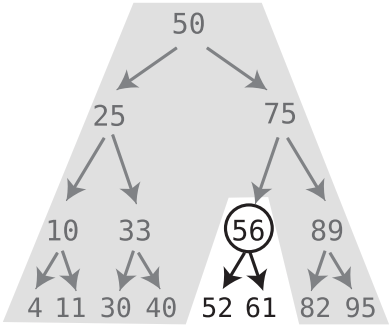


Next, the computer asks itself: is the number we're searching for (61) greater or less than the value of this node? If the number we're looking for is less than the current node, look for it in the left child. If it's greater than the current node, look for it in the right child.

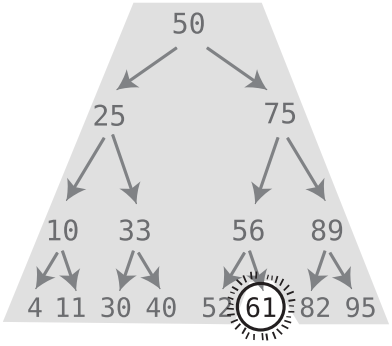
In this example, because 61 is greater than 50, we know it must be somewhere to the right, so we search the right child. In the following picture, we've shaded out all the nodes we've eliminated from our search, since we know that the 61 cannot possibly be there:



"Are you my mother?" asks the algorithm. Since the 75 is not the 61 we're looking for, we need to move down to the next level. And because 61 is less than 75, we'll check the left child, since the 61 could only be in that subtree as shown in the diagram on page 253.



The 56 is not the 61 we're looking for, so we continue our search. Since 61 is greater than 56, we search for the 61 in the right child of the 56:



We found it! In this example, it took us four steps to find our desired value.

The Efficiency of Searching a Binary Search Tree

If you take another look at the steps we just walked through, you'll notice that each step eliminates half of the remaining nodes from our search. For example, when we begin our search, we start at the root node, and our desired value may be found among any of the root's descendants. However, when we then decide to continue the search with, say, the root's right child, we eliminate the left child and *all of its descendants* from the search.

We'd say, then, that searching in a binary search tree is O(log N), which is the apt description for any algorithm that eliminates half of the remaining values with each step. (We'll see soon, though, that this is only for a perfectly balanced binary search tree, which is a best-case scenario.)

Log(N) Levels

Here's yet another way of describing why search in a binary search tree is O(log N), which will reveal another property about binary trees in general: *if there are N nodes in a balanced binary tree, there will be about log N levels (that is, rows).*

To understand this, let's assume each row in the tree is completely filled with nodes, and that there aren't any empty positions. If you think about it, each time we add a new full level to the tree, we end up roughly doubling the number of nodes that the tree has. (Really, we're doubling the nodes and adding one.)

For example, a binary tree with four complete levels has 15 nodes. (Go ahead, count them.) If we add a fifth complete level, that means we add two children to each of the eight nodes in the fourth level. This means we add 16 new nodes, roughly doubling the size of the tree.

It emerges that each new level doubles the size of the tree. Accordingly, *a tree containing N nodes will require log(N) levels* to hold all the nodes.

In the context of binary search, we noted that the pattern of log(N) is that with each step of the search, we can eliminate half of the remaining data. The number of levels needed in a binary tree follows this pattern as well.

Let's take a binary tree that needs to hold 31 nodes. With our fifth level, we can hold 16 of those nodes. This took care of roughly half of the data, leaving us with just 15 nodes that we still need to find room for. With the fourth level, we take care of eight of those nodes, leaving us with seven unaccounted for. With the third level, we take care of four of those nodes, and so on.

Indeed, log 31 is (approximately) 5. So, we've now concluded that a balanced tree with N nodes will have log(N) levels.

Since this is the case, it makes a lot of sense as to why searching a binary search tree takes up to log(N) steps: because each step of the search causes us to move down a level, we take up to as many steps as there are levels in the tree.

However you prefer to think about it, searching a binary search tree takes O(log N).

Now, while search in a binary search tree is O(log N), so is binary search within an ordered array, in which each number we select also eliminates half of the remaining possible values. In this regard, then, searching a binary search tree has the same efficiency as binary search within an ordered array.

Where binary search trees really shine over ordered arrays, though, is with insertion. We'll get to that soon.

Code Implementation: Searching a Binary Search Tree

To implement the search operation, as well as the other binary search tree operations, we're going to make heavy use of recursion. You learned back in *Recursively Recurse with Recursion* that recursion is key when dealing with data structures that have an arbitrary number of levels of depth. A tree is such a data structure, as it can have an infinite number of levels.

Here's how we can use recursion to implement search with Python:

```
def search(searchValue, node):
    # Base case: If the node is nonexistent
    # or we've found the value we're looking for:
    if node is None or node.value == searchValue:
        return node

    # If the value is less than the current node, perform
    # search on the left child:
    elif searchValue < node.value:
        return search(searchValue, node.leftChild)

    # If the value is greater than the current node, perform
    # search on the right child:
    else:  # searchValue > node.value
        return search(searchValue, node.rightChild)
```

This `search` function accepts the `searchValue` we're searching for and a `node` that we'll use as the base for our search. The first time we call `search`, the `node` will be the root node. However, in the subsequent recursive calls, the `node` may be another node within the tree.

Our function deals with four possible cases, two of which are the base cases:

```
if node is None or node.value == searchValue:
    return node
```

One base case is when the node contains the `searchValue` we're looking for, in which case we can return the node and not make any recursive calls.

The other base case is when `node` is `None`. This will make more sense after we've examined the other cases, so let's come back to this shortly.

The next case is when the `searchValue` is less than the value of the current node:

```
elif searchValue < node.value:
    return search(searchValue, node.leftChild)
```

In this case, we know that if it exists in the tree, the searchValue will have to be found somewhere among this node's left descendants. So, we recursively call the search function on this node's left child.

The next case is the inverse; it's when the searchValue is greater than the current node:

```
else:  # searchValue > node.value
    return search(searchValue, node.rightChild)
```

In this case, we recursively call search on the current node's right child.

Now, when we make these recursive calls on the current node's children, note that we didn't check whether the current node even has any children. That's where the first base case comes in:

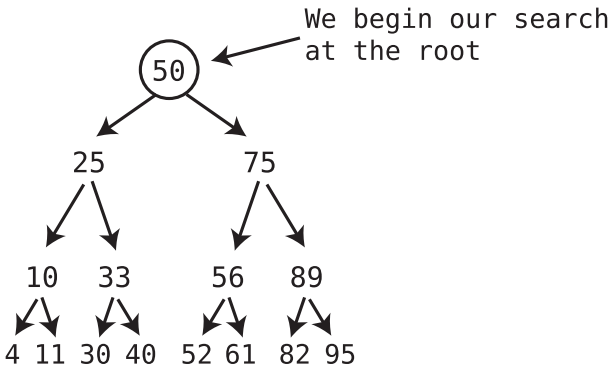
```
if node is None
```

That is to say, if it turns out that we called search on a child node that doesn't actually exist, we end up returning None (since the node will actually contain None). This case will happen if the searchValue does not exist within our tree, as we'll try to access the node where the searchValue *should* be found, but our search hits a dead end. In this case, it's appropriate that we return None, indicating that the searchValue is not within the tree.

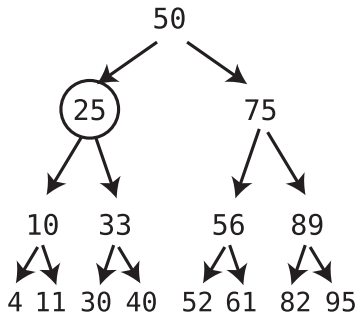
Insertion

As I mentioned earlier, binary search trees are at their best when it comes to insertion. Now we'll see why.

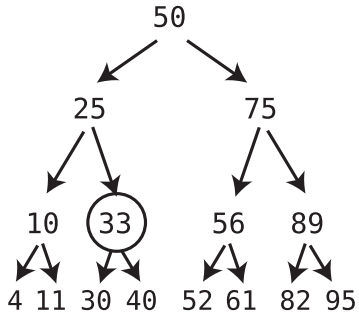
Say we want to insert the number 45 into our example tree. The first thing we'd have to do is find the correct node to attach the 45 to. To begin our search, we start at the root:



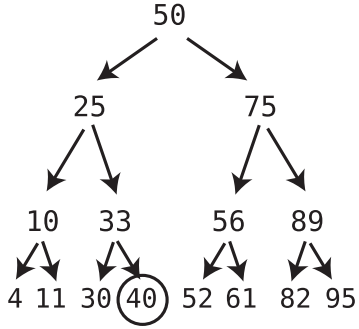
Since 45 is less than 50, we drill down to the left child:



Since 45 is greater than 25, we must inspect the right child:

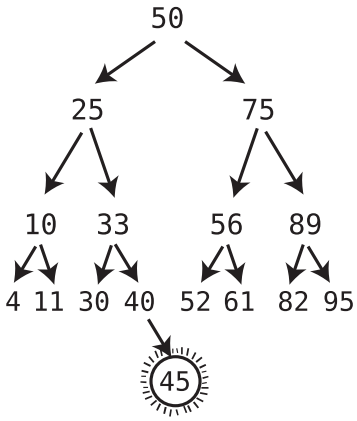


Since 45 is greater than 33, we check the 33's right child:



At this point, we've reached a node that has no children, so we have nowhere to go. This means we're ready to perform our insertion.

Since 45 is greater than 40, we insert it as a right child of the 40:



In this example, insertion took five steps, consisting of four search steps plus one insertion step. Insertion always takes just one extra step beyond a search, which means insertion takes (log N) + 1 steps. In Big O Notation, which ignores constants, this is O(log N).

In an ordered array, by contrast, insertion takes O(N), because in addition to search, we must shift a lot of data to the right to make room for the value we're inserting.

This is what makes binary search trees so efficient. While ordered arrays have O(log N) search and O(N) insertion, binary search trees have O(log N) search *and* O(log N) insertion. This becomes critical in an application in which you anticipate a lot of changes to your data.

Code Implementation: Binary Search Tree Insertion

Here's a Python implementation of inserting a new value into a binary search tree. Like the search function, it is recursive:

```
def insert(value, node):
    if value < node.value:

        # If the left child does not exist, we want to insert
        # the value as the left child:
        if node.leftChild is None:
            node.leftChild = TreeNode(value)
        else:
            insert(value, node.leftChild)
```

```
elif value > node.value:

    # If the right child does not exist, we want to insert
    # the value as the right child:
    if node.rightChild is None:
        node.rightChild = TreeNode(value)
    else:
        insert(value, node.rightChild)
```

The insert function accepts a value that we're going to insert, and a node that serves as the ancestor node for which our value will become a descendant.

First, we check whether the value is less than the value of the current node:

```
if value < node.value:
```

If the value is less than the node, we know that we need to insert the value somewhere among the left descendants of the node.

We then check to see whether the current node has a left child. If the node doesn't have a left child, we make the value into the left child, since that's exactly where the value belongs:

```
if node.leftChild is None:
    node.leftChild = TreeNode(value)
```

This is the base case, since we don't need to make any recursive calls.

However, if the node already has a left child, we can't place the value there. Instead, we recursively call insert on the left child so that we continue to search for the spot in which we'll place the value:

```
else:
    insert(value, node.leftChild)
```

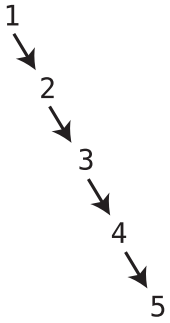
Eventually, we'll hit a descendant node that doesn't have its own child, and that's where the value is going to go.

The rest of the function is the exact inverse; it handles cases where the value is greater than the current node.

The Order of Insertion

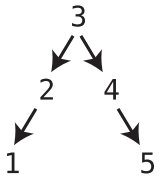
It is important to note that only when creating a tree out of randomly sorted data do trees usually wind up being well-balanced. However, if we insert *sorted* data into a tree, it can become imbalanced and less efficient. For example, if we

were to insert the following data in this order—1, 2, 3, 4, 5—we'd end up with a tree that looks like this:



This tree is completely linear, so searching for the 5 within this tree would take O(N).

However, if we inserted the same data in the following order—3, 2, 4, 1, 5—the tree would be evenly balanced:



Only with a balanced tree does search take O(log N).

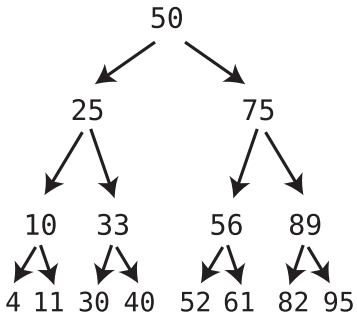
Because of this, if you ever want to convert an ordered array into a binary search tree, you'd better first randomize the order of the data.

It emerges that in a worst-case scenario, when a tree is completely *imbalanced,* search is O(N). In a best-case scenario, when it is perfectly balanced, search is O(log N). In the typical scenario, in which data is inserted in random order, a tree will be pretty well balanced and search will take about O(log N).

Deletion

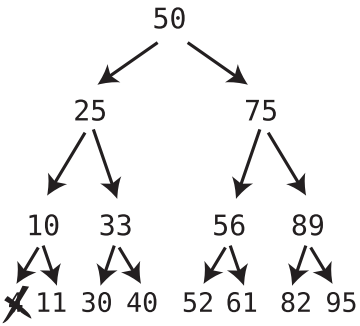
Deletion is the least straightforward operation within a binary search tree and requires some careful maneuvering.

Let's say we want to delete the 4 from this binary search tree:

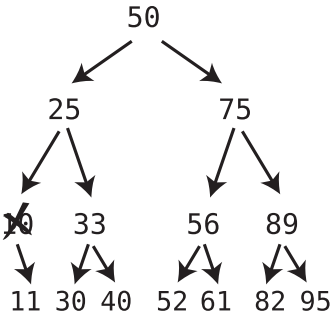


First, we perform a search to find the 4. We won't visualize this search again, since you've already got that down.

Once we find the 4, we can delete it one step:

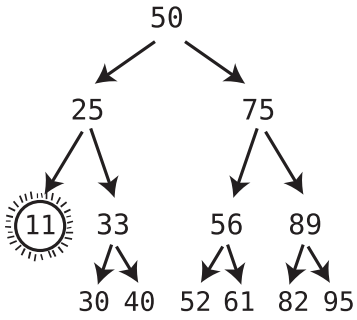


Well, that was simple. But let's see what happens when we try to delete the 10:



We end up with an 11 that isn't connected to the tree anymore. And we can't have that, because we'd lose the 11 forever.

However, there's a solution to this problem: we can plug the 11 into where the 10 used to be:

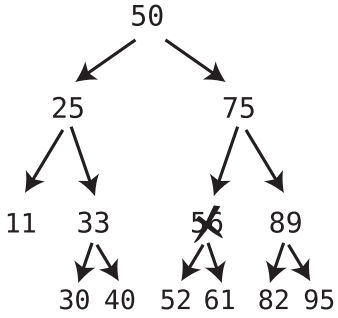


So far, our deletion algorithm follows these rules:

- If the node being deleted has no children, simply delete it.
- If the node being deleted has one child, delete the node and plug the child into the spot where the deleted node was.

Deleting a Node with Two Children

Deleting a node that has two children is the most complex scenario. Let's say we want to delete the 56 in this tree:



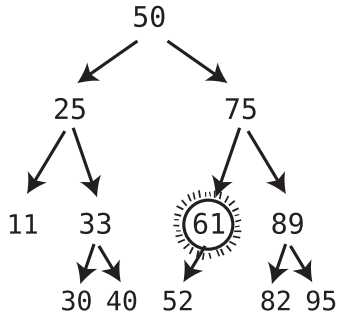
What are we going to do with its former children, 52 and 61? We cannot move *both* of them to where the 56 was. This is where the next rule of the deletion algorithm comes into play:

- When deleting a node with two children, replace the deleted node with the *successor* node. The successor node is the child node whose value is the *least of all values that are greater than the deleted node.*

That was a tricky sentence. To put in other words: if we were to put the deleted node and all of its descendants in ascending order, the successor node would be the next number after the one we just deleted.

In this case, it's easy to figure out which node is the successor, since the deleted node had only two descendants. If we put the numbers 52-56-61 in ascending order, the next number after 56 is 61.

Once we find the successor node, we plug it into where the deleted node was. So, we replace the 56 with the 61:



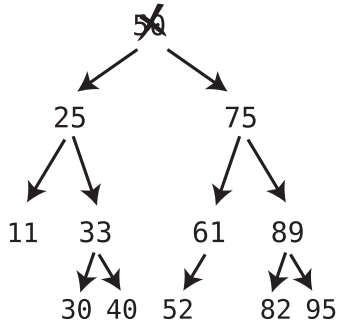
Finding the Successor Node

How does the computer find the successor node? This can be tricky when we delete a node high up in the tree.

Here's the algorithm for finding the successor node:

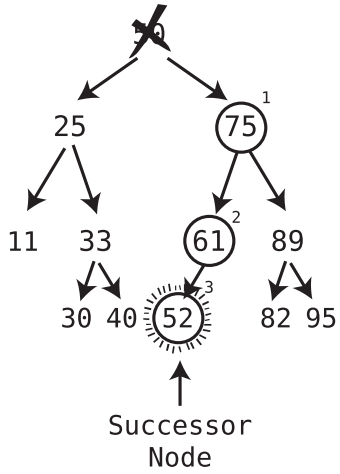
- Visit the right child of the deleted value, and then keep on visiting the left child of each subsequent child until there are no more left children. The bottom value is the successor node.

Let's see this again in action in a more complex example. Let's delete the root node:



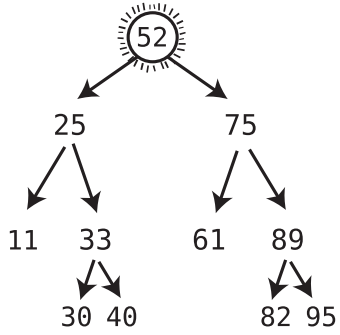
We now need to plug the successor node into where the 50 was, and turn it into the root node. So, let's find the successor node.

To do this, we first visit the *right* child of the deleted node, and then keep descending *leftward* until we reach a node that doesn't have a left child:



It turns out that the 52 is the successor node.

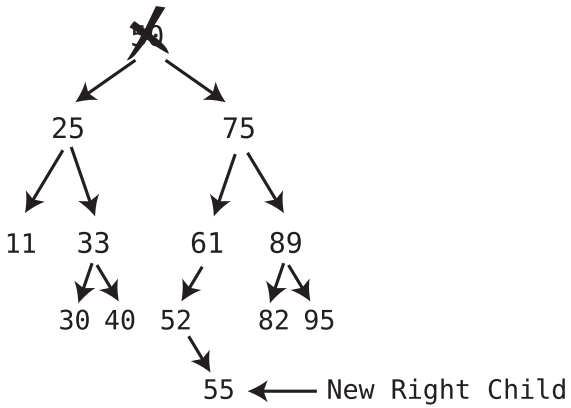
Now that we've found the successor node, we plug it into the node we deleted:



And we're done!

Successor Node with a Right Child

However, there is one case we haven't accounted for yet, and that's where the successor node has a right child of its own. Let's recreate the preceding tree, but add a right child to the 52 as shown in the diagram on page 265.

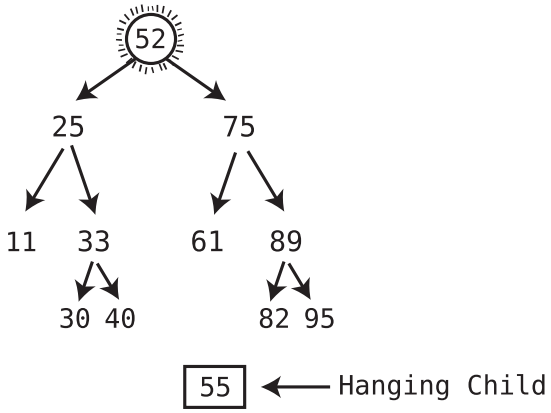


In this case, we can't simply plug the successor node—the 52—into the root, since we'd leave its child of 55 hanging. Because of this, there's one more rule to our deletion algorithm:

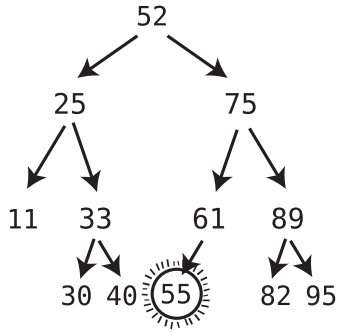
- If the successor node has a right child, after plugging the successor node into the spot of the deleted node, take the former right child of the successor node and turn it into the *left child of the former parent of the successor node.*

That was another tricky sentence, so let's walk through the steps.

First, we plug the successor node (52) into the root. This leaves the 55 dangling without a parent:



Next, we turn the 55 into the left child of the former parent of the successor node. In this case, 61 used to be the parent of the successor node, so we make the 55 the left child of the 61:



And now we're *really* done.

The Complete Deletion Algorithm

Putting all the steps together, the algorithm for deletion from a binary search tree is:

- If the node being deleted has no children, simply delete it.
- If the node being deleted has one child, delete the node and plug the child into the spot where the deleted node was.
- When deleting a node with two children, replace the deleted node with the *successor* node. The successor node is the child node whose value is the *least of all values that are greater than the deleted node.*
- To find the successor node: visit the right child of the deleted value, and then keep on visiting the left child of each subsequent child until there are no more left children. The bottom value is the successor node.
- If the successor node has a right child, after plugging the successor node into the spot of the deleted node, take the former right child of the successor node and turn it into the *left child of the former parent of the successor node.*

Code Implementation: Binary Search Tree Deletion

Here's a recursive Python implementation of deletion from a binary search tree:

```
def delete(valueToDelete, node):

    # The base case is when we've hit the bottom of the tree,
    # and the parent node has no children:
    if node is None:
        return None

    # If the value we're deleting is less or greater than the current node,
    # we set the left or right child respectively to be
    # the return value of a recursive call of this
    # very method on the current
    # node's left or right subtree.
    elif valueToDelete < node.value:
        node.leftChild = delete(valueToDelete, node.leftChild)

        # We return the current node (and its subtree if existent) to
        # be used as the new value of its parent's left or right child:
        return node
    elif valueToDelete > node.value:
        node.rightChild = delete(valueToDelete, node.rightChild)
        return node

    # If the current node is the one we want to delete:
    elif valueToDelete == node.value:

        # If the current node has no left child, we delete it by
        # returning its right child (and its subtree if existent)
        # to be its parent's new subtree:
        if node.leftChild is None:
            return node.rightChild

            # (If the current node has no left OR right child, this ends up
            # being None as per the first line of code in this function.)

        elif node.rightChild is None:
            return node.leftChild

        # If the current node has two children, we delete the current node
        # by calling the lift function (below),
        # which changes the current node's
        # value to the value of its successor node:
        else:
            node.rightChild = lift(node.rightChild, node)
            return node

def lift(node, nodeToDelete):

    # If the current node of this function has a left child,
    # we recursively call this function to continue down
    # the left subtree to find the successor node.
    if node.leftChild:
        node.leftChild = lift(node.leftChild, nodeToDelete)
        return node
```

```
# If the current node has no left child, that means the current node
# of this function is the successor node, and we take its value
# and make it the new value of the node that we're deleting:
else:
    nodeToDelete.value = node.value
    # We return the successor node's right child to be now used
    # as its parent's left child:
    return node.rightChild
```

Admittedly, this code is a bit involved, so let's break it down a bit.

The function accepts two arguments:

```
def delete(valueToDelete, node):
```

The valueToDelete is the value we're deleting from the tree, and the node is the root of the tree. When we first call this function, the node will be the actual root node, but as this function calls itself recursively, the node may be lower down in the tree and simply be the root of a smaller subtree. In any case, the node is the root of either the whole tree or one of its subtrees.

The base case is when the node does not actually exist:

```
if node is None:
    return None
```

This will happen when a recursive call to this function tries to access a child node that does not exist. In this case, we return None.

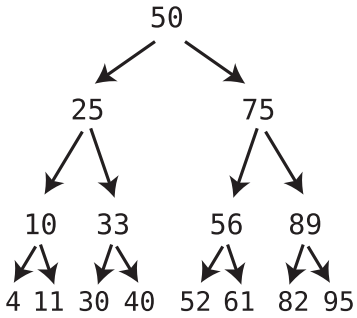
Next, we check whether the valueToDelete is less than or greater than the value of the current node:

```
elif valueToDelete < node.value:
    node.leftChild = delete(valueToDelete, node.leftChild)
    return node
elif valueToDelete > node.value:
    node.rightChild = delete(valueToDelete, node.rightChild)
    return node
```

This snippet may not be immediately intuitive, but here's how it works. If the valueToDelete is less than the value of the current node, we know that if the valueToDelete is to be found somewhere in the tree, it'll be found among the current node's left descendants.

Here's the clever part: we then *overwrite* the current node's left child to be the result of recursively calling the delete function on the current node's left child. The delete function itself eventually returns a node, so we take this result and make it the current node's left child.

Often, however, this "overwrite" doesn't actually change the left child, because calling delete on the left child may return that same left child. To make sense of this, imagine we're trying to delete the 4 from this example tree:



Originally, node is the root node, which has the value of 50. Since 4 (the valueToDelete) is less than 50, we say that the 50's left child should now be the result of calling delete on the 50's current left child, which is 25.

So, what will the 50's left child be? Let's see.

When we recursively call delete on 25, we again determine that 4 is less than 25 (the current node), so we continue the recursion with the 25's left child, which is the 10. However, no matter what the 25's left child will end up being, since we say return node, we *return the 25 node* at the end of the current function call.

This means that back when I said that the 50's left child should be the result of calling delete on the 25, we ended getting back the 25 itself, so the 50's left child actually remains unchanged.

The current node's left or right child *will* change, however, if the result of the next recursive call involves an actual deletion.

Let's examine that bit of code next:

```
elif valueToDelete == node.value:
```

This is to say, the current node is the one we want to delete. In order to delete it properly, we first need to determine whether the current node has any children, as this will affect the deletion algorithm.

We begin by checking whether the node we're going to delete has any left children:

```
if node.leftChild is None:
    return node.rightChild
```

If the current node has no left children, we can return the current node's *right* child as a result of this function. Remember, *whichever node we return will become either the left or right child of the previous node in the call stack.* So, let's imagine that the current node has a right child. In this case, by returning the right child, we make it a child node of the *previous* node in the call stack, effectively deleting the current node from the tree.

Now, if the current node happens to not have a right child, that's okay too, since we'll end up passing None as a result of this function, and that will also effectively remove the current node from the tree.

Moving on in our code, if the current node does have a left child but doesn't have a right child, we can still easily delete the current node:

```
elif node.rightChild is None:
    return node.leftChild
```

In this case, we delete the current node by returning its left child to become the child node of the previous node in the call stack.

Finally, we encounter the most complex case, which is when the node we're deleting has *two* children:

```
else:
    node.rightChild = lift(node.rightChild, node)
    return node
```

In this case, we call the lift function, take its result, and make it the current node's right child.

What does the lift function do?

When we call it, we pass in the current node's right child in addition to the node itself. The lift function accomplishes four things:

1. It finds the successor node.
2. It overwrites the value of the nodeToDelete and makes it the value of the successor node. This is how we get the successor node into the correct spot. Note that we don't move the actual successor node *object* anywhere; we simply copy its value into the node we're "deleting."
3. To eliminate the original successor node object, the function then turns the original successor node's right child into its parent's left child.
4. After all the recursion is said and done, it finally returns either the original rightChild passed into it in the first place, or None if the original rightChild

ended up serving as the successor node (which would happen if it had no left children of its own).

We then take return value of `lift` and make it the current `node`'s right child. This would either leave the right child alone, or change it to `None` if the right child was used as the successor node.

It's not just you; the `delete` function is one of the most complex pieces of code in the book. It may take careful study to follow the entire process, even after this walk-through.

The Efficiency of Binary Search Tree Deletion

Like search and insertion, deleting from trees is also typically O(log N). This is because deletion requires a search plus a few extra steps to deal with any hanging children. Contrast this with deleting a value from an ordered array, which is O(N) due to shifting elements to the left to close the gap of the deleted value.

Binary Search Trees in Action

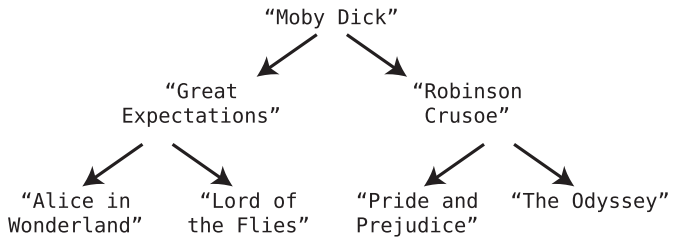
We've seen that binary search trees boast efficiencies of O(log N) for search, insertion, and deletion, making it an efficient choice for scenarios in which we need to store and manipulate ordered data. This is particularly true if we will be modifying the data often, because while ordered arrays are just as fast as binary search trees when searching data, binary search trees are significantly faster when it comes to inserting and deleting data.

For example, let's say we're creating an application that maintains a list of book titles. We'd want our application to have the following functionality:

- Our program should be able to print the list of book titles in alphabetical order.
- Our program should allow for constant changes to the list.
- Our program should allow the user to search for a title within the list.

If we didn't anticipate that our book list would be changing that often, an ordered array would be a suitable data structure to contain our data. However, we're building an app that should be able to handle many changes in real time. If our list had millions of titles, a binary search tree may be a better choice.

Such a tree might look something like this:



Here, the titles are positioned based on their alphabetical order. A title that comes earlier in the alphabet is considered a "lower" value, while titles that come later are "greater" values.

Binary Search Tree Traversal

Now, we've already seen how to search, insert, and delete data from a binary search tree. I mentioned, though, that we also want to be able to print the entire list of book titles in alphabetical order. How can we do that?

First, we need the ability to *visit* every single node in the tree. *Visiting* nodes is just another term for accessing them. The process of visiting every node in a data structure is known as *traversing* the data structure.

Second, we need to make sure we traverse the tree in alphabetically ascending order so that we can print the list in that order. There are multiple ways to traverse a tree, but for this application, we will perform what is known as *inorder traversal,* so that we can print each title in alphabetical order.

Recursion is a great tool for performing traversal. We'll create a recursive function called `traverse` that can be called on a particular node. The function then performs the following steps:

1. Call itself (`traverse`) recursively on the node's left child. The function will keep getting called until we hit a node that does not have a left child.
2. "Visit" the node. (For our book title app, we print the value of the node at this step.)
3. Call itself (`traverse`) recursively on the node's right child. The function will keep getting called until we hit a node that does not have a right child.

For this recursive algorithm, the base case is when we call `traverse` on a child that does not actually exist, in which case we `return` without doing anything further.

Here's a Python traverse_and_print function that works for our list of book titles. Note how concise it is:

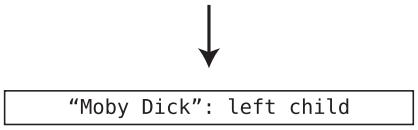
```
def traverse_and_print(node):
    if node is None:
        return
    traverse_and_print(node.leftChild)
    print(node.value)
    traverse_and_print(node.rightChild)
```

Let's walk through the inorder traversal step by step.

We first call traverse_and_print on *Moby Dick*. This, in turn, calls traverse_and_print on the left child of *Moby Dick*, which is *Great Expectations:*

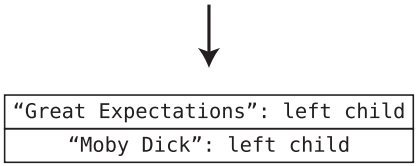
```
traverse_and_print(node.leftChild)
```

Before we move on to that, though, we're going to add to the call stack the fact that we're in the middle of the function in *Moby Dick*, and the fact that we're in the middle of traversing its left child:



We then proceed with traverse_and_print("Great Expectations"), which calls traverse_and_print on the left child of *Great Expectations*, which is *Alice in Wonderland.*

Let's add traverse_and_print("Great Expectations") to the call stack before moving on:



The traverse_and_print("Alice in Wonderland") calls traverse_and_print on the left child of *Alice in Wonderland.* However, there *isn't* any left child (the base case), so nothing happens. The next of line of traverse_and_print is:

```
print(node.value)
```

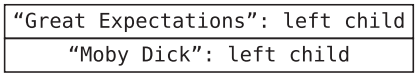
which prints "Alice in Wonderland".

Next, the function attempts to traverse_and_print the *right* child of *Alice in Wonderland:*

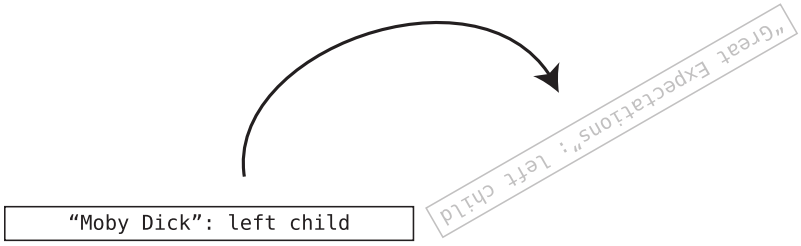
```
traverse_and_print(node.rightChild)
```

However, there is no right child (the base case), so the function returns without doing anything further.

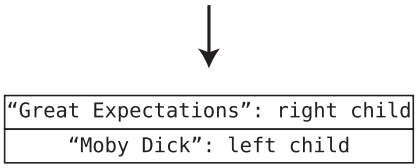
Since we've completed the function traverse_and_print("Alice in Wonderland"), we check the call stack to see where we're up to in this recursive soup:



Ah, that's right. We were in the middle of traverse_and_print("Great Expectations"), and we had just completed calling traverse_and_print on its left child. Let's pop this from the call stack:

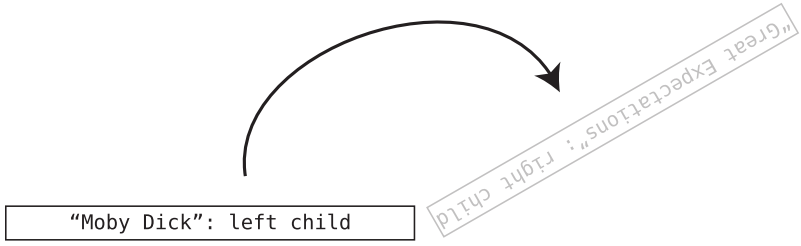


And let's continue. The function next prints "Great Expectations" and then calls traverse_and_print on the right child, which is *Lord of the Flies.* Before moving on to that, though, let's hold our place within this function in the call stack:

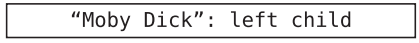


We now execute traverse_and_print("Lord of the Flies"). First, we call traverse_and_print on its left child, but it doesn't have one. Next, we print *Lord of the Flies.* Finally, we call traverse_and_print on its right child, but that doesn't exist either, so the function is now done.

We look at our call stack and see that we were in the process of executing traverse_and_print on the right child of *Great Expectations.* We can pop that from the stack and continue as shown in the diagram on page 275.

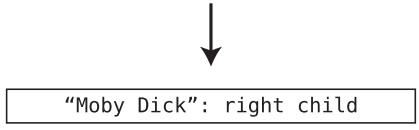


Now, it just so happens that we've also now completed everything we have to do in traverse_and_print("Great Expectations"), so we can go back to the call stack to see what to do next:



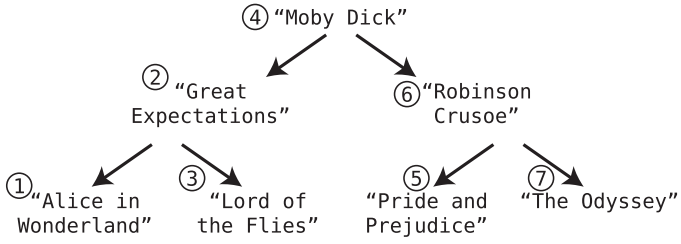
We can see that we were in the middle of traverse_and_print of the left child of *Moby Dick*. We can pop that from the call stack (which leaves the stack empty for now), and continue with the next step within traverse_and_print("Moby Dick"), which is to print *Moby Dick*.

Then, we call traverse_and_print on the right child of *Moby Dick*. We'll add this to the call stack:



For the sake of brevity (although it's probably too late for that), I'll let you walk through the rest of traverse_and_print function from here.

By the time our function has finished executing, we'll have printed the nodes in this order:



And that's how we achieve our goal of printing the book titles in alphabetical order. Note that tree traversal is O(N), since by definition, traversal visits all N nodes of the tree.

Wrapping Up

The binary search tree is a powerful node-based data structure that provides order maintenance, while also offering fast search, insertion, and deletion. It's more complex than its linked list cousin, but it offers tremendous value.

However, the binary search tree is just one type of tree. There are many different kinds of trees, and each brings unique advantages to specialized situations. In the next chapter, we're going to discover another tree that will bring unique speed advantages to a specific, but common scenario.

Exercises

The following exercises provide you with the opportunity to practice with binary search trees. The solutions to these exercises are found in the section, Chapter 15, on page 456.

1. Imagine you were to take an empty binary search tree and insert the following sequence of numbers in this order: [1, 5, 9, 2, 4, 10, 6, 3, 8].

   Draw a diagram showing what the binary search tree would look like. Remember, the numbers are being inserted in the order presented here.

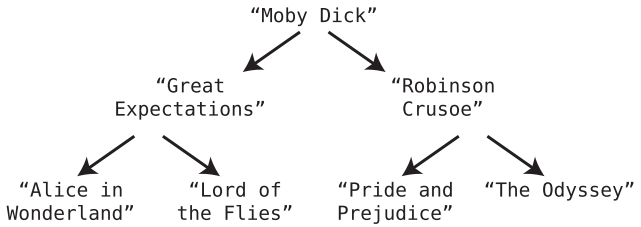
2. If a well-balanced binary search tree contains 1,000 values, what is the maximum number of steps it would take to search for a value within it?

3. Write an algorithm that finds the greatest value within a binary search tree.

4. In the text I demonstrated how to use *inorder* traversal to print a list of all the book titles. Another way to traverse a tree is known as *preorder* traversal. Here is the code for it as applied to our book app:

   ```
   def traverse_and_print(node):
       if node is None:
           return
       print(node.value)
       traverse_and_print(node.leftChild)
       traverse_and_print(node.rightChild)
   ```

   For the example tree in the text (the one with *Moby Dick* and the other book titles), write out the order in which the book titles are printed with preorder traversal. As a reminder, here is the example tree as shown in the diagram on page 277.



5. There is yet another form of traversal called *postorder* traversal. Here is the code as applied to our book app:

```
def traverse_and_print(node):
    if node is None:
        return
    traverse_and_print(node.leftChild)
    traverse_and_print(node.rightChild)
    print(node.value)
```

For the example tree in the text (which also appears in the previous exercise), write out the order in which the book titles are printed with postorder traversal.

CHAPTER 16

Keeping Your Priorities Straight with Heaps

Now that we've discovered the tree, we've unlocked many new data structures. In the previous chapter, we focused specifically on the binary search tree, but there are many other types of trees as well. Like all data structures, each type of tree comes with its own benefits and drawbacks, and the trick is knowing which one to harness in a specific situation.

In this chapter, we'll explore the heap, a type of tree data structure that has special powers that can be leveraged for specific scenarios, in particular when we want to constantly keep tabs on the greatest or least data element in a dataset.

To appreciate what a heap can do, let's take a look at a completely different data structure: the priority queue.

Priority Queues

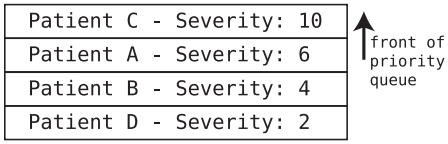
You learned about Queues, on page 144, and discovered that the queue is a list in which items are processed First In, First Out (FIFO). Essentially, this means that data is inserted only at the *end* of the queue, and data is accessed and removed only from the *front* of the queue. In accessing the queue's data, we give precedence to the order in which the data was inserted.

A *priority queue* is a list whose deletions and access are just like a classic queue, but whose insertions are like an ordered array. That is, we only delete and access data from the *front* of the priority queue, but when we insert data, we always make sure the data remains sorted in a specific order.

One classic example of where a priority queue is helpful is an application that manages the triage system for a hospital emergency room. In the ER, we don't treat people strictly in the order in which they arrived. Instead, we treat people in the order of the severity of their symptoms. If someone suddenly arrives

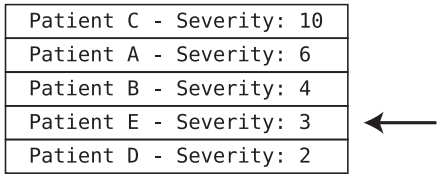
with a life-threatening injury, that patient will be placed at the front of the queue, even if the person with the flu had arrived hours earlier.

Let's say our triage system ranked the severity of a patient's condition on a scale of 1 to 10, with 10 being the most critical. Our priority queue may look like this:



In determining the next patient to treat, we will always select the patient at the front of the priority queue, since that person's need is the most urgent. In this case, the next patient we'd treat is Patient C.

If a new patient now arrives with a condition severity of 3, we'll initially place this patient at the appropriate spot within the priority queue. We'll call this person Patient E:



The priority queue is an example of an abstract data type. That is, it can be implemented using other, more fundamental data structures. One straightforward way we can implement a priority queue is by using an ordered array. That is, we use an array and apply the following constraints:

- When we insert data, we ensure we always maintain proper order.
- Data can only be removed from the end of the array. (This will represent the front of the priority queue.)

While this approach is straightforward, let's analyze its efficiency.

The priority queue has two primary operations: deletions and insertions.

We've seen in *Why Data Structures Matter* that deleting from the front of an array is O(N), since we have to shift all the data over to fill the gap created at index 0. However, we've cleverly tweaked our implementation so that we consider the *end* of array to be the *front* of the priority queue. This way, we're always deleting from the end of the array, which is O(1).

With O(1) deletions, our priority queue is in good shape so far. But what about insertions?

You learned that inserting into an ordered array is O(N), since we have to inspect up to all N elements of the array to determine where our new data should go. (And even if we find the correct spot early, we need to then shift all the remaining data over to the right.)

Our array-based priority queue, then, has deletions that are O(1) and insertions that are O(N). If we expect there to be many items in our priority queue, the O(N) insertions may cause some real unwanted drag in our application.

Because of this, computer scientists discovered another data structure that serves as a more efficient foundation for the priority queue. And this data structure is the heap.

Heaps

There are several different types of heaps, but we're going to focus on the *binary heap.*

The binary heap is a specific kind of binary tree. As a reminder, a binary tree is a tree where each node has a maximum of two child nodes. (The binary *search* tree from the last chapter was one specific type of binary tree.)

Now, even binary heaps come in two flavors: the max-heap and the min-heap. We're going to work with the max-heap for now, but as you'll see later, the difference between the two is trivial.

Going forward, I'm going to refer to this data structure simply as a heap, even though we're specifically working with a binary max-heap.

The *heap* is a binary tree that maintains the following conditions:

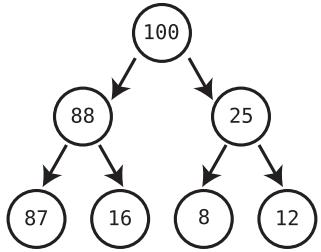
- The value of each node must be greater than each of its descendant nodes. This rule is known as the *heap condition.*
- The tree must be *complete.* (I'll explain the meaning of this shortly.)

Let's break down both of these conditions, starting with the heap condition.

The Heap Condition

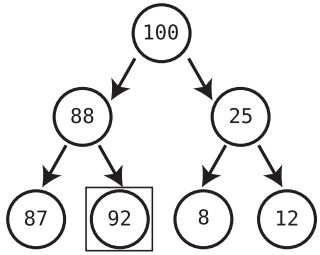
The *heap condition* says that each node's value must be greater than each and every one of its descendants.

For example, the following tree meets the heap condition since each node is greater than any of its descendants as shown in the diagram on page 282.



In this example, the root node of 100 has no descendant that is greater than it. Similarly, the 88 is greater than both of its children, and so is the 25.

The following tree isn't a valid heap, because it doesn't meet the heap condition:



As you can see, the 92 is greater than its parent of 88. This violates the heap condition.

Note how the heap is structured very differently than the binary search tree. In a binary search tree, each node's right child is greater than it. In a heap, however, a node *never* has any descendant that is greater than it. As they say, "A binary search tree doth not a heap make." (Or something like that.)

We can also construct a heap that has the opposite heap condition, which is each node must contain a *smaller* value than any of its descendants. Such a heap is known as the min-heap, which I mentioned earlier. We're going to continue to focus on the max-heap, where each node must be *greater* than all of its descendants. Ultimately, whether a heap is a max-heap or a min-heap is pretty trivial, as everything else about both heaps is identical; they only have reversed heap conditions. Otherwise, the fundamental idea is the same.

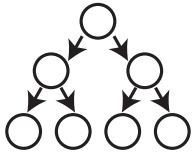
Complete Trees

Now, let's get to the second rule of heaps—that the tree needs to be complete.

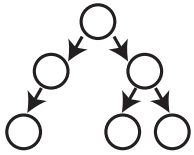
A *complete tree* is a tree that is completely filled with nodes; no nodes are missing. So, if you read each level of the tree from left to right, all of the nodes are there. However, the bottom row *can* have empty positions, as long as there

aren't any nodes to the right of these empty positions. This is best demonstrated with examples.

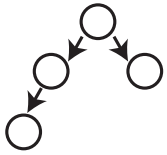
The following tree is complete, since each level (meaning, each row) of the tree is completely filled in with nodes:



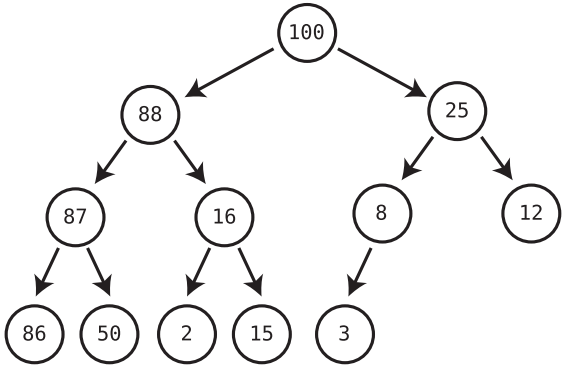
The following tree is *not* complete, since it's missing a node on the third level:



Now, the next tree is actually considered complete, since its empty positions are limited to the bottom row, and there aren't any nodes found to the right of the empty positions:



A heap, then, is a tree that meets the heap condition *and* is also complete. Here is just one more example of a heap:



This is a valid heap since each node is greater than any of its descendants, and the tree is also complete. While it does have some gaps in the bottom row, these gaps are limited to the very right of the tree.

Heap Properties

Now that you know what a heap is, let's examine some of its interesting properties.

While the heap condition dictates that the heap be ordered a certain way, this ordering is still useless when it comes to searching for a value within the heap.

For example, let's say that in the above heap, we want to search for the value 3. If we start at the root node of 100, should we search among its left or right descendants? In a binary search tree, we'd know that the 3 must be among the 100's left descendants. In a heap, however, all we know is that the 3 has to be a descendant of the 100, and can't be its ancestor. But we'd have no idea as to which child node to search next. Indeed, the 3 happens to be among the 100's *right* descendants, but it could have also easily been among its left descendants.

Because of this, heaps are said to be *weakly ordered* as compared to binary search trees. While heaps have *some* order, as descendants cannot be greater than their ancestors, this isn't enough order to make heaps worth searching through.

There's another property of heaps that may be obvious by now, but is worth calling attention to: in a heap, the root node will always have the *greatest* value. (In a min-heap, the root will contain the smallest value.) This will be the key as to why the heap is a great tool for implementing priority queues. In the priority queue, we always want to access the value with the greatest priority. With a heap, we always know that we can find this in the root node. Thus, the root node will represent the item with the highest priority.

The heap has two primary operations: inserting and deleting. As we noted, searching within a heap would require us to inspect each node, so search is not an operation usually implemented in the context of heaps. (A heap can also have an optional "read" operation, which would simply look at the value of the root node.)

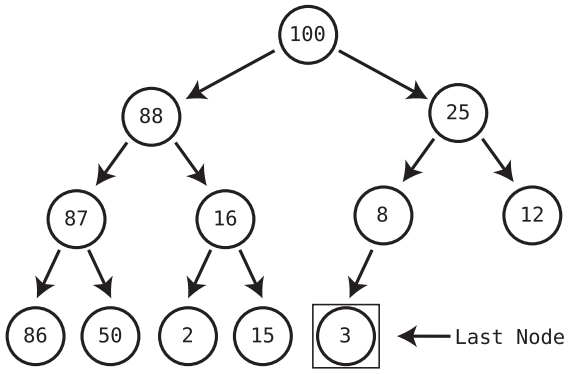
Before we move on to how the heap's primary operations work, let me define one more term, since it'll be used heavily in the upcoming algorithms.

The heap has something called a *last node*. A heap's *last node* is the rightmost node in its bottom level.

Take a look at the heap on page 285.

In this heap, the 3 is the last node, since it's the rightmost node in the bottom row.

Next, let's get into the heap's primary operations.



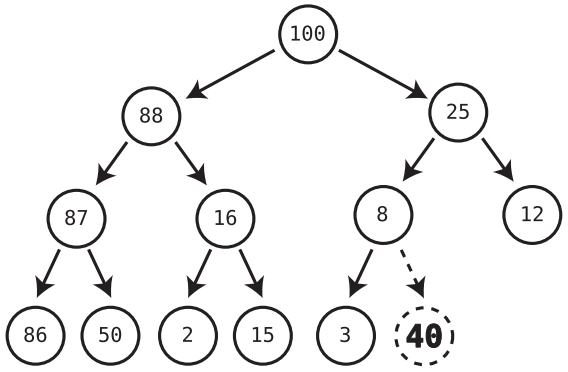
Heap Insertion

To insert a new value into the heap, we perform the following algorithm:

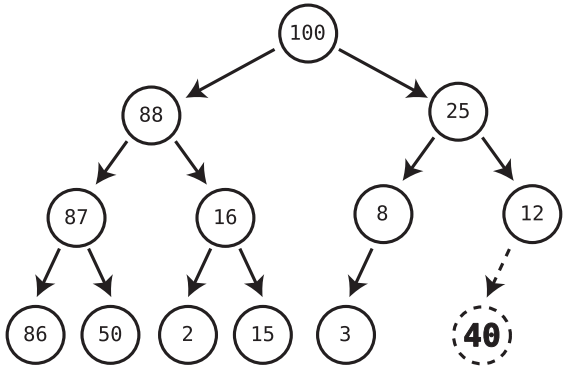
1. We create a node containing the new value and insert it at the next available rightmost spot in the bottom level. Thus, this value becomes the heap's last node.
2. Next, we compare this new node with its parent node.
3. If the new node is greater than its parent node, we swap the new node with the parent node.
4. We repeat Step 3, effectively moving the new node up through the heap, until it has a parent whose value is greater than it.

Let's see this algorithm in action. Here's what would happen if we were to insert a 40 into the heap:

Step 1: We add the 40 as the heap's last node:

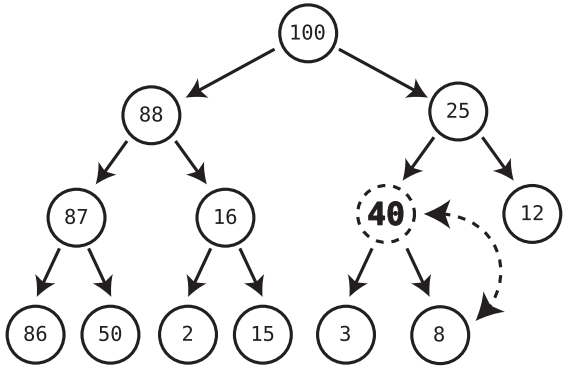


Note that doing the following would have been incorrect:

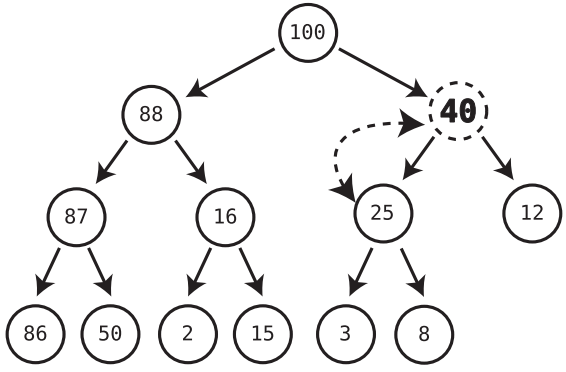


Placing the 40 as a child of the 12 node makes the tree *incomplete*, since we'd now have a node to the right of an empty position. In order for a heap to remain a heap, it must always be complete.

Step 2: We compare the 40 with its parent node, which happens to be the 8. Since the 40 is greater than the 8, we swap the two nodes:



Step 3: We compare the 40 with its new parent, the 25. Since the 40 is greater than the 25, we swap them:



Step 4: We compare the 40 to its parent, which is the 100. Since the 40 is smaller than the 100, we're done!

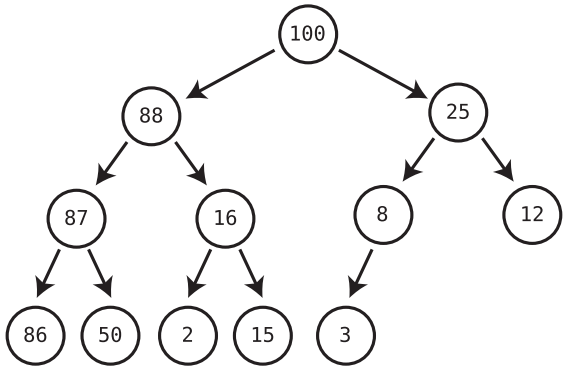
This process of moving the new node up the heap, is called *trickling* the node up through the heap. Sometimes it moves up to the right, and sometimes it moves up to the left, but it always moves up until it settles into the correct position.

The efficiency of inserting into a heap is O(log N). As you saw in the previous chapter, for N nodes in any binary tree, the tree is organized into about log(N) rows. Since at most we'd have to trickle the new value up to the top row, this will take log(N) steps at most.

Looking for the Last Node

While the insertion algorithm seems pretty straightforward, there's one little snag. The first step has us place the new value as the heap's last node. But this begs the question: how do we find the spot that will be the last node?

Let's take a look again at the heap before we inserted the 40:

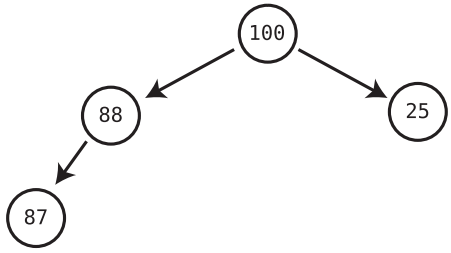


We know by looking at the diagram that to make the 40 into the last node, we'd make it the right child of the 8, as that's the next available spot in the bottom row.

However, a computer doesn't have eyeballs, and doesn't see the heap as a bunch of rows. All it sees is the root node, and can follow links to child nodes. So, how do we create an algorithm for the computer to find the spot for the new value?

Take our example heap. When we start at the root node of 100, do we tell the computer to look among the 100's right descendants to find the next available spot for the new last node?

While it's true that in our example heap the next available spot is among the 100's right descendants, take a look at the following alternative heap:



In this heap, the next available spot for the new last node would be the 88's right child, which is among the 100's *left* descendants.

Essentially, then, just as it's impossible to search through a heap, it's impossible to efficiently find the heap's last node (or next available spot to hold a new last node) without having to inspect each and every node.

So, how *do* we find the next available node? I'll explain this later on, but for now, let's call this issue the Problem of the Last Node. I promise we'll come back to it.

In the meantime, let's explore the heap's other primary operation, which is deletion.

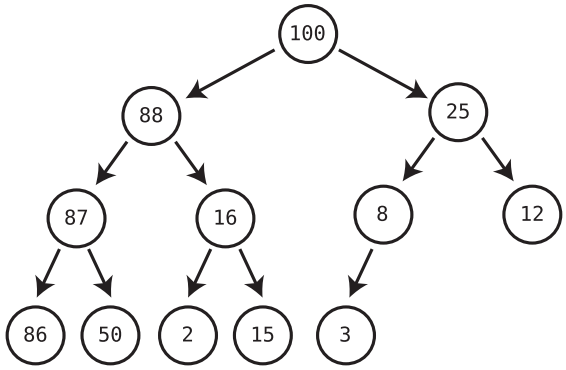
Heap Deletion

The first thing to know about deleting a value from a heap is that *we only ever delete the root node.* This is right in line with the way a priority queue works, in that we only access and remove the highest-priority item.

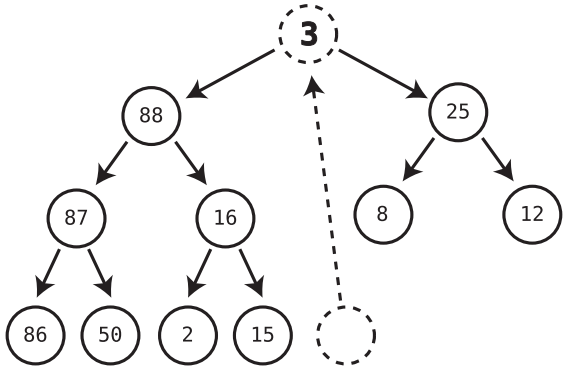
The algorithm for deleting the root node of a heap is as follows:

1. Move the *last node* into where the root node was, effectively removing the original root node.
2. Trickle the root node down into its proper place. I'll explain how trickling down works shortly.

Let's say we're going to remove the root node from the heap on page 289.



In this example, the root node is the 100. To delete it, we overwrite the root by placing the last node there instead. In this case, the last node is the 3. So, we move the 3 and place it where the 100 was:



Now, we can't leave the heap as is, since the heap condition has been violated and since the 3 is currently less than some (actually, most) of its descendants. To make things right again, we need to trickle the 3 down until its heap condition has been restored.

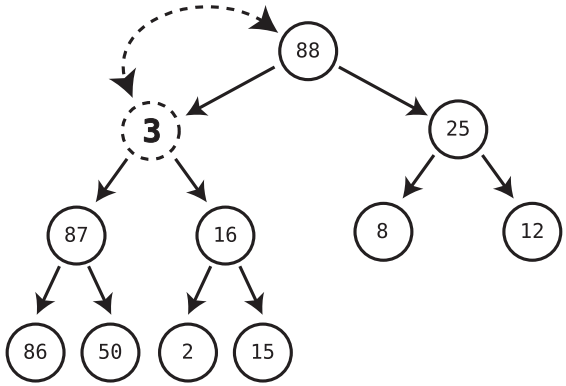
Trickling down is a tad more complex than trickling up, since each time we trickle a node down, we have two possible directions as to where we will trickle it down. That is, we can either swap it with its left child or its right child. (When trickling up, on the other hand, each node has only one parent to swap with.)

Here's the algorithm for trickling *down*. For the sake of clarity, we're going to call the node we're trickling the "trickle node." (Sounds gross, I know.)

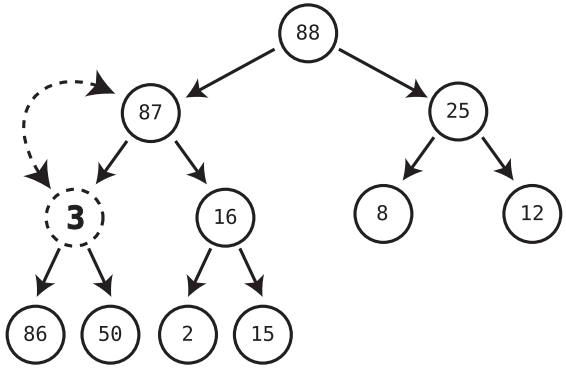
1. We check both children of the trickle node and see which one is larger.
2. If the trickle node is smaller than the larger of the two child nodes, we swap the trickle node with that larger child.
3. We repeat Steps 1 and 2 until the trickle node has no children who are greater than it.

Let's see this in action.

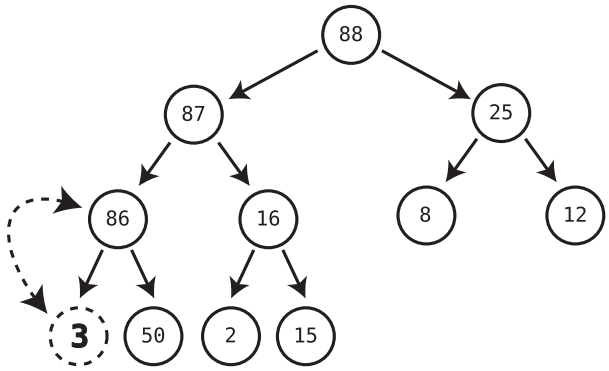
Step 1: The 3, which is the trickle node, currently has two children, the 88 and the 25. The 88 is larger of the two, and since the 3 is smaller than the 88, we swap the trickle node with the 88:



Step 2: The trickle node now has two new children, the 87 and the 16. The 87 is the larger one, and it is greater than the trickle node. So, we swap the trickle node with the 87:



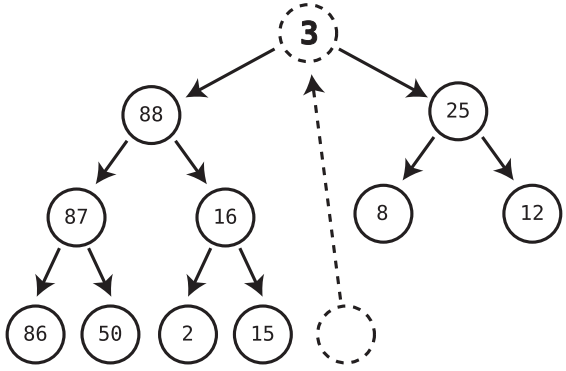
Step 3: The trickle node's children are currently the 86 and the 50. The 86 is the larger of the two, and it is also greater than the trickle node, so we swap the 86 with the trickle node:



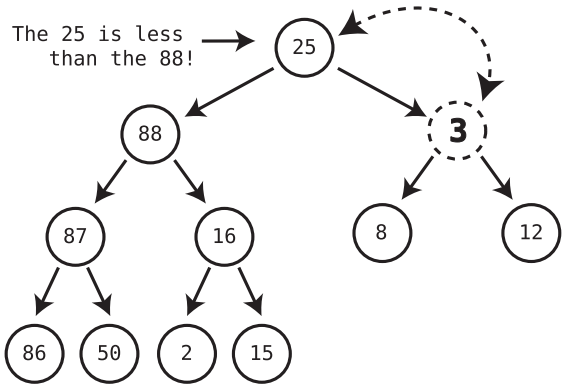
At this point, the trickle node has no children that are greater than it. (In fact, it has no children at all.) So, we're done, as the heap condition has been restored.

The reason why we always swap the trickle node with the *greater* of its two children is because if we swap it the with the smaller one, we'd end up violating the heap condition immediately. Watch what happens when we try to swap the trickle node with smaller child.

Let's start again with the trickle node of 3 as our root:



Let's swap the 3 with the 25, which is the smaller child:



We've now placed the 25 in a situation where it is a parent of the 88. Since the 88 is greater than its parent, the heap condition has been broken.

Like insertion, the time complexity of deletion from a heap is O(log N), as we have to trickle a node from the root down through all log(N) levels of the heap.

Heaps vs. Ordered Arrays

Now that you know the efficiency of heaps, let's see why it's a great choice for implementing priority queues.

Here's a side-by-side comparison of ordered arrays versus heaps:

	Ordered Array	Heap
Insertion	O(N)	O(log N)
Deletion	O(1)	O(log N)

At first glance, it seems that it's a wash. Ordered arrays are slower than heaps when it comes to insertion, but faster than heaps for deletion.

However, heaps are considered to be the better choice, and here's why.

While O(1) is extremely fast, O(log N) is still *very* fast. And O(N), by comparison, is slow. With this in mind, we can rewrite the earlier table this way:

	Ordered Array	Heap
Insertion	Slow	Very fast
Deletion	Extremely fast	Very fast

In this light, it becomes clearer as to why the heap is considered the better choice. We'd rather use a data structure that is consistently very fast than a data structure that is sometimes extremely fast and sometimes slow.

It's worth pointing out that priority queues generally perform insertions and deletions in about equal proportion. Think about the emergency room example, where we expect to treat everyone who comes in. So, we want both our insertions and deletions to be fast. If either operation is slow, our priority queue will be inefficient.

With a heap, then, we ensure that both of the priority queue's primary operations—insertion and deletion—perform at a very fast clip.

The Problem of the Last Node…Again

While the heap deletion algorithm seems straightforward, it once again raises the Problem of the Last Node.

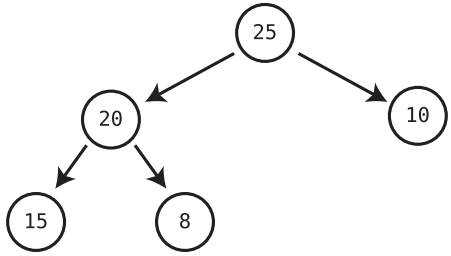
I explained that the first step of deletion requires us to move the last node and turn it into the root node. But, how do we find the last node in the first place?

Before we solve the Problem of the Last Node, let's first explore why insertion and deletion are so dependent on the last node anyway. Why couldn't we insert new values elsewhere in the heap? And why, when deleting, can't we replace the root node with some other node other than the last node?

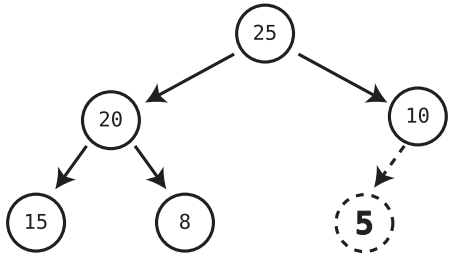
Now, if you think about it, you'll realize that if we were to use other nodes, the heap would become incomplete. But this begs the next question: why is completeness *important* for the heap?

The reason why completeness is important is because we want to ensure our heap remains *well-balanced.*

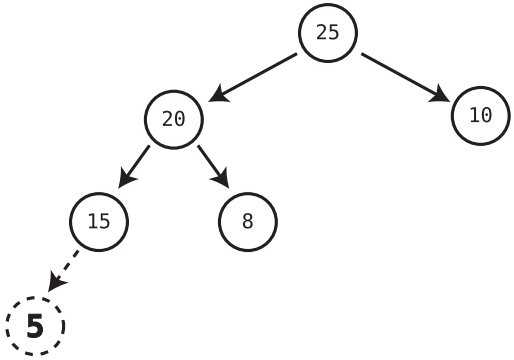
To see this clearly, let's take another look at insertion. Let's say we have the following heap:



If we want to insert a 5 into this heap, the only way to keep the heap well-balanced is by making the 5 the last node, in this case, making it a child of the 10:

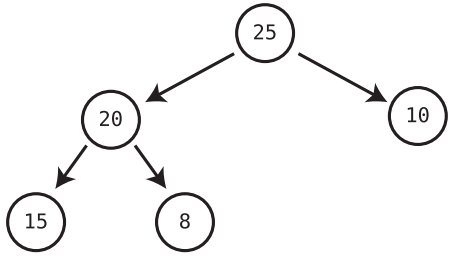


Any alternative to this algorithm would cause imbalance. Say, in an alternative universe, the algorithm was to insert the new node into the bottom leftmost node, which we could easily find by traversing the left children until we hit the bottom. This would make the 5 a child of the 15:



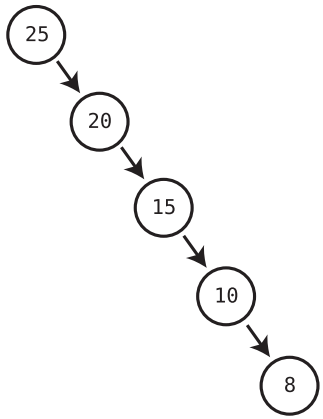
Our heap is now somewhat imbalanced, and it's easy to see how much more imbalanced it would become if we kept inserting new nodes at the bottom leftmost spot.

Similarly, when deleting from a heap, we always turn the last node into the root because, otherwise, the heap can become imbalanced. Take again our example heap:



If in our alternative universe we always moved the bottom rightmost node into the root position, the 10 would become the root node, and we'd end up with an imbalanced heap with a bunch of left descendants and zero right descendants.

Now, the reason why this balance is so important is because it's what allows us to achieve O(log N) operations. In a severely imbalanced tree like the following one, traversing it could take O(N) steps instead:



But this brings us back to the Problem of the Last Node. What algorithm would allow us to consistently find the last node of any heap? (Again, without having to traverse all N nodes.)

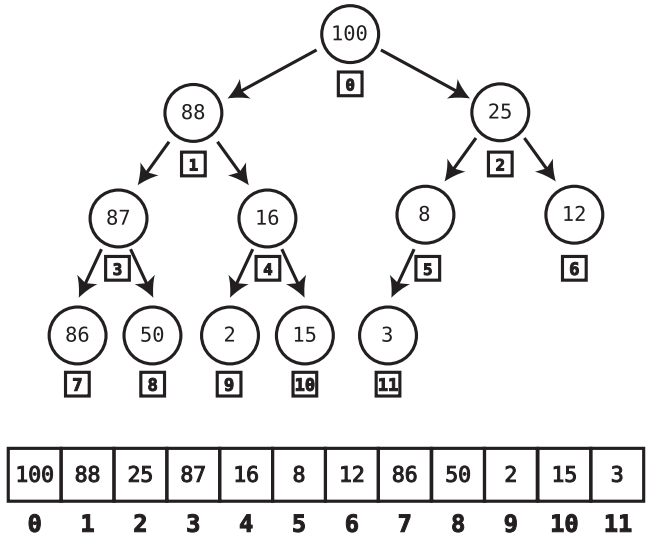
And this is where our plot takes a sudden twist.

Arrays as Heaps

Because finding the last node is so critical to the heap's operations, and we want to make sure that finding the last node is efficient, heaps are *usually implemented using arrays.*

While until now we always assumed that every tree consists of independent nodes connected to each other with links (just like a linked list), you will now see that we can also use an array to implement a heap. That is, the heap itself can be an abstract data type that really uses an array under the hood.

The diagram on page 296 shows how an array is used to store the values of a heap.



The way this works is that we assign each node to an index within the array. In the previous diagram, the index of each node is found in a square below the node. If you look carefully, you'll see that we assign the index of each node according to a specific pattern.

The root node is always stored at index 0. We then move down a level and go from left to right, assigning each node to the next available index in the array. So, on the second level, the left node (88) becomes index 1, and the right node (25) becomes index 2. When we reach the end of a level, we move down to the next level and repeat this pattern.

Now, the reason why we're using an array to implement the heap is because it solves the Problem of the Last Node. How?

When we implement the heap in this fashion, *the last node will always be the final element of the array.* Since we move top down and left to right when assigning each value to the array, the last node will always be the final value in the array. In the previous example, you can see that the 3, which is the last node, is the last value in the array.

Because the last node will always be found at the end of the array, it becomes trivial to find the last node: we just need to access the final element. Additionally, when we insert a new node into the heap, we do so at the end of the array in order to make it the last node.

Before we get into the other details of how an array-based heap works, we can already begin to code its basic structure. Here is the beginning of our heap implementation in Ruby:

```
class Heap
  def initialize
    @data = []
  end

  def root_node
    return @data.first
  end

  def last_node
    return @data.last
  end
end
```

As you can see, we initialize the heap as an empty array. We have a `root_node` method, which returns the first item of this array, and we also have a `last_node` method that returns the last value of this array.

Traversing an Array-Based Heap

As you've seen, the heap's insertion and deletion algorithms require us to be able to trickle our way through the heap. Trickling, in turn, requires us to be able to traverse the heap by accessing a node's parent or children. But how do we move from node to node when all the values are merely stored in an array? Traversing a heap would have been straightforward if we could have simply followed each node's links. But now that the heap is an array under the hood, how do we know which nodes are connected to each other?

There's an interesting solution to this. It turns out that when we assign the indexes of the heap's nodes according to the pattern described earlier, the following traits of a heap are always true:

- To find the left child of any node, we can use the formula, `(index * 2) + 1`
- To find the right child of any node, we can use the formula, `(index * 2) + 2`

Take another look at the previous diagram and focus on the 16, which is at index 4. To find its left child, we multiply its index (4) by 2 and add 1, which yields 9. This means that index 9 is the left child of the node at index 4.

Similarly, to find the right child of index 4, we multiply the 4 by 2 and add 2, which yields 10. This means that index 10 is the right child of index 4.

Because these formulas always work, we're able to treat our array as a tree.

Let's add these two methods to our Heap class:

```
def left_child_index(index)
  return (index * 2) + 1
end

def right_child_index(index)
  return (index * 2) + 2
end
```

Each of these methods accept an index within the array and return the left or right child index, respectively.

Here's another important trait of array-based heaps:

- To find a node's parent, we can use the formula, (index - 1) / 2

Note that this formula uses integer division, meaning we throw away any numbers beyond the decimal point. For example, 3 / 2 is considered 1, rather than the more accurate 1.5.

Again, in our example heap, focus on index 4. If we take that index, subtract 1, and then divide by 2, we get 1. And as you can see in the diagram, the parent of the node at index 4 is found at index 1.

So, now we can add another method to our Heap class:

```
def parent_index(index)
  return (index - 1) / 2
end
```

This method accepts an index and calculates the index of its parent node.

Code Implementation: Heap Insertion

Now that we have the essential elements of our Heap in place, let's implement the insertion algorithm:

```
def insert(value)
  # Turn value into last node by inserting it at the end of the array:
  @data << value

  # Keep track of the index of the newly inserted node:
  new_node_index = @data.length - 1

  # The following loop executes the "trickle up" algorithm.

  # If the new node is not in the root position,
  # and it's greater than its parent node:
  while new_node_index > 0 &&
  @data[new_node_index] > @data[parent_index(new_node_index)]
```

```
    # Swap the new node with the parent node:
    @data[parent_index(new_node_index)], @data[new_node_index] =
    @data[new_node_index], @data[parent_index(new_node_index)]

    # Update the index of the new node:
    new_node_index = parent_index(new_node_index)
  end
end
```

As usual, let's break this thing down.

Our insert method accepts the value we're inserting into our heap. The first thing we do is make this new value the last node by adding it to the very end of the array:

```
@data << value
```

Next, we keep track of the index of the new node, as we'll need it later. Right now, the index is the last index in the array:

```
new_node_index = @data.length - 1
```

Next, we trickle up the new node to its proper place using a while loop:

```
while new_node_index > 0 &&
@data[new_node_index] > @data[parent_index(new_node_index)]
```

This loop runs as long as two conditions are met. The main condition is that the new node is greater than its parent node. We also make a condition that the new node must have an index greater than 0, as funny things can happen if we try to compare the root node with its nonexistent parent.

Each time this loop runs, we swap the new node with its parent node, since the new node is currently greater than the parent:

```
@data[parent_index(new_node_index)], @data[new_node_index] =
@data[new_node_index], @data[parent_index(new_node_index)]
```

We also then update the index of the new node appropriately:

```
new_node_index = parent_index(new_node_index)
```

Since this loop only runs while the new node is greater than its parent, the loop ends once the new node is in its proper place.

Code Implementation: Heap Deletion

Here is a Ruby implementation of deleting an item from a heap. The main method is the delete method, but to make the code simpler, we've created two helper methods, has_greater_child and calculate_larger_child_index:

```
def delete
  # We only ever delete the root node from a heap, so we
  # pop the last node from the array and make it the root node:
  @data[0] = @data.pop
  # Track the current index of the "trickle node":
  trickle_node_index = 0

  # The following loop executes the "trickle down" algorithm:

  # We run the loop as long as the trickle node has a child
  # that is greater than it:
  while has_greater_child(trickle_node_index)
    # Save larger child index in variable:
    larger_child_index = calculate_larger_child_index(trickle_node_index)

    # Swap the trickle node with its larger child:
    @data[trickle_node_index], @data[larger_child_index] =
    @data[larger_child_index], @data[trickle_node_index]

    # Update trickle node's new index:
    trickle_node_index = larger_child_index
  end
end

def has_greater_child(index)
  # We check whether the node at index has left and right
  # children and if either of those children are greater
  # than the node at index:
  (@data[left_child_index(index)] &&
   @data[left_child_index(index)] > @data[index]) ||
  (@data[right_child_index(index)] &&
    @data[right_child_index(index)] > @data[index])
end

def calculate_larger_child_index(index)
  # If there is no right child:
  if !@data[right_child_index(index)]
    # Return left child index:
    return left_child_index(index)
  end

  # If right child value is greater than left child value:
  if @data[right_child_index(index)] > @data[left_child_index(index)]
    # Return right child index:
    return right_child_index(index)
  else # If the left child value is greater or equal to right child:
    # Return the left child index:
    return left_child_index(index)
  end
```

Let's first dive into the delete method.

The delete method doesn't accept any arguments, since we only ever delete the root node. Here's how the method works.

First, we remove the last value from the array and make it the first value:

```
@data[0] = @data.pop
```

This simple line effectively deletes the original root node, as we are overwriting the root node's value with the last node's value.

Next, we need to trickle the new root node down to its proper place. We called this the "trickle node" earlier, and our code reflects this as well.

Before we start the actual trickling, we keep track of the trickle node's index, as we'll need it later. Currently, the trickle node is at index 0:

```
trickle_node_index = 0
```

We then use a while loop to execute the trickle-down algorithm. The loop runs as long as the trickle node has any children that are greater than it:

```
while has_greater_child(trickle_node_index)
```

This line uses the has_greater_child method, which returns whether a given node has any children who are greater than that node.

Within this loop, we first find the index of the greater of the trickle node's children:

```
larger_child_index = calculate_larger_child_index(trickle_node_index)
```

This line uses the method calculate_larger_child_index, which returns the index of the trickle node's greater child. We store this index in a variable called larger_child_index.

Next, we swap the trickle node with its greater child:

```
@data[trickle_node_index], @data[larger_child_index] =
@data[larger_child_index], @data[trickle_node_index]
```

Finally, we update the index of the trickle node, which will be the index it was just swapped with:

```
trickle_node_index = larger_child_index
```

Alternate Heap Implementations

Our heap implementation is now complete. It's worth noting that while we did use an array to implement the heap under the hood, it *is possible* to implement a heap using linked nodes as well. (This alternative implementation uses a different trick to solve the Problem of the Last Node, one that involves binary numbers.)

However, the array implementation is the more common approach, so that is what I presented here. It's also really interesting to see how an array can be used to implement a tree.

Indeed, it's possible to use an array to implement *any* sort of binary tree, such as the binary search tree from the previous chapter. However, the heap is the first case of a binary tree where an array implementation provided an advantage, as it helps us find the last node easily.

Heaps as Priority Queues

Now that you understand how heaps work, we can circle back to our old friend, the priority queue.

Again, the primary function of a priority queue is to allow us immediate access to the highest-priority item in the queue. In our emergency room example, we want to first address the person with the most serious problem.

It's for this reason that a heap is a natural fit for priority queue implementations. The heap gives us immediate access to the highest-priority item, which can always be found at the root node. Each time we take care of the highest-priority item (and then remove it from the heap), the next-highest item floats to the top of the heap and is on deck to be addressed next. And the heap accomplishes this while maintaining very fast insertions and deletions, both of which are O(log N).

Contrast this to the ordered array, which requires much slower insertions of O(N) in order to ensure that each new value is in its proper place.

It turns out that the heap's weak ordering *is its very advantage.* The fact that it doesn't have to be perfectly ordered allows us to insert new values in O(log N) time. At the same time, the heap is *ordered just enough* so that we can always access the one item we need at any given time—the heap's greatest value.

Wrapping Up

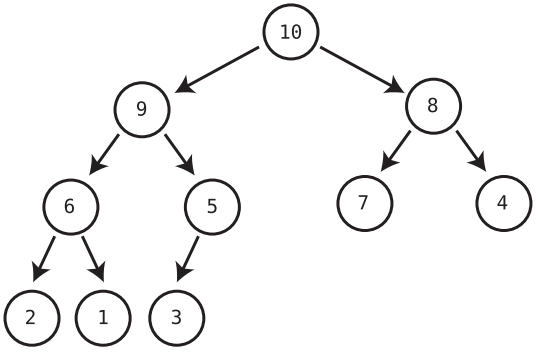
So far, we've seen how different types of trees can optimize different types of problems. Binary search trees kept search fast while minimizing the cost of insertions, and heaps were the perfect tool for building priority queues.

In the next chapter, we'll explore another tree that is used to power some of the most common text-based operations you use on a day-to-day basis.

Exercises

The following exercises provide you with the opportunity to practice with heaps. The solutions to these exercises are found in the section, Chapter 16, on page 458.

1. Draw what the following heap would look like after we insert the value 11 into it:



2. Draw what the previous heap would look like after we delete the root node.

3. Imagine you've built a brand-new heap by inserting the following numbers into the heap in this particular order: 55, 22, 34, 10, 2, 99, 68. If you then pop them from the heap one at a time and insert the numbers into a new array, in what order would the numbers now appear?

CHAPTER 17

It Doesn't Hurt to Trie

Did you ever wonder how your smartphone's autocomplete feature works? Autocomplete is the feature you see when you start typing "catn" and your phone suggests that the word you're about to type is either "catnip" or "catnap." (Yes, I text my friends about catnip all the time.)

For this to work, your phone has access to an entire dictionary of words. But what data structure are these words stored in?

Let's imagine, for a moment, that all the words in the English language are stored within an array. If the array were unsorted, we'd have to search *every word* in the dictionary to find all words that start with "catn." This is O(N), and a very slow operation considering that N is a pretty large number in this case, as it's the number all of the words in the dictionary.

A hash table wouldn't help us either, as it hashes the *entire word* to determine where in memory the value should be stored. As the hash table wouldn't have a "catn" key, we'd have no easy way to locate "catnip" or "catnap" within the hash table.

Things improve greatly if we store the words inside an ordered array. That is, if the array contained all the words in alphabetical order, we could use binary search to find a word beginning with "catn" in O(log N) time. And while O(log N) isn't bad, we can do even better. In fact, if we use a special tree-based data structure, we can approach O(1) speeds in finding our desired words.

The examples in this chapter show how *tries* can be used for applications dealing with text, as tries allow for important features such as autocomplete and autocorrect. However, tries have other uses as well, including applications involving IP addresses or phone numbers.

Tries

The *trie* is a kind of tree that is ideal for text-based features such as autocomplete. Before we get into how the trie works, though, let's first address its pronunciation.

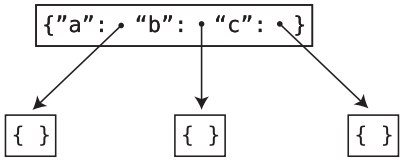
In my (unasked for) opinion, the trie is one of the most unfortunately-named data structures out there. The word trie is actually derived from the word *retrieval.* So technically, it should be pronounced "tree." But since that would be confused with the word *tree,* which is the general term for all tree-based data structures, most people pronounce trie as "try." Some resources refer to this same data structure as the prefix tree or the digital tree, but trie, amazingly, remains the most popular name. So, there you have it.

Here's one last comment before we dive into the details. The trie is not as well documented as the other data structures in this book, and many different resources implement the trie in slightly different ways. I've chosen a particular implementation that I find to be the most straightforward and understandable, but you'll find other implementations out there. In any case, the general ideas behind most of the implementations remain the same.

The Trie Node

Like most other trees, the trie is a collection of nodes that point to other nodes. However, the trie is *not* a binary tree. Whereas a binary tree doesn't allow any node to have more than two child nodes, a trie node can have *any number* of child nodes.

In our implementation, each trie node contains a hash table, where the keys are English characters and the values are other nodes of the trie. Take a look at the following diagram:



Here, the root node contains a hash table with the keys `"a"`, `"b"`, and `"c"`. The values are other trie nodes, which are the children of this node. These children also contain hash tables, which will in turn point to *their* children. (For now, we left the children's hash tables empty, but they'll contain data in future diagrams.)

The implementation of the actual trie node itself is very simple. Here is our Python version of the TrieNode class:

```
class TrieNode:

  def __init__(self):
    self.children = {}
```

As you can see, the TrieNode just contains a hash table.

If we print (to the console) the data from our root node in the previous example, we'd get something like this:

```
{'a': <__main__.TrieNode instance at 0x108635638>,
 'b': <__main__.TrieNode instance at 0x108635878>,
 'c': <__main__.TrieNode instance at 0x108635ab8>}
```

Again, in this hash table, the keys are individual character strings, and the values are instances of other TrieNodes.

The Trie Class

To fully create our trie, we will also need a separate Trie class, which will keep track of the root node:

```
class Trie:

  def __init__(self):
    self.root = TrieNode()
```

This class keeps track of a self.root variable that points to the root node. In this implementation, when we create a new Trie, it begins with an empty TrieNode as its root.

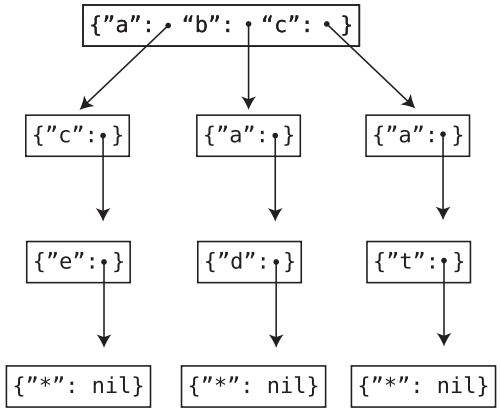
As we progress through this chapter, we're going to add our trie operation methods to this Trie class as well.

Storing Words

Now, the point of our trie is to store words. Let's see how the following trie stores the words, "ace," "bad," and "cat" as shown in the diagram on page 308.

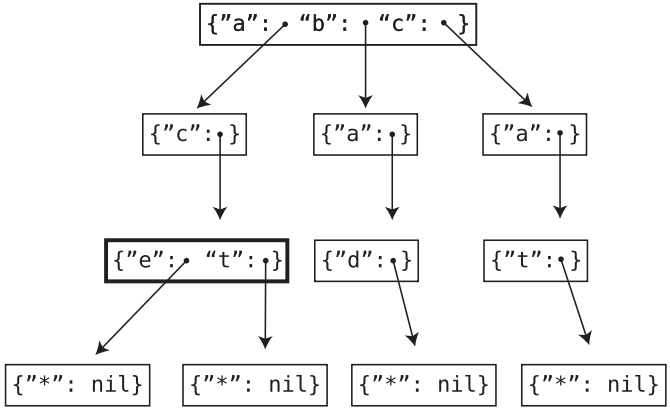
This trie stores the three words by turning each character of each word into its own trie node. If you start with the root node and follow its "a" key, for example, it points to a child node containing a key of "c". The "c" key, in turn, points to a node that contains a key of "e". When we string these three characters together, we get the word "ace".

With this pattern, you can see how the trie also stores the words "bad" and "cat".



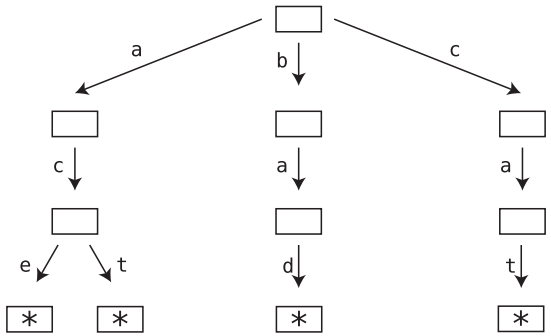
You'll note that the final characters in these words actually have children nodes of their own. If you look at the `"e"` node from the word `"ace"`, for example, you can see that the `"e"` points to a child node that contains a hash table with a key of `"*"`, an asterisk. (The value doesn't actually matter, so it can simply be null.) This indicates we've reached the end of a word, and that `"ace"`, therefore, is a complete word. The need for this `"*"` key will become more apparent shortly.

Now, here's where tries become even more interesting. Let's say we want to also store the word `"act"`. To do this, we keep the existing `"a"` and `"c"` keys that already exist, but we add one new node containing the key of `"t"`. Look at the following diagram:



As you can see, the bolded node's hash table now contains *two* children nodes, an "e" and a "t". By doing this, we indicate that both "ace" and "act" are valid dictionary words.

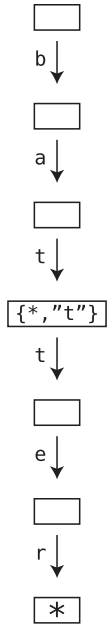
To make things easier to visualize going forward, we are going to represent our tries using a simpler diagram. Here is the same trie using this new format:



In this diagram style, we place each hash table key next to an arrow pointing to its child node.

The Need for the Asterisk

Let's say we want to store the words "bat" and "batter" in a trie. This is an interesting case, since the word "batter" actually contains the word "bat." Here's how we handle this:

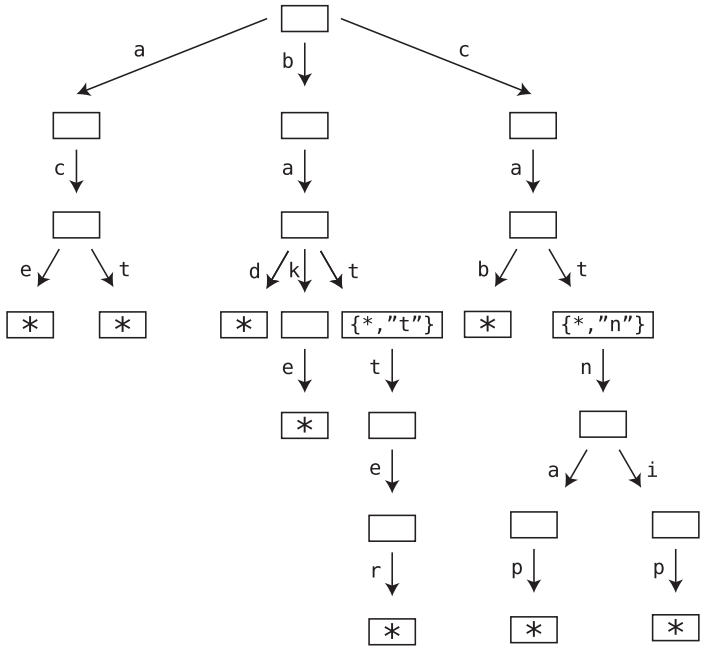


The first "t" points to a node that has *two* keys. One key is a "*" key (with a null value), while the other is a "t" whose value points to another node. This indicates that "bat" is itself a word, even though it's also a prefix of the longer word, "batter".

Note that in this diagram, we're not using classic hash table syntax anymore, but a condensed syntax so as to save space. We've used curly braces to indicate that the node contains a hash table. However, the {*, "t"} is not a key-value pair, but simply two keys. The "*" key has a null value, and the "t" key has the next node as its value.

This is why those "*" are critical. We need them to indicate when parts of a word are also words themselves.

Let's bring this all together with a more complex example. Here is a trie that contains the words "ace," "act," "bad," "bake," "bat," "batter," "cab," "cat," "catnap," and "catnip":



Tries used in real applications may contain thousands of words. If they don't contain the entire English language, they at least contain the most common words.

In order to build our autocomplete feature, let's first analyze the basic trie operations.

Trie Search

The most classic trie operation is search, namely, determining whether a string is found in the trie. There are two flavors of search: we can search to see whether the string is a *complete* word, or we can search to see whether the string is at least a word *prefix* (that is, the beginning of a word). These two versions are similar, but we'll implement the latter one, where our search will look for prefixes. This search will end up finding complete words as well, since a complete word is at least as good as a prefix.

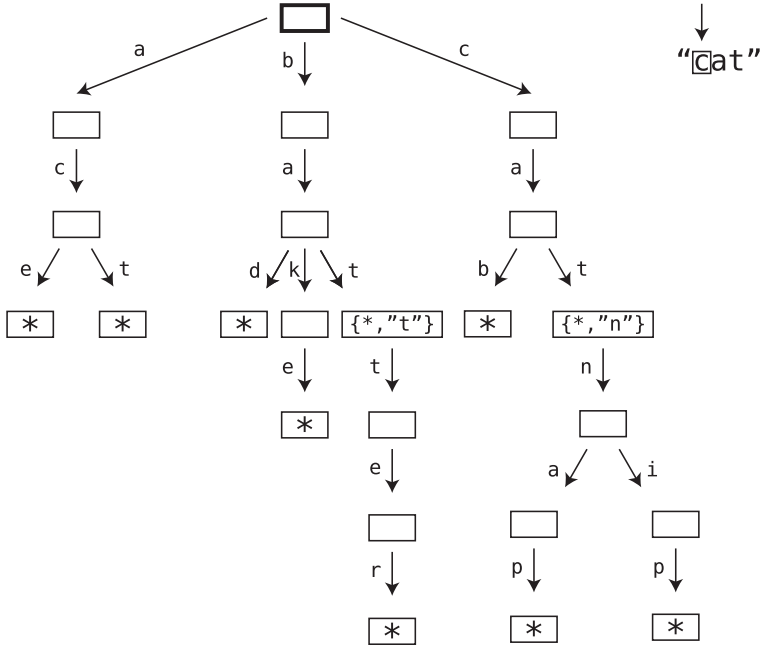
The algorithm for prefix search performs the following steps (they'll become clearer when we walk though an example that follows):

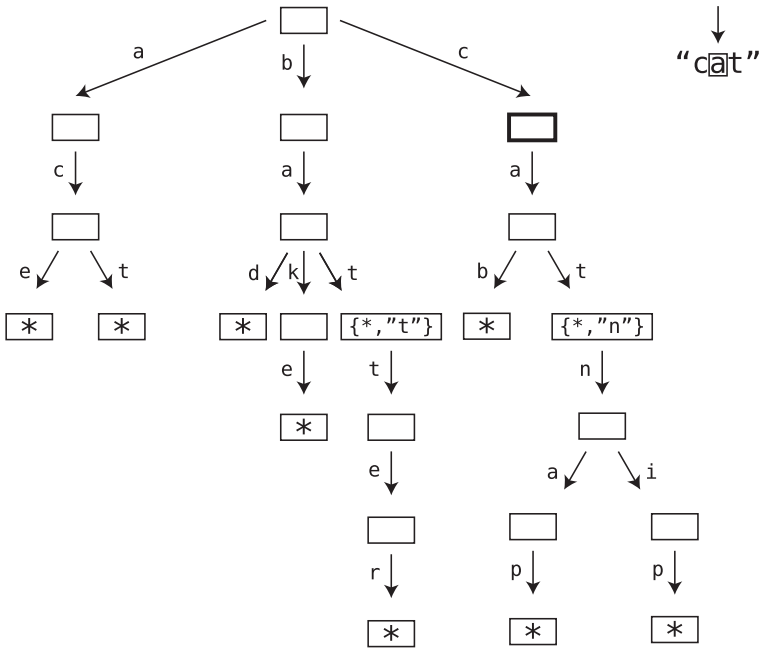
1. We establish a variable called `currentNode`. At the beginning of our algorithm, this points to the root node.
2. We iterate over each character of our search string.
3. As we point to each character of our search string, we look to see if the `currentNode` has a child with that character as a key.
4. If it does not, we return `None`, as it means our search string does not exist in the trie.
5. If the `currentNode` *does* have a child with the current character as the key, we update the `currentNode` to become that child. We then go back to Step 2, continuing to iterate over each character in our search string.
6. If we get to the end of our search string, it means we've found our search string.

Let's see this in action by searching for the string `"cat"` in our trie from earlier.

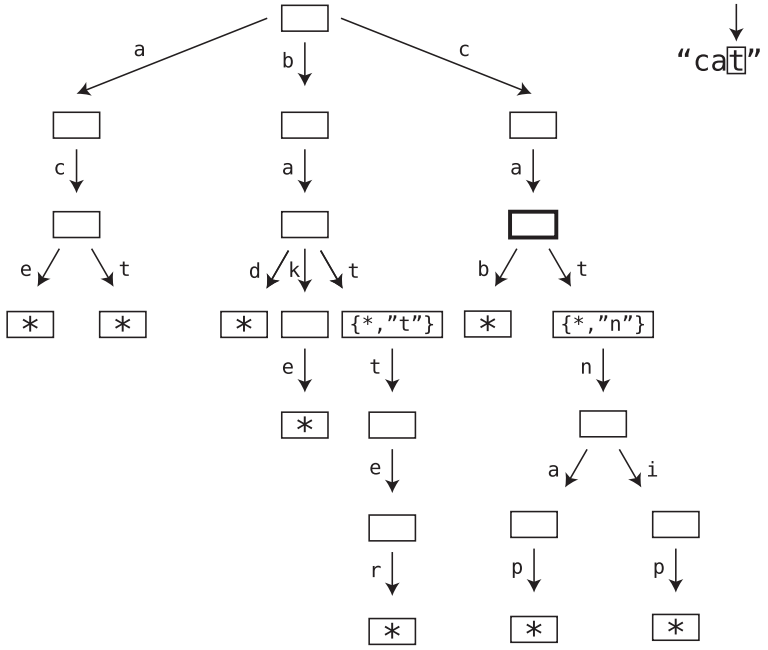
Setup: We set the `currentNode` to be the root node. (The `currentNode` is indicated in bold in the diagrams on the following pages.) We also point to the first character of our string, which is the `"c"` as shown in the top diagram on page 312.

Step 1: Since the root node has `"c"` as a child key, we update the `currentNode` to become that key's value. We also continue iterating through the characters in our search string, so we point to the next character, which is the `"a"` as shown in the bottom diagram on page 312.


"cat"
a
b
c
c
a
a
e
t
d
k
t
b
t
*
*
*
{*,"t"}
*
{*,"n"}
e
t
n
e
a
i
r
p
p
*
*
*


"cat"
a
b
c
c
a
a
e
t
d
k
t
b
t
*
*
*
{*,"t"}
*
{*,"n"}
e
t
n
e
a
i
r
p
p
*
*
*

Step 2: We inspect the currentNode for a child with the key of "a". It has one, so we make that child the new currentNode. We then proceed to search for the next character in our string, which is the "t":



Step 3: We're now pointing to the "t" of our search string. Since the currentNode has a "t" child, we follow it as shown in the diagram on page 314.

Since we've reached the end of our search string, it means we've found "cat" in our trie.

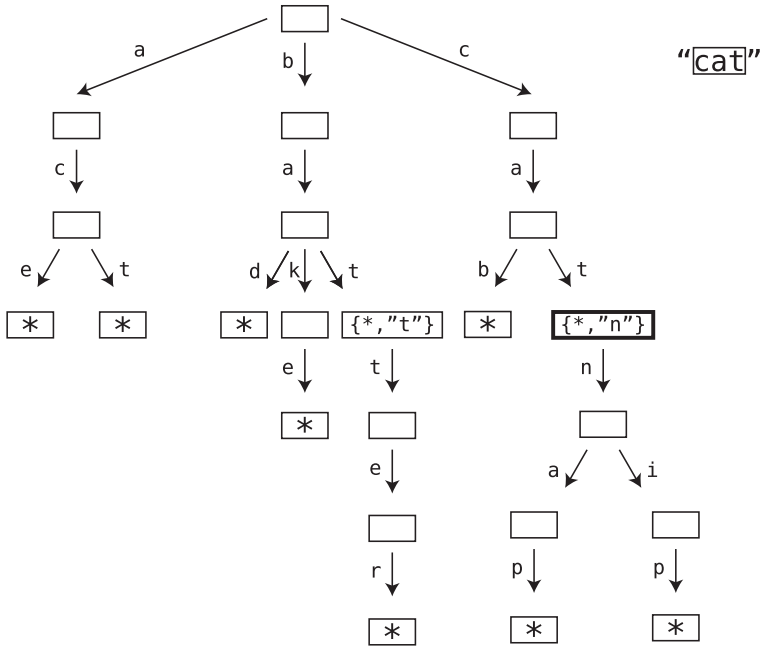
Code Implementation: Trie Search

Let's implement trie search by adding a search method to our Trie class:

```
def search(self, word):
  currentNode = self.root

  for char in word:
    # If the current node has child key with current character:
    if currentNode.children.get(char):
      # Follow the child node:
      currentNode = currentNode.children[char]
    else:
      # If the current character isn't found among
      # the current node's children, our search word
      # must not be in the trie:
      return None

  return currentNode
```



Our search method accepts a string that represents the word (or prefix) we're searching for.

First, we establish the root node as our currentNode:

```
currentNode = self.root
```

Then, we iterate over each character of our search word:

```
for char in word:
```

Within each round of the loop we check whether the current node has any children with the current character as the key. If there is such a child, we update the current node to be the child node:

```
if currentNode.children.get(char):
  currentNode = currentNode.children[char]
```

If there is no such child, we return None, as it means we've hit a dead end, and our search word is not contained in the trie.

If we get past the end of the loop, it means we've found the entire word in our trie. In this case, we return the currentNode. The reason we return the current

node, as opposed to just returning True, is to help us with the autocomplete feature, as I'll explain when we get there.

The Efficiency of Trie Search

The great thing about trie search is that it's incredibly efficient.

Let's analyze the number of steps it takes.

In our algorithm, we focus on each character of our search string one at a time. As we do so, we use each node's hash table to find the appropriate child node in one step. As you know, a hash table lookup takes just O(1) time. It turns out, then, that our algorithm takes as many steps *as there are characters in our search string.*

This can be much faster than using binary search on an ordered array. Binary search is O(log N), with N being the number of words in our dictionary. Trie search, on the other hand, takes only as many steps as the number of characters in our search term. For a word like "cat," that's just three steps.

Expressing trie search in terms of Big O is slightly tricky. We can't quite call it O(1), since the number of steps isn't constant, as it all depends on the search string's length. And O(N) can be misleading, since N normally refers to the amount of data in the data structure. This would be the number of nodes in our trie, which is a much greater number than the number of characters in our search string.

Most references have decided to call this O(K), where K is the number of characters in our search string. Any letter other than N would have worked here, but K it is.

Even though O(K) isn't constant, as the size of the search string can vary, O(K) is similar to constant time in one important sense. Most *non*-constant algorithms are tied to the amount of data at hand. That is, as N data increases, the algorithm slows down. With an O(K) algorithm, though, our trie can grow tremendously, but that will have no affect on the speed of our search. An O(K) algorithm on a string of three characters will always take three steps, no matter how large the trie is. The only factor that affects our algorithm's speed is the size of our input, rather than all the available data. This makes our O(K) algorithm extremely efficient.

While search is the most common type of operation performed on tries, it's difficult to test drive it without populating our trie with data, so let's tackle insertion next.

Trie Insertion

Inserting a new word into a trie is similar to searching for an existing word.

Here's how the algorithm goes:

1. We establish a variable called currentNode. At the beginning of our algorithm, this points to the root node.
2. We iterate over each character of our search string.
3. As we point to each character of our search string, we look to see if the currentNode has a child with that character as a key.
4. If it does, we update the currentNode to become that child node and we go back to Step 2, moving on to the next character of our search string.
5. If the currentNode does *not* have a child node that matches the current character, we create such a child node and update the currentNode to be this new node. We then go back to Step 2, moving on to the next character of our search string.
6. After we insert the final character of our new word, we add a "*" child to the last node to indicate the word is complete.

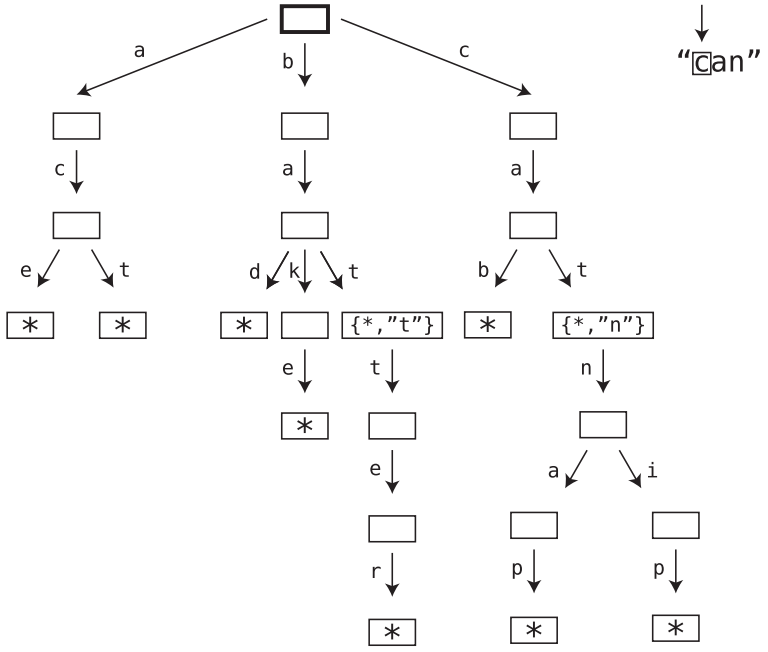
Let's see this in action by inserting the word "can" into our earlier example trie.

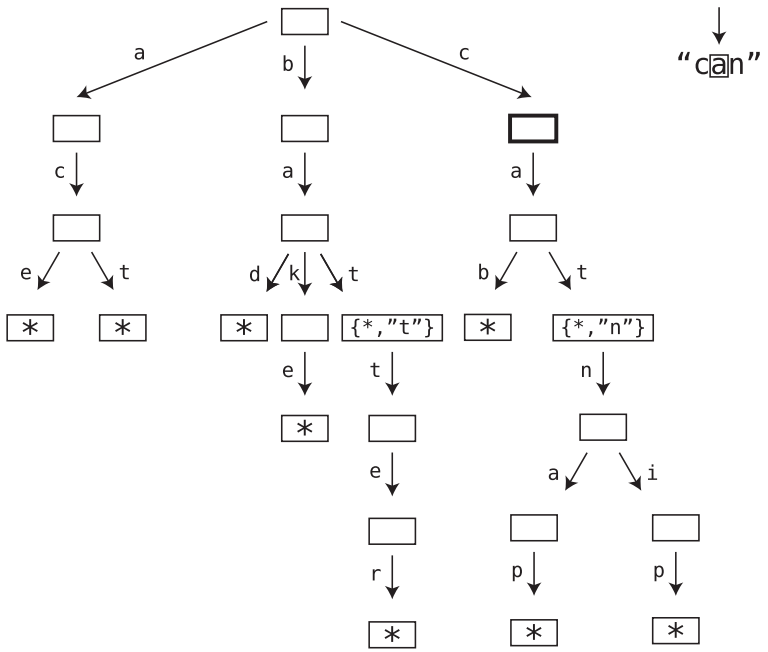
Setup: We set the currentNode to be the root node. We also point to the first character of our string, which is the "c" as shown in the top diagram on page 317.

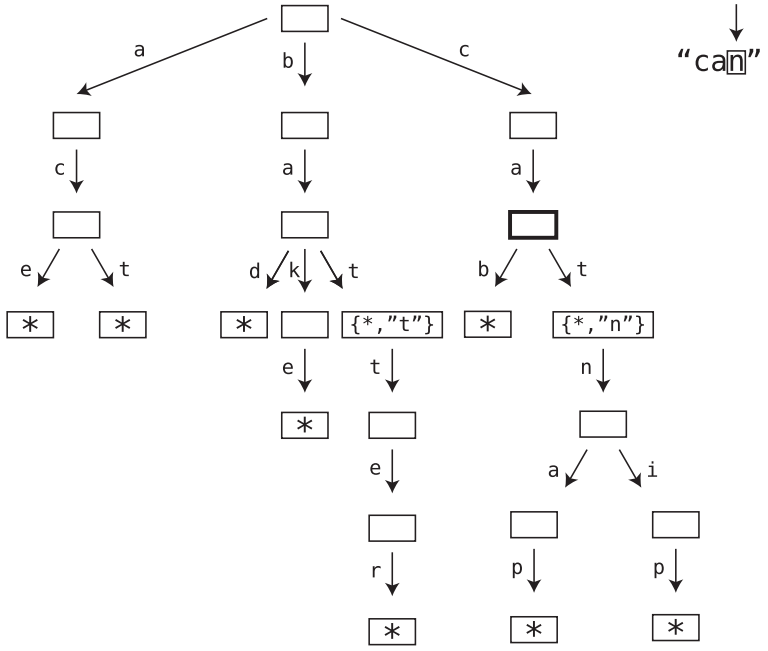
Step 1: The root node has a "c" child key, so we turn that key's value into the currentNode. We also point to the next character of our new word, the "a" as shown in the bottom diagram on page 317.

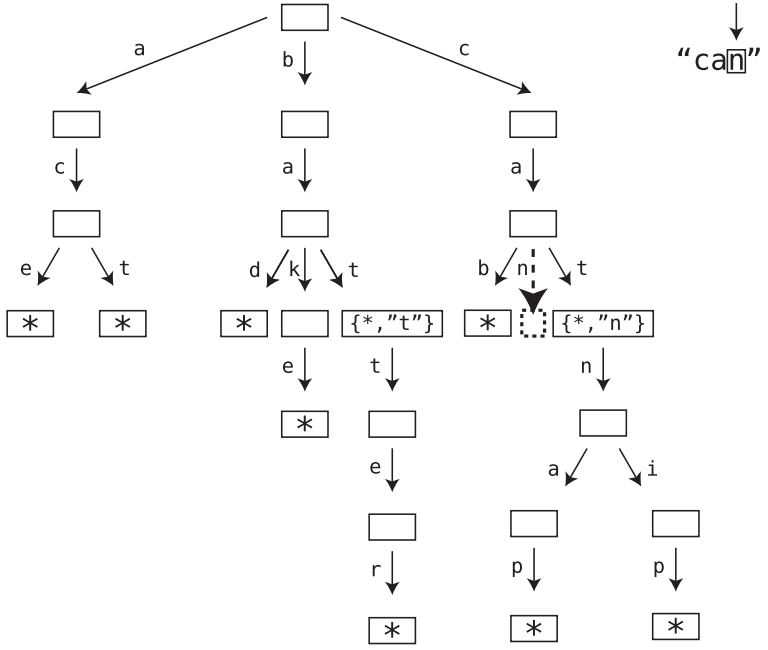
Step 2: We inspect the currentNode for a child with the key of "a". There is one, so we make that the currentNode and point to the next character of our string, which is the "n" as shown in the top diagram on page 318.

Step 3: The currentNode does *not* have an "n", so we need to create that child as shown in the bottom diagram on page 318.

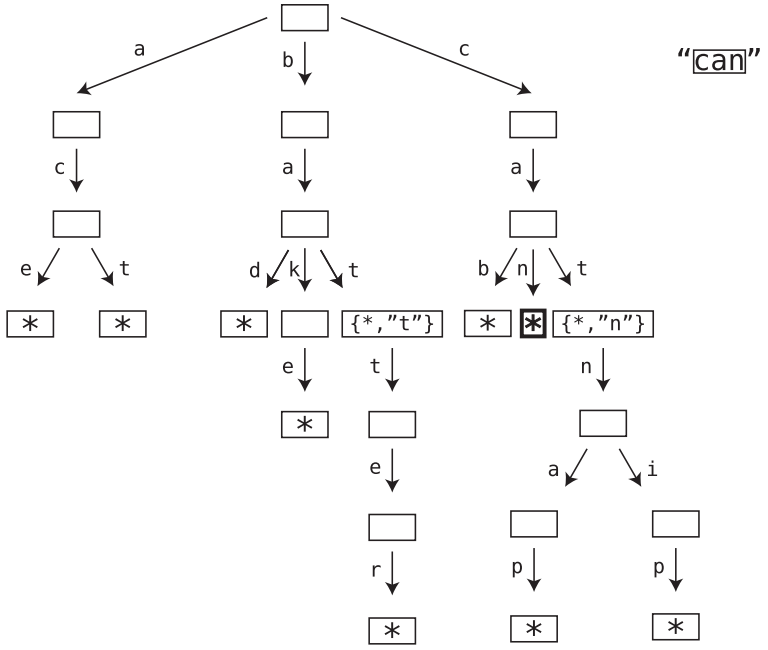

a
b
c
"can"
c
a
a
e
t
d
k
t
b
t
*
*
*
*
{*,"t"}
*
{*,"n"}
e
t
n
e
a
i
r
p
p
*
*
*


a
b
c
"can"
c
a
a
e
t
d
k
t
b
t
*
*
*
*
{*,"t"}
*
{*,"n"}
e
t
n
e
a
i
r
p
p
*
*
*


a
b
c
"can"
c
a
a
e
t
d
k
t
b
t
*
*
*
{*,"t"}
*
{*,"n"}
e
t
n
e
a
i
r
p
p
*
*
*


a
b
c
"can"
c
a
a
e
t
d
k
t
b
n
t
*
*
*
{*,"t"}
*
{*,"n"}
e
t
n
e
a
i
r
p
p
*
*
*

Step 4: We're done inserting "can" into our trie, so we cap it off with a child of "*":



And we're done!

Code Implementation: Trie Insertion

Here is the insert method for our Trie class. You'll note that most of it looks the same as the earlier search method:

```
def insert(self, word):
  currentNode = self.root

  for char in word:
    # If the current node has child key with current character:
    if currentNode.children.get(char):
      # Follow the child node:
      currentNode = currentNode.children[char]
    else:
      # If the current character isn't found among
      # the current node's children, we add
      # the character as a new child node:
      newNode = TrieNode()
      currentNode.children[char] = newNode

      # Follow this new node:
      currentNode = newNode

  # After inserting the entire word into the trie,
  # we add a * key at the end:
  currentNode.children["*"] = None
```

The first part of this method is the same as search. It diverges when the currentNode doesn't have the child that matches the current character. When this is the case, we add a new key-value pair to the currentNode's hash table, with the key being the current character, and the value being a new TrieNode:

```
newNode = TrieNode()
currentNode.children[char] = newNode
```

We then update the currentNode to be this new node:

```
currentNode = newNode
```

We then repeat the loop until we're done inserting our new word. Once we're done, we add a "*" key to the final node's hash table, with the value as None:

```
currentNode.children["*"] = None
```

Like search, trie insertion takes about O(K) steps. If we count the adding of the "*" at the end, it's technically K + 1 steps, but because we drop the constants, we express the speed as O(K).

Building Autocomplete

We're just about ready to build a real autocomplete feature. To make this a tad easier, let's first build a slightly simpler function that we'll use to help us with this feature.

Collecting All the Words

The next method we're going to add to our Trie class is a method that returns an array of *all* the words in the trie. Now, it's rare that we'd actually want to list the *entire* dictionary. However, we're going to allow this method to accept any node of the trie as an argument so that it can list all the words that start from that node.

The following method, called collectAllWords, collects a list of all the trie's words starting from a particular node:

```
def collectAllWords(self, node=None, word="", words=[]):
  # This method accepts three arguments. The first is the
  # node whose descendants we're collecting words from.
  # The second argument, word, begins as an empty string,
  # and we add characters to it as we move through the trie.
  # The third argument, words, begins as an empty array,
  # and by the end of the function will contain all the words
  # from the trie.
```

```
# The current node is the node passed in as the first parameter,
# or the root node if none is provided:
currentNode = node or self.root

# We iterate through all the current node's children:
for key, childNode in currentNode.children.items():
  # If the current key is *, it means we hit the end of a
  # complete word, so we can add it to our words array:
  if key == "*":
    words.append(word)
  else: # If we're still in the middle of a word:
    # We recursively call this function on the child node.
    self.collectAllWords(childNode, word + key, words)

return words
```

This method relies heavily on recursion, so let's break it down carefully.

The method accepts three primary arguments: `node`, `word`, and `words`. The `node` argument allows us to specify which node in the trie to start collecting the words from. If we don't pass in this argument, our method will start from the root node, collecting every word in the entire trie.

The `word` and `words` arguments are used as part of the method's recursion, so we aren't expected to specify them at the outset. By default, the `words` argument is an empty array. As we discover complete words in our trie, we add those strings to this array, which is what we return at the end of our function.

The `word` argument defaults to an empty string. As we move through the trie, we add characters to this `word`. When we reach a `"*"`, the `word` is considered complete and we add it to the `words` array.

Let's now break down each line of code.

The first thing we do is set the `currentNode`:

```
currentNode = node or self.root
```

By default, the `currentNode` will be the root node, unless we passed in some other node as the method's first parameter. Let's assume for now that the `currentNode` is indeed the root node.

Next, we begin a loop that iterates over all the key-value pairs in the `currentNode`'s children hash table:

```
for key, childNode in currentNode.children.items():
```

In each iteration of the loop, the key is always a single-character string, and the value, childNode, is another instance of TrieNode.

Let's skip to the else clause, as this is where the magic happens:

```
self.collectAllWords(childNode, word + key, words)
```

This line recursively calls the collectAllWords function, with the childNode passed in as the first argument. This is our way of traversing the trie and collecting the words as we go.

The second argument we pass in is word + key, making it so that as we move through each node of the trie, we add the key to the current word, building up the word as we go.

The third argument is the words array. By passing this array along in each recursive call, we're able to collect the complete words into it and build this list as we traverse the trie.

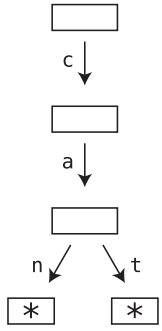
The base case is when we reach a "*" key, indicating we've completed a word. At this point, we can add the word to the words array:

```
if key == "*":
  words.append(word)
```

At the end of the function, we return the words array. If we called this function without passing in a specific node, this will return the complete list of words in the trie.

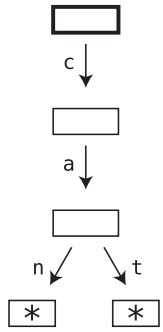
Recursion Walk-Through

Let's run through a quick visual example of this using a simple trie. This trie holds two words, "can" and "cat":



Call 1: In the very first call of collectAllWords, the currentNode starts out at the root, word is an empty string, and the words array is empty:

```
word = " "
words = [ ]
```

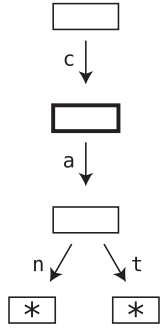


We iterate over the root node's children. The root node happens to only have one child key, the "c", which points to a child node. Before we recursively call collectAllWords on this child node, we need to add the current call to the call stack.

We then recursively call collectAllWords on the "c" child node. We also pass in word + key as the word argument. word + key is the string "c", since word was empty, and the key is "c". And we also pass in the words array, which is still empty. The next image shows where we are once we make this recursive call:

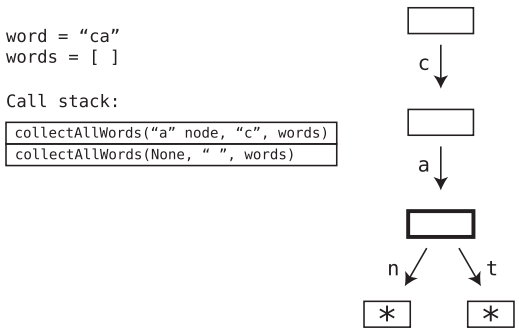
```
word = "c"
words = [ ]

Call stack:
collectAllWords(None, " ", words)
```



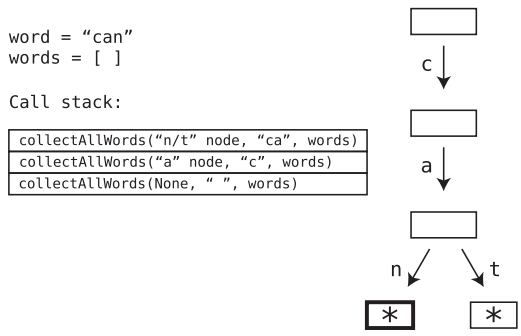
Call 2: We iterate over the current node's children. It has just one child key, the "a". Before recursively calling collectAllWords on the respective child node, we will add the current call to the call stack. In the diagram that follows, we call this current node the "a" node, meaning it's the node that has "a" as a child.

We then recursively call collectAllWords. We pass in the child node, "ca" (which is word + key), and the still-empty words array:



Call 3: We iterate over the current node's children, which are "n" and "t". We'll start with the "n" first. Before making any recursive calls, though, we need to first add the current call to our call stack. In the diagram that follows, we call this current node the "n/t" node, meaning it's the node that has both "n" and "t" as children.

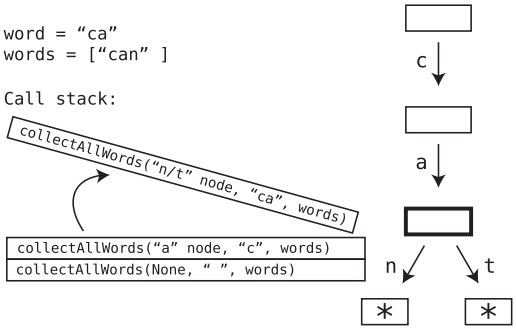
When we then call collectAllWords on the "n" child, and we also pass in "can" as the word argument, as well as the empty words array:



Call 4: We iterate over the current node's children. In this case, it has only one child, which is the "*". This is our base case. We add the current word, which is "can", to the words array:

```
words = ["can"]
```

Call 5: We now pop the top call from our call stack, which was the call of collectAllWords on the node with the children keys of "n" and "t", and where word was "ca". This means we now return to that call (as we return to whatever call we pop from the call stack):



There's a subtle but important point to make here. In the current call, word is back to "ca", since that was the word argument when we first initiated this call. However, the words array now contains the word "can", even though the array was empty when we originally made this call.

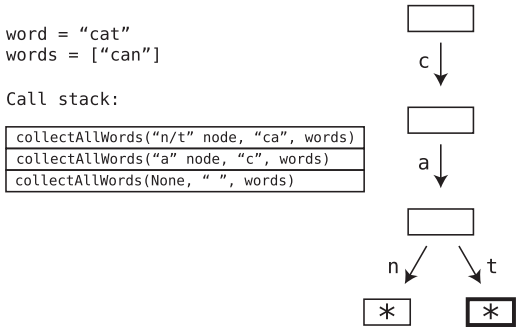
Here's the reason why this works. In many programming languages, an array can be passed up and down a call stack because the array remains the same object in memory even when we add new values to it. (The same concept applies to hash tables as well, which was how we were able to pass it along as part of the memoization technique you learned about in *Dynamic Programming*).

When a string is modified, on the other hand, the computer actually creates a new string instead of truly modifying the original string object. Therefore, when we updated word by changing it from "ca" to "can", the previous call still only has access to the original string, "ca". (In some languages, this may work slightly differently. For our purposes, though, this is the general idea.)

In any case, we're in the middle of a call where words contains the word "can", and word is "ca".

Call 6: At this point, we've already iterated over the "n" key, so the loop is now up to the "t" key. Before recursively calling collectAllWords on the "t" child node, we need to add the current call to the call stack again. (This actually will be the second time we add this call to the call stack. We previously popped it off, but now we're going to add it again.)

When we call collectAllWords on the "t" child, we pass in "cat" as the word argument (since that is word + key) , and the words array:



Call 7: We iterate over the current node's children. The only child here is "*", so we add the current word of "cat" to our words array:

```
words = ["can", "cat"]
```

At this point, we can unwind the call stack, popping off each call and completing their execution, which all end by returning the words array. The final call we complete—which was the first call to kick this all off— returns words as well. Because this contains the strings "can" and "cat", we've successfully returned the trie's entire word list.

Completing Autocomplete

We're finally ready to implement our autocomplete feature. In fact, we've pretty much done all the legwork already. All we need to do is put the pieces together.

Here is a basic autocomplete method that we can drop into our Trie class:

```
def autocomplete(self, prefix):
  currentNode = self.search(prefix)
  if not currentNode:
    return None
  return self.collectAllWords(currentNode)
```

Yes, that's it. By using our search method and collectAllWords method together, we can autocomplete any prefix. Here's how this works.

The autocomplete method accepts the prefix parameter, which is the string of characters the user begins typing in.

First, we search the trie for the existence of the prefix. If the search method doesn't find the prefix in the trie, it returns None. However, if the prefix is found in the trie, the method returns *the node in the trie that represents the final character in the prefix.*

We noted earlier that we could have simply had the search method return True once it finds the word. The reason why we had it return the final node was so that we could use the search method to help us with the autocomplete feature.

Our autocomplete method continues by calling the collectAllWords method on the node returned by the search method. This finds and collects all words that stem from that final node, which represent all the complete words that can be appended to the original prefix to form a word.

Our method finally returns an array of all possible endings to the user's prefix, which we could then display to the user as possible autocomplete options.

Tries with Values: A Better Autocomplete

If we think about it, a good autocomplete feature isn't necessarily going to display *every* possible word the user might be intending to type. Showing, say, sixteen options would be overwhelming to the user, so we'd rather show just the most popular words from the available list.

For example, if the user starts typing "bal," the user might be intending to type "ball," "bald," or "balance." Now, it's also possible that the user is typing the obscure word "balter" (which means to dance clumsily, if you wanted to know). However, odds are that "balter" is *not* the intended word, since it's not a common word.

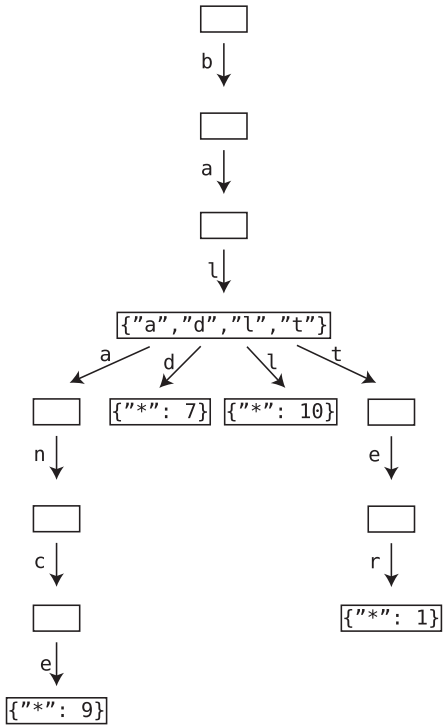
In order to display the most popular word options, we somehow need to store the word popularity data in our trie as well. And we're in luck, since we only need to make a slight modification to our trie to accomplish this.

In our current trie implementation, every time we've set a "*" key, we've made its value null. This is because we've only ever paid attention to the "*" key; the value carried no significance.

However, we can take advantage of these values to store additional data about the words themselves, such as how popular they are. To keep things simple,

let's work with a simple range of 1 to 10, with 1 being the rarest kind of word, and 10 being an extremely popular word.

Let's say "ball" is a very popular word and has a popularity score of 10. The word "balance" may be slightly less popular, and have a score of 9. The word "bald" is used even less often, and has a score of 7. And since "balter" is a hardly-known word, we'll give it a score of 1. We can store these scores in the trie this way:



By using a number as the value of each "*" key, we can effectively store each word's popularity score. When we now collect all the words of the trie, we can collect their scores along with them and then sort the words in order of their popularity. We can then choose to only display the most popular word options.

Wrapping Up

We've now covered three types of trees: binary search trees, heaps, and tries. There are *many* other types of trees as well, such as AVL trees, Red-Black trees, 2-3-4 trees, and plenty of others. Each tree has unique traits and behaviors that can be leveraged for specific situations. I encourage you to

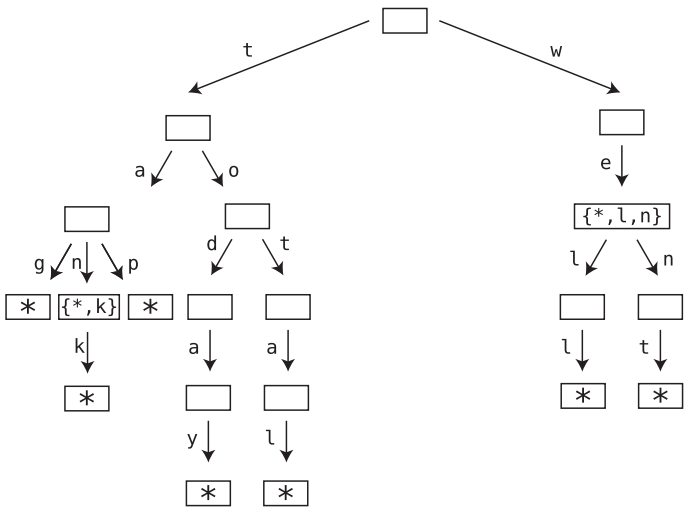
learn more about these various trees, but in any case, you now have a taste for how different trees can solve different problems.

It's now time for the final data structure of the book. Everything you've learned about trees will help you understand graphs. Graphs are helpful in so many different situations, and that's why they're so popular. So, let's dive in.

Exercises

The following exercises provide you with the opportunity to practice with tries. The solutions to these exercises are found in the section, Chapter 17, on page 459.

1. List all the words stored in the following trie:



2. Draw a trie that stores the following words: "get," "go," "got," "gotten," "hall," "ham," "hammer," "hill," and "zebra."

3. Write a function that traverses each node of a trie and prints each key, including all "*" keys.

4. Write an *autocorrect* function that attempts to replace a user's typo with a correct word. The function should accept a string that represents text a user typed in. If the user's string is *not* in the trie, the function should return an alternative word that shares the longest possible prefix with the user's string.

   For example, let's say our trie contained the words "cat," "catnap," and "catnip." If the user accidentally types in "catnar," our function should

return "catnap," since that's the word from the trie that shares the longest prefix with "catnar." This is because both "catnar" and "catnap" share a prefix of "catna," which is five characters long. The word "catnip" isn't as good, since it only shares the shorter, four-character prefix of "catn" with "catnar."

One more example: if the user types in "caxasfdij," the function could return any of the words "cat," "catnap," and "catnip," since they all share the same prefix of "ca" with the user's typo.

If the user's string is found in the trie, the function should just return the word itself. This should be true even if the user's text is not a complete word, as we're only trying to correct typos, not suggest endings to the user's prefix.

CHAPTER 18

Connecting Everything with Graphs

Let's say we're building a social network that allows people to be friends with one another. These friendships are mutual, so if Alice is friends with Bob, then Bob is also friends with Alice.

How can we best organize this data?

One basic approach might be to use a two-dimensional array that stores the list of friendships:

```
friendships = [
  ["Alice", "Bob"],
  ["Bob", "Cynthia"],
  ["Alice", "Diana"],
  ["Bob", "Diana"],
  ["Elise", "Fred"],
  ["Diana", "Fred"],
  ["Fred", "Alice"]
]
```

Here, each subarray containing a pair of names represents a "friendship" between two people.

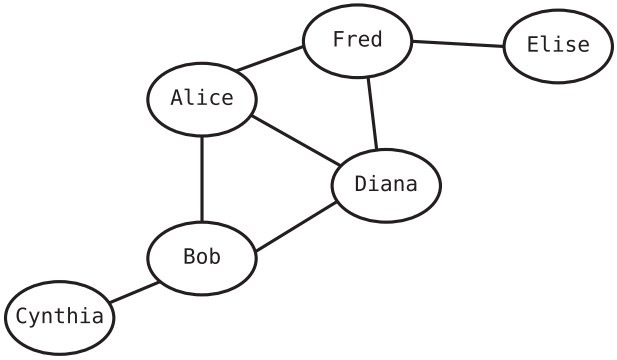
Unfortunately, with this approach, there's no quick way to see who Alice's friends are. If we look carefully, we can see that that Alice is friends with Bob, Diana, and Fred. But for the computer to determine this, it would have to comb through all the relationships in the list, since Alice can be present in any of them. This is O(N), which is very slow.

Thankfully, we can do much, *much* better than this. With a data structure known as a *graph,* we can find Alice's friends in *just O(1) time.*

Graphs

A *graph* is a data structure that specializes in relationships, as it easily conveys how data is connected.

Here is a visualization of our social network, displayed as a graph:



Each person is represented by a node, and each line indicates a friendship with another person. If you look at Alice, for example, you can see that she is friends with Bob, Diana, and Fred, since her node has lines that connect to their nodes.

Graphs vs. Trees

You might have noticed that graphs look similar to trees, which we've dealt with in the past few chapters. Indeed, *trees are a type of graph.* Both data structures consist of nodes connected to each other.

So, what's the difference between a graph and a tree?

Here's the deal: while all trees are graphs, not all graphs are trees.

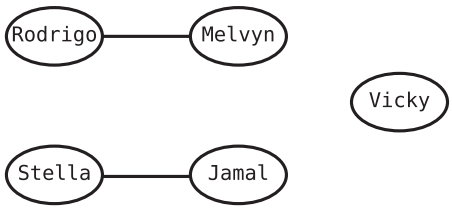
Specifically, for a graph to be considered a tree, it cannot have *cycles,* and all nodes must be *connected.* Let's see what that means.

A graph may have nodes that form what is known as a *cycle,* that is, nodes that reference each other circularly. In the previous example, Alice is friends with Diana, and Diana is connected to Bob, and Bob is connected…to Alice. These three nodes form a cycle.

Trees, on the other hand, are not "allowed" to have cycles. If a graph has a cycle, then it's not a tree.

Another characteristic specific to trees is that every node is somehow connected to every other node, even if the connections are indirect. However, it's possible for a graph to not be fully connected.

See the following graph:



In this social network, we have two pairs of friends. However, no one from either pair are friends with anyone from the other pair. Additionally, we can see that Vicky has no friends yet, as perhaps she signed up for this social network just minutes ago. With trees, however, there's never a node that's disconnected from the rest of the tree.

Graph Jargon

Graphs have a bit of their own technical jargon. We're used to calling each piece of data a *node,* but in "graph speak," each node is called a *vertex.* The lines between nodes, um—vertices, have their own name as well, and are called *edges.* And vertices that are connected by an edge are said to be *adjacent* to each other. Some also refer to adjacent vertices as *neighbors.*

In our first graph, then, the vertices of "Alice" and "Bob" are adjacent to each other, since they share an edge.

I mentioned earlier that it's possible for a graph to have a vertex that isn't connected at all to other vertices. However, a graph where *all* the vertices are connected in some way is said to be a *connected graph.*

The Bare-Bones Graph Implementation

For the sake of code organization, we're going to use object-oriented classes to represent our graphs, but it's worth noting that we can also use a basic hash table (see *Blazing Fast Lookup with Hash Tables*) to represent a rudimentary graph. Here's a bare-bones Ruby implementation of our social network using a hash table:

```
friends = {
  "Alice" => ["Bob", "Diana", "Fred"],
  "Bob" => ["Alice", "Cynthia", "Diana"],
  "Cynthia" => ["Bob"],
  "Diana" => ["Alice", "Bob", "Fred"],
  "Elise" => ["Fred"],
  "Fred" => ["Alice", "Diana", "Elise"]
}
```

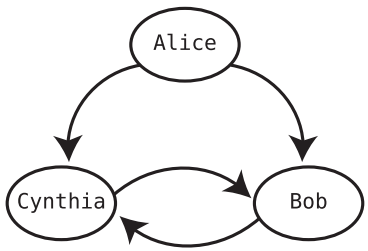
With a graph, we can look up Alice's friends in O(1), because we can look up the value of any key in the hash table with one step:

```
friends["Alice"]
```

This immediately returns the array containing all of Alice's friends.

Directed Graphs

In some social networks, relationships are *not* mutual. For example, a social network may allow Alice to "follow" Bob, but Bob doesn't have to follow Alice back. Let's construct a new graph that demonstrates who follows whom:



This is known as a *directed graph.* In this example, the arrows indicate the *direction* of the relationship. Alice follows both Bob and Cynthia, but no one follows Alice. We can also see that Bob and Cynthia follow each other.

We can still use our simple hash-table implementation to store this data:

```
followees = {
  "Alice" => ["Bob", "Cynthia"],
  "Bob" => ["Cynthia"],
  "Cynthia" => ["Bob"]
}
```

The only difference here is that we are using the arrays to represent the people each person *follows.*

Object-Oriented Graph Implementation

I demonstrated how a hash table can be used to implement a graph, but going forward, we'll work with an object-oriented approach.

Here's the beginning of an object-oriented graph implementation, using Ruby:

```
class Vertex
  attr_accessor :value, :adjacent_vertices

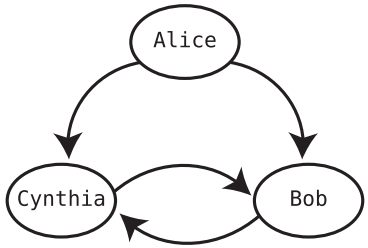
  def initialize(value)
    @value = value
    @adjacent_vertices = []
  end

  def add_adjacent_vertex(vertex)
    @adjacent_vertices << vertex
  end
end
```

The Vertex class has two primary attributes, the value and an array of adjacent_vertices. In our social network example, each vertex represents a person, and the value might be a string containing the person's name. With a more complex application, we'd probably want to store multiple pieces of data inside a vertex, such as the person's additional profile information.

The adjacent_vertices array contains all the vertices this vertex connects to. We can add a new adjacent vertex to a given vertex using the add_adjacent_vertex method.

Here's how we can use this class to build a directed graph representing who follows whom in this image:




```
alice = Vertex.new("alice")
bob = Vertex.new("bob")
cynthia = Vertex.new("cynthia")

alice.add_adjacent_vertex(bob)
alice.add_adjacent_vertex(cynthia)
bob.add_adjacent_vertex(cynthia)
cynthia.add_adjacent_vertex(bob)
```

Now, if we were building an *undirected* graph for the social network (where all friendships are mutual), it would make sense if we add Bob to Alice's list of friends, we should automatically add Alice to Bob's list of friends as well.

To do this, we can modify our add_adjacent_vertex method as follows:

```
def add_adjacent_vertex(vertex)
  return if adjacent_vertices.include?(vertex)
  @adjacent_vertices << vertex
  vertex.add_adjacent_vertex(self)
end
```

Let's say we're calling this method on Alice, and adding Bob to her list of friends. As with the previous version, we use @adjacent_vertices << vertex to add Bob to Alice's list of @adjacent_vertices. However, we then call this very method on Bob's vertex, with vertex.add_adjacent_vertex(self). This adds Alice to Bob's list of friends as well.

Now, this would end up in an infinite loop, since Alice and Bob would continue to call add_adjacent_vertex on each other forever. Because of this, we added the line, return if adjacent_vertices.include?(vertex), which cuts the method short if, say, Alice already has Bob in her list of friends.

To keep things simple going forward, we're going to work with graphs that are connected (again, meaning all vertices are connected to each other in some way). With such graphs, we can use this one Vertex class to achieve all the algorithms going forward. The general idea is if we have access to just one vertex, we can find all other vertices from there, since all the vertices are connected.

However, it's important to point out that if we are dealing with a disconnected graph, it may be impossible to discover all the vertices just from one vertex. In this case, we may need to store all the graph vertices in some additional data structure, such as an array, so that we have access to all of them. (It's common to see graph implementations use a separate Graph class to contain this array.)

Adjacency List vs. Adjacency Matrix

Our implementation of the graph uses a simple list (in the form of an array) to store a vertex's adjacent vertices. This approach is known as the *adjacency list* implementation.

However, it's good to know that there is another implementation that uses two-dimensional arrays instead of lists. This alternative approach is known as the *adjacency matrix*, and it can provide advantages in specific situations.

Both approaches are popular, and I've decided to stick with the adjacency list because I find it to be more intuitive. However, I recommend you research the adjacency matrix as well, as it can be useful and is particularly interesting.

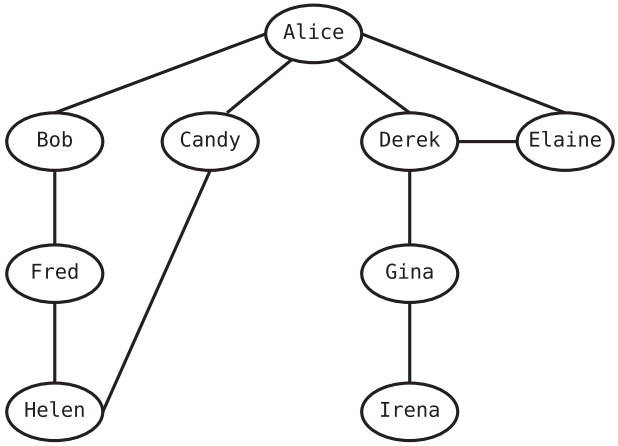
Graph Search

One of the most common graph operations is searching for a particular vertex.

Now, when dealing with graphs, the term "search" can have several connotations. In the simplest sense, to "search" a graph means to find a particular vertex somewhere within the graph. This would be similar to searching for a value within an array or a key-value pair inside a hash table.

However, when applied to graphs, the term *search* usually has a more specific connotation, and that is: *if we have access to one vertex in the graph, we must find another particular vertex that is somehow connected to this vertex.*

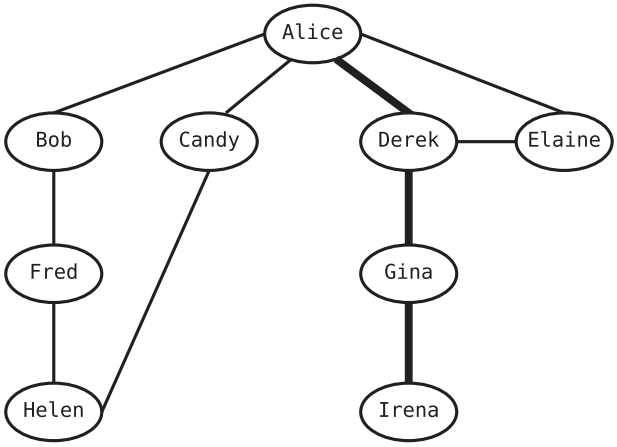
For example, take a look at this example social network:



Let's say we currently have access to Alice's vertex. If we said that we will search for Irena, it would mean that we are trying to find our way from Alice to Irena.

Interestingly, you can see that there are two different *paths* we can take to get from Alice to Irena.

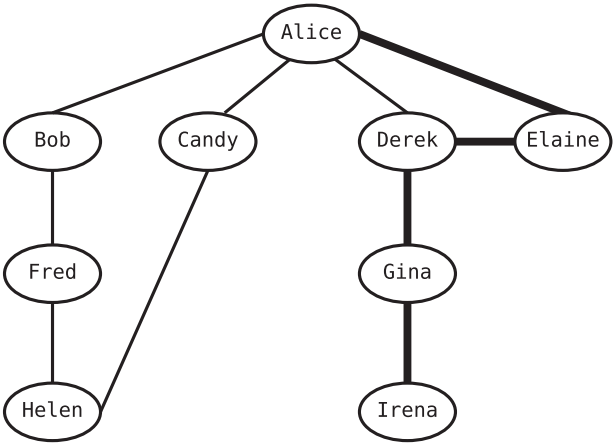
The shorter path is pretty obvious as shown in the graph on page 338.



That is, we can get from Alice to Irena in this sequence:

Alice -> Derek -> Gina -> Irena

However, we can take a slightly longer path to get to Irena as well:



This longer path is:

Alice -> Elaine -> Derek -> Gina -> Irena

The term *path* is an official graph term, and it means the specific sequence of edges to get from one vertex to another.

Now, searching a graph (which you now know means getting from one vertex to another) can be useful for a variety of use cases.

Perhaps the most obvious application for graph search is searching for a particular vertex within a connected graph. When this is the case, search can

be used to find *any* vertex within the entire graph even if we have access to just one random vertex.

Another use for graph search is to discover whether two vertices are connected. For example, we may want to know whether Alice and Irena are somehow connected to each other in this network. A search would give us the answer.

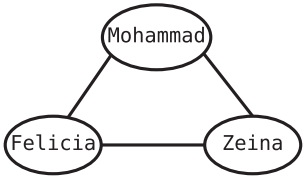
Search can also be used even if we aren't looking for one particular vertex. That is, we can use graph search to merely traverse a graph, which can be useful if we want to perform an operation on every vertex in the graph. You'll see shortly how this works.

Depth-First Search

There are two well-known approaches for graph search: *depth-first search* and *breadth-first search*. Both approaches can get the job done, but each provide unique advantages in particular situations. We're going to start with depth-first search, also referred to as DFS, because it's actually quite similar to the algorithm for binary tree traversal that we discussed back in Binary Search Tree Traversal, on page 272. In fact, it's *also* the same essential algorithm that we saw in Filesystem Traversal, on page 156.

As mentioned earlier, graph search can be used to either find a particular vertex, or it can be used to simply traverse the graph. We're going to begin by using depth-first search to traverse the graph, since that algorithm is slightly simpler.

The key to any graph search algorithm is keeping track of which vertices we've visited so far. If we don't do this, we can end up in an infinite cycle. Take the following graph, for example:



Here, Mohammed is friends with Felicia. And Felicia also happens to be friends with Zeina. But Zeina is friends with Mohammed. So, our code would end up going in circles unless we keep track of which vertices we've already traversed.

This problem didn't occur when we dealt with trees (or filesystem traversal), since trees can't have cycles. But since a graph *can* have a cycle, we need to address this issue now.

One way to keep track of our visited vertices is by using a hash table. As we visit each vertex, we add the vertex (or its value) as a key in the hash table and assign it an arbitrary value, such as the boolean `true`. If a vertex is present in the hash table, it means we've already visited it.

With this in mind, the depth-first search algorithm works as follows.

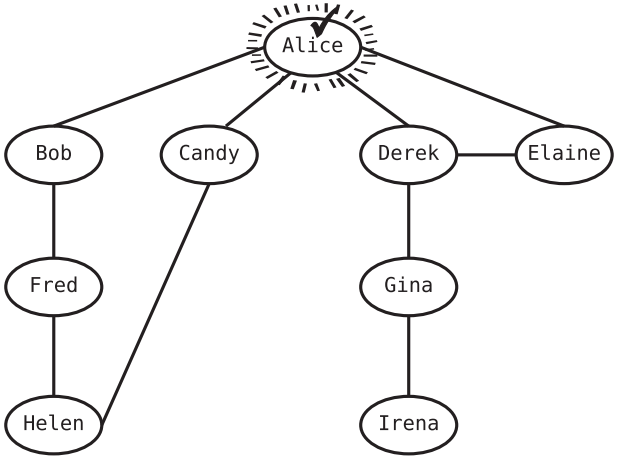
1. Start at any random vertex within the graph.
2. Add the current vertex to the hash table to mark it as having been visited.
3. Iterate through the current vertex's adjacent vertices.
4. For each adjacent vertex, if the adjacent vertex has already been visited, ignore it.
5. If the adjacent vertex has *not* yet been visited, recursively perform depth-first search on that vertex.

Depth-First Search Walk-Through

Let's see this in action.

In this walk-through, we're going to start with Alice. In the following diagrams, the vertex with lines around it is the current vertex. A check mark means we've officially marked the vertex as having been visited (and added it to the hash table).

Step 1: We start with Alice and give her a check mark to indicate that we've officially visited her vertex:



Next up, we'll iterate over Alice's neighbors using a loop. These will be Bob, Candy, Derek, and Elaine.

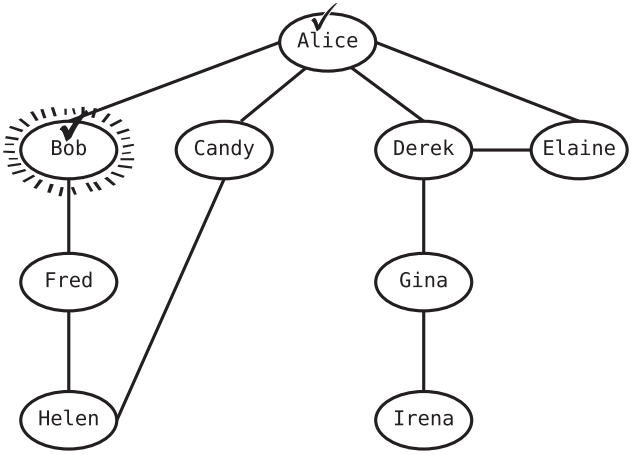
The order of which neighbor to visit first doesn't matter, so let's just start with Bob. He seems nice.

Step 2: We now perform depth-first search on Bob. Note this is making a recursive call, as we're already in the middle of a depth-first search of Alice.

As with all recursion, the computer needs to remember which function calls it's still in the middle of, so it first adds Alice to the call stack:



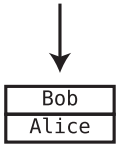
We can now begin the depth-first search on Bob, which makes Bob the current vertex. We mark him as visited:



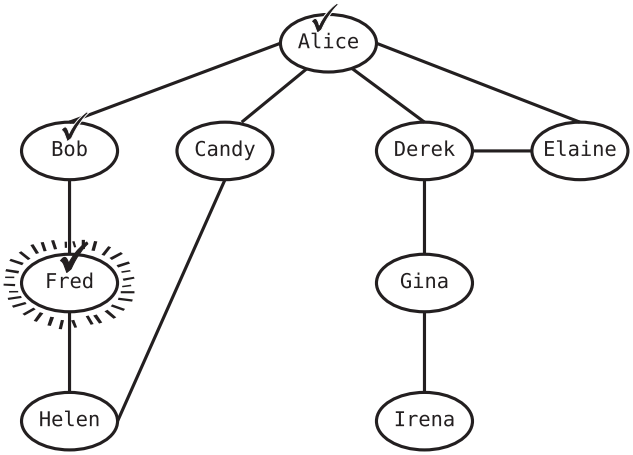
We then iterate over Bob's adjacent vertices. These are Alice and Fred.

Step 3: Alice has already been visited, so we can ignore her.

Step 4: The only other neighbor, then, is Fred. We call the depth-first search function on Fred's vertex. The computer first adds Bob to the call stack to remember that it's actually still in the middle of searching Bob:



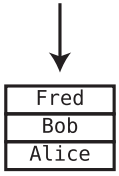
We now perform depth-first search on Fred. He is now the current vertex, so we mark him as visited as shown in the graph on page 342.



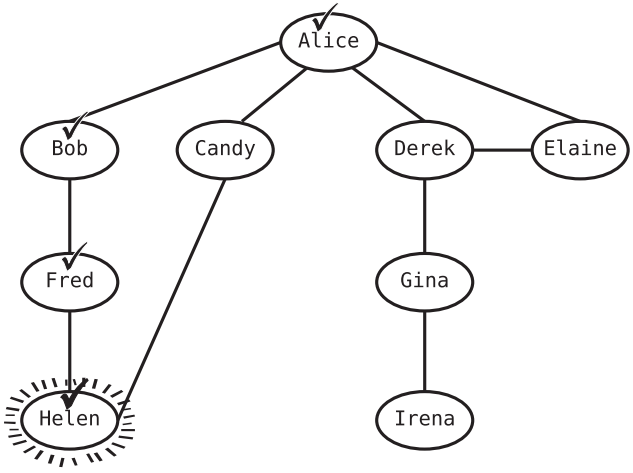
Next, we iterate over Fred's adjacent vertices, which are Bob and Helen.

Step 5: Bob has already been visited, so we ignore him.

Step 6: The only remaining adjacent vertex is Helen. We recursively perform depth-search first on Helen, so the computer first adds Fred to the call stack:



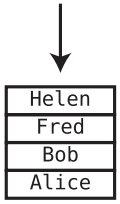
We now begin depth-first search on Helen. She's the current vertex, so we mark her as visited:



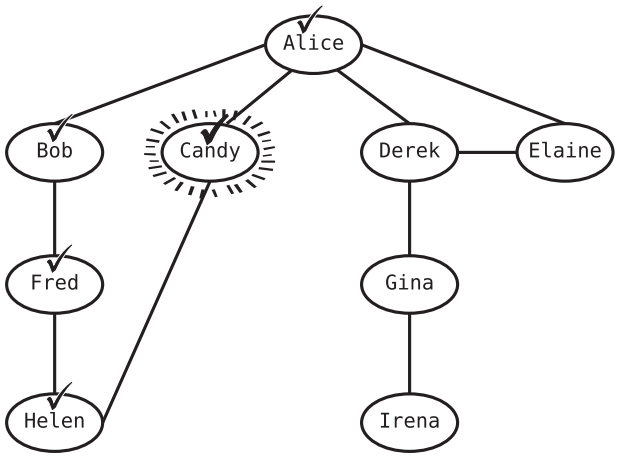
Helen has two adjacent vertices: Fred and Candy.

Step 7: We've already visited Fred, so we can ignore him.

Step 8: Candy has *not* yet been visited, so we recursively perform depth-search on Candy. First, though, Helen gets added to the call stack:



We perform depth-first search on Candy. She is now the current vertex, and we mark her as visited:



Candy has two adjacent vertices: Alice and Helen.

Step 9: We've already visited Alice, so we can ignore her.

Step 10: We've already visited Helen, so we can ignore her as well.

Since Candy has no other neighbors, we're done performing depth-first search on Candy. At this point, then, the computer begins to unwind the call stack.

First, it pops off Helen from the call stack. We've already iterated over all her neighbors, so the depth-first search on Helen is complete.

The computer pops off Fred. We've iterated over all his neighbors too, so we're done searching him as well.

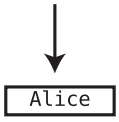
The computer pops off Bob, but we're done with him as well.

The computer then pops Alice off the call stack. Within our search of Alice, we were in the middle of looping through all of Alice's neighbors. Now, this loop already iterated over Bob. (This was Step 2.) This leaves Candy, Derek, and Elaine.

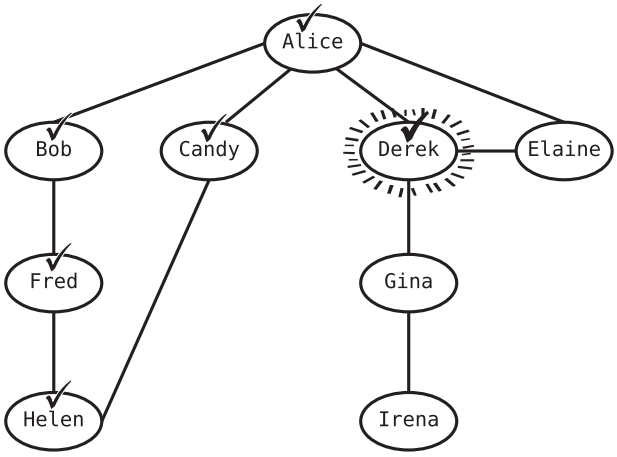
Step 11: Candy has already been visited, so there's no need to perform search on her.

However, we have not yet visited Derek or Elaine.

Step 12: Let's proceed by recursively performing depth-first search on Derek. The computer adds Alice to the call stack once again:



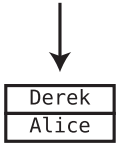
The depth-first search of Derek now begins. Derek is the current vertex, so we mark him as visited:



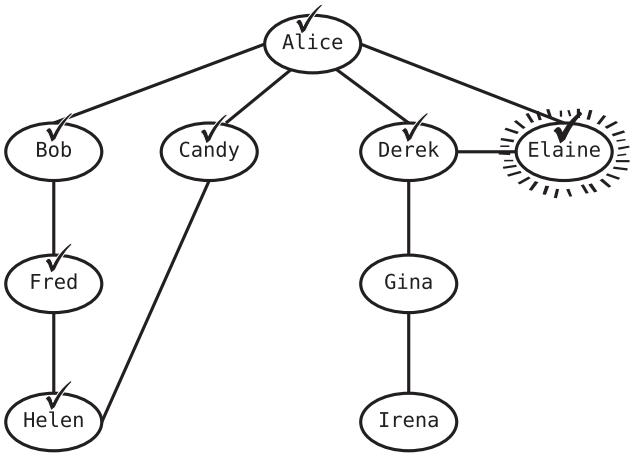
Derek has three adjacent vertices: Alice, Elaine, and Gina.

Step 13: Alice has already been visited, so we don't need to perform another search on her.

Step 14: Let's visit Elaine next, by recursively performing depth-first search on her vertex. Before we do, the computer adds Derek to the call stack:



We now perform depth-first search on Elaine. We mark Elaine as visited:



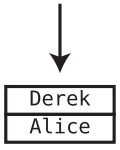
Elaine has two adjacent vertices: Alice and Derek.

Step 15: Alice has already been visited, so there's no need to perform another search on her.

Step 16: Derek, too, has already been visited.

Since we iterated over all Elaine's neighbors, we're done searching Elaine. The computer now pops Derek from the call stack and loops over his remaining adjacent vertices. In this case, Gina is the final neighbor to visit.

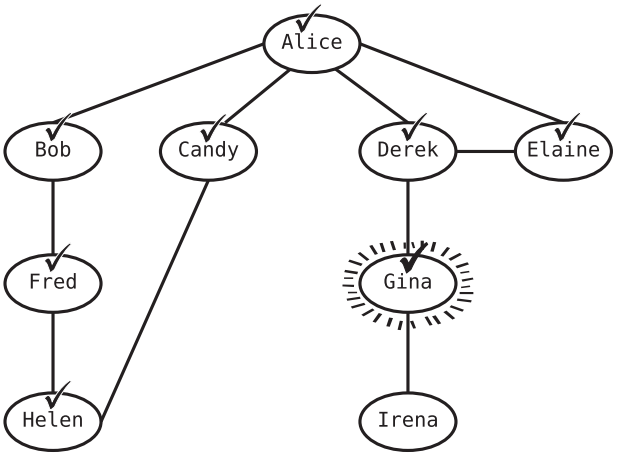
Step 17: We never visited Gina before, so we recursively perform depth-first search on her vertex. First, though, the computer adds Derek to the call stack again:



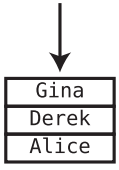
We begin our depth-first search of Gina, and mark her as visited as shown in the graph on page 346.

Gina has two neighbors: Derek and Gina.

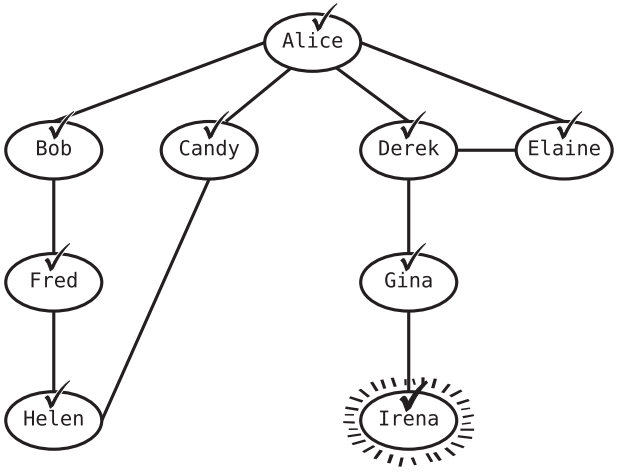
Step 18: Derek has already been visited.



Step 19: Gina has one unvisited adjacent vertex, namely, Irena. Gina gets added to the call stack so that we can recursively perform depth-first search on Irena:



We begin search on Irena, and mark her as visited:



We iterate over Irena's neighbors. Irena has only one neighbor: Gina.

Step 20: Gina has already been visited.

The computer then unwinds the call stack, popping off each vertex one by one. However, since each vertex on the call stack has already iterated over all of its neighbors, there's nothing more for the computer to do with each vertex.

This means we're done!

Code Implementation: Depth-First Search

Here is an implementation of depth-first traversal:

```
def dfs_traverse(vertex, visited_vertices={})
  # Mark vertex as visited by adding it to the hash table:
  visited_vertices[vertex.value] = true

  # Print the vertex's value, so we can make sure our traversal really works:
  puts vertex.value

  # Iterate through the current vertex's adjacent vertices:
  vertex.adjacent_vertices.each do |adjacent_vertex|

    # Ignore an adjacent vertex if we've already visited it:
    next if visited_vertices[adjacent_vertex.value]

    # Recursively call this method on the adjacent vertex:
    dfs_traverse(adjacent_vertex, visited_vertices)
  end
end
```

Our dfs_traverse method accepts a single vertex and optionally can accept the visited_vertices hash table. The first time we call this function, the visited_vertices starts out as empty. As we visit vertices, though, we populate this hash table with the vertices we've visited, and pass the hash table with each recursive call.

The first thing we do within the function is mark the current vertex as visited. We do this by adding the vertex's value to the hash:

```
visited_vertices[vertex.value] = true
```

We then optionally print the vertex's value just to get feedback that we've truly traversed it:

```
puts vertex.value
```

Next, we iterate over all the adjacent vertices of the current vertex:

```
vertex.adjacent_vertices.each do |adjacent_vertex|
```

If the adjacent vertex we're iterating over has already been visited, we just skip to the next round of the loop:

```
next if visited_vertices[adjacent_vertex.value]
```

Otherwise, we recursively call dfs_traversal on the adjacent vertex:

```
dfs_traverse(adjacent_vertex, visited_vertices)
```

Again, we also pass in the visited_vertices hash table so the ensuing call has access to it.

If we want to use depth-first search to actually search for a particular vertex, we can use a modified version of the previous function:

```
def dfs(vertex, search_value, visited_vertices={})
  # Return the original vertex if it happens to
  # be the one we're searching for:
  return vertex if vertex.value == search_value

  visited_vertices[vertex.value] = true

  vertex.adjacent_vertices.each do |adjacent_vertex|
    next if visited_vertices[adjacent_vertex.value]

    # If the adjacent vertex is the vertex we're searching for, just return
    # that vertex:
    return adjacent_vertex if adjacent_vertex.value == search_value

    # Attempt to find the vertex we're searching for by recursively calling
    # this method on the adjacent vertex:
    vertex_were_searching_for =
      dfs(adjacent_vertex, search_value, visited_vertices)

    # If we were able to find the correct vertex using the above recursion,
    # return the correct vertex:
    return vertex_were_searching_for if vertex_were_searching_for
  end

  # If we haven't found the vertex we're searching for:
  return nil
end
```

This implementation also recursively calls itself for each vertex, but returns the vertex_were_searching_for if it finds the correct vertex.

Breadth-First Search

Breadth-first search, often abbreviated BFS, is another way to search a graph. Unlike depth-first search, breadth-first search does *not* use recursion. Instead, the algorithm revolves around our old friend, the queue. As you'll recall, the queue is a FIFO data structure, and whatever goes in first, comes out first.

Here is the algorithm for breadth-first search. As with our walkthrough of depth-first search, we're going to focus on graph *traversal* using breadth-first search. That is, we're going to visit each vertex from our example social network.

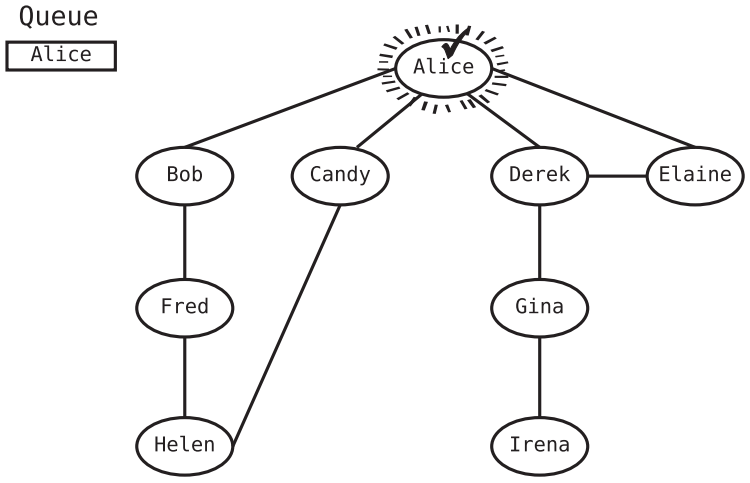
Here is the algorithm for breadth-first traversal:

1. Start at any vertex within the graph. We'll call this the "starting vertex."
2. Add the starting vertex to the hash table to mark it as having been visited.
3. Add the starting vertex to a queue.
4. Start a loop that runs while the queue isn't empty.
5. Within this loop, remove the first vertex from the queue. We'll call this the "current vertex."
6. Iterate over all the adjacent vertices of current vertex.
7. If the adjacent vertex was already visited, ignore it.
8. If the adjacent vertex has *not* yet been visited, mark it as visited by adding it to a hash table, and add it to the queue.
9. Repeat this loop (starting from Step 4) until the queue is empty.

Breadth-First Search Walk-Through

This isn't as complex as it seems. Let's walk through the traversal step by step.

To set things up, let's work with Alice as our starting vertex. We'll mark her as visited and add her to the queue:

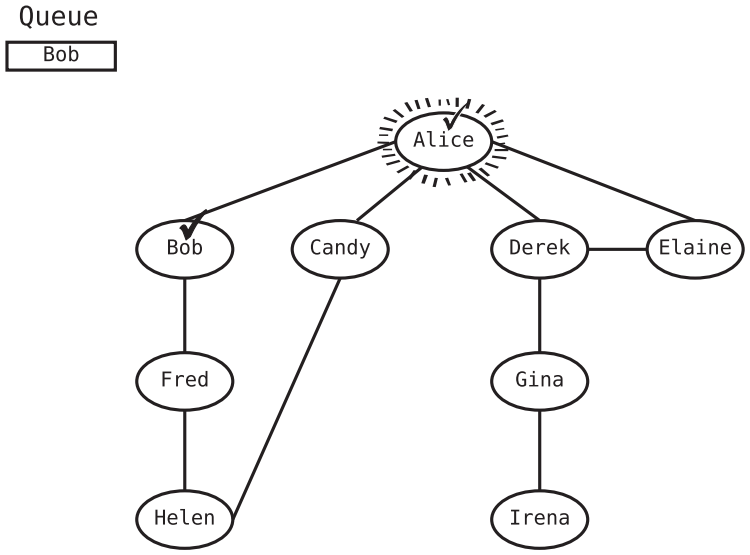


We now begin the core algorithm.

Step 1: We remove the first vertex from the queue and it make it the current vertex. This would be Alice, as she's currently the *only* item in the queue. So, at this point, the queue is actually empty.

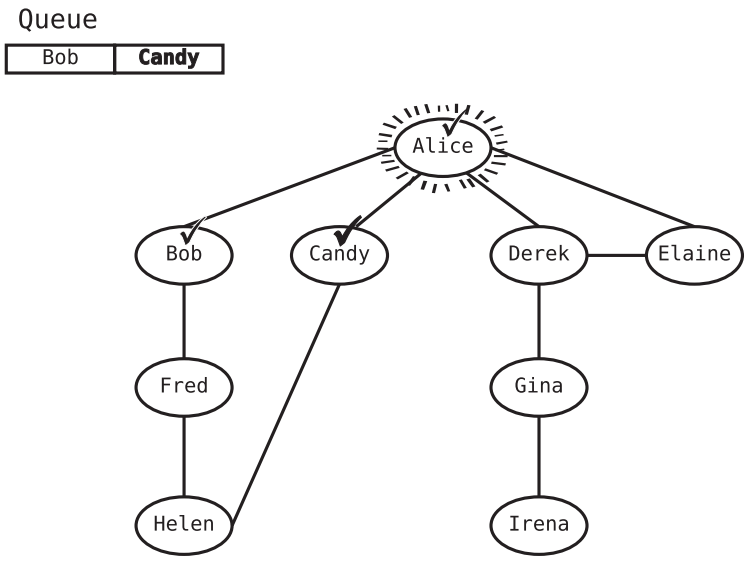
Since Alice is the current vertex, we proceed to iterate over Alice's adjacent vertices.

Step 2: We'll start with Bob. We'll mark him as visited, and add him to the queue:



Note that Alice *is still the current vertex*, as indicated by the lines around her. However, we've still marked Bob as visited and added him to the queue.

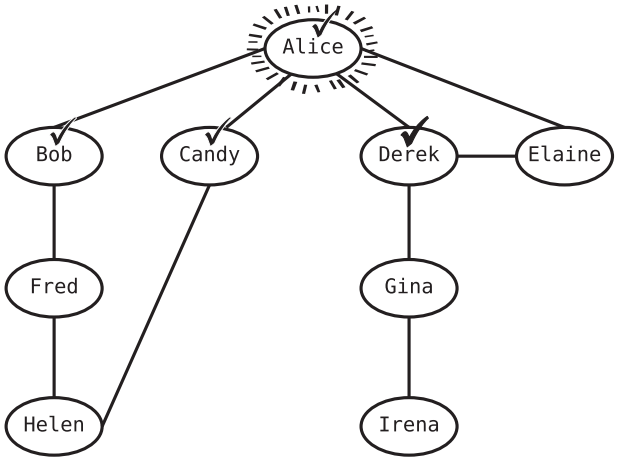
Step 3: We move on to Alice's other adjacent vertices. Let's choose Candy, so we'll mark her as visited and add her to the queue:



Step 4: We then mark Derek as visited and add him to the queue:

Queue

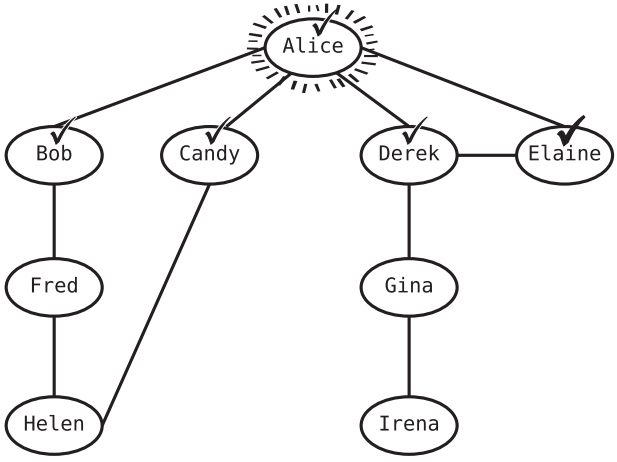
Bob	Candy	**Derek**



Step 5: We do the same with Elaine:

Queue

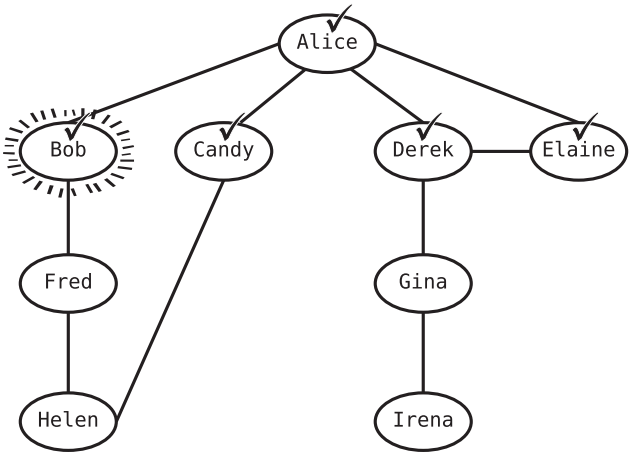
Bob	Candy	Derek	**Elaine**



Step 6: Now that we've iterated through all the neighbors of the current vertex (Alice), we remove the first item from the queue and make it the current vertex. In our case, Bob is at the front of the queue, so we dequeue him and make him the current vertex as shown in the graph on page 352.

Queue

Candy	Derek	Elaine



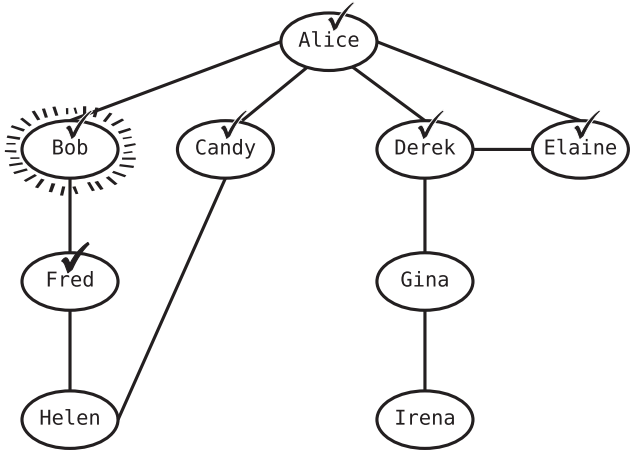
Since Bob is the current vertex, we iterate over all of his adjacent vertices.

Step 7: Alice has already been visited, so we ignore her.

Step 8: Fred has not yet been visited, so we mark him as visited and add him to the queue:

Queue

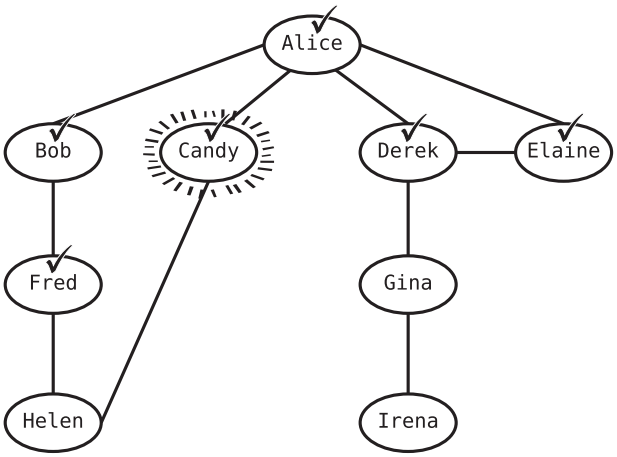
Candy	Derek	Elaine	**Fred**



Step 9: Bob has no more adjacent vertices. This means we pull the first item out of the queue and make it the current vertex. This would be Candy:

Queue

Derek	Elaine	Fred



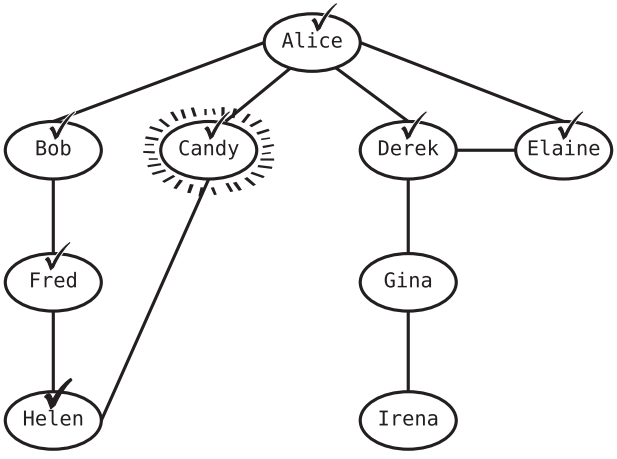
We iterate over Candy's adjacent vertices.

Step 10: Alice has already been visited, so we ignore her again.

Step 11: Helen, on the other hand, has not yet been visited. We mark Helen as visited and add her to the queue:

Queue

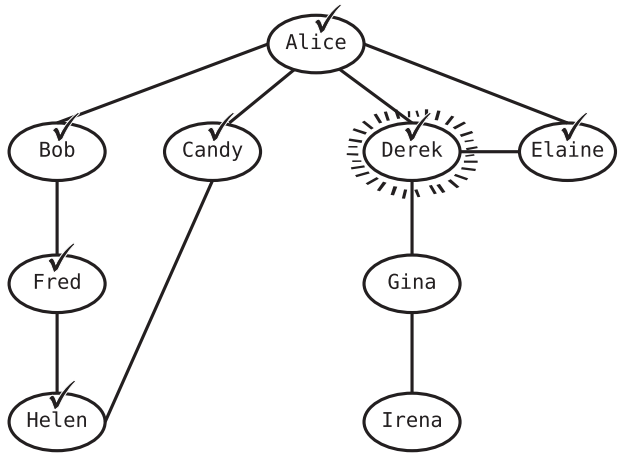
Derek	Elaine	Fred	**Helen**



Step 12: We're done iterating over Candy's adjacent vertices, so we pull the first item from the queue (Derek), and make it the current vertex:

Queue

Elaine	Fred	Helen



Derek has three adjacent vertices, so we iterate over them.

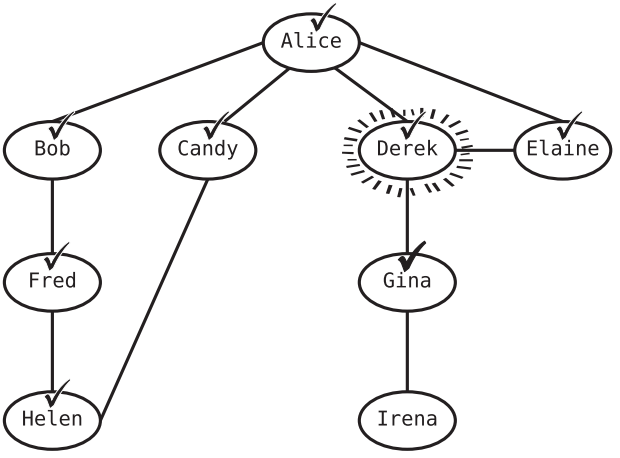
Step 13: Alice has already been visited, so we ignore her.

Step 14: The same goes for Elaine.

Step 15: This leaves Gina, so we mark her as visited and add her to the queue:

Queue

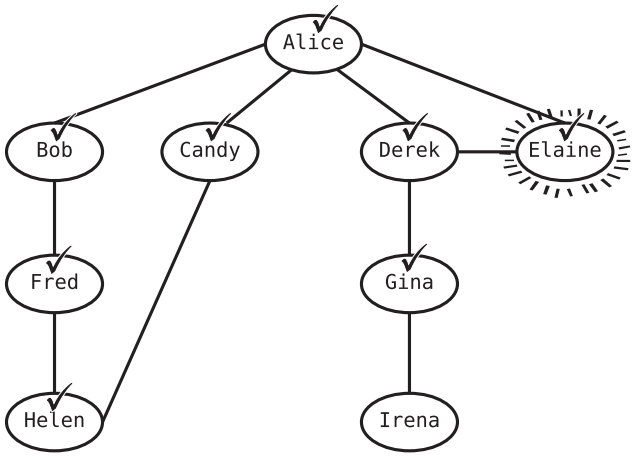
Elaine	Fred	Helen	**Gina**



Step 16: We've visited all of Derek's immediate friends, so we take Elaine off the queue and designate her as the current vertex:

Queue

Fred	Helen	Gina



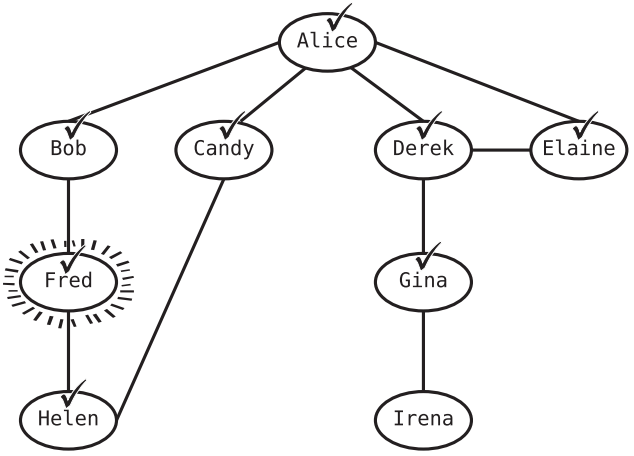
Step 17: We iterate over Elaine's adjacent vertices, starting with Alice. She's already been visited, though.

Step 18: We've already visited Derek, too.

Step 19: We pull the next person off the queue (Fred), and turn him into the current vertex:

Queue

Helen	Gina

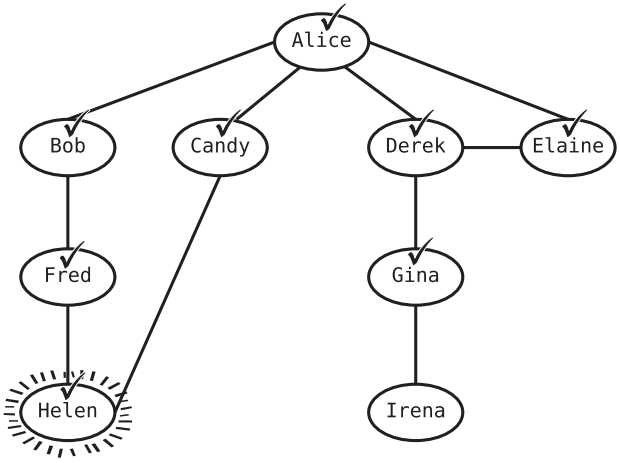


Step 20: We iterate over Fred's neighbors. Bob has already been visited.

Step 21: Helen also has already been visited.

Step 22: Since Helen is at the front of the queue, we dequeue Helen and make her the current vertex:

```
Queue
┌──────┐
│ Gina │
└──────┘
```

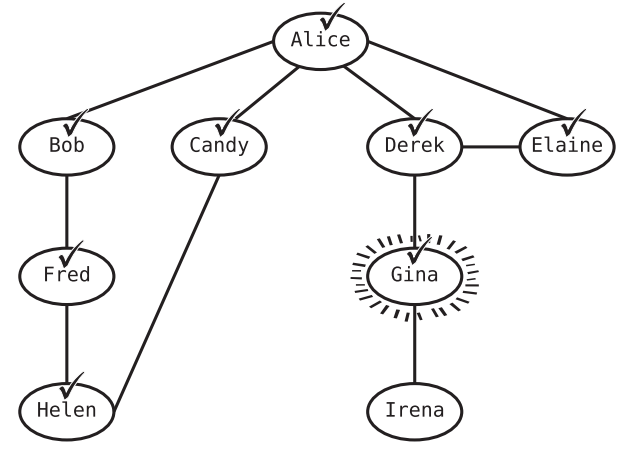


Step 23: Helen has two adjacent vertices. We've already visited Fred.

Step 24: We've also already visited Candy.

Step 25: We remove Gina from the queue and make her the current vertex:

```
Queue
(Empty)
```

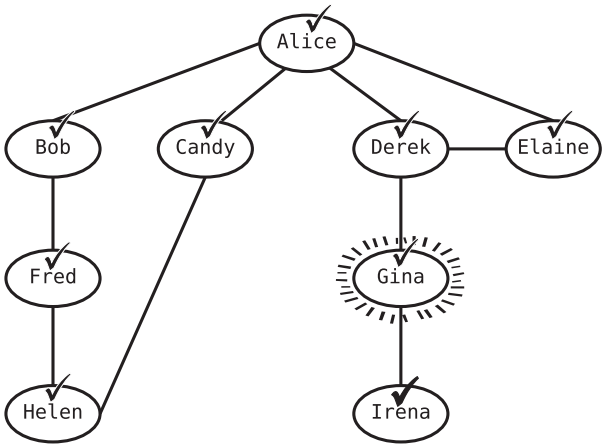


Step 26: We iterate over Gina's neighbors. Derek has already been visited.

Step 27: Gina has one unvisited adjacent friend, Irena, so we visit Irena and add her to the queue:

Queue

Irena

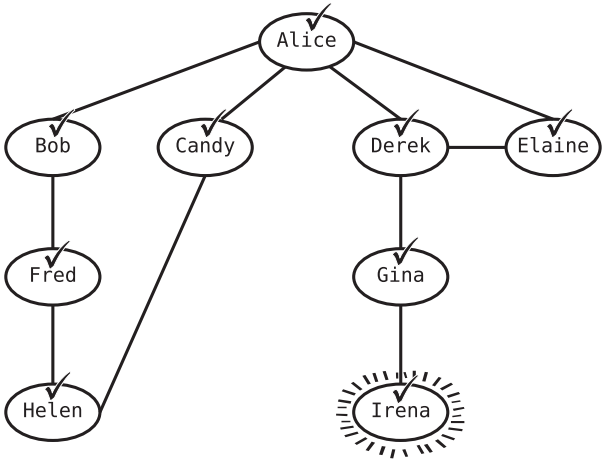


We're now done iterating over Gina's neighbors.

Step 28: We remove the first (and only) person from the queue, which is Irena. She becomes the current vertex:

Queue

(Empty)



Step 29: Irena has just one adjacent vertex, Gina, but Gina has already been visited.

We should now remove the next item from the queue, but the queue is empty! This means our traversal is complete.

Code Implementation: Breadth-First Search

Here is our code for breadth-first traversal:

```
def bfs_traverse(starting_vertex)
  queue = Queue.new

  visited_vertices = {}
  visited_vertices[starting_vertex.value] = true
  queue.enqueue(starting_vertex)

  # While the queue is not empty:
  while queue.read

    # Remove the first vertex off the queue and make it the current vertex:
    current_vertex = queue.dequeue

    # Print the current vertex's value:
    puts current_vertex.value

    # Iterate over current vertex's adjacent vertices:
    current_vertex.adjacent_vertices.each do |adjacent_vertex|

      # If we have not yet visited the adjacent vertex:
      if !visited_vertices[adjacent_vertex.value]

        # Mark the adjacent vertex as visited:
        visited_vertices[adjacent_vertex.value] = true

        # Add the adjacent vertex to the queue:
        queue.enqueue(adjacent_vertex)
      end
    end
  end
end
```

The bfs_traverse method accepts a starting_vertex, which is the vertex we begin our search from.

We start by creating the queue that fuels our algorithm:

```
queue = Queue.new
```

We also create the visited_vertices hash table in which we keep track of which vertices we've already visited:

```
visited_vertices = {}
```

We then mark the starting_vertex as visited and add it to the queue:

```
visited_vertices[starting_vertex.value] = true
queue.enqueue(starting_vertex)
```

We begin a loop that runs as long as the queue isn't empty:

```
while queue.read
```

We remove the first item from the queue and make it the current vertex:

```
current_vertex = queue.dequeue
```

Next, we print the vertex's value just to see in our console that our traversal is working correctly:

```
puts current_vertex.value
```

We then iterate over all the adjacent vertices of our current vertex:

```
current_vertex.adjacent_vertices.each do |adjacent_vertex|
```

For each adjacent vertex that hasn't been visited, we add it to the hash table to mark it as visited and then add it to the queue:

```
if !visited_vertices[adjacent_vertex.value]
  visited_vertices[adjacent_vertex.value] = true
  queue.enqueue(adjacent_vertex)
end
```

And that's the gist of it.

DFS vs. BFS

If you look carefully at the order of breadth-first search, you'll notice that we first traverse all of Alice's immediate connections. We then spiral outward and gradually move farther and farther from Alice. With depth-first search, though, we immediately move as far away from Alice as we possibly can until we're forced to return to her.

So, we have two methods of searching a graph: depth-first and breadth-first. Is one approach better than the other?

As you've probably caught on by now, it depends on your situation. In some scenarios, depth-first may be faster, while in others, breadth-first might be the better choice.

Usually, one of the main factors in determining which algorithm to use is the nature of the graph you're searching and what you're searching for. The key here, as mentioned earlier, is that breadth-first search traverses all the vertices closest to the starting vertex before moving farther away. Depth-first search,

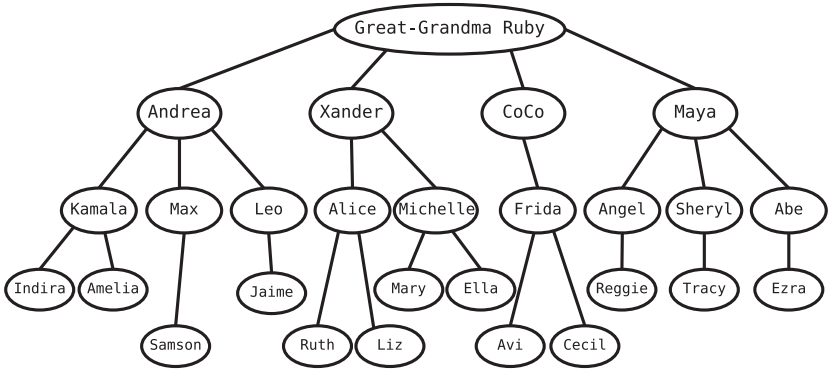
on the other hand, immediately moves as far away from the starting vertex as it can. Only when the search hits a dead end does it return back to the starting vertex.

So, let's say we want to find all the *direct* connections of a person in a social network. For example, we may want to find all of Alice's actual friends in our earlier example graph. We're not interested in who her friends' friends are—we only want a list of her direct connections.

If you look at the breadth-first approach, you'll see that we immediately find all of Alice's direct friends (Bob, Candy, Derek, and Elaine) before moving on to her "second-degree" connections.

However, when we traversed the graph with the depth-first algorithm, we ended up touching Fred and Helen (two people who aren't Alice's friends) before finding Alice's other friends. In a larger graph, we could waste even more time traversing many unnecessary vertices.

But, let's take a different scenario. Say our graph represents a family tree, which may look as follows:



This family tree shows all the descendants of Great-Grandma Ruby, the proud matriarch of a wonderful family. Let's say we know that Ruth is a great-grandchild of Ruby, and we want to find Ruth in the graph.

Now, here's the thing. If we used breadth-first search, we'd end up traversing all of Ruby's children and grandchildren before reaching even the first great-grandchild.

However, if we use depth-first search, we'd move down the graph right away, reaching the first great-grandchild in just a few steps. While it's possible that we'd have to traverse the entire graph before finding Ruth, we at least have

a shot at finding her quickly. With breadth-first search, though, we have no choice but to traverse all the non–great-grandchildren before we can start inspecting the great-grandchildren.

The question to always ask, then, is whether we want to stay close to the starting vertex during our search, or do we specifically want to move far away. Breadth-first search is good for staying close, and depth-first search is ideal for moving far away quickly.

The Efficiency of Graph Search

Let's analyze the time complexity of graph search using Big O Notation.

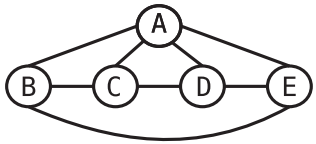
In both depth-first search and breadth-first search, we traverse all the vertices in a worst-case scenario. The worst-case scenario may be that we're intending to do a full-graph traversal, or we may be searching for a vertex that doesn't exist in the graph. Or, the vertex we're searching for may just happen to be the last vertex in the graph that we check.

In any case, we touch all vertices in the graph. At first glance, this would seem to be O(N), with N being the number of vertices.

However, in both search algorithms, for each vertex we traverse, *we also iterate over all of its adjacent vertices.* We may ignore an adjacent vertex if it has already been visited, but we still spend a step checking that vertex to see whether we've visited it.

So, for each vertex we visit, we also spend steps checking each of the vertex's adjacent neighbors. This would seem tough to peg down using Big O Notation, since each vertex may have a different number of adjacent vertices.

Let's analyze a simple graph to make this clear:



Here, Vertex A has four neighbors. By contrast, B, C, D, and E each have three neighbors. Let's count the number of steps it takes to search this graph.

At the very least, we have to visit each of the five vertices. This alone takes five steps.

Then, for *each* vertex, we iterate over each of its neighbors.

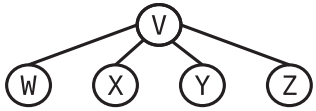
This would add the following steps:

A: 4 steps to iterate over 4 neighbors
B: 3 steps to iterate over 3 neighbors
C: 3 steps to iterate over 3 neighbors
D: 3 steps to iterate over 3 neighbors
E: 3 steps to iterate over 3 neighbors

This yields 16 iterations.

So, we have the visiting of the five vertices, plus 16 iterations over adjacent neighbors. That's a total of 21 steps.

But, here's another graph with five vertices:



This graph has five vertices, but the count of iterations over adjacent neighbors is as follows:

V: 4 steps to iterate over 4 neighbors
W: 1 step to iterate over 1 neighbor
X: 1 step to iterate over 1 neighbor
Y: 1 step to iterate over 1 neighbor
Z: 1 step to iterate over 1 neighbor

This is a total of eight iterations.

We have the five vertices, plus eight iterations over adjacent neighbors. This is a total of 13 steps.

So, we have two graphs, each containing five vertices. However, searching one takes 21 steps, while searching the other takes 13 steps.

It emerges that we can't just count how many vertices are in the graph. Instead, *we also need to consider how many adjacent neighbors each vertex has.*

To effectively describe the efficiency of graph search, then, we're going to need to use *two* variables. We need one to represent the number of vertices in the graph, and another to be the total number of adjacent neighbors each vertex has.

O(V + E)

Interestingly enough, Big O Notation doesn't use the variable N to describe either of these things. Instead, it uses the variables V and E.

The V is the easier one. V stands for *vertex*, and represents the number of vertices in the graph.

E, interestingly, stands for *edge,* meaning the number of edges in the graph.

Now, computer scientists describe the efficiency of graph search as O(V + E). This means that the number of steps is the number vertices in the graph *plus* the number of edges in the graph. Let's see why this is the efficiency of graph search, as it's not immediately intuitive.

Specifically, if you look at our two earlier examples, you'll notice that V + E doesn't seem to be accurate.

In the A-B-C-D-E graph, there are five vertices and eight edges. This would be a total of 13 steps. However, we noted that there is actually a total of 21 steps.

And in the V-W-X-Y-Z graph, there are five vertices and four edges. O(V + E) says that graph search would have nine steps. But we saw that there are actually 13.

The reason for this discrepancy is that while O(V + E) only counts the number of edges once, in reality, graph search touches each edge *more* than once.

In the V-W-X-Y-Z graph, for example, there are only four edges. However, the edge between V and W is used twice. That is, when V is the current vertex, we find its adjacent neighbor W using that edge. But when W is the current vertex, we find its adjacent vertex V using that same edge.

With this in mind, the most accurate way to describe the efficiency of graph search in the V-W-X-Y-Z graph would be to count the five vertices, plus:

2 * edge between V and W
2 * edge between V and X
2 * edge between V and Y
2 * edge between V and Z

So, this comes out to be V + 2E, since V is 5, and each edge is used twice.

The answer, though, to why we just call this O(V + E), is because *Big O drops the constants.* While in reality the number of steps is V + 2E, we reduce this to O(V + E).

So, while O(V + E) is ultimately just an approximation, it's good enough, as are all expressions of Big O.

What is definitely clear, though, is that increasing the number of edges *will* increase the number of steps. After all, both the A-B-C-D-E and V-W-X-Y-Z

graphs have five vertices, but because the A-B-C-D-E graph has more edges, it takes considerably more steps.

At the end of the day, graph search is O(V + E) in a worst-case scenario, where the vertex we're searching for is the last one we find (or isn't present in the graph at all). And this is true for both breadth-first search as well as depth-first search.

However, we saw earlier that depending on the shape of the graph and the data we're searching for, our choice of breadth-first versus depth-first can optimize our search where we'd hope to find our vertex at some point *before* having to traverse the entire graph. That is, the right method of search can help us increase the odds that we won't end up in a worst-case scenario, and that we'll find the vertex early.

In the next section, you're going to learn about a specific type of graph that comes with its own set of search methods that can be used to solve some really complex but useful problems.

Graph Databases

Because graphs are so efficient at working with data involving relationships (such as friends in a social network), special *graph databases* are often used to store this kind of data in real-world software applications. These databases use the concepts you're learning about in this chapter, as well as other elements of graph theory, to optimize efficiency of operations around this kind of data. Indeed, many social networking applications are powered by graph databases under the hood.

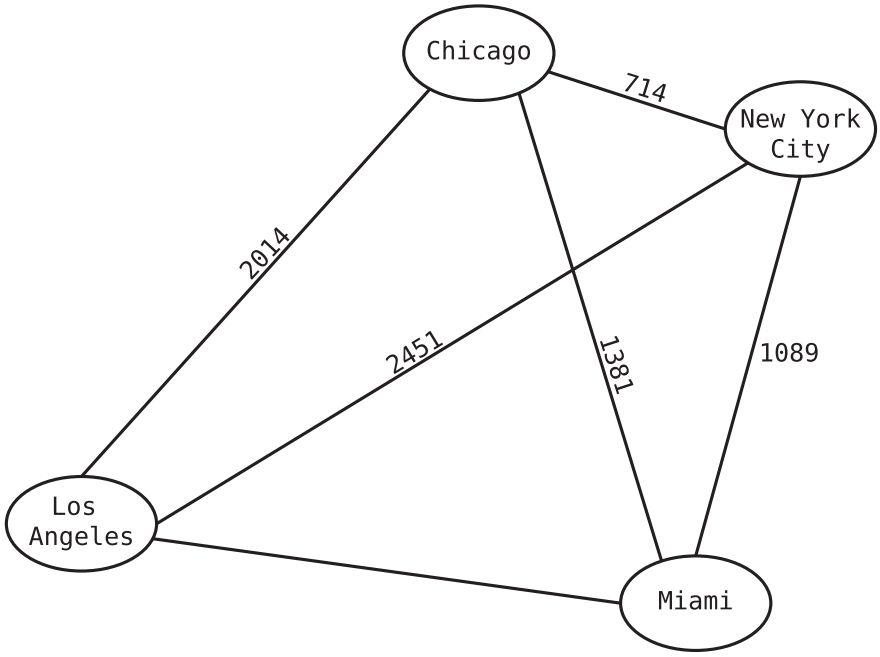
Some examples of graph databases include Neo4j,[a] ArangoDB,[b] and Apache Giraph.[c] These websites are a good place to start if you're interested in learning more about how graph databases work.

a. http://neo4j.com
b. https://www.arangodb.com
c. http://giraph.apache.org

Weighted Graphs

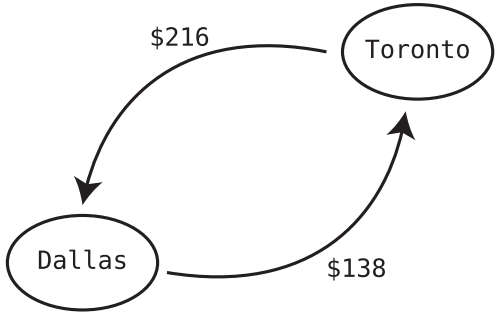
We've already seen that graphs can come in a number of different flavors. Another useful type of graph, known as a *weighted graph,* adds additional information to the *edges* of the graph.

Here's a weighted graph that represents a basic map of several major cities in the United States:



In this graph, each edge is accompanied by a number that represents the distance in miles between the cities the edge connects. For example, there are 714 miles between Chicago and New York City.

It's also possible to have weighted graphs that are also directional. In the following example, we can see that although a flight from Dallas to Toronto is $138, a flight from Toronto back to Dallas is $216:



Weighted Graphs in Code

We need to make a slight modification to our code if we want to add weights to our graph. One way to do this is to use a hash table to represent the adjacent vertices rather than an array:

```
class WeightedGraphVertex
  attr_accessor :value, :adjacent_vertices

  def initialize(value)
    @value = value
    @adjacent_vertices = {}
  end

  def add_adjacent_vertex(vertex, weight)
    @adjacent_vertices[vertex] = weight
  end
end
```

As you can see, `@adjacent_vertices` is now a hash table instead of an array. The hash table will contain key-value pairs where in each pair, the adjacent vertex is the key and the weight (of the edge from this vertex to the adjacent vertex) is the value.

When we use the `add_adjacent_vertex` method to add an adjacent vertex, we now pass in both the adjacent vertex as well as the weight.

So, if we want to create the Dallas–Toronto flight price graph from earlier, we can run the following code:

```
dallas = City.new("Dallas")
toronto = City.new("Toronto")

dallas.add_adjacent_vertex(toronto, 138)
toronto.add_adjacent_vertex(dallas, 216)
```

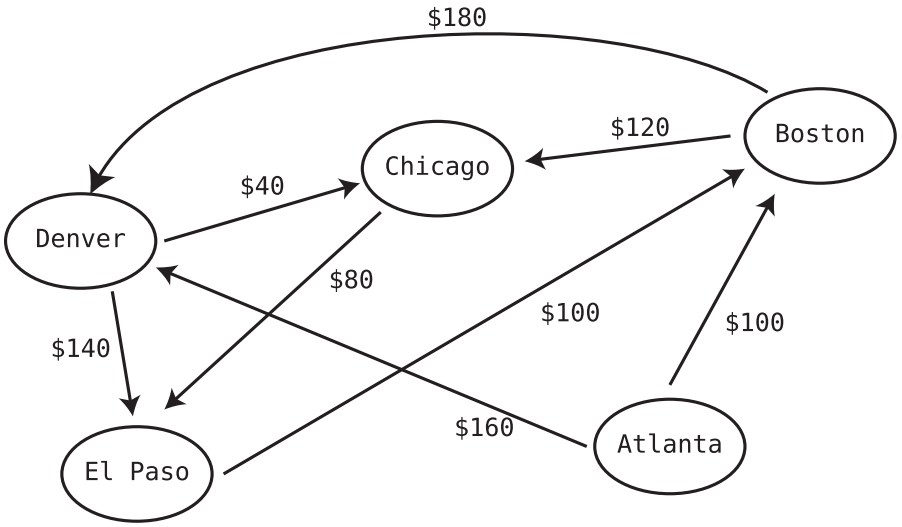
The Shortest Path Problem

Weighted graphs can be very useful in modeling all sorts of datasets, and they also come with some powerful algorithms that help us make the most out of that data.

Let's harness one of these algorithms to save us a bit of money.

The graph on page 367 demonstrates the costs of available flights between five different cities.

Now, say I'm in Atlanta and want to fly to El Paso. Unfortunately, we can see in this graph that there's no direct route from Atlanta to El Paso at this time. However, I can get there if I'm willing to stop over in other cities along the



way. For example, I can fly from Atlanta to Denver, and then from Denver to El Paso. But there are other paths as well, and each path has a different price. The Atlanta–Denver–El Paso path will set me back $300, but the Atlanta–Denver–Chicago–El Paso path only costs $280.

The puzzle, now, is this: how do we create an algorithm that finds the *cheapest* price I'd have to shell out to get to my destination? Let's assume we don't care about how many stops we have to make; we're just out to get the cheapest fare.

This kind of puzzle is known as the *shortest path problem.* This problem can take other forms as well. For example, if the graph showed us distances between cities, we might want to find the path that has the shortest distance. But here, the shortest path we're looking for is the cheapest one, since the weights represent flight prices.

Dijkstra's Algorithm

Numerous algorithms can solve the shortest path problem, and one of the most famous is one discovered by Edsger Dijkstra (pronounced "dike' struh") in 1959. Unsurprisingly, this algorithm is known as *Dijkstra's algorithm.*

In the following section, we're going to use Dijkstra's algorithm to find the cheapest path in our city flights example.

Dijkstra's Algorithm Setup

The first thing to note is that Dijkstra's algorithm comes with a free bonus. By the time we're done, we're not just going to find the cheapest price from Atlanta to El Paso, but also we're going to find the cheapest prices from Atlanta to *all* known cities. As you'll see, the algorithm simply works this way; we end up gathering all of this data. So, we'll know the cheapest price from Atlanta to Chicago, and the cheapest price from Atlanta to Denver, and so on.

To set things up, we'll create a way to store the cheapest known prices from our starting city to all other known destinations. In our code that follows, we'll use a hash table for this. For our example walk-through, though, we'll use a visual table that looks like this:

From Atlanta To:	City #1	City #2	City #3	Etc.
	?	?	?	?

The algorithm will begin at the Atlanta vertex, as it's the only city we're currently aware of. As we discover new cities, we'll add them to our table and record the cheapest price from Atlanta to each of these cities.

By the time the algorithm is complete, the table will look like this:

Cheapest Price from Atlanta To:	Boston	Chicago	Denver	El Paso
	$100	$200	$160	$280

In code, this will be represented with a hash table that looks like this:

```
{"Atlanta" => 0, "Boston" => 100, "Chicago" => 200,
"Denver" => 160, "El Paso" => 280}
```

(Note that Atlanta is in the hash table as well with a value of 0. We'll need this to get the algorithm to work, but it also makes sense, as it costs nothing to get to Atlanta from Atlanta, since you're already there!)

In our code and going forward, we will call this table the `cheapest_prices_table`, as it stores all the cheapest prices from the starting city to all other destinations.

Now, if all we wanted to do is figure out the cheapest price to get to a particular destination, the `cheapest_prices_table` would contain all the data we need. But we probably also want to know the actual path that would get us the cheapest price. That is, if we want to get from Atlanta to El Paso, we don't want to just know the cheapest price is $280; we also want to know that to get this price, we need to fly the specific path of Atlanta–Denver–Chicago–El Paso.

To achieve this, we'll also need *another* table, which we'll call the `cheapest_previous_stopover_city_table`. The purpose of this table will only become clear once we

jump into the algorithm, so I'll hold off on the explanation until then. For now, though, it'll suffice to show what it'll look like by the end of the algorithm.

Cheapest Previous Stopover City from Atlanta:	Boston	Chicago	Denver	El Paso
	Atlanta	Denver	Atlanta	Chicago

(Note that this table, too, will be implemented using a hash table.)

Dijkstra's Algorithm Steps

Now that everything is set up, here are the steps for Dijkstra's algorithm. For clarity, I'm going to describe the algorithm in terms of cities, but you can replace the word "city" with "vertex" to make it work for any weighted graph. Also note that these steps will become clearer when we walk through an example, but here we go:

1. We visit the starting city, making it our "current city."
2. We check the prices from the current city to each of its adjacent cities.
3. If the price to an adjacent city from the starting city is cheaper than the price currently in `cheapest_prices_table` (or the adjacent city isn't yet in the `cheapest_prices_table` at all):

   a. We update the `cheapest_prices_table` to reflect this cheaper price.

   b. We update the `cheapest_previous_stopover_city_table`, making the adjacent city the key and the current city the value.
4. We then visit whichever unvisited city has the cheapest price from the starting city, making it the current city.
5. We repeat the Steps 2 through 4 until we've visited every known city.

Again, this will all make more sense when we walk through an example.

Dijkstra's Algorithm Walk-Through

Let's walk through Dijkstra's algorithm step by step.

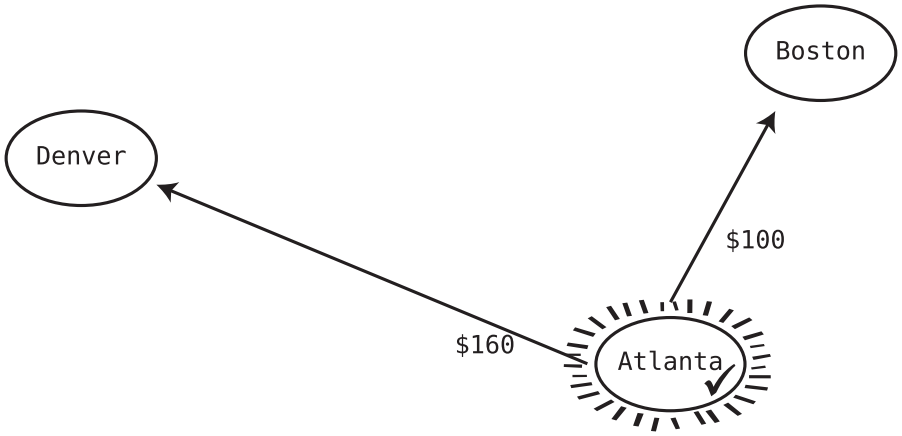
To start, our `cheapest_prices_table` only contains Atlanta:

From Atlanta To:
$0

At the start of the algorithm, Atlanta is the only city we have access to; we have not yet "discovered" the other cities.

Step 1: We officially visit Atlanta and make it the `current_city`.

To indicate that it's the current_city, we'll surround it with lines. And to add to the record that we've visited it, we'll add a check mark:



In the next steps, we'll proceed to inspect each of the current_city's adjacent cities. This is how we "discover" new cities; if a city we have access to has adjacent cities we weren't aware of before, we can add them to our map.

Step 2: One city adjacent to Atlanta is Boston. As we can see, the price from Atlanta to Boston is $100. We then check the cheapest_prices_table to see whether this is the cheapest known price from Atlanta to Boston, but it turns out we haven't recorded *any* prices from Atlanta to Boston yet. That means this is the cheapest *known* flight from Atlanta to Boston (as of now), so we add that to the cheapest_prices_table:

From Atlanta To:	***Boston***
$0	*$100*

Since we've made a change to the cheapest_prices_table, we now also need to modify the cheapest_previous_stopover_city_table, making the adjacent city (Boston) the key and the current_city the value:

Cheapest Previous Stopover City from Atlanta:	***Boston***
	Atlanta

Adding this data to this table means that to earn the cheapest known price from Atlanta to Boston ($100), the city we need to visit *immediately before Boston* is Atlanta. At this point, this is obvious, since Atlanta is the *only* way to get to Boston that we know of. However, as we proceed, we'll see why this second table becomes useful.

Step 3: We've checked out Boston, but Atlanta has another adjacent city, Denver. We check whether the price ($160) is the cheapest known route from Atlanta to Denver, but Denver isn't the cheapest_prices_table at all yet, so we add it as the cheapest known flight:

From Atlanta To:	Boston	*Denver*
$0	$100	*$160*

We then also add Denver and Atlanta as a key-value pair to the cheapest_previous_stopover_city_table:

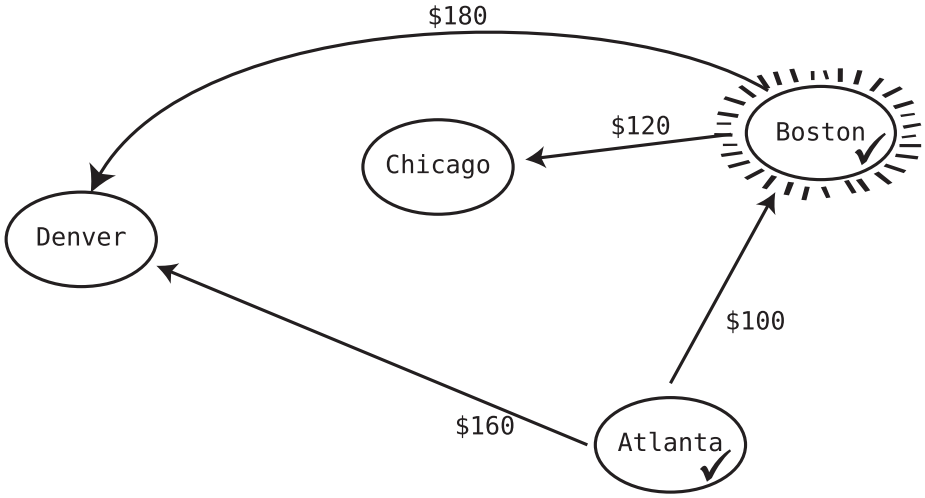
Cheapest Previous Stopover City from Atlanta:	Boston	*Denver*
	Atlanta	*Atlanta*

Step 4: By this point, we've inspected all of Atlanta's adjacent cities, so it's time to visit our next city. But we need to figure out which city to visit next.

Now, as stated in the algorithm steps earlier, we only proceed to visit cities we haven't yet visited so far. Furthermore, among the unvisited cities, we always choose to *first* visit the city that has the cheapest known route *from the starting city.* We can get this data from the cheapest_prices_table.

In our example, the only cities we know about that we haven't visited yet are Boston or Denver. By looking at the cheapest_prices_table, we can see that it's cheaper to get from Atlanta to Boston than it is to get from Atlanta to Denver, so we're going to visit Boston next.

Step 5: We visit Boston and designate it as the current_city:



Next, we're going to inspect Boston's adjacent cities.

Step 6: Boston has two adjacent cities, Chicago and Denver. (Atlanta is not considered adjacent, since we can't fly from Boston to Atlanta.)

Which city should we visit first—Chicago or Denver? Again, we want to first visit the city whose price is cheapest if we were flying to it *from Atlanta.* So, let's do the math.

The price from Boston to Chicago alone is $120. When looking at the cheapest_prices_table, we can see that the cheapest route from Atlanta to Boston is $100. So, this means that the cheapest flight from Atlanta to Chicago *with Boston as the previous stopover city* would be $220.

Since at this point, this is the only known price from Atlanta to Chicago, we'll add it to the cheapest_prices_table. We'll insert it in the middle of the table to keep the cities alphabetized:

From Atlanta To:	Boston	*Chicago*	Denver
$0	$100	*$220*	$160

Again, because we made a change to that table, we'll also modify the cheapest_previous_stopover_city_table. The adjacent city always becomes the key, and the current_city always becomes the value, so the table becomes:

Cheapest Previous Stopover City from Atlanta:	Boston	*Chicago*	Denver
	Atlanta	*Boston*	Atlanta

In our quest to find the city to visit next, we analyzed Chicago. We will inspect Denver next.

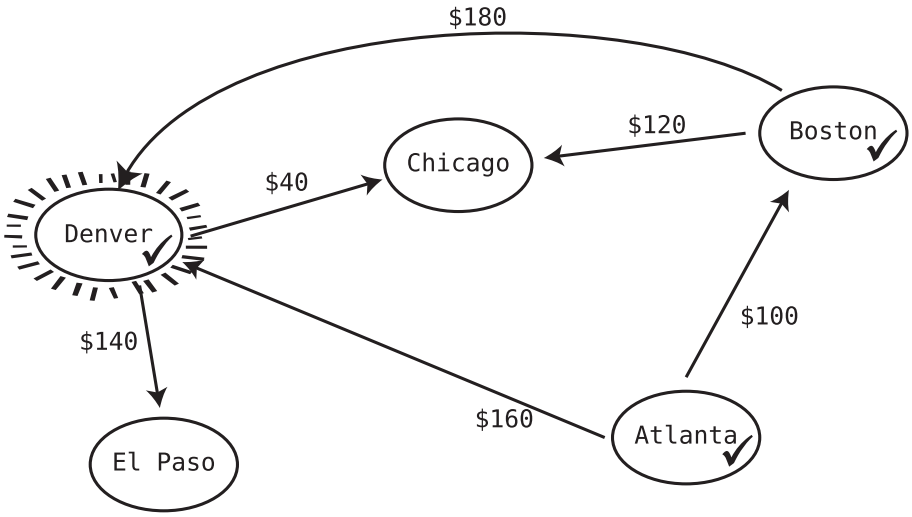
Step 7: Let's now look at the edge between Boston and Denver. We can see that the price is $180. Since the cheapest flight from Atlanta to Boston, again, is $100, that would mean the cheapest flight from Atlanta to Denver *through Boston as the previous stopover city* is $280.

This gets a little interesting, because when we inspect our cheapest_prices_table, we can see that the cheapest route from Atlanta to Denver is $160, *which is cheaper* than the Atlanta–Boston–Denver route. Accordingly, *we do not modify* either of our tables. That is, we want to leave $160 as the cheapest known route from Atlanta to Denver.

We're done with this step, and since we've looked at all of Boston's adjacent cities, we can now visit our next city.

Step 8: The current known unvisited cities are Chicago and Denver. Again, the one we visit next—and pay careful attention to this—is the city with the cheapest known path *from our starting city* (Atlanta).

Looking at our cheapest_prices_table, we can see that it's cheaper to go from Atlanta to Denver ($160) than it is to go to from Atlanta to Chicago ($220), so that means that we visit Denver next:



Next up, we'll look at Denver's adjacent cities.

Step 9: Denver has two adjacent cities, Chicago and El Paso. Which of these cities will we visit next? To find out, we need to analyze the prices to each city. Let's start with Chicago.

It costs just $40 to go from Denver to Chicago (a good deal!), which means the cheapest flight from Atlanta to Chicago *through Denver as the previous stopover city* would be $200, since the cheapest route from Atlanta to Denver is $160.

When looking at the cheapest_prices_table, we can see that the current cheapest price from Atlanta to Chicago is $220. That means this new route we just found to Chicago through Denver is even cheaper, so we can update the cheapest_prices_table accordingly:

From Atlanta To:	Boston	Chicago	Denver
$0	$100	*$200*	$160

Whenever we update the cheapest_prices_table, we also have to update the cheapest_previous_stopover_city_table. We set the adjacent city (Chicago) as the key, and the current_city (Denver) as the value. Now, in this case, Chicago already exists as a key. This means we'll be overwriting its value from Boston to Denver:

Cheapest Previous Stopover City from Atlanta:	Boston	Chicago	Denver
	Atlanta	*Denver*	Atlanta

What this means is that to nab the cheapest flight path from Atlanta to Chicago, we need to stop over at Denver as the city immediately prior to Chicago. That is, Denver should be our second-to-last stop before we proceed to Chicago. Only then will we save the most money.

This information will be useful in determining the cheapest path from Atlanta to our destination city, as you'll see in a little bit. Hang on, we're almost there!

Step 10: Denver has another adjacent city, El Paso. The price from Denver to El Paso is $140. We can now construct our first known price from Atlanta to El Paso. The cheapest_prices_table tells us the cheapest price from Atlanta to Denver is $160. This means if we then go from Denver to El Paso, we incur another $140, making the total price from Atlanta to El Paso $300. We can add this to the cheapest_prices_table:

From Atlanta To:	Boston	Chicago	Denver	*El Paso*
$0	$100	$200	$160	*$300*

We must then also add the key-value pair of El Paso-Denver to our cheapest_previous_stopover_city_table:

Cheapest Previous Stopover City from Atlanta:	Boston	Chicago	Denver	*El Paso*
	Atlanta	Denver	Atlanta	*Denver*

Again, this means that to save the most money when flying from Atlanta to El Paso, our second-to-last stop should be Denver.

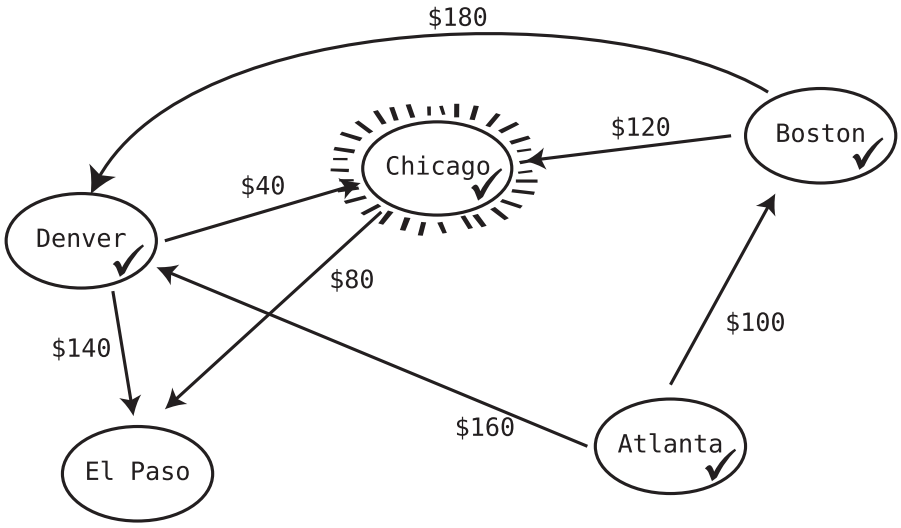
We've seen all our current_city's adjacent cities, so it's time to visit our next city.

Step 11: We have two known unvisited cities, Chicago and El Paso. Since it's cheaper to get from Atlanta to Chicago ($200) than it is to get from Atlanta to El Paso ($300), we visit Chicago next as shown in the graph on page 375.

Step 12: Chicago has only one adjacent city, El Paso. The price from Chicago to El Paso is $80 (not bad). With this information, we can now calculate the cheapest price from Atlanta to El Paso when assuming that Chicago is our second-to-last stop.

The cheapest_prices_table shows us that the cheapest path from Atlanta to Chicago is $200. Adding the $80 to this means the cheapest price from Atlanta to El Paso *with Chicago as the second-to-last stop* would cost $280.

Wait! This is cheaper than the currently known cheapest path from Atlanta to El Paso. In our cheapest_prices_table, we see that the cheapest known price is $300. But when we fly through Chicago, the price is $280, which is cheaper.



Accordingly, we need to update the cheapest_prices_table to indicate our newly-found cheapest path to El Paso:

From Atlanta To:	Boston	Chicago	Denver	El Paso
$0	$100	$200	$160	*$280*

We also need to update the cheapest_previous_stopover_city_table, with El Paso as the key and Chicago as the value:

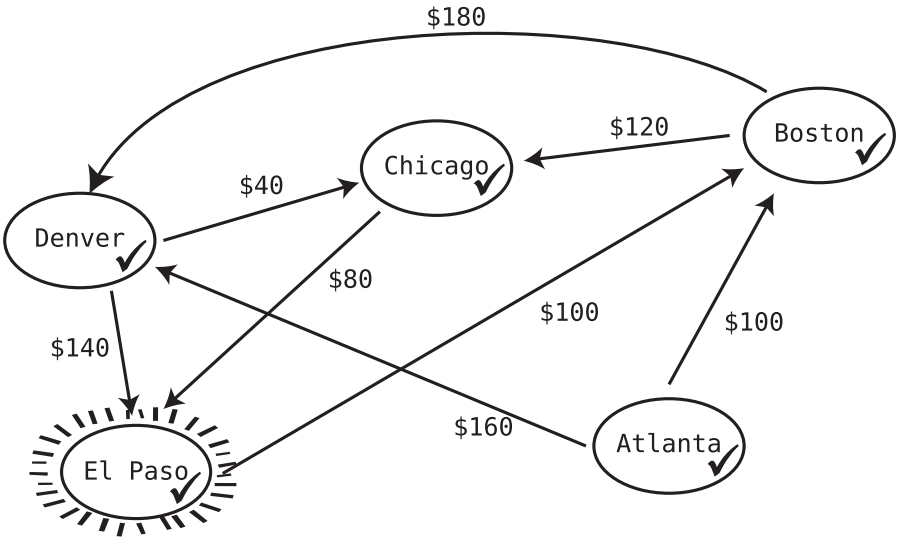
Cheapest Previous Stopover City from Atlanta:	Boston	Chicago	Denver	El Paso
	Atlanta	Denver	Atlanta	*Chicago*

Chicago has no more adjacent cities, so we can now visit our next city.

Step 13: El Paso is the only known unvisited city, so let's make it our current_city as shown in the graph on page 376.

Step 14: El Paso has only one outbound flight, which is to Boston. That flight costs $100. Now, the cheapest_prices_table reveals that the cheapest price from Atlanta to El Paso is $280. So, if we travel from Atlanta to Boston *with El Paso* as the second-to-last stop, our grand total will be $380. This is more expensive than the cheapest-known price from Atlanta to Boston ($100), so we don't update any of our tables.

Since we've visited every known city, we now have all the information we need to find the cheapest path from Atlanta to El Paso.



Finding the Shortest Path

If we just want to know the cheapest price from Atlanta to El Paso, we can look in our cheapest_prices_table and see that it's $280. But if we want to figure out the *exact* path to fly to snag that low price, we have one last thing to do.

Remember the cheapest_previous_stopover_city_table? It's now time to actually use that data.

Currently, the cheapest_previous_stopover_city_table looks like this:

Cheapest Previous Stopover City from Atlanta:	Boston	Chicago	Denver	El Paso
	Atlanta	Denver	Atlanta	Chicago

We can use this table to draw the shortest path from Atlanta to El Paso—if we go backward.

Let's look at El Paso. Its corresponding city is Chicago. This means the cheapest route from Atlanta to El Paso involves stopping over in Chicago as the *immediate step* before flying to El Paso. Let's write this down:

```
Chicago -> El Paso
```

Now, if we look up Chicago in the cheapest_previous_stopover_city_table, we can see that its corresponding value is Denver. This means the cheapest route from Atlanta to *Chicago* involves stopping in Denver right before Chicago. Let's add this to our figure:

```
Denver -> Chicago -> El Paso
```

If we then look up Denver in the cheapest_previous_stopover_city_table, we can see that the cheapest flight to get from Atlanta to Denver is to fly directly from Atlanta to Denver:

```
Atlanta -> Denver -> Chicago -> El Paso
```

Now, Atlanta happens to be our starting city, so this route is the exact path we'd take from Atlanta to El Paso to get the cheapest price.

Let's review the logic we're using to chain together the cheapest path.

Remember, the cheapest_previous_stopover_city_table contains, for each destination, the second-to-last stop before that destination to earn the cheapest price if you're flying from Atlanta.

So, from the cheapest_previous_stopover_city_table, we can see that the cheapest price from Atlanta to El Paso means…

- …we need to fly directly from Chicago to El Paso, and…
- …we need to fly directly from Denver to Chicago, and…
- …we need to fly directly from Atlanta to Denver…

which means our cheapest path is:

```
Atlanta -> Denver -> Chicago -> El Paso
```

And…that's it. Whew!

Code Implementation: Dijkstra's Algorithm

Before we implement the actual algorithm in Ruby, we'll first implement a City class, which is similar to our WeightedGraphVertex class from earlier, but uses terms like routes and price. This will make the code that follows (somewhat) easier to understand:

```
class City
  attr_accessor :name, :routes

  def initialize(name)
    @name = name
    @routes = {}
  end

  def add_route(city, price)
    @routes[city] = price
  end
end
```

To set up our example from earlier, you can run this code:

```
atlanta = City.new("Atlanta")
boston = City.new("Boston")
chicago = City.new("Chicago")
denver = City.new("Denver")
el_paso = City.new("El Paso")

atlanta.add_route(boston, 100)
atlanta.add_route(denver, 160)
boston.add_route(chicago, 120)
boston.add_route(denver, 180)
chicago.add_route(el_paso, 80)
denver.add_route(chicago, 40)
denver.add_route(el_paso, 140)
```

Finally, here's the code for Dijkstra's algorithm. It doesn't make for light reading, and it's probably the most complex piece of code in this book. However, if you're ready to study it carefully, read on.

In our implementation here, this method does not live inside the City class, but outside it. The method accepts two City instances, and returns the shortest path between them:

```
def dijkstra_shortest_path(starting_city, final_destination)
  cheapest_prices_table = {}
  cheapest_previous_stopover_city_table = {}

  # To keep our code simple, we'll use a basic array to keep track of
  # the known cities we haven't yet visited:
  unvisited_cities = []

  # We keep track of the cities we've visited using a hash table.
  # We could have used an array, but since we'll be doing lookups,
  # a hash table is more efficient:
  visited_cities = {}

  # We add the starting city's name as the first key inside the
  # cheapest_prices_table. It has a value of 0, since it costs nothing
  # to get there:
  cheapest_prices_table[starting_city.name] = 0

  current_city = starting_city

  # This loop is the core of the algorithm. It runs as long as we can
  # visit a city that we haven't visited yet:
  while current_city

    # We add the current_city's name to the visited_cities hash to record
    # that we've officially visited it. We also remove it from the list
    # of unvisited cities:
    visited_cities[current_city.name] = true
    unvisited_cities.delete(current_city)
```

```
  # We iterate over each of the current_city's adjacent cities:
  current_city.routes.each do |adjacent_city, price|

    # If we've discovered a new city,
    # we add it to the list of unvisited_cities:
    unvisited_cities <<
      adjacent_city unless visited_cities[adjacent_city.name]

    # We calculate the price of getting from the STARTING city to the
    # ADJACENT city using the CURRENT city as the second-to-last stop:
    price_through_current_city =
      cheapest_prices_table[current_city.name] + price

    # If the price from the STARTING city to the ADJACENT city is
    # the cheapest one we've found so far...
    if !cheapest_prices_table[adjacent_city.name] ||
     price_through_current_city <
       cheapest_prices_table[adjacent_city.name]

      # ... we update our two tables:
      cheapest_prices_table[adjacent_city.name] =
        price_through_current_city
      cheapest_previous_stopover_city_table[adjacent_city.name] =
        current_city.name
    end
  end

  # We visit our next unvisited city. We choose the one that is cheapest
  # to get to from the STARTING city:
  current_city = unvisited_cities.min do |city|
    cheapest_prices_table[city.name]
  end
end

# We have completed the core algorithm. At this point, the
# cheapest_prices_table contains all the cheapest prices to get to each
# city from the starting city. However, to calculate the precise path
# to take from our starting city to our final destination,
# we need to move on.

# We'll build the shortest path using a simple array:
shortest_path = []

# To construct the shortest path, we need to work backwards from our
# final destination. So, we begin with the final destination as our
# current_city_name:
current_city_name = final_destination.name

# We loop until we reach our starting city:
while current_city_name != starting_city.name

  # We add each current_city_name we encounter to the shortest path array:
  shortest_path << current_city_name
```

```
    # We use the cheapest_previous_stopover_city_table to follow each city
    # to its previous stopover city:
    current_city_name =
      cheapest_previous_stopover_city_table[current_city_name]
  end

  # We cap things off by adding the starting city to the shortest path:
  shortest_path << starting_city.name

  # We reverse the output so we can see the path from beginning to end:
  return shortest_path.reverse
end
```

There's a decent amount of code here, so let's break it down.

The dijkstra_shortest_path function accepts two vertices, representing the starting_city and final_destination.

Eventually, our function will return an array of strings that represent the cheapest path. For our example, this function would return:

```
["Atlanta", "Denver", "Chicago", "El Paso"]
```

The first thing our function does is set up the two primary tables that fuel the entire algorithm:

```
cheapest_prices_table = {}
cheapest_previous_stopover_city_table = {}
```

We then set up ways to track which cities we've visited and the ones we have yet to visit:

```
unvisited_cities = []
visited_cities = {}
```

It may seem odd that unvisited_cities is an array, while visited_cities is a hash table. The reason we've made visited_cities a hash table is because in the remainder of our code, we only use it for lookups, for which a hash table is an ideal choice in terms time complexity.

The choice of the best data structure for the unvisited_cities is less simple. In our code that follows, the next city we visit is always the cheapest unvisited city to reach from the starting city. Ideally, then, we always want immediate access to the cheapest option from among the unvisited cities. It's a bit easier to get this access with an array than with a hash table.

In truth, a priority queue would be a perfect fit for this, as its whole function is to provide ready access to the least (or greatest) value from a collection of items. As you learned in *Keeping Your Priorities Straight with Heaps*, a heap is generally the best data structure for implementing a priority queue.

However, I've instead chosen to use a simple array for this implementation only to keep the code as simple and as small as possible, as Dijkstra's algorithm is complex enough on its own. However, I encourage you to try replacing the array with a priority queue.

Next, we add the first key-value pair to the cheapest_prices_table with the starting_city as the key and 0 as the value. Again, this makes sense because it costs nothing to get to the starting_city as we're already there:

```
cheapest_prices_table[starting_city.name] = 0
```

As the last bit of setup, we designate the starting_city to be our current_city:

```
current_city = starting_city
```

We now begin the core of the algorithm, which takes the form of a loop that runs as long as we can access a current_city. Within this loop, we mark the current_city as having been visited by adding its name to the visited_cities hash table. And in case the current_city is currently in the list of unvisited_cities, we delete it from there as well:

```
while current_city
  visited_cities[current_city.name] = true
  unvisited_cities.delete(current_city)
```

Next, within the while loop we begin another loop, iterating over all of the adjacent cities of the current_city:

```
current_city.routes.each do |adjacent_city, price|
```

Within this inner loop, we first add each adjacent city to the array of unvisited_cities if it's a city we've never visited before:

```
unvisited_cities << adjacent_city unless visited_cities[adjacent_city.name]
```

In this particular implementation, it's possible for a city to exist multiple times within the unvisited_cities array, and that's okay, as we delete all those instances with the line, unvisited_cities.delete(current_city). We could, alternatively, also make sure the current_city doesn't already exist within the unvisited_cities before adding it.

Next, we calculate the cheapest possible price to get from the starting city to the adjacent city, assuming that the current_city is the second-to-last stop. We do this by using the cheapest_prices_table to look up the cheapest known route to the current_city and then adding that to the price of the route from the current_city to the adjacent city. This calculation then gets stored in a variable called price_through_current_city:

```
price_through_current_city = cheapest_prices_table[current_city.name] + price
```

Then, we look within the cheapest_prices_table to see whether this price_through_current_city is now the cheapest known flight from the starting city to the adjacent city. If the adjacent city isn't yet in the cheapest_prices_table, this price is, by definition, the cheapest known price:

```
if !cheapest_prices_table[adjacent_city.name] ||
 price_through_current_city < cheapest_prices_table[adjacent_city.name]
```

If the price_through_current_city *is* now the cheapest route from the starting city to the adjacent city, we update the two main tables. That is, we store the new price for the adjacent city in the cheapest_prices_table. And we also update the cheapest_previous_stopover_city_table with the adjacent city's name as the key and the current_city's name as the value:

```
cheapest_prices_table[adjacent_city.name] = price_through_current_city
cheapest_previous_stopover_city_table[adjacent_city.name] = current_city.name
```

After iterating over all the adjacent cities of the current_city, it's time to visit the next city. We visit the cheapest unvisited city we can reach from the starting city by making it the new current_city:

```
current_city = unvisited_cities.min do |city|
  cheapest_prices_table[city.name]
end
```

If there are no more known unvisited cities, the current_city will become nil and the while loop will end.

At this point, the two tables have been fully populated with all the data we need. If we so chose, we could at this point simply return the cheapest_prices_table and see all the cheapest prices to all known cities from the starting_city.

Instead, though, we proceed to find the precise cheapest path to get to our final_destination.

To set things up for this, we create an array called shortest_path, which is what we'll return at the end of the function:

```
shortest_path = []
```

We also create a variable called current_city_name, which starts out as the name of the final_destination:

```
current_city_name = final_destination.name
```

We then begin a while loop that populates the shortest_path. This loop works backward from the final_destination until we reach the starting_city:

```
while current_city_name != starting_city.name
```

Within the loop, we add the current_city_name to the shortest_path array and then use the cheapest_previous_stopover_city_table to find the city that should be the stop immediately preceding the current_city_name. This previous city now becomes the new current_city_name:

```
shortest_path << current_city_name
current_city_name = cheapest_previous_stopover_city_table[current_city_name]
```

For code readability, we had the loop end as soon as it hit the starting_city, so we now manually add the starting_city's name to the end of the shortest_path:

```
shortest_path << starting_city.name
```

The shortest_path now contains the backward path from the final_destination to the starting_city. So, we return the reversed version of this array to provide the shortest path from the starting_city to the final_destination:

```
return shortest_path.reverse
```

Although our implementation deals with cities and prices, all the variable names can be changed to handle the shortest path for *any* weighted graph.

The Efficiency of Dijkstra's Algorithm

Dijkstra's algorithm is a general description of the approach for finding the shortest path within a weighted graph, but it doesn't specify the precise code implementation. In fact, there are a number of variations out there in how this algorithm can be written.

In our code walk-through, for example, we used a simple array for the unvisited_cities data structure, but I noted that a priority queue could be used instead.

It turns out that the precise implementation has a considerable effect on the algorithm's time complexity. But let's at least analyze *our* implementation.

When we use a simple array for keeping track of the cities we haven't visited yet (unvisited_cities), our algorithm can take up to $O(V^2)$ steps. This is because the worst-case scenario for Dijkstra's algorithm is when each vertex has an edge leading to every other vertex within the graph. In this case, for every vertex we visit, we check the weight of the path from that vertex to every other vertex. This is V vertices multiplied by V vertices, which is $O(V^2)$.

Other implementations, such as using a priority queue instead of an array, lead to faster speeds. Again, there are several variations of Dijkstra's algorithm, and each variation needs its own analysis to determine its precise time complexity.

Whatever implementation of the algorithm you choose, though, is a big win over the alternative, which would be to find *every* possible path through the

graph and then select the fastest one. Dijkstra's algorithm gives a sure way to proceed thoughtfully through the graph, and zero in on the shortest path.

Wrapping Up

We're almost at the end of our journey, as this chapter represents the last significant data structure you'll encounter in this book. You've seen graphs are extremely powerful tools for dealing with data involving relationships, and in addition to making our code fast, they can also help solve tricky problems.

In truth, I could fill a book just discussing graphs. There are so many interesting and useful algorithms surrounding this data structure, such as the minimum spanning tree, topological sort, bidirectional search, the Floyd–Warshall algorithm, the Bellman–Ford algorithm, and graph coloring, just to name a few. However, this chapter should serve as the foundation for you to explore these additional topics.

Along our travels, our primary focus has been on how fast our code will run. That is, we've been measuring how efficient our code performs in terms of time, and we've been measuring that in terms of counting the number of steps our algorithms take.

However, efficiency can be measured in other ways than speed alone. In particular, we might care about how much *memory* a data structure or algorithm might consume. In the next chapter, you'll learn how to analyze the efficiency of our code in terms of *space.*

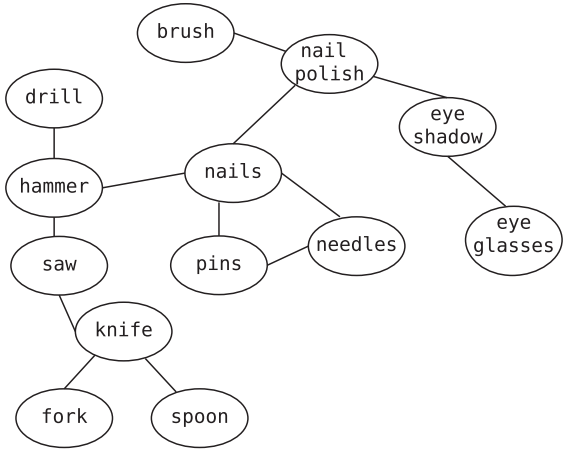
Exercises

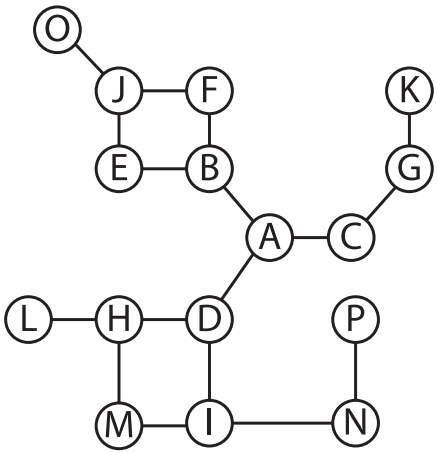
The following exercises provide you with the opportunity to practice with graphs. The solutions to these exercises are found in the section, Chapter 18, on page 461.

1. The first graph on page 385 powers an e-commerce store's recommendation engine. Each vertex represents a product available on the store's website. The edges connect each product to other "similar" products the site will recommend to the user when browsing a particular item.

   If the user is browsing "nails," what other products will be recommended to the user?

2. If we perform *depth*-first search on the second graph on page 385 starting with the "A" vertex, what is the order in which we'll traverse all the vertices? Assume that when given the choice to visit multiple adjacent vertices, we'll first visit the node that is earliest in the alphabet.





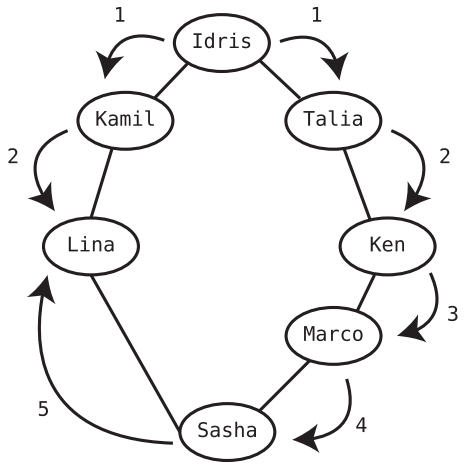
3. If we perform *breadth*-first search on the previous graph starting with the "A" vertex, what is the order in which we'll traverse all the vertices? Assume that when given the choice to visit multiple adjacent vertices, we'll first visit the node that is earliest in the alphabet.

4. In this chapter, I only provided the code for breadth-first *traversal*, as discussed in Breadth-First Search, on page 348. That is, the code simply printed the value of each vertex. Modify the code so that it performs an actual *search* for a vertex value provided to the function. (We did this for depth-first search.) That is, if the function finds the vertex it's searching for, it should return that vertex's value. Otherwise, it should return null.

5. In Dijkstra's Algorithm, on page 367, we saw how Dijkstra's algorithm helped us find the shortest path within a weighted graph. However, the concept of a shortest path exists within an unweighted graph as well. How?

   The shortest path in a classic (unweighted) graph is the path that takes the fewest number of vertices to get from one vertex to another.

   This can be especially useful in social networking applications. Take the example network that follows:



   If we want to know how Idris is to connected Lina, we'd see that she's connected to her from two different directions. That is, Idris is a second-degree connection to Lina through Kamil, but she is also a fifth-degree connection through Talia. Now, we're probably interested in *how closely* Idris is connected to Lina, so the fact that she's a fifth-degree connection is unimportant given that they're also second-degree connections.

   Write a function that accepts two vertices from a graph and returns the shortest path between them. The function should return an array containing the precise path, such as `["Idris", "Kamil", "Lina"]`.

   Hint: The algorithm may contain elements of both breadth-first search and Dijkstra's algorithm.

CHAPTER 19

Dealing with Space Constraints

When analyzing the efficiency of various algorithms throughout this book, we've focused exclusively on how fast they run—that is, their time complexity. However, another measure of efficiency can be useful as well, which is how much *memory* an algorithm consumes. This measure is known as *space complexity*.

Space complexity becomes an important factor when memory is limited. If you have an enormous amount of data, or are programming for a small device with limited memory, space complexity can matter a lot.

In a perfect world, we'd always use algorithms that are both fast *and* memory-efficient. However, there are times where we can't have both, and we need to choose between the two. Each situation requires a careful analysis to know when we need to prioritize speed over memory, and memory over speed.

Big O of Space Complexity

Interestingly, computer scientists use Big O Notation to describe space complexity just as they do for time complexity.

Back when I introduced Big O Notation in *O Yes! Big O Notation*, I described Big O in terms of what I called the "key question." For time complexity, the key question was: *if there are N data elements, how many steps will the algorithm take?*

To use Big O for space complexity, we just need to reframe the key question. When it comes to memory consumption, the key question is: *if there are N data elements, how many units of memory will the algorithm consume?*

Here's a simple example.

Let's say we're writing a JavaScript function that accepts an array of strings and returns an array of those strings in ALL CAPS. For example, the function would accept an array like ["tuvi", "leah", "shaya", "rami"] and return ["TUVI", "LEAH", "SHAYA", "RAMI"]. Here's one way we can write this function:

```
function makeUppercase(array) {
  let newArray = [];
  for(let i = 0; i < array.length; i++) {
    newArray[i] = array[i].toUpperCase();
  }
  return newArray;
}
```

In this makeUpperCase() function, we accept an array. We then create a *brand-new array* called newArray and fill it with uppercase versions of each string from the original array.

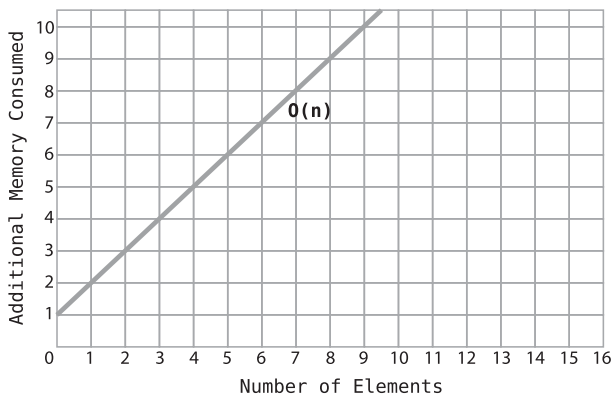
By the time this function is complete, we will have two arrays floating around in our computer's memory. We have the original array, which contains ["tuvi", "leah", "shaya", "rami"], and we have newArray, which contains ["TUVI", "LEAH", "SHAYA", "RAMI"].

When we analyze this function in terms of space complexity, we can see that this function *creates* a brand-new array that contains N elements. This is *in addition* to the original array which also held N elements.

So, let's return to our key question: *if there are N data elements, how many units of memory will the algorithm consume?*

Because our function generated an additional N data elements (in the form of newArray), we'd say that this function has a *space efficiency of O(N).*

The way this appears on the following graph should look quite familiar:



Note that this graph is identical to the way we've depicted O(N) in graphs in previous chapters, with the exception that the vertical axis now represents *memory consumed* rather than time.

Now, let's present an alternative makeUppercase() function that is more memory-efficient:

```
function makeUppercase(array) {
  for(let i = 0; i < array.length; i++) {
    array[i] = array[i].toUpperCase();
  }
  return array;
}
```

In this second version, we do not create any new arrays. Instead, we modify each string within the original array *in place*, transforming them into uppercase one at a time. We then return the modified array.

This is a drastic improvement in terms of memory consumption, since our new function *doesn't consume any additional memory at all.*

How do we describe this in terms of Big O Notation?

Recall that with time complexity, an O(1) algorithm was one whose speed remained constant no matter how large the data. Similarly, with space complexity, O(1) means that the memory consumed by an algorithm is constant no matter how large the data.

Our revised makeUppercase function consumes a constant amount of additional space (zero!) no matter whether the original array contains four elements or one hundred. Because of this, this function is said to have a space efficiency of O(1).

It's worth emphasizing that when using Big O to describe space complexity, we're only counting the *new data* the algorithm is generating. Even our second makeUppercase function deals with N elements of data in the form of the array passed into the function. However, we're not factoring those N elements into our Big O description, since the original array exists in any case, and we're only focused on the *extra* space the algorithm consumes. This extra space is more formally known as *auxiliary space.*

However, it's good to know that there are some references that include the original input when calculating the space complexity, and that's fine. We're not including it, and whenever you see space complexity described in another resource, you need to determine whether it's including the original input.

Let's now compare the two versions of makeUppercase() in both time and space complexity:

Version	Time Complexity	Space Complexity
Version #1	O(N)	O(N)
Version #2	O(N)	O(1)

Both versions are O(N) in time complexity, since they take N steps for N data elements. However, the second version is more memory-efficient, as it is O(1) in space complexity compared to the first version's O(N).

It turns out that Version #2 is more efficient than Version #1 in terms of space while not sacrificing any speed, which is a nice win.

Trade-Offs Between Time and Space

Here's a function that accepts an array and returns whether it contains any duplicate values (you may recognize this function from *Speeding Up Your Code with Big O*):

```
function hasDuplicateValue(array) {
    for(let i = 0; i < array.length; i++) {
        for(let j = 0; j < array.length; j++) {
            if(i !== j && array[i] === array[j]) {
                return true;
            }
        }
    }
    return false;
}
```

This algorithm uses nested loops and has a time complexity of $O(N^2)$. We'll call this implementation Version #1.

Here's a second implementation, Version #2, that employs a hash table and just a single loop:

```
function hasDuplicateValue(array) {
    let existingValues = {};
    for(let i = 0; i < array.length; i++) {
        if(!existingValues[array[i]]) {
            existingValues[array[i]] = true;
        } else {
            return true;
        }
    }
    return false;
}
```

Version #2 starts out with an empty hash table called existingValues. We then iterate over each item from the array, and as we encounter each new item, we store it as a key in the existingValues hash table. (We set the value arbitrarily to true.) If, however, we encounter an item that is already a key the hash table, we return true, as it means we found a duplicate value.

Now, which of these two algorithms is more efficient? Well, it all matters whether you consider time or space. As far as time is concerned, Version #2 is much more efficient, as it's only O(N), compared with Version #1's $O(N^2)$.

However, when it comes to *space*, Version #1 is actually more efficient than Version #2. Version #2 consumes up to O(N) space, as it creates a hash table that may end up containing all N values from the array passed to the function. Version #1, however, does not consume any additional memory beyond the original array, and therefore has a space complexity of O(1).

Let's look at the complete contrast between the two versions of hasDuplicateValue():

Version	Time Complexity	Space Complexity
Version #1	$O(N^2)$	O(1)
Version #2	O(N)	O(N)

We can see that Version #1 more efficient when it comes to memory, but Version #2 is faster in terms of raw speed. So, how do we decide which algorithm to choose?

The answer, of course, is that it depends on the situation. If we need our application to be blazing fast and we have enough memory to handle it, then Version #2 might be preferable. If, on the other hand, we're dealing with a hardware/data combination where we need to consume memory sparingly and speed isn't our biggest need, then Version #1 might be the right choice. Like all technology decisions, when there are trade-offs, we need to look at the big picture.

Let's look at a third version of this same function, and see where it falls compared to the first two versions:

```
function hasDuplicateValue(array) {
    array.sort((a, b) => (a < b) ? -1 : 1);

    for(let i = 0; i < array.length - 1; i++) {
        if (array[i] === array[i + 1]) {
            return true;
        }
    }
    return false;
}
```

This implementation, which we'll call Version #3, begins by sorting the array. It then iterates over each value within the array and checks to see whether it's the same as the next value. If it is, we've found a duplicate value. However, if we get to the end of the array and there are no two consecutive values that are the same, we know that the array contains no duplicates.

Let's analyze the time and space efficiency of Version #3.

In terms of time complexity, this algorithm is O(N log N). We can assume that JavaScript's sorting algorithm is one that takes O(N log N), as the fastest known sorting algorithms are known to do. The additional N steps of iterating over the array are trivial beside the sorting steps, so O(N log N) is the grand total when it comes to speed.

Space is a slightly more complex matter, as various sorting algorithms consume varying amounts of memory. Some of the earliest algorithms we've encountered in the book, like Bubble Sort and Selection Sort, consume no extra space because all the sorting happens in place. Interestingly, though, the faster sorts do take up some space for reasons you'll see shortly. Most implementations of Quicksort actually take up O(log N) space.

So, let's see where Version #3 lands in comparison with the previous two versions:

Version	Time Complexity	Space Complexity
Version #1	$O(N^2)$	O(1)
Version #2	O(N)	O(N)
Version #3	O(N log N)	O(log N)

It turns out Version #3 strikes an interesting balance between time and space. In terms of time, Version #3 is faster than Version #1, but slower than Version #2. When it comes to space, it's more efficient than Version #2, but less efficient than Version #1.

So, when may we want to use Version #3? Well, if we're concerned about both time *and* space, this might be our fix.

Ultimately, in each given situation, we need to know what our minimum acceptable speeds and bounds of memory are. Once we understand our constraints, we can then pick and choose from various algorithms so that we can eke out acceptable efficiency for our speed and memory needs.

Up until this point, you've seen how our algorithms can consume extra space when they create additional pieces of data, such as new arrays or hash tables.

However, it's possible for an algorithm to consume space even if it's not doing any of those things. And this can come to bite us if we're not expecting it.

The Hidden Cost of Recursion

We've dealt quite a bit with recursive algorithms in this book. Let's look at a simple recursive function:

```
function recurse(n) {
  if (n < 0) { return; }

  console.log(n);
  recurse(n - 1);
}
```

This function accepts a number `n` and counts down to 0, printing each decrementing number along the way.

It's a pretty straightforward bit of recursion, and seems pretty harmless. Its speed is O(N), as the function will run as many times as the argument `n`. And it doesn't create any new data structures, so it doesn't seem to take up any additional space.

Or does it?

I explained how recursion works under the hood back in *Recursively Recurse with Recursion*. You learned that each time a function recursively calls itself, an item is added to the call stack so that the computer can come back to the outer function once the inner function is complete.

If we pass the number 100 into our `recurse` function, it would add `recurse(100)` before proceeding to execute `recurse(99)`. And it would then add `recurse(99)` to the call stack before executing `recurse(98)`.

The call stack would be at its peak when `recurse(-1)` is called, and there would be 101 items in the call stack, from `recurse(100)` down to `recurse(0)`.

Now, even though the call stack will eventually get unwound, we do need enough memory to store these 100 items in our call stack in the first place. It emerges, then, that our recursive function *takes up O(N) space.* In this case, N is the number passed into the function; if we pass in the number 100, we need to temporarily store 100 function calls in the call stack.

An important principle emerges from all of this: *A recursive function takes up a unit of space for each recursive call it makes.* This is the sneaky way recursion can gobble up memory; even if our function does not explicitly create new data, the recursion itself adds data to the call stack.

To properly calculate how much space a recursive function takes, we always need to figure out how big the call stack would be at its peak.

For our `recurse` function, the call stack will be about as large as whatever number `n` is.

At first, this may seem a bit trivial. After all, our modern computers can handle a few items on the call stack, right? Well, let's see.

When I pass in the number 20,000 into the `recurse` function on my sleek, up-to-date laptop, *my computer cannot process it.* Now, 20,000 doesn't seem like a very large number. But this is what happens when I run `recurse(20000)`:

My computer prints the numbers from `20000` until `5387` and then terminates with the message:

```
RangeError: Maximum call stack size exceeded
```

Because the recursion lasted from `20000` until about `5000` (I'm rounding `5387` down), we can figure out that the call stack reached a size of about 15,000 when the computer ran out of memory. It turns out my computer will not tolerate a call stack that contains more than 15,000 items.

This is a *huge* limitation on recursion, as I can't use my beautiful `recurse` function on a number much greater than 15,000!

Let's contrast this to a simple loop approach:

```
function loop(n) {
  while (n >= 0) {
    console.log(n);
    n--;
  }
}
```

This function accomplishes the same goal using a basic loop instead of recursion.

Because this function does not use recursion, and does not take up any additional memory, it can run with huge number without *ever* causing the computer to run out of space. The function may take some time on huge numbers, but it'll get the job done without giving up prematurely as the recursive function did.

With this in mind, we can now understand why Quicksort is said to take up O(log N) space. Quicksort makes O(log N) recursive calls, so at its peak, has a call stack that is the size of log(N).

When we can implement a function using recursion, then, we need to weigh recursion's benefits against its drawbacks. Recursion allows us to use the "magical" top-down mindset, as you learned about in *Learning to Write in Recursive*, but we also need our function to get the job done. And if we're processing a lot of data, or even just a number like 20,000, recursion may not cut it.

Again, this isn't to discredit recursion. It just means we need to weigh all the pros and cons of each algorithm in every situation.

Wrapping Up

You've now learned how to measure the efficiency of our algorithms from all angles, including time *and* space. You are now armed with the analytical abilities to weigh each algorithm against the next and make informed decisions on which approach to use for our own applications.

As you are now able to make your own decisions, it's time to move on to the final chapter of our journey. In it, I am going to provide a few last pieces of advice on how to optimize your own code, while taking you through some realistic scenarios that we'll optimize together.

Exercises

The following exercises provide you with the opportunity to practice with space constraints. The solutions to these exercises are found in the section, Chapter 19, on page 464.

1. Following is the "Word Builder" algorithm we encountered in Word Builder, on page 97. Describe its *space* complexity in terms of Big O:

```
function wordBuilder(array) {
  let collection = [];

  for(let i = 0; i < array.length; i++) {
    for(let j = 0; j < array.length; j++) {
      if (i !== j) {
        collection.push(array[i] + array[j]);
      }
    }
  }

  return collection;
}
```

2. Following is a function that reverses an array. Describe its *space* complexity in terms of Big O:

```
function reverse(array) {
  let newArray = [];

  for (let i = array.length - 1; i >= 0; i--) {
    newArray.push(array[i]);
  }

  return newArray;
}
```

3. Create a new function to reverse an array that takes up just O(1) extra space.

4. Following are three different implementations of a function that accepts an array of numbers and returns an array containing those numbers multiplied by 2. For example, if the input is `[5, 4, 3, 2, 1]`, the output will be `[10, 8, 6, 4, 2]`.

```
function doubleArray1(array) {
  let newArray = [];

  for(let i = 0; i < array.length; i++) {
    newArray.push(array[i] * 2);
  }

  return newArray;
}

function doubleArray2(array) {
  for(let i = 0; i < array.length; i++) {
    array[i] *= 2;
  }

  return array;
}

function doubleArray3(array, index=0) {
  if (index >= array.length) { return; }

  array[index] *= 2;
  doubleArray3(array, index + 1);

  return array;
}
```

Fill in the table that follows to describe the efficiency of these three versions in terms of both time and space:

Version	Time Complexity	Space Complexity
Version #1	?	?
Version #2	?	?
Version #3	?	?

CHAPTER 20

Techniques for Code Optimization

We've come quite a long way. You now have the tools to analyze the time and space complexity of algorithms across a wide variety of data structures. With these concepts, you're now able to write code that is fast, memory-efficient, and beautiful.

In this final chapter of our journey, I'd like to leave you with some additional techniques for optimizing code. Sometimes it's difficult to see how an algorithm can be improved. Over the years, I've found the following mental strategies help me see how I can make my code more efficient. I hope they will be helpful to you as well.

Prerequisite: Determine Your Current Big O

Before we get into the optimization techniques, though, it's important to stress that there is something you need to do *before* you start optimizing your algorithm.

The prerequisite to optimization is *determining the efficiency of your current code.* After all, it's impossible to make an algorithm faster if you don't know how fast it is now.

By now, you have a thorough understanding of Big O Notation and the various categories of Big O. Once you know what category of Big O your algorithm belongs to, you can begin optimizing.

Throughout the rest of this chapter, I'll refer to this step of determining your current algorithm's Big O as the "prereq."

Start Here: The Best-Imaginable Big O

While all the techniques in this chapter are useful, you'll find that some come in handy for certain scenarios, while others are effective for other scenarios.

However, this first technique applies to *all* algorithms, and should be the first step of your optimization process.

And here it is.

Once you've determined the efficiency of your current algorithm (the prereq), come up with what you believe to be what I call the "best-imaginable Big O." (I've seen others refer to this as the "best-conceivable runtime" when applied to speed.)

Essentially, the best-imaginable Big O is the absolute best Big O you could dream of for the problem at hand. This is the Big O you know is absolutely impossible to beat.

For example, if we were to write a function that prints every item from an array, we'd probably say that the best-imaginable Big O for this task is O(N). Given that we have to print each of the N items in the array, *we have no choice* but to process each of the N items. There's just no way around this fact, as we need to "touch" each item in order to print it. Because of this, O(N) is the best-imaginable Big O we could imagine for this scenario.

When optimizing an algorithm, then, we need to determine *two* Big Os. We need to know the Big O our algorithm *currently* takes (the prereq), and we need to come up with the best-imaginable Big O the task could *possibly* take.

If these two Big Os are not the same, it means we have something to optimize. If, say, my current algorithm has a runtime of $O(N^2)$, but the best-imaginable Big O is O(N), we now have an improvement to strive for. The gap between the two Big Os shows us the potential gains we can make through optimization.

Let's summarize these steps:

1. Determine the Big O category of your current algorithm. (This is the prereq.)
2. Determine the best-imaginable Big O you could dream of for the problem at hand.
3. If the best-imaginable Big O is faster than your current Big O, you can now try to optimize your code, with the goal of bringing your algorithm as close to the best-imaginable Big O as possible.

It's important to stress that it's *not always possible to achieve the best-imaginable Big O.* After all, just because you can dream of something doesn't mean it can become reality.

In fact, it might turn out that your current implementation cannot be optimized further at all. However, the best-imaginable Big O is still a tool for giving us a goal to shoot for with our optimization.

Often, I find that I can successfully optimize an algorithm to a speed that is *in between* my current Big O and the best-imaginable Big O.

For example, if my current implementation is $O(N^2)$, and my best-imaginable Big O is O(log N), I will aim to optimize my algorithm to become O(log N). If, in the end, my optimizations speed up my code to "just" O(N), that's still a great success, and the best-imaginable Big O will have served a useful purpose.

Stretching the Imagination

As you've seen, the benefit of coming up with the best-imaginable Big O is that it gives us an optimization goal to shoot for. To really make the most of this, it's worth stretching the imagination a bit to come up with a best-imaginable Big O that is *amazing.* In fact, I recommend making your best-imaginable Big O the fastest Big O you can think of that isn't outright impossible.

Here's another mental trick I use for stoking my imagination. I pick a *really* fast Big O for the problem at hand—let's call it "Amazing Big O." I then ask myself, "If someone told me that they know how to pull off the Amazing Big O for this problem, would I believe them?" If I'd believe someone who said that they figured out how to solve this problem with the efficiency of Amazing Big O, I then make the Amazing Big O my best-imaginable Big O.

Once we know the Big O of our current algorithm and the best-imaginable Big O that we're aiming for, we are now primed for optimization.

In the remainder of this chapter, we are going to explore additional optimization techniques and mental strategies that can help us boost the efficiency of our code.

Magical Lookups

One of my favorite go-to optimization techniques is to ask myself, "If I could magically find a desired piece of information in O(1) time, can I make my algorithm faster?" If the answer to this is yes, I then use a data structure (often a hash table) to make that magic happen. I call this technique "magical lookups."

Let me clarify this technique with an example.

Magically Looking Up Authors

Let's say we're writing library software and we have data about books and their authors contained in two separate arrays.

Specifically, the array of authors looks like this:

```
authors = [
  {"author_id" => 1, "name" => "Virginia Woolf"},
  {"author_id" => 2, "name" => "Leo Tolstoy"},
  {"author_id" => 3, "name" => "Dr. Seuss"},
  {"author_id" => 4, "name" => "J. K. Rowling"},
  {"author_id" => 5, "name" => "Mark Twain"}
]
```

As you can see, it's an array of hash tables, with each hash table containing an author's name and ID.

We also have a separate array containing data about books:

```
books = [
  {"author_id" => 3, "title" => "Hop on Pop"},
  {"author_id" => 1, "title" => "Mrs. Dalloway"},
  {"author_id" => 4, "title" => "Harry Potter and the Sorcerer's Stone"},
  {"author_id" => 1, "title" => "To the Lighthouse"},
  {"author_id" => 2, "title" => "Anna Karenina"},
  {"author_id" => 5, "title" => "The Adventures of Tom Sawyer"},
  {"author_id" => 3, "title" => "The Cat in the Hat"},
  {"author_id" => 2, "title" => "War and Peace"},
  {"author_id" => 3, "title" => "Green Eggs and Ham"},
  {"author_id" => 5, "title" => "The Adventures of Huckleberry Finn"}
]
```

Like the authors array, the books array contains a number of hash tables. Each hash table contains a book's title and the author_id, which can allow us to determine the book's author using the data from the authors array. "Hop on Pop," for example, has the author_id of 3. This means the author of "Hop on Pop" is Dr. Seuss, since he is the author whose ID is 3, as indicated in the authors array.

Now, let's say we want to write code that combined this information together to create an array in the following format:

```
books_with_authors = [
  {"title" => "Hop on Pop", "author" => "Dr. Seuss"}
  {"title" => "Mrs. Dalloway", "author" => "Virginia Woolf"}
  {"title" => "Harry Potter and the Sorcerer's Stone",
    "author" => "J. K. Rowling"}
  {"title" => "To the Lighthouse", "author" => "Virginia Woolf"}
  {"title" => "Anna Karenina", "author" => "Leo Tolstoy"}
```

```
  {"title" => "The Adventures of Tom Sawyer", "author" => "Mark Twain"}
  {"title" => "The Cat in the Hat", "author" => "Dr. Seuss"}
  {"title" => "War and Peace", "author" => "Leo Tolstoy"}
  {"title" => "Green Eggs and Ham", "author" => "Dr. Seuss"}
  {"title" => "The Adventures of Huckleberry Finn", "author" => "Mark Twain"}
]
```

To do this, we'd probably need to iterate through the array of `books` and connect each book to its respective author. How would we go about this specifically?

One solution may be to use nested loops. The outer loop would iterate over each book, and for each book, we'd run an inner loop that would check each author until it found the one with the connecting ID. Here's a Ruby implementation of this approach:

```
def connect_books_with_authors(books, authors)
  books_with_authors = []

  books.each do |book|
    authors.each do |author|
      if book["author_id"] == author["author_id"]
        books_with_authors <<
          {title: book["title"],
          author: author["name"]}
      end
    end
  end

  return books_with_authors
end
```

Before we can optimize our code, we need to fulfill our prereq and determine our current algorithm's Big O.

This algorithm has a time complexity of O(N * M), since for each of the N books, we need to loop through M authors to find the book's author.

Now, let's see if we can do better.

Again, the first thing we need to do is come up with the best-imaginable Big O. In this case, we definitely need to iterate over all N books, so it would seem impossible to beat O(N). Since O(N) is the fastest speed I can think of that isn't downright impossible, we'll say that O(N) is our best-imaginable Big O.

We're now ready to use the new "magical lookups" technique. To do this, I'll ask myself the question mentioned at the start of this section: "If I could magically find a desired piece of information in O(1) time, can I make my algorithm faster?"

Let's apply this to our scenario. We currently run an outer loop that iterates over all the books. Currently, for each book, we run an inner loop that tries to find the book's author_id in the authors array.

But what if we had the magical ability to find an author *in just O(1) time?* That is, what if we didn't have to loop through *all* the authors each time we wanted to look one up, and we could instead find the author immediately? That would bring a huge speed boost to our algorithm, as we could potentially eliminate our inner loop and bring our code's speed up to the vaunted O(N).

Now that we've determined that this magical finding ability could help us, the next step is to try to make this magic come alive.

Bringing in the Extra Data Structure

One of the easiest ways we can achieve this magical lookup ability is to bring an additional data structure into our code. We'll use this data structure to specifically store data in such a way that allows us to look that data up quickly. In many cases, the hash table is the perfect data structure for this, since it has O(1) lookups, as you learned in *Blazing Fast Lookup with Hash Tables*.

Right now, because the author hash tables are stored in an array, it will always take us O(M) steps (M being the number of authors) to find any given author_id within that array. But if we store that same information in a hash table, we'd now gain our "magical" ability to find each author in just O(1) time.

Here's one possibility of what this hash table could look like:

```
author_hash_table =
{1 => "Virginia Woolf", 2 => "Leo Tolstoy", 3 => "Dr. Seuss",
4 => "J. K. Rowling", 5 => "Mark Twain"}
```

In this hash table, each key is the author's ID, and the value of each key is the author's name.

So, let's optimize our algorithm by first moving the authors data into this hash table, and only then run our loop through the books:

```
def connect_books_with_authors(books, authors)
  books_with_authors = []
  author_hash_table = {}

  # Convert author data into author hash table:
  authors.each do |author|
    author_hash_table[author["author_id"]] = author["name"]
  end
```

```
  books.each do |book|
    books_with_authors <<
      {"title" => book["title"],
        "author" => author_hash_table[book["author_id"]]}
  end

  return books_with_authors
end
```

In this version, we first iterate through the authors array and use that data to create the author_hash_table. This takes M steps, with M being the number of authors.

We then iterate through the list of books and use the author_hash_table to "magically" find each author in a single step. This loop takes N steps, with N being the number of books.

This optimized algorithm takes a grand total of O(N + M) steps, since we run a single loop through the N books, and a single loop through the M authors. This is drastically faster than our original algorithm, which took O(N * M).

It's worth noting that by creating the extra hash table, we're using up an additional O(M) space, whereas our initial algorithm didn't take up any extra space at all. However, this is great optimization if we're willing to sacrifice the memory for the sake of speed.

We made this magic happen by first dreaming what magical O(1) lookups could do for us, and we then granted ourselves our own wish by using a hash table to store our data in an easy-to-find way.

The fact that we can look up hash table data in O(1) time isn't new, as we looked at this back in *Blazing Fast Lookup with Hash Tables*. The specific tip I'm sharing here, though, is to constantly *imagine* that you can perform O(1) lookups on any kind of data and notice whether that would speed up your code. Once you have the vision of how O(1) lookups would help you, you can then try to use a hash table or other data structure to turn that dream into reality.

The Two-Sum Problem

Let's look at another scenario in which we can benefit from magical lookups. This is one my favorite optimization examples.

The Two-Sum Problem is a well-known coding exercise. The task is to write a function that accepts an array of numbers and return true or false depending on whether there are any two numbers in the array that add up to 10 (or another given number). For simplicity, let's assume there will never be duplicate numbers in the array.

Let's say our array is:

```
[2, 0, 4, 1, 7, 9]
```

Our function would return `true`, since the `1` and `9` add up to 10.

If the array was:

```
[2, 0, 4, 5, 3, 9]
```

we'd return `false`. Even though the three numbers `2`, `5`, and `3` add up to 10, we specifically need *two* numbers to add up to 10.

The first solution that comes to mind is to use nested loops to compare each number to every other number and see if they add up to 10. Here is a JavaScript implementation:

```
function twoSum(array) {
  for(let i = 0; i < array.length; i++) {
    for(let j = 0; j < array.length; j++) {
      if(i !== j && array[i] + array[j] === 10) {
        return true;
      }
    }
  }
  return false;
}
```

As always, before attempting an optimization, we need to satisfy our prereq and figure out the current Big O of our code.

As is typical in a nested-loop algorithm, this function has a runtime of $O(N^2)$.

Next, to see if our algorithm is worth optimizing, we need to see if the best-imaginable Big O would be any better.

In this case, it would seem that we absolutely have to visit each number in the array at least once. So, we couldn't beat O(N). And if someone told me that there's an O(N) solution to this problem, I suppose I'd believe them. So, let's make O(N) our best-imaginable Big O.

Now, let's ask ourselves the magical-lookup question: "If I could magically find a desired piece of information in O(1) time, can I make my algorithm faster?"

Sometimes it helps to begin walking through our current implementation while asking this question along the way, so let's do that.

Let's mentally walk through our outer loop with the example array of `[2, 0, 4, 1, 7, 9]`. This loop begins with the first number, which is the number 2.

Now, what piece of information might we desire to look up while we're looking at the 2? Again, we want to know if this 2 could be added to another number in the array to provide a sum of 10.

Thinking about it further, while looking at the 2, I'd want to know whether *there's an 8 somewhere* in this array. If we could, magically, do an O(1) lookup and know that there's an 8 in the array, we could immediately return true. Let's call the 8 the 2's *counterpart,* since the two numbers add up to 10.

Similarly, when we move on to the 0, we'd want to do an O(1) lookup to find its counterpart—a 10—in the array, and so on.

With this approach, we can iterate through the array just once and do magical O(1) lookups along the way to see whether each number's counterpart exists in the array. As soon as we find any number's counterpart, we return true, but if we get to the end of the array without finding any numerical counterparts, we return false.

Now that we've determined we'd benefit from these magical O(1) lookups, let's try to pull off our magic trick by bringing in an extra data structure. Again, the hash table is usually the default option for magical lookups because of its O(1) reads. (It's uncanny how often hash tables can be used to speed up algorithms.)

Since we want to be able to look up any number from the array in O(1) time, we'll store those numbers as keys in a hash table. The hash table may look like this:

```
{2: true, 0: true, 4: true, 1: true, 7: true, 9: true}
```

We can use any arbitrary item to serve as the values; let's decide to use true.

Now that we can look up any number in O(1) time, how do we look up a number's counterpart? Well, we noticed that when we iterated over the 2, we knew that the counterpart should be 8. We knew this because we know intuitively that 2 + 8 = 10.

Essentially, then, we can calculate any number's counterpart by subtracting it from 10. Because 10 - 2 = 8, that means 8 is the 2's counterpart.

We now have all the ingredients to create a really fast algorithm:

```
function twoSum(array) {
  let hashTable = {};
```

```
  for(let i = 0; i < array.length; i++) {
    // Check if the hash table contains a key which, when added
    // to the current number, would add up to 10:
    if(hashTable[10 - array[i]]) {
      return true;
    }

    // Store each number as a key in the hash table:
    hashTable[array[i]] = true;
  }

  // Return false if we get to the end of the array without
  // finding any number's counterpart:
  return false;
}
```

This algorithm iterates once through each number in the array.

As we visit each number, we check whether the hash table contains a key that is the counterpart of the current number. We calculate this is as `10 - array[i]`. (For example, if `array[i]` is 3, the counterpart would be 7, since 10 - 3 = 7.)

If we find any number's counterpart, we immediately return `true`, as that means we've found two numbers that add up to 10.

Additionally, as we iterate over each number, we insert the number as a key into the hash table. This is how we populate the hash table with the numbers as we proceed through the array.

With this approach, we drastically increased the algorithm's speed to O(N). We pulled this off by storing all of the data elements in a hash table for the express purpose of being able to perform O(1) lookups throughout the loop.

Consider the hash table your magical wand, and become the programming wizard you were destined to be. (Okay, enough of that.)

Recognizing Patterns

One of the most helpful strategies for both code optimization and algorithm development in general is to find patterns within the problem at hand. Often, the discovery of a pattern can help you cut through all the complexity of a problem and develop an algorithm that is actually quite simple.

The Coin Game

Here's a great example. There's a game I call "the coin game" in which two players compete in the following way. They start with a pile of coins, and each player has the choice of removing either one or two coins from the pile. The player who removes the last coin *loses.* Fun, right?

It turns out that this isn't a game of random chance, and with the right strategy, you can *force* your opponent to take the last coin and lose the game. To make this clear, let's start with some really small coin piles and see how the game plays out.

If there's just one coin in the pile, the player whose turn it is loses, since they have no choice but to take the last coin.

If there are two coins left, the player whose turn it is can force a win. This is because they can take just one coin, and thereby force their opponent to take the final coin.

When there are three coins remaining, the player whose turn it is can also force a win, since they can remove two coins, forcing their opponent to take the final coin.

Now, when there are four coins left, the current player is in trouble. If they remove one coin, the opponent is given a pile of three coins, which we established earlier can allow that player to force a win. Similarly, if the current player removes two coins, the opponent is left with two coins, which can also allow the opponent to force a win.

If we were to write a function that calculated whether you can win the game when presented with a coin pile of a given size, what approach should we take? If we think about this carefully, we may realize that we can use subproblems to help calculate an accurate result for any number of coins. This would make top-down recursion a natural fit for solving this problem.

Here's a Ruby implementation of a recursive approach:

```
def game_winner(number_of_coins, current_player="you")
  if number_of_coins <= 0
    return current_player
  end

  if current_player == "you"
    next_player = "them"
  elsif current_player == "them"
    next_player = "you"
  end

  if game_winner(number_of_coins - 1, next_player) == current_player ||
    game_winner(number_of_coins - 2, next_player) == current_player
    return current_player
  else
    return next_player
  end
end
```

This game_winner function is given a number of coins and the player whose turn it is (either "you" or "them"). The function then returns either "you" or "them" as the winner of the game. When the function is first called, the current_player is "you".

We define our base case as when the current_player is dealt 0 or fewer coins. This means the other player took the last coin and the current player, by default, won the game.

We then define a next_player variable, which keeps track of which player will go next.

Then, we do our recursion. We recursively call our game_winner function on piles of coins that are both one and two coins smaller than the current pile, and see if the next player would win or lose in those scenarios. If the next_player loses in both scenarios, that means the current_player will win.

This isn't an easy algorithm, but we pulled it off. Now, let's see if we can optimize it.

To satisfy our prereq, we first need to figure out our algorithm's current speed.

You may have noticed that this function makes multiple recursive calls. If alarm bells are going off in your head, that's for good reason. The time complexity of this function is a whopping $O(2^N)$, which can be unbearably slow.

Now, we can improve this by using the memoization technique you learned about in *Dynamic Programming*, which could bring the speed up to O(N), with N being the number of coins in the starting pile. That's a huge improvement.

But let's see if we can push our algorithm's speed even further.

To determine whether we can optimize our algorithm further, we need to ask ourselves what we think the best-imaginable Big O is.

Because N is just a single number, I could conceive that we can make an algorithm that takes just O(1) time. Since we don't actually have to touch N items in an array or anything like that, if someone told me they figured out an algorithm for the coin game that was just O(1), I'd believe them. So, let's strive for O(1).

But how do we get there? This is where finding a pattern can help.

Generating Examples

While each problem has a unique pattern, I found a technique for finding patterns that helps across *all* problems. And that is to *generate numerous*

examples. This means we should take a whole bunch of example inputs, calculate their respective outputs, and see if we can detect a pattern.

Let's apply this to our case.

If we map out who wins for coin piles of size 1 through 10, we get this table:

Number of Coins	Winner
1	Them
2	You
3	You
4	Them
5	You
6	You
7	Them
8	You
9	You
10	Them

The pattern becomes clear when we lay it out this way. Basically, starting with 1 coin, every third number gives victory to the opponent. Otherwise, you are the winner.

So, if we take the number of coins and subtract 1, each `"them"` ends up at a number that is divisible by 3. At this point, then, we can determine who will win based on a single division calculation:

```
def game_winner(number_of_coins)
  if (number_of_coins - 1) % 3 == 0
    return "them"
  else
    return "you"
  end
end
```

This code is saying that if after subtracting 1, the `number_of_coins` is divisible by 3, the winner is `"them"`. Otherwise, `"you"` are the winner.

Because this algorithm consists of a single mathematical operation, it's O(1) in both time and space. It's also a lot simpler! This is a real win-win-win.

By generating many examples of coin piles (as inputs) and seeing who'd win the game (as outputs), we were able to identify a pattern in how the coin game works. We were then able to use this pattern to cut to the heart of the problem and turn a slow algorithm into an instantaneous one.

The Sum Swap Problem

Here's an example where we can use both pattern recognition and magical lookups *together* to optimize an algorithm.

The next problem, known as the "sum swap" problem, goes like this:

We want to write a function that accepts two arrays of integers. As an example, let's say our arrays are:

```
array_1 = [5, 3, 2, 9, 1]     Sum: 20
array_2 = [1, 12, 5]          Sum: 18
```

Currently, the numbers in `array_1` add up to 20, while the numbers in `array_2` add up to 18.

Our function needs to find one number from each array that can be swapped to cause the two array sums to be equal.

In this example, if we swapped the 2 from `array_1` and the 1 from `array_2`, we'd get:

```
array_1 = [5, 3, 1, 9, 1]     Sum: 19
array_2 = [2, 12, 5]          Sum: 19
```

And both arrays would now have the same sum, namely 19.

To keep things simple, our function won't actually perform the swap, but will return the two indexes that we'd have to swap. We can do this as an array containing the two indexes. So, in this case, we swapped index `2` of `array_1` with index `0` of `array_2`, so we'll return an array of `[2, 0]`. In a case where there is no possible swap that makes the two arrays equal, we'll return `nil`.

One way we can write this algorithm is to use nested loops. That is, as our outer loop points to each number from `array_1`, an inner loop can iterate over each number from `array_2` and test the sums of each array if we were to swap the two numbers.

To begin optimizing this, we must first satisfy our prereq of knowing the Big O of our current algorithm.

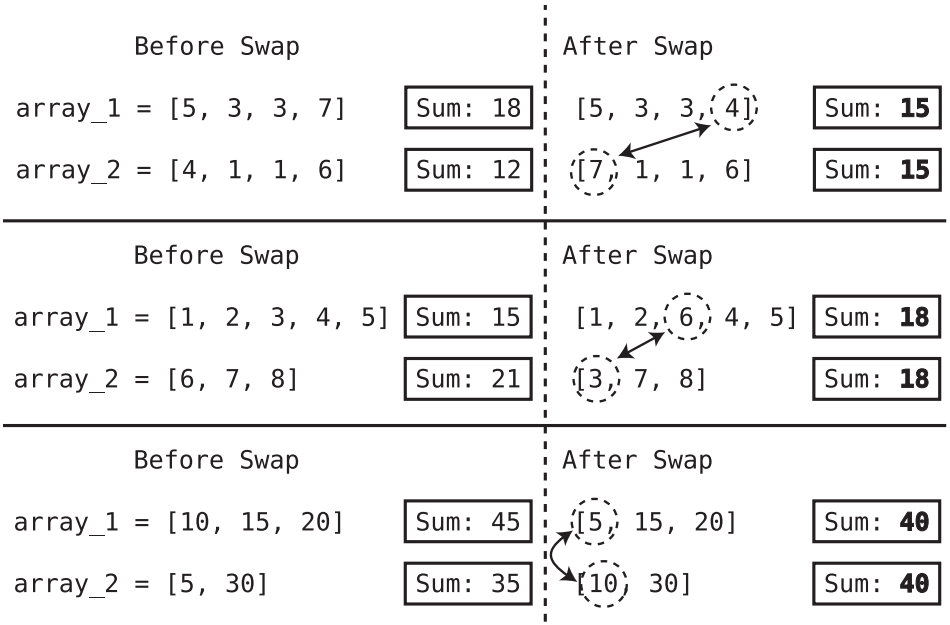
Because our nested-loops approach visits M numbers from the second array for each of the N numbers of the first array, this algorithm is O(N * M). (I'm discussing N and M because the arrays may be two different sizes.)

Can we do better? To find out, let's determine what we think the best-imaginable Big O may be.

It would seem that we absolutely have to visit each number from the two arrays at least once, since we need to be aware of what all the numbers are. But it's possible that this may be *all* we need to do. If so, this would be O(N + M). Let's make this our best-imaginable Big O and aim for that.

Next, we need to try to dig up any patterns hidden within the problem. Again, the best technique to dig up patterns is to come up with numerous examples and look for patterns among them.

So, let's look at a number of different examples where swapping numbers will cause the two arrays to have equal sums:



In looking at these examples, a few patterns begin to emerge. Some of these patterns may seem obvious, but let's look at them anyway.

One pattern is that to achieve equal sums, the larger array needs to trade a larger number with a smaller number from the smaller array.

A second pattern is that with a single swap, each array's sum changes by the same amount. For example, when we swap a 7 with a 4, one array's sum *decreases* by 3, while the other array's sum *increases* by 3.

A third interesting pattern is that the swaps always cause the two array sums to fall out *exactly in the middle* of where the two array sums began.

In the first case, for example, array_1 was 18 and array_2 was 12. When making a correct swap, the two arrays land at 15, which is exactly in the middle between 18 and 12.

When we think about it further, this third pattern is a logical outgrowth of the other patterns. Since a swap causes the two arrays to shift their sums by the same amount, the *only* way to make their sums equal is to meet in the middle.

Based on this, if we know the sums of the two arrays, we should be able to look at any number in one of the arrays and calculate what number it should be swapped with.

Let's take this example again:

```
array_1 = [5, 3, 3, 7]   Sum: 18

array_2 = [4, 1, 1, 6]   Sum: 12
```

We know that for a swap to work successfully, we'll need the two arrays' sums to land in the middle. The exact middle between 18 and 12 is 15.

Let's look at different numbers from array_1 and figure out what number we'd want to swap it with. We can call this other number its "counterpart." Let's start with the first number from array_1, which is the number 5.

What number would we want to swap the 5 with? Well, we know that we want array_1 to decrease by 3, and array_2 to increase by 3, so we'd need to swap the 5 with a number 2. It just so happens that array_2 doesn't contain a 2, so the 5 cannot be successfully swapped with any number from array_2.

If we look at the next number from array_1, it's a 3. We'd have to swap this with a 0 from array_2 to get the two sums to be equal. Alas, a 0 doesn't exist in array_2.

The last number from array_1, though, is a 7. We can calculate that we'd want to swap the 7 with a 4 to make the sums both land at 15. Luckily, array_2 does contain a 4, so we can make a successful swap.

So, how do we express this pattern in code?

Well, we can first determine how much an array sum needs to shift using this calculation:

```
shift_amount = (sum_1 - sum_2) / 2
```

Here, sum_1 is the sum of array_1, and sum_2 is the sum of array_2. If sum_1 is 18 and sum_2 is 12, we end up with a difference of 6. We then divide that by 2 to determine how much each array needs to shift. This is the shift_amount.

In this case, the shift_amount is 3, indicating that array_2 needs to increase by 3 in order to hit the target sum. (Likewise, array_1 needs to *decrease* by 3.)

So, we can start building our algorithm by first calculating the sums of the two arrays. We can then loop through all the numbers in one of the arrays and look for the counterpart in the other.

If we were to iterate over each number in array_2, for example, we know that the current number would have to be swapped with its counterpart, which would be the current number plus the shift_amount. For example, if the current number is 4, to find its counterpart, we add the shift_amount to it (3) and get 7. This means we need to find a 7 in array_1 to swap with our current number.

So, we've figured out that we can look at any number in either array and know exactly what its counterpart from the other array should be. But how does this help? Don't we still need to use nested loops and have an algorithm that is O(N * M)? That is, for each number in one array, we have to search the entire other array for the counterpart.

This is where we can invoke magical lookups and ask ourselves, "If I could magically find a desired piece of information in O(1) time, can I make my algorithm faster?"

Indeed, if we could find a number's counterpart from the other array in just O(1) time, our algorithm would be much faster. And we can achieve those quick lookups by following the usual technique of bringing in our good ol' hash table.

If we first store the numbers from one array in a hash table, we can then immediately find any number from it in O(1) time as we iterate through the other array.

Here's the complete code:

```
def sum_swap(array_1, array_2)
  # Hash table to store values of first array:
  hash_table = {}
  sum_1 = 0
  sum_2 = 0

  # Get sum of first array, while storing its values
  # in a hash table, together with an index
  array_1.each_with_index do |num, index|
    sum_1 += num
    hash_table[num] = index
  end
```

```
  # Get sum of second array:
  array_2.each do |num|
    sum_2 += num
  end

  # Calculate how much a number in the second array needs
  # to shift by:
  shift_amount = (sum_1 - sum_2) / 2

  # Iterate over each number in second array:
  array_2.each_with_index do |num, index|

    # Check hash table for the number's counterpart in the
    # first array, which is calculated as the current number
    # plus the amount it has to shift by:
    if hash_table[num + shift_amount]
      return [hash_table[num + shift_amount], index]
    end
  end

  return nil
end
```

This approach is much faster than our original O(N * M) one. If we consider `array_1` to be N and `array_2` to be M, we could say that this algorithm runs in O(N + M) time. While we do iterate over `array_2` twice and is technically 2M, it becomes M since we drop the constants.

This approach takes up an extra O(N) space, since we copy all N numbers from `array_1` into the hash table. Again, we're sacrificing space to gain time, but this is a big win if speed is our primary concern.

In any case, this is another example of where discovering patterns allows us to cut to the heart of the problem and develop a simple and fast solution.

Greedy Algorithms

This next tactic can speed up some of the most stubborn algorithms. It doesn't work in every situation, but when it does, it can be a game changer.

Let's talk about writing greedy algorithms.

This may sound like a strange term, but here's what it means. A *greedy algorithm* is one that, in each step, chooses what appears to be the best option *at that moment in time.* This will make sense with a basic example.

Array Max

Let's write an algorithm that finds the greatest number in an array. One way we can do this is to use nested loops and check each number against every

other number in the array. When we find the number that is greater than every other number, it means we've found the greatest number in the array.

As is typical for such algorithms, this approach takes $O(N^2)$ time.

Another approach would be to sort the array in ascending order and return the final value from the array. If we use a fast sorting algorithm like Quicksort, this would take O(N log N) time.

There's a third option, which is the greedy algorithm. Here it is:

```
def max(array)
  greatest_number = array[0]

  array.each do |number|
    if number > greatest_number
      greatest_number = number
    end
  end

  return greatest_number
end
```

As you can see, the first line of our function assumes the first number in the array is the greatest_number. Now, this is a "greedy" assumption. That is, we're declaring the first number to be the greatest_number because it's the greatest number we've encountered so far. Of course, it's also the *only* number we've encountered so far! But that's what a greedy algorithm does—it chooses what appears to be the best option based on the information available at that moment in time.

Next, we iterate over all the numbers in the array. As we find any number that is greater than the greatest_number, we make this new number the greatest_number. Here too, we're being greedy; each step selects the best option based on what we know at that moment in time.

We're basically like a child in a candy shop grabbing the first candy we see, but as soon as we see a bigger candy, we drop the first one and grab the bigger one.

Yet, this seemingly naive greediness actually works. By the time we're done with the function, our greatest_number will indeed be the greatest number in the entire array.

And while being greedy isn't a virtue in a societal context, it can do wonders for algorithm speed. This algorithm takes just O(N) time, as we touch each number in the array just once.

Largest Subsection Sum

Let's see another example of how greed pays off.

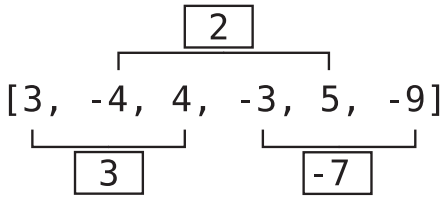
We're going to write a function that accepts an array of numbers and returns the largest sum that could be computed from any "subsection" of the array.

Here's what I mean. Let's take the following array:

```
[3, -4, 4, -3, 5, -9]
```

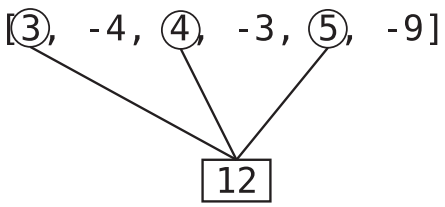
If we computed the sum of all the numbers in this array, we'd get -4.

But we can also compute the sum of *subsections* of the array:

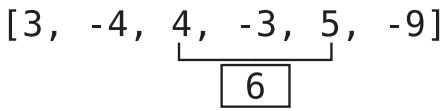


When I refer to subsections, I mean *contiguous subsections.* That is, a subsection is a section of the array that contains a series of numbers *in a row.*

The following is *not* a contiguous subsection, since the numbers are not in a row:



Our job is to find the largest sum that can be computed from *any* subsection within the array. In our example, the largest sum is 6, derived from the following subsection:



To make the discussion simpler, let's assume the array contains at least one positive number.

Now, how can we write the code to calculate the largest subsection sum?

One approach would be to calculate the sum of every subsection within the array and pick out the greatest one. However, there are about $N^2 / 2$ subsections for N items in an array, so the mere generation of the different subsections would take $O(N^2)$ time.

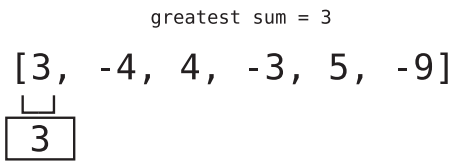
Again, let's start by dreaming up the best-imaginable Big O. We definitely need to inspect each number at least once, so we can't beat O(N). So, let's make O(N) our goal.

At first glance, O(N) seems beyond our reach. How can we add up multiple subsections by iterating over the array a single time?

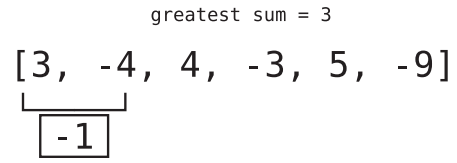
Let's see what happens if we get a little greedy…

A greedy algorithm in this context would attempt to "grab" the greatest sum at each step as we iterate over the array. Here's what this might look like as we iterate over the earlier example array.

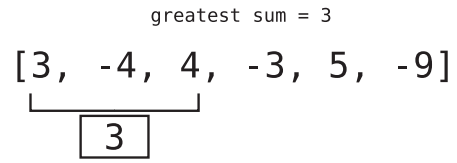
Starting at the front of the array, we encounter a 3. In perfect greedy fashion, we'll say that our greatest sum is 3:



Next, we reach the -4. When we add this to the previous number of 3, we get a current sum -1. So, 3 is still our greatest sum:

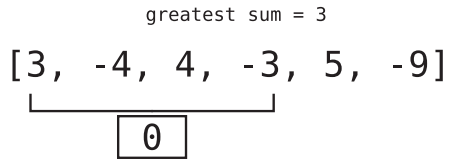


We then hit the 4. If we add this to our current sum, we get 3:



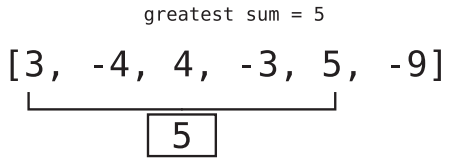
As of now, 3 is still the greatest sum.

The next number we reach is a -3. This puts our current sum at 0:

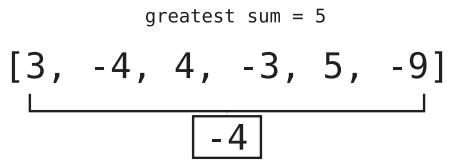


Again, while 0 is our *current* sum, 3 is still our *greatest* sum.

Next, we reach the 5. This makes our current sum 5. In our greed, we'll declare this to be the greatest sum, as it's the greatest sum we've encountered so far:

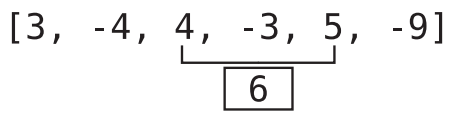


We then reach the last number, which is -9. This deflates our current number to -4:



By the time we get to the end of the array, our greatest sum is 5. So, if we follow this pure-greed approach, it would appear that our algorithm should return 5.

However, 5 is actually *not* the greatest subsection sum. There's a subsection in the array that yields a sum of 6:

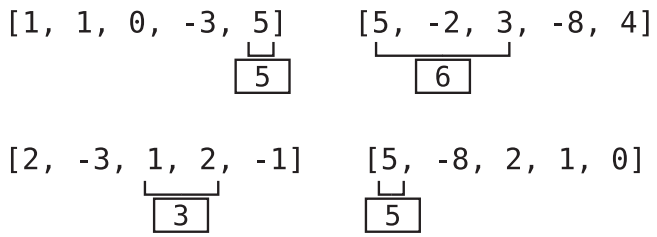


The problem with our algorithm is that we only calculated the largest sum based on subsections that always begin with the first number in the array. But there are other subsections that begin with numbers later on in the array as well. And we haven't accounted for those.

Our greedy algorithm, then, didn't pan out as we'd hoped.

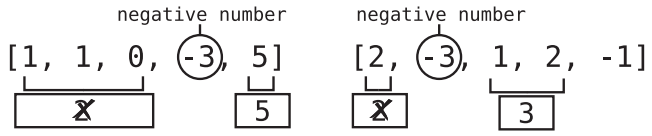
But we shouldn't give up yet! Often, we need to tweak greedy algorithms a bit to get them to work.

Let's see if finding a pattern may help. (It usually does.) As we've seen before, the best way to find a pattern is to generate lots of examples. So, let's come up with some examples of arrays with their largest subsection sums and see if we discover anything interesting:



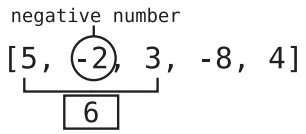
When analyzing these cases, an interesting question emerges: why is it that in some cases, the greatest sum comes from a subsection that starts at the beginning of the array, and in other cases it doesn't?

In looking at these cases, we can see that when the greatest subsection *doesn't* start at the beginning, it's because a negative number broke the streak:



That is, the greatest subsection *would've* been derived from the beginning of the array, but a negative number killed the streak, and the greatest subsection has to start later on in the array.

But wait a second. In some cases, the greatest subsection *includes* a negative number, and the negative number didn't break the streak:



So, what's the difference?

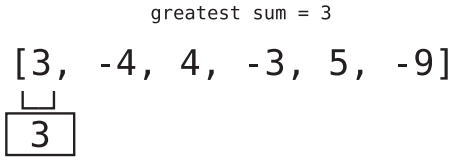
The pattern is: if the negative number causes the preceding subsection's sum to sink to a negative number, the streak is broken. But if the negative number simply lowers the current subsection's sum, and the sum remains a positive number, the streak isn't broken.

If we think about it, this makes sense. If as we're iterating through the array, our current subsection's sum becomes less than 0, *we're best off just resetting*

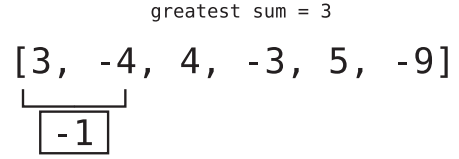
the current sum to 0. Otherwise, the current negative sum will just detract from the greatest sum we're trying to find.

So, let's use this insight to tweak our greedy algorithm.

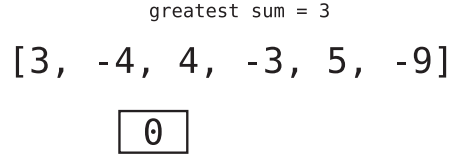
Again, let's start with the 3. The greatest sum is currently 3:



Next, we encounter the -4. This would make our current sum -1:



Since we're trying to find the subsection with the greatest sum, and our current sum is a negative number, we need to reset the current sum to 0 before continuing on to the next number:

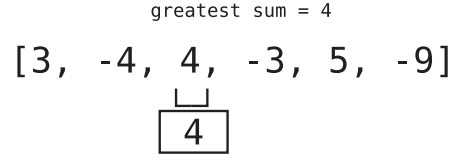


We'll also start a brand-new subsection beginning with the next number.

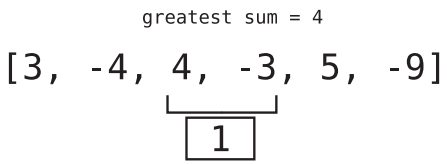
Again, the reasoning is that if the next number is positive, we may as well just start the next subsection from there, without letting the current negative number drag down the sum. Instead, we're going to perform a reset by setting the current sum to 0 and considering the next number to be the *beginning* of a new subsection.

So, let's continue.

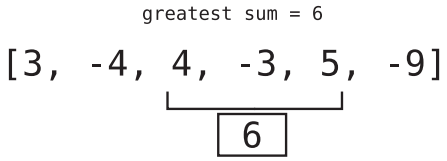
We now reach a 4. Again, this is the beginning of a new subsection, so the current sum is 4, which also becomes the greatest sum we've seen yet:



Next, we encounter the -3. The current sum is now 1:



We next come upon a 5. This makes the current sum 6, which is the greatest sum as well:



Finally, we reach the -9. This would make the current sum -4, in which case we'd reset it to 0. However, we've also reached the end of the array, and we can conclude the greatest sum is 6. And, indeed, that is the correct result.

Here is the code for this approach:

```
def max_sum(array)
  current_sum = 0
  greatest_sum = 0

  array.each do |num|
    # If current sum is a negative number, reset current sum to zero:
    if current_sum + num < 0
      current_sum = 0
    else
      current_sum += num

      # Greedily assume the current sum is the greatest sum
      # if it's the greatest sum we've encountered so far:
      greatest_sum = current_sum if current_sum > greatest_sum
    end
  end

  return greatest_sum
end
```

Using this greedy algorithm, we were able to solve this thorny problem in just O(N) time, as we loop through the array of numbers just once. That's a great improvement over our initial $O(N^2)$ approach. In terms of space, this algorithm is O(1), as we don't generate any extra data.

While the discovery of a pattern helped us discover the precise solution, by adopting the greedy mindset, we knew what kind of pattern we were looking for in the first place.

Greedy Stock Predictions

Let's look at one more greedy algorithm.

Say we're writing financial software that makes stock predictions. The particular algorithm we're working on now looks for a positive trend for a given stock.

Specifically, we're writing a function that accepts an array of stock prices and determines whether there are any *three* prices that create an upward trend.

For example, take this array of stock prices that represents the price of a given stock over time:

```
[22, 25, 21, 18, 19.6, 17, 16, 20.5]
```

Although it may be difficult to spot at first, there are three prices that form an upward trend:



That is, as we go from left to right, there are three prices where a "right hand" price is greater than a "middle" price, which in turn is greater than a "left hand" price.

The following array, on the other hand, does not contain a three-point upward trend:

```
[50, 51.25, 48.4, 49, 47.2, 48, 46.9]
```

Our function should return `true` if the array contains an upward trend of three prices, and `false` if it does not.

So, how do we go about this?

One way we can do this is with three nested loops. That is, as one loop iterates over each stock price, a second loop iterates over all the stock prices that follow. And for each round of the second loop, a third nested loop checks all the prices that follow the second price. As we point to each set of three stock prices, we check whether they are in ascending order. As soon as we find such a set, we return `true`. But if we complete the loops without finding any such trend, we return `false`.

The time complexity of this algorithm is $O(N^3)$. That's pretty slow! Is there any way we can optimize this?

Let's first think about the best-imaginable Big O. We definitely need to inspect each stock price to find a trend, so we know that our algorithm cannot be faster than O(N). Let's see if we can optimize for such a speed.

Once again, it's time to get greedy.

To apply the greedy mentality to our case, we'd want to somehow keep "grabbing" what we think is the lowest point of our three-price upward trend. It would also be cool if we can use the same greedy approach to constantly "grab" what we think are the middle and highest points of that trend.

Here's what we'll do:

We'll assume the first price from the array is the lowest point in the three-price upward trend.

As far as the middle price, we'll initialize it to a number that is guaranteed to be greater than even the highest stock price in the array. To do this, we'll set it to infinity. (Many programming languages support the concept of infinity.) This particular step might be the least intuitive at first glance, but you'll see shortly why we need to do this.

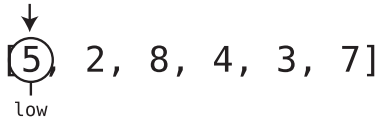
We'll then make a single pass through the entire array, according to the following steps:

1. If the current price is lower than the lowest price we've encountered so far, this price becomes the new lowest price.
2. If the current price is higher than the lowest price, but lower than the middle price, we update the middle price to be the current price.
3. If the current price is higher than the lowest and middle price, it means we've found a three-price upward trend!

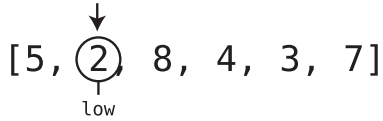
Let's see this in action. First, we'll start with a simple example, working with this array of stock prices:

```
[5, 2, 8, 4, 3, 7]
```

We begin iterating through the array, starting with the 5. We start out of the gates with pure greed, and assume that this 5 is the smallest price in the three-point trend as shown in the array on page 424.



Next, we proceed to the 2. Because the 2 is lower than the 5, we get even greedier and assume that the 2 is now the lowest price in the trend:



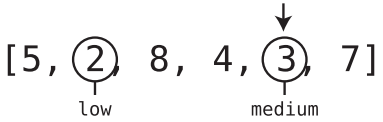
We arrive at the next number in the array, which is 8. This is higher than our lowest point, so we keep the lowest point at 2. However, it's less than the current middle price, which is infinity, so we now greedily assign the 8 to be our *middle* point in the three-point trend:



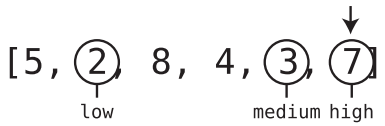
Next up, we reach the 4. This is higher than the 2, so we continue to assume that the 2 is the lowest point in our trend. However, because the 4 is less than the 8, we make the 4 our middle point instead of the 8. This, too, is out of greed, as by making our middle point lower, we increase our chances of finding a higher price later on, forming the trend that we're seeking. So, the 4 is our new middle point:



The next number in the array is the 3. We'll leave our lowest price at 2, since the 3 is greater than it. But we will make it our new middle point, since it's less than the 4:



Finally, we reach the 7, which is the last value in the array. Because the 7 is greater than our middle price (which is 3), this means the array contains an upward three-point trend, and our function can return true:



Note that two such trends exist in the array. There's 2-3-7, but there's also 2-4-7. Ultimately, though, this doesn't matter to us, since we're just trying to determine whether this array contains *any* upward trend, so finding a single instance is enough to return true.

Here's an implementation of this algorithm:

```
def increasing_triplet?(array)
  lowest_price = array[0]
  middle_price = Float::INFINITY

  array.each do |price|
    if price <= lowest_price
      lowest_price = price

    # if current price is higher than lowest price
    # but lower than middle price:
    elsif price <= middle_price
      middle_price = price

    # if the current price is higher than the middle price:
    else
      return true
    end
  end

  return false
end
```

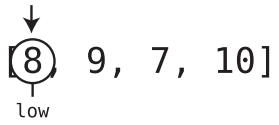
One counterintuitive aspect of this algorithm is worth pointing out. Specifically, in some scenarios, it would appear this algorithm wouldn't work, yet it does.

Let's take a look at this scenario:

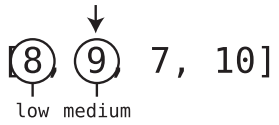
```
[8, 9, 7, 10]
```

Let's see what happens when we apply our algorithm to this array.

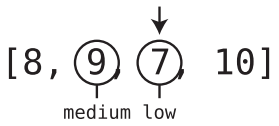
At first, the 8 becomes our lowest point:



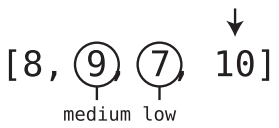
Then, the 9 becomes our middle point:



Next, we reach the 7. Because this is lower than our lowest point, we update the lowest point to be 7:



We then reach the 10:



Because the 10 is greater than the current middle point (9) our function returns `true`. Now, this is the correct response, since our array indeed contains the trend of 8-9-10. However, by the time our function is done, our lowest point variable is actually the pointing to the 7. But the 7 is not part of the upward trend!

Despite this being the case, our function still returned the correct response. And this is because all our function needs to do is reach a number that is higher than the middle point. Because the middle point was only established once we already found a lower point before it, as soon as we reach a number higher than the middle point, it still means an upward trend is present in the array. This is true even though we ended up overwriting the lower point to be some other number later on.

In any case, our greedy approach paid off, as we only iterated over our array a single time. This is an astounding improvement, as we turned an algorithm that ran at $O(N^3)$ into one of $O(N)$.

Of course, a greedy approach doesn't *always* work. But it's another tool you can try out when optimizing your algorithms.

Change the Data Structure

Another helpful optimization technique is to imagine what would happen if we stored our given data in an alternative data structure.

For example, we may be working on a problem where the data is given to us in the form of an array. However, reimagining that same data stored as a hash table, tree, or other data structure can sometimes reveal clever optimization opportunities.

Our using of a hash table for the magical lookup technique earlier is a specific example of this. However, we're about to see that changing the data structure can be useful for other scenarios as well.

The Anagram Checker

Here's one example. Let's say we're writing a function that determines whether two given strings are anagrams of one another. We encountered an anagram function before in Anagram Generation, on page 177, but there we dealt with a function that produced every anagram of a string. Here, we're just going to compare two strings side by side. We'll return `true` if they're anagrams of each other, and `false` if they're not.

Now, we could actually use the anagram-generating function to solve this problem. That is, we can produce all the anagrams of the first string and see if the second string matches any of those anagrams. However, since for N characters in the string there will always be N! anagrams, our algorithm will take at least O(N!) time. This is disastrously slow.

You know the drill. Before we proceed to optimize our code, we need to come up with our best imaginable-Big O.

Now, we certainly need to visit each character from both strings at least once. And since the strings may be of different sizes, touching each character just once would be O(N + M). As I couldn't imagine a faster speed for the task at hand, this is what we'll aim for.

Let's work our way there.

A second possible approach to our problem is to run nested loops to compare the two strings. Specifically, as an outer loop iterates over each character from the first string, we compare that character to every character of the second string. Each time we find a match, we delete a character from the

second string. The idea here is that if every character from the first string is also present in the second string, we'd end up deleting every character from the second string by the time we complete our outer loop.

So, if by the time we finish looping, there are still characters remaining from the second string, it means the strings aren't anagrams. Also, if we're still iterating over the first word, but we've already deleted the entire second string, it also means the strings weren't anagrams. But if we make it to end of the loop and the second string has been completely deleted, we can conclude that the two strings are indeed anagrams.

Here's a Python implementation of this:

```
def areAnagrams(firstString, secondString):
  # Convert secondString into an array so we can delete characters from it,
  # as strings are immutable in Python:
  secondStringArray = list(secondString)

  for i in range(0, len(firstString)):

    # If we're still iterating through the firstString, but the
    # secondStringArray is already empty:
    if len(secondStringArray) == 0:
      return False

    for j in range(0, len(secondStringArray)):

      # If we find the same character in both the firstString
      # and secondStringArray:
      if firstString[i] == secondStringArray[j]:

        # Delete the character from the second array and
        # go back to the outer loop:
        del secondStringArray[j]
        break

  # The two strings are only anagrams if the secondStringArray
  # has no characters remaining by the time we're done
  # iterating over the firstString:
  return len(secondStringArray) == 0
```

Now, it just so happens that deleting items from an array while you're looping through it can be error prone; if you don't do it right, it's like sawing off the tree branch you're sitting on. But even though we've handled that correctly, our algorithm runs at O(N * M). This is way faster than O(N!), but much slower than the O(N + M) we're shooting for.

An even faster approach would be to sort the two strings. If after sorting the two strings they're exactly the same, it means they're anagrams; otherwise, they're not.

This approach will take O(N log N) for each string using a fast sorting algorithm like Quicksort. Since we may have two strings of different sizes, this adds up to be O(N log N + M log M). This is a nice improvement over O(N * M), but let's not stop now—we're aiming for O(N + M), remember?

This is where using an alternative data structure can be extremely helpful. We're dealing with strings, but let's imagine that we'd store the string data in other types of data structures.

We *could* store a string as an array of single characters. However, this doesn't help us any.

Let's next imagine the string as a hash table. What would that even look like?

One possibility is to create a hash table where each character is a key, and the value is the number of occurrences of that character within the word. For example, the string `"balloon"` would become:

```
{"b" => 1, "a" => 1, "l" => 2, "o" => 2, "n" => 1}
```

This hash table indicates that the string has one `"b"`, one `"a"`, two `"l"`s, two `"o"`s, and one `"n"`.

Now, this doesn't tell us *everything* about the string. Namely, we couldn't tell from the hash table the *order* of characters within the string. So, there's a bit of data loss in this regard.

However, that data loss is *exactly* what we need to help us determine whether the two strings are anagrams. That is, two strings turn out to be anagrams if they have the same number of each character, no matter what the order is.

Take the words "rattles," "startle," and "starlet." They all have two `"t"`s, one `"a"`, one `"l"`, one `"e"`, and one `"s"`—and that's what allows them to be anagrams and easily reordered to become each other.

We can now write an algorithm that converts each string into a hash table that tallies the count of each type of character. Once we've converted the two strings into two hash tables, all that's left is to compare the two hash tables. If they're identical, it means the two strings are anagrams.

Here's an implementation of this:

```
def areAnagrams(firstString, secondString):
  firstWordHashTable = {}
  secondWordHashTable = {}
```

```
# Create hash table out of first string:
for char in firstString:
  if firstWordHashTable.get(char):
    firstWordHashTable[char] += 1
  else:
    firstWordHashTable[char] = 1

# Create hash table out of second string:
for char in secondString:
  if secondWordHashTable.get(char):
    secondWordHashTable[char] += 1
  else:
    secondWordHashTable[char] = 1

# The two strings are anagrams only if the two hash tables are identical:
return firstWordHashTable == secondWordHashTable
```

In this algorithm, we iterate over each character from both strings just once, which is N + M steps.

Checking whether the hash tables are equal may take up to another N + M steps. This is especially true in a language like JavaScript, in which we can only check for hash table equality by manually iterating over each key-value pair of the two hash tables. However, this is still just 2(N + M) steps, which reduces to O(N + M). This is much faster than any of our previous approaches.

To be fair, we're taking up some extra space with the creation of these hash tables. Our previous suggestion of sorting the two strings and comparing them would take up no extra space if we did the sorting in place. But if speed is what we're after, we can't beat the hash table approach, as we touch each character from the strings *just once.*

By converting the strings into another data structure (in this case, hash tables), we were able to access the original data in such a way that allowed our algorithm to become blazing fast.

It's not always obvious what new data structure to use, so it's good to imagine how the current data may look if it were converted into a variety of formats and see if that reveals any optimizations. That being said, hash tables turn out very often to be a great choice, so that's a good place to start.

Group Sorting

Here's another example of how changing the data structure can allow us to optimize our code. Let's say we have an array containing several different values and we want to reorder the data so that the same values are grouped together. However, we don't necessarily care what order the *groups* are in.

For example, let's say we have the following array:

```
["a", "c", "d", "b", "b", "c", "a", "d", "c", "b", "a", "d"]
```

Our goal is to sort this into groups, such as:

```
["c", "c", "c", "a", "a", "a", "d", "d", "d", "b", "b", "b"]
```

Again, we don't care about the order of the groups, so these results would also be acceptable:

```
["d", "d", "d", "c", "c", "c", "a", "a", "a", "b", "b", "b"]
["b", "b", "b", "c", "c", "c", "a", "a", "a", "d", "d", "d"]
```

Now, any classic sorting algorithm would accomplish our task, since we'd end up with this:

```
["a", "a", "a", "b", "b", "b", "c", "c", "c", "d", "d", "d"]
```

As you know, the fastest sorting algorithms clock in at O(N log N). But can we do better?

Let's begin by coming up with the best-imaginable Big O. Since we know that there aren't sorting algorithms faster than O(N log N), it may be difficult to imagine how we can sort something in a faster time.

However, since we're not doing a precise sort, if someone told me that our task can be done in O(N) time, I suppose I'd believe them. We certainly can't beat O(N), since we need to visit each value at least once. So, let's aim for O(N).

Let's employ the technique we've been discussing and imagine our data in the form of another data structure.

We may as well start with a hash table. What would our array of strings look like if it were a hash table?

If we took a similar approach as we did with the anagrams, we could represent our array in the following way:

```
{"a" => 3, "c" => 3, "d" => 3, "b" => 3}
```

As with our previous example, there's some data loss. That is, we couldn't convert this hash table back into our original array, as we wouldn't know the original order of all the strings.

However, for our purposes of grouping, this data loss doesn't matter. In fact, the hash table contains all the data we need to create the grouped array we're looking for.

Specifically, we can iterate over each key-value pair within the hash table and use that data to populate an array with the correct number of each string. Here's the code for this:

```
def group_array(array)
  hash_table = {}
  new_array = []

  # Store the tallies of each string in a hash table:
  array.each do |value|
    if hash_table[value]
      hash_table[value] += 1
    else
      hash_table[value] = 1
    end
  end

  # Iterate over the hash table and populate a new array
  # with the correct number of each string:
  hash_table.each do |key, count|
    count.times do
      new_array << key
    end
  end

  return new_array
end
```

Our group_array function accepts an array and then begins by creating an empty hash_table and an empty new_array.

We first collect the tallies of each string and store them in the hash table:

```
array.each do |value|
  if hash_table[value]
    hash_table[value] += 1
  else
    hash_table[value] = 1
  end
end
```

This creates the hash table that looks like this:

```
{"a" => 3, "c" => 3, "d" => 3, "b" => 3}
```

Then, we proceed to iterate over each key-value pair and use this data to populate the new_array:

```
hash_table.each do |key, count|
  count.times do
    new_array << key
  end
end
```

That is, when we reach the pair, "a" => 3, we add three "a"s to the new_array. And when we reach "c" => 3, we add three "c"s to the new_array, and so on. By the time we're done, our new_array' will contain all the strings organized in groups.

This algorithm takes just O(N) time, which is a significant optimization over the O(N log N) that sorting would have taken. We do use up O(N) space with the extra hash table and new_array, although we could choose to overwrite the original array to save additional memory. That being said, the space taken up by the hash table would still be O(N) in the worst case, where each string in the array is different.

But again, if speed is our goal, we achieved our best-imaginable Big O, which is a fantastic win.

Wrapping Up

The techniques presented here can be useful in optimizing your code. Again, you'll always want to start by determining your current Big O and the best-imaginable Big O. After that, you have all the other techniques at your disposal.

You'll find that some techniques work better in certain situations than others, but they're all worth thinking about in a given situation to see if they may be the right tool for the job.

With experience, you'll hone your optimization sense and probably develop additional techniques of your own!

Parting Thoughts

You've learned a lot in this journey.

You've learned that the right choices in algorithm design and data structures can dramatically affect the performance of our code.

You've learned how to determine how efficient our code is.

And you've learned how to optimize our code to make it faster, more memory-efficient, and more elegant.

What you can take away from this book is a framework for making educated technology decisions. Creating great software involves evaluating the trade-offs of the available options, and you are now armed with the ability to see the pros and cons of each option and make the best choice for the task at hand. And you also have the ability to think of *new* options that may not have been immediately obvious at first glance.

It's important to note that it's always best to *test* your optimizations with benchmarking tools. Using these tools that test the actual speed of your code is a great sanity check to ensure your optimizations are indeed helpful. There are many excellent software applications out there that can measure the speed and memory consumption of your code. The knowledge in this book, though, will point you in the right direction, and the benchmarking tools can confirm whether you've made the right choice.

I hope you also take away from this book the knowledge that topics like these—which seem so complex and esoteric—are just a combination of simpler, easier concepts that are within your grasp. Don't be intimidated by resources that make a concept seem difficult simply because they don't explain it well—a concept can always be broken down in a way so that it can be understood.

The topic of data structures and algorithms is broad and deep, and we've only just scratched the surface. With the foundation we've built, you are now able to explore and master new concepts of computer science, and I hope that you go out there and continue to learn new things and grow in your technical proficiency.

Good luck!

Exercises

The following exercises provide you with the opportunity to practice with optimizing your code. The solutions to these exercises are found in the section, Chapter 20, on page 465.

1. You're working on software that analyzes sports players. Following are two arrays of players of different sports:

```
basketball_players = [
  {first_name: "Jill", last_name: "Huang", team: "Gators"},
  {first_name: "Janko", last_name: "Barton", team: "Sharks"},
  {first_name: "Wanda", last_name: "Vakulskas", team: "Sharks"},
  {first_name: "Jill", last_name: "Moloney", team: "Gators"},
  {first_name: "Luuk", last_name: "Watkins", team: "Gators"}
]

football_players = [
  {first_name: "Hanzla", last_name: "Radosti", team: "32ers"},
  {first_name: "Tina", last_name: "Watkins", team: "Barleycorns"},
  {first_name: "Alex", last_name: "Patel", team: "32ers"},
  {first_name: "Jill", last_name: "Huang", team: "Barleycorns"},
  {first_name: "Wanda", last_name: "Vakulskas", team: "Barleycorns"}
]
```

If you look carefully, you'll see that there are some players who participate in more than one kind of sport. Jill Huang and Wanda Vakulskas play both basketball *and* football.

You are to write a function that accepts two arrays of players and returns an array of the players who play in *both* sports. In this case, that would be:

```
["Jill Huang", "Wanda Vakulskas"]
```

While there are players who share first names and players who share last names, we can assume there's only one person who has a particular *full* name (meaning first *and* last name).

We can use a nested-loops approach, comparing each player from one array against each player from the other array, but this would have a runtime of O(N * M). Your job is to optimize the function so that it can run in just O(N + M).

2. You're writing a function that accepts an array of distinct integers from 0, 1, 2, 3...up to N. However, the array will be missing one integer, and your function is to *return the missing one.*

   For example, this array has all the integers from 0 to 6, but is missing the 4:

   ```
   [2, 3, 0, 6, 1, 5]
   ```

   Therefore, the function should return `4`.

   The next example has all the integers from 0 to 9, but is missing the 1:

   ```
   [8, 2, 3, 9, 4, 7, 5, 0, 6]
   ```

   In this case, the function should return the `1`.

   Using a nested-loops approach would take up to $O(N^2)$. Your job is to optimize the code so that it has a runtime of O(N).

3. You're working on some more stock-prediction software. The function you're writing accepts an array of predicted prices for a particular stock over the course of time.

   For example, this array of seven prices:

   ```
   [10, 7, 5, 8, 11, 2, 6]
   ```

   predicts that a given stock will have these prices over the next seven days. (On Day 1, the stock will close at $10; on Day 2, the stock will close at $7; and so on.)

Your function should calculate the greatest profit that could be made from a single "buy" transaction followed by a single "sell" transaction.

In the previous example, the most money could be made if we bought the stock when it was worth $5 and sold it when it was worth $11. This yields a profit of $6 per share.

Note that we could make even more money if we buy and sell multiple times, but for now, this function focuses on the most profit that could be made from just *one* purchase followed by *one* sale.

Now, we could use nested loops to find the profit of every possible buy-and-sell combination. However, this would be $O(N^2)$ and too slow for our hotshot trading platform. Your job is to optimize the code so that the function clocks in at just O(N).

4. You're writing a function that accepts an array of numbers and computes the highest product of any two numbers in the array. At first glance, this is easy, as we can just find the two greatest numbers and multiply them. However, our array can contain negative numbers and look like this:

   ```
   [5, -10, -6, 9, 4]
   ```

   In this case, it's actually the product of the two *lowest* numbers, -10 and -6 that yield the highest product of 60.

   We could use nested loops to multiply every possible pair of numbers, but this would take $O(N^2)$ time. Your job is to optimize the function so that it's a speedy O(N).

5. You're creating software that analyzes the data of body temperature readings taken from hundreds of human patients. These readings are taken from healthy people and range from 97.0 degrees Fahrenheit to 99.0 degrees Fahrenheit. An important point: within this application, *the decimal point never goes beyond the tenths place.*

   Here's a sample array of temperature readings:

   ```
   [98.6, 98.0, 97.1, 99.0, 98.9, 97.8, 98.5, 98.2, 98.0, 97.1]
   ```

   You are to write a function that sorts these readings from lowest to highest.

   If you used a classic sorting algorithm such as Quicksort, this would take O(N log N). *However, in this case, it's actually possible to write a faster sorting algorithm.*

   Yes, that's right. Even though you've learned that the fastest sorts are O(N log N), this case is different. Why? In this case, there's a *limited*

number of possibilities of what the readings will be. In such a case, we can sort these values in O(N). It may be N multiplied by a constant, but that's still considered O(N).

6. You're writing a function that accepts an array of unsorted integers and returns the length of the *longest consecutive sequence* among them. The sequence is formed by integers that increase by 1. For example, in the array:

   ```
   [10, 5, 12, 3, 55, 30, 4, 11, 2]
   ```

   the longest consecutive sequence is 2-3-4-5. These four integers form an increasing sequence because each integer is one greater than the previous one. While there's also a sequence of 10-11-12, it's only a sequence of three integers. In this case, the function should return `4`, since that's the length of the *longest* consecutive sequence that can be formed from this array.

   One more example:

   ```
   [19, 13, 15, 12, 18, 14, 17, 11]
   ```

   This array's longest sequence is 11-12-13-14-15, so the function would return `5`.

   If we sorted the array, we can then traverse the array just once to find the longest consecutive sequence. However, the sorting itself would take O(N log N). Your job is to optimize the function so that it takes O(N) time.

APPENDIX 1

Exercise Solutions

Chapter 1

These are the solutions to the exercises found in the section, Exercises, on page 19.

1. Let's break down each of the cases:

   a. Reading from an array always takes just one step.

   b. Searching for a nonexistent element within an array of size 100 will take 100 steps, as the computer needs to inspect each element of the array before determining the element cannot be found.

   c. The insertion will take 101 steps: 100 shifts of each element to the right, and one step to insert the new element at the front of the array.

   d. Insertion at the end of an array always takes one step.

   e. The deletion will take 100 steps: first the computer deletes the first element and then shifts the remaining 99 elements to the left, one at a time.

   f. Deletion at the end of an array always takes one step.

2. Let's break down each of the cases:

   a. Like the array, reading from the array-based set will take just one step.

   b. Like the array, searching the array-based set will take 100 steps, as we inspect each element before concluding that the element isn't there.

   c. To insert into the set, we first need to conduct a full search to make sure the value doesn't already exist within the set. This search will

take 100 steps. Then, we need to shift all 100 elements to the right to make room for the new value. Finally, we drop the new value at the beginning of the set. This is a total of 201 steps.

d. This insertion takes 101 steps. Again, we need to conduct a full search before inserting, which takes 100 steps. We then conclude with the final step of inserting the new value at the end of the set.

e. The deletion will take 100 steps, just like a classic array.

f. The deletion will take one step, just like a classic array.

3. If the array contains N elements, searching for all instances of the string "apple" in an array will take N steps. When searching for just one instance, we can cut our search short as soon as we find it. But if we need to find all instances, we have no choice but inspect each element of the entire array.

Chapter 2

These are the solutions to the exercises found in the section, Exercises, on page 34.

1. Linear search on this array would take four steps. We start at the beginning of the array and check each element from left to right. Because the 8 is the fourth number, we'll find it in four steps.

2. Binary search would take just one step in this case. We start the binary search at the middlemost element, and the 8 just happens to be the middlemost element!

3. To solve this, we need to count how many times we halve 100,000 until we get down to 1. If we keep dividing 100,000 by 2, we see that it takes us 16 times until we get down to about 1.53.

   This means a worst-case scenario would take about 16 times.

Chapter 3

These are the solutions to the exercises found in the section, Exercises, on page 45.

1. This is O(1). We can consider N to be the year passed into the function. But no matter what the year is, the algorithm doesn't vary in how many steps it takes.

2. This is O(N). For N elements in the array, the loop will run N times.

3. This is O(log N). In this case, N is the number numberOfGrains, which is passed into the function. The loop runs as long as placedGrains < numberOfGrains, but placedGrains starts at 1 and *doubles* each time the loop runs. If, for example, numberOfGrains was 256, we'd keep doubling the placedGrains nine times until we reach 256, meaning that our loop would run nine times for an N of 256. If numberOfGrains was 512, our loop would run 10 times, and if numberOfGrains was 1024, the loop would run 11 times. Because our loop runs only one more time each time N is doubled, this is O(log N).

4. This is O(N). N is the number of strings within the array, and the loop will take N steps.

5. This is O(1). We can consider N to be the size of the array, but the algorithm takes a fixed number of steps no matter what N is. The algorithm does account for whether N is even or odd, but in either case, it takes the same number of steps.

Chapter 4

These are the solutions to the exercises found in the section, Exercises, on page 60. The solutions provided here are in Python, but you can find the solutions in JavaScript and Ruby in the code download.[1]

1. Here is the completed table:

N Elements	O(N)	O(log N)	$O(N^2)$
100	100	About 7	10,000
2000	2000	About 11	4,000,000

2. The array would have 16 elements, since 16^2 is 256. (Another way of saying this is that the square root of 256 is 16.)

3. The algorithm has a time complexity of $O(N^2)$. N, in this case, is the size of the array. We have an outer loop that iterates over the array N times, and for each of those times, an inner loop iterates over the same array N times. This results in N^2 steps.

1. https://pragprog.com/titles/jwdsal2/source_code

4. The following version is O(N), as we only iterate through the array once:

```
def greatestNumber(array):
  greatestNumberSoFar = array[0]

  for i in array:
    if i > greatestNumberSoFar:
      greatestNumberSoFar = i

  return greatestNumberSoFar
```

Chapter 5

These are the solutions to the exercises found in the section, Exercises, on page 76.

1. After dropping the constants, we can reduce the expression to O(N).
2. After dropping the constant, we can reduce the expression to $O(N^2)$.
3. This algorithm is one of O(N), with N being the size of the array. While there are two distinct loops that process the N elements, this is simply 2N, which is reduced to O(N) after dropping the constants.
4. This algorithm is one of O(N), with N being the size of the array. Within the loop, we run three steps, which means our algorithm takes 3N steps. However, this is reduced to O(N) when we eliminate the constants.
5. This algorithm is one of $O(N^2)$, with N being the size of the array. While we only run the inner loop half of the time, this simply means that the algorithm runs for $N^2 / 2$ steps. However, the division by 2 is a constant, so we express this simply as $O(N^2)$.

Chapter 6

These are the solutions to the exercises found in the section, Exercises, on page 93. The solutions provided here are in JavaScript, but you can find the solutions in Python and Ruby in the code download.[2]

1. In Big O Notation, $2N^2 + 2N + 1$ gets reduced to $O(N^2)$. After getting rid of all the constants, we're left with $N^2 + N$, but we also drop the N since it's a lower order than N^2.
2. log N is a lower order than N, so it's simply reduced to O(N).

2. https://pragprog.com/titles/jwdsal2/source_code

3. The important thing to note here is that the function ends as soon as we find a pair that sums to 10. The best-case scenario, then, is when the first two numbers add up to 10, since we can end the function before the loops even get underway. An average-case scenario may be when the two numbers are somewhere in the middle of the array. The worst-case scenarios are when there aren't any two numbers that add up to 10, in which case we must exhaust both loops completely. This worst-case scenario is $O(N^2)$, where N is the size of the array.

4. This algorithm has an efficiency of O(N), as the size of the array is N, and the loop iterates through all N elements.

   This algorithm continues the loop even if it finds an `"X"` before the end of the array. We can make the code more efficient if we return `true` as soon as we find an `"X"`:

```
function containsX(string) {

  for(let i = 0; i < string.length; i++) {
    if (string[i] === "X") {
      return true;
    }
  }

  return false;
}
```

Chapter 7

These are the solutions to the exercises found in the section, Exercises, on page 109.

1. Here, N is the size of the array. Our loop runs for N / 2 times, as it processes two values each round of the loop. However, this is expressed as O(N) because we drop the constant.

2. It's slightly tricky to define N in this case, since we are dealing with two distinct arrays. However, the algorithm only processes each value once, so we could decide to call N the total number of values between both arrays, and the time complexity would be O(N). If we want to be more literal, and call one array N and the other M, we could alternatively express the efficiency as O(N + M). However, because we're simply adding N and M together, it's simpler to just use N to refer to the total number of data elements across both arrays and call it O(N).

3. In a worst-case scenario, this algorithm runs for the number of characters in the "needle" multiplied by the number of characters in the "haystack." Since the needle and haystack may have different numbers of characters, this is O(N * M).

4. N is the size of the array, and the time complexity is $O(N^3)$, as it's processed through triply-nested loops. Really, the middle loop runs N / 2 times, and the innermost loop runs N / 4 times, so this is N * (N / 2) * (N / 4), which is N^3 / 8 steps. But we drop the constant, leaving us with $O(N^3)$.

5. N is the size of the resumes array. Since in each round of the loop we eliminate half of the resumes, we have an algorithm of O(log N).

Chapter 8

These are the solutions to the exercises found in the section, Exercises, on page 131. The solutions provided here are in JavaScript, but you can find the solutions in Python and Ruby in the code download.[3]

1. The following implementation first stores the values of the first array in a hash table and then checks each value of the second array against that hash table:

```
function getIntersection(array1, array2) {
  let intersection = [];
  let hashTable = {};

  for(let i = 0; i < array1.length; i++) {
    hashTable[array1[i]] = true;
  }

  for(let j = 0; j < array2.length; j++) {
    if(hashTable[array2[j]]) {
      intersection.push(array2[j]);
    }
  }

  return intersection;
}
```

   This algorithm has an efficiency of O(N).

2. The following implementation checks each string in the array. If the string isn't yet in the hash table, the string gets added. If the string *is* in the hash table, that means it's been added before, which means it's a duplicate! This algorithm has a time complexity of O(N):

3. https://pragprog.com/titles/jwdsal2/source_code

```
function findDuplicate(array) {
  let hashTable = {};

  for(let i = 0; i < array.length; i++) {
    if(hashTable[array[i]]) {
      return array[i];
    } else {
      hashTable[array[i]] = true;
    }
  }

}
```

3. The following implementation begins by creating a hash table out of all the characters we encounter in the string. Next, we iterate over each character of the alphabet and check to see whether the character is contained within our hash table. If it isn't, it means the character is missing from the string, so we return it:

```
function findMissingLetter(string) {
    // Store all encountered letters in hash table:
    let hashTable = {};
    for(let i = 0; i < string.length; i++) {
        hashTable[string[i]] = true;
    }

    // Report first unseen letter:
    let alphabet = "abcdefghijklmnopqrstuvwxyz";
    for(let i = 0; i < alphabet.length; i++) {
        if(!hashTable[alphabet[i]]) {
            return alphabet[i];
        }
    }
}
```

4. The following implementation begins by iterating over each character in the string. If the character does not yet exist in the hash table, the character is added to the hash table as a key with the value of 1, indicating the character has been found once so far. If the character is already in the hash table, we simply increment the value by 1. So, if the character "e" has the value of 3, it means the "e" exists three times within the string.

   Then, we iterate over the characters again and return the first character that only exists once within the string. This algorithm is O(N):

```
function firstNonDuplicate(string) {
  let hashTable = {};

  for(let i = 0; i < string.length; i++) {
    if(hashTable[string[i]]) {
      hashTable[string[i]]++;
```

```
    } else {
      hashTable[string[i]] = 1;
    }
  }

  for(let j = 0; j < string.length; j++) {
    if(hashTable[string[j]] == 1) {
      return string[j];
    }
  }
}
```

Chapter 9

These are the solutions to the exercises found in the section, Exercises, on page 148. The solutions provided here are in Ruby, but you can find the solutions in JavaScript and Python in the code download.[4]

1. Presumably, we'd want to be nice to the callers and answer their phone calls in the order in which they were received. For this, we'd use a queue, which processes data FIFO (First In, First Out).

2. We'd be able to read the 4, which is now the top element in the stack. This is because we'll have popped the 6 and the 5, which were previously sitting on top of the 4.

3. We'd be able to read the 3, which is now at the front of the queue, after having dequeued the 1 and the 2.

4. We can take advantage of the stack because of the fact that we pop each item in the reverse order of which they were pushed onto the stack. So, we'll first push each character of the string onto the stack. Then, we'll pop each one off while adding them to the end of a new string:

```
def reverse(string)
  stack = Stack.new

  string.each_char do |char|
    stack.push(char)
  end

  new_string = ""

  while stack.read
    new_string += stack.pop
  end

  return new_string
end
```

4. https://pragprog.com/titles/jwdsal2/source_code

Chapter 10

These are the solutions to the exercises found in the section, Exercises, on page 159. The solutions provided here are in Python, but you can find the solutions in JavaScript and Ruby in the code download.[5]

1. The base case is `if low > high`—that is, we want to stop the recursion once `low` has exceeded `high`. Otherwise, we'd end up printing numbers even greater than the `high` number, and onto infinity.

2. We'd have infinite recursion! `factorial(10)` calls `factorial(8)`, which calls `factorial(6)`, which calls `factorial(4)`, which calls `factorial(2)`, which calls `factorial(0)`. Since our base case is `if n == 1`, we never end up with `n` ever being 1, so the recursion continues. `factorial(0)` calls `factorial(-2)`, and so on.

3. Let's say `low` is 1 and `high` is 10. When we call `sum(1, 10)`, that in turn returns `10 + sum(1, 9)`. That is, we return the sum of 10 plus whatever the `sum` of 1 through 9 is. `sum(1, 9)` ends up calling `sum(1, 8)`, which in turn calls `sum(1, 7)`, and so on.

   We want the last call to be `sum(1, 1)`, in which we simply want to return the number 1. This becomes our base case:

   ```
   def sum(low, high):
     # Base case:
     if high == low:
       return low

     return high + sum(low, high - 1)
   ```

4. This approach is similar to the file directory example from the text:

   ```
   def print_all_items(array):
     for value in array:
       # if the current item is a "list", i.e. array:
       if isinstance(value, list):
         print_all_items(value)
       else:
         print(value)
   ```

   We iterate over each item within the outer array. If the value is itself another array, we recursively call the function on that subarray. Otherwise, it's the base case where we simply print the value to the screen.

5. https://pragprog.com/titles/jwdsal2/source_code

Chapter 11

These are the solutions to the exercises found in the section, Exercises, on page 181. The solutions provided here are in Ruby, but you can find the solutions in JavaScript and Python in the code download.[6]

1. Let's call our function `character_count`. The first step is to pretend that the `character_count` function has already been implemented.

   Next, we need to identify the subproblem. If our problem is the array, `["ab", "c", "def", "ghij"]`, then our subproblem can be the same array missing one string. Let's specifically say that our subproblem is the array minus the *first* string, which would be `["c", "def", "ghij"]`.

   Now, let's see what happens when we apply the "already implemented" function on the subproblem. If we were to call `character_count(["c", "def", "ghij"])`, we'd get a return value of `8`, since there are eight characters in total.

   So, to solve our original problem, all we have to do is add the length of the first string (`"ab"`) to the result of calling the `character_count` function on the subproblem.

   Here's one possible implementation:

   ```
   def character_count(array)
     # Alternative base case:
     # return array[0].length if array.length == 1

     # Base case: when the array is empty:
     return 0 if array.length == 0

     return array[0].length + character_count(array[1, array.length - 1])
   end
   ```

   Note that we made our base case to be when the array is empty, in which case there are zero string characters. We mention an alternative and equally viable base case in the comments, which is when the array contains one string. In that case, we return the length of that single string.

2. First, let's pretend the `select_even` function already works. Next, let's identify the subproblem. If we try to select all the even numbers in the example array, `[1, 2, 3, 4, 5]`, we could say that the subproblem is all the numbers in the array beside the first one. So, let's imagine `select_even([2, 3, 4, 5])` already works and returns `[2, 4]`.

6. https://pragprog.com/titles/jwdsal2/source_code

Since the first number in the array is 1, we actually don't want to do anything other than return the [2, 4]. However, if the first number in the array was a 0, we'd want to return the [2, 4] with the 0 added to it.

The base case can be when the array is empty.

Here's one possible implementation:

```
def select_even(array)
  return [] if array.empty?

  if array[0].even?
    return [array[0]] + select_even(array[1, array.length - 1])
  else
    return select_even(array[1, array.length - 1])
  end
end
```

3. The definition of a triangular number was given as n plus the previous number from the pattern. If the name of our function is triangle, we can express this simply as n + triangle(n - 1). The base case is when n is 1.

```
def triangle(n)
  return 1 if n == 1
  return n + triangle(n - 1)
end
```

4. Let's pretend our function, index_of_x, has already been implemented. Next, let's say the subproblem is our string minus its first character. For example, if our input string is "hex", the subproblem is "ex".

   Now, index_of_x("ex") would return 1. To calculate the index of the "x" for the original string, we would add 1 to this since the additional "h" at the front of the string moves the "x" down one index.

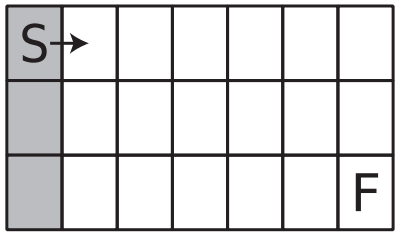
```
def index_of_x(string)
  return 0 if string[0] == 'x'
  return index_of_x(string[1, string.length - 1]) + 1
end
```

5. This exercise is similar to the "Staircase Problem." Let's break this down:

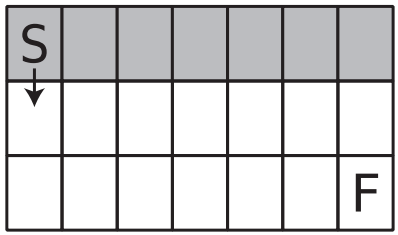
   From the starting position, we have only two choices of movement. We can either move one space to the right or one space downward.

   What this means is that the total number of unique shortest paths will be the umber of paths from space to the right of S + the number of paths from space below S.

The number of paths from the space to the right of S is the same as calculating the paths in a grid of six columns and three rows, as you can see here:



The number of paths from the space below the S is the equivalent of the paths in a grid of seven columns and two rows:



Recursion allows us to express this beautifully:

```
return unique_paths(rows - 1, columns) + unique_paths(rows, columns - 1)
```

All we need to do now is add the base case. Possible base cases include when we have just one row or one column, since in such cases, there's only one path available to us.

Here's the complete function:

```
def unique_paths(rows, columns)
  return 1 if rows == 1 || columns == 1
  return unique_paths(rows - 1, columns) + unique_paths(rows, columns - 1)
end
```

Chapter 12

These are the solutions to the exercises found in the section, Exercises, on page 197. The solutions provided here are in Ruby, but you can find the solutions in JavaScript and Python in the code download.[7]

7. https://pragprog.com/titles/jwdsal2/source_code

1. The problem here is that we have *two* recursive calls to the function within itself. We can easily reduce it to one:

```
def add_until_100(array)
  return 0 if array.length == 0
  sum_of_remaining_numbers = add_until_100(array[1, array.length - 1])
  if array[0] + sum_of_remaining_numbers > 100
    return sum_of_remaining_numbers
  else
    return array[0] + sum_of_remaining_numbers
  end
end
```

2. Here is the memoized version:

```
def golomb(n, memo={})
  return 1 if n == 1

  if !memo[n]
    memo[n] = 1 + golomb(n - golomb(golomb(n - 1, memo), memo), memo)
  end

  return memo[n]
end
```

3. To accomplish memoization here, we need to make a key out of both the number of rows and number of columns. We can make our key a simple array of [rows, columns]:

```
def unique_paths(rows, columns, memo={})
  return 1 if rows == 1 || columns == 1
  if !memo[[rows, columns]]
    memo[[rows, columns]] = unique_paths(rows - 1, columns, memo) +
    unique_paths(rows, columns - 1, memo)
  end
  return memo[[rows, columns]]
end
```

Chapter 13

These are the solutions to the exercises found in the section, Exercises, on page 224. The solutions provided here are in JavaScript, but you can find the solutions in Ruby and Python in the code download.[8]

1. If we sort the numbers, we know that the three greatest numbers will be at the end of the array, and we can just multiply them together. The sorting will take O(N log N):

8. https://pragprog.com/titles/jwdsal2/source_code

```
function greatestProductOf3(array) {
  array.sort((a, b) => (a < b) ? -1 : 1);

  return array[array.length - 1] * array[array.length - 2] *
    array[array.length - 3];
}
```

(This code takes for granted that there are at least three values in the array. You can add code to handle arrays where this is not the case.)

2. If we presort the array, we can then expect each number to be at its own index. That is, the 0 should be at index 0, the 1 should be at index 1, and so on. We can then iterate through the array looking for a number that doesn't equal the index. Once we find it, we know that we just skipped over the missing number:

```
function findMissingNumber(array) {
  array.sort((a, b) => (a < b) ? -1 : 1);

  for(let i = 0; i < array.length; i++) {
    if(array[i] !== i) {
      return i;
    }
  }

  return null;
}
```

The sorting takes N log N steps, and the loop afterward takes N steps. However, we reduce the expression (N log N) + N to O(N log N) since the added N is a lower order compared to N log N.

3. This implementation uses nested loops and is $O(N^2)$:

```
function max(array) {
  for(let i = 0; i < array.length; i++) {
    iIsGreatestNumber = true;

    for(let j = 0; j < array.length; j++) {
      if(array[j] > array[i]) {
        iIsGreatestNumber = false;
      }
    }

    if(iIsGreatestNumber) {
      return array[i];
    }
  }
}
```

This implementation simply sorts the array and returns the last number. The sorting is O(N log N):

```
function max(array) {
  array.sort((a, b) => (a < b) ? -1 : 1);

  return array[array.length - 1];
}
```

This implementation is O(N) as we loop once through the array:

```
function max(array) {
  let greatestNumberSoFar = array[0];

  for(let i = 0; i < array.length; i++) {
    if(array[i] > greatestNumberSoFar) {
      greatestNumberSoFar = array[i];
    }
  }

  return greatestNumberSoFar;
}
```

Chapter 14

These are the solutions to the exercises found in the section, Exercises, on page 244. The solutions provided here are in Ruby, but you can find the solutions in JavaScript and Python in the code download.[9]

1. One way we can do this is with a simple while loop:

```
def print
  current_node = first_node

  while current_node
    puts current_node.data
    current_node = current_node.next_node
  end
end
```

2. With a doubly linked list, we have immediate access to the last nodes and can follow their "previous node" links to access the previous nodes. This code is simply the inverse of the previous exercise:

```
def print_in_reverse
  current_node = last_node

  while current_node
    puts current_node.data
    current_node = current_node.previous_node
  end
end
```

9. https://pragprog.com/titles/jwdsal2/source_code

3. Here, we use a while loop to move through each node. However, before we move forward, we check ahead using the node's link to ensure that there *is* a next node:

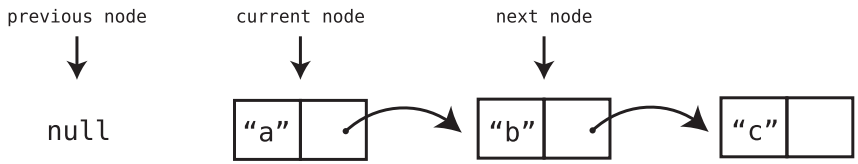
```
def last
  current_node = first_node

  while current_node.next_node
    current_node = current_node.next_node
  end

  return current_node.data
end
```

4. One way to reverse a classic linked list is to iterate through the list while keeping track of three variables.

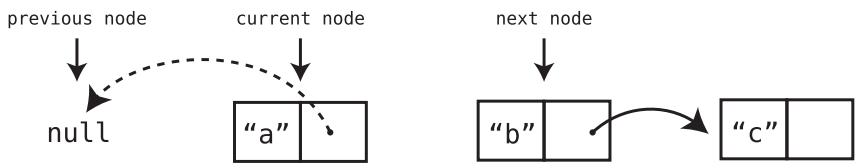
   The primary variable is the current_node, which is the primary node we're iterating over. We also keep track of the next_node, which is the node immediately after the current_node. And we also keep track of the previous_node, which is the node immediately before the current_node. See the following diagram:



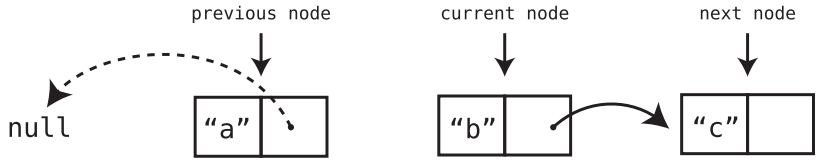
   Note that when we first begin and the current_node is the first node, the previous_node points to null; there are no nodes before the first node.

   Once we have our three variables set up, we proceed with our algorithm, which begins a loop.

   Inside the loop, we first change the current_node's link to point to the previous_node:



Then, we shift all our variables to the right:



We begin the loop again, repeating this process of changing the current_node's link to the point to the previous_node, until we reach the end of the list. Once we reach the end, the list will have been fully reversed.

Here is the actual implementation for this algorithm:

```
def reverse!
  previous_node = nil
  current_node = first_node

  while current_node
    next_node = current_node.next_node

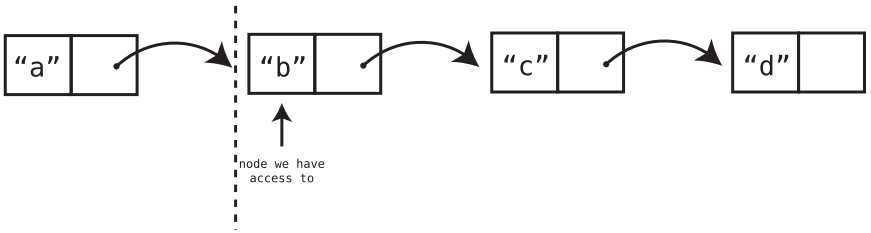
    current_node.next_node = previous_node

    previous_node = current_node
    current_node = next_node
  end

  self.first_node = previous_node
end
```

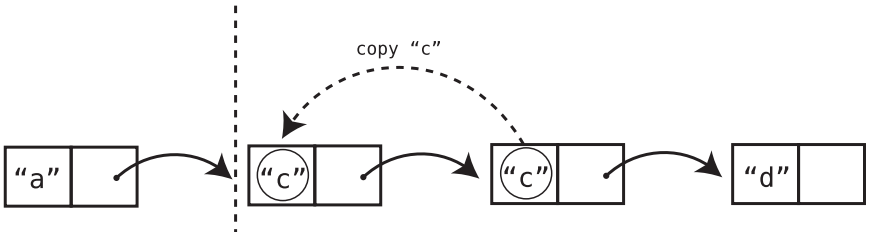
5. Believe it or not, we can delete a middle node without having access to any of the nodes that precede it.

   Following is a diagram of an example situation. There are four nodes, but we only have access to node "b". This means we don't have access to node "a", since links only point *forward* in a classic linked list. We've indicated this using a dashed line—that is, we don't have access to any node to the left of the dashed line:

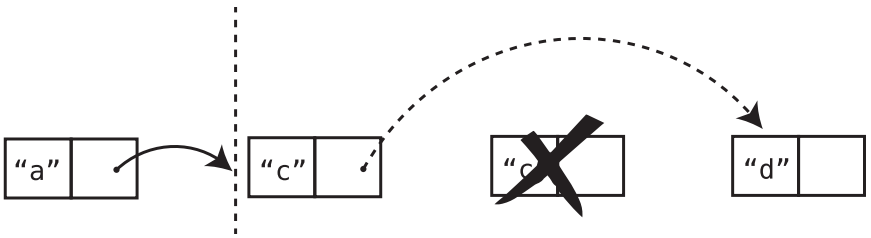


Now, here's how we can delete node "b" (even though we don't have access to node "a"). For the sake of clarity, we're going to call this node the "access node," since it's the first node we have access to.

First, we take *next* node beyond the access node and copy its data into the access node, overwriting the access node's data. In our example, this means copying the string "c" into our access node:



We then change the link of the access node and have it point to the node that is *two* nodes to the right of it. This effectively deletes the original "c" node:



The actual code for this is short and sweet:

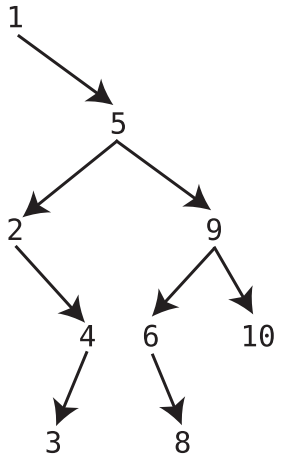
```
def delete_middle_node(node)
  node.data = node.next_node.data
  node.next_node = node.next_node.next_node
end
```

Chapter 15

These are the solutions to the exercises found in the section, Exercises, on page 276. The solutions provided here are in Python, but you can find the solutions in JavaScript and Ruby in the code download.[10]

10. https://pragprog.com/titles/jwdsal2/source_code

1. The tree should look like this. Note that it's not well balanced, as the root node only has a right subtree and no left one:

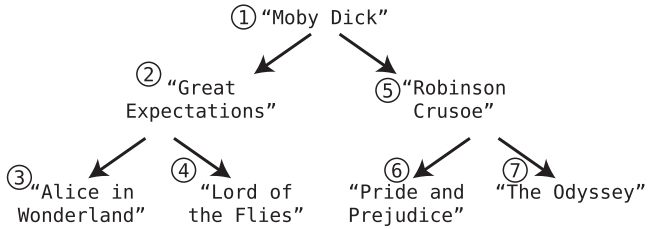


2. Search within a balanced binary search tree takes a maximum of about log(N) steps. So, if N is 1,000, search should take a maximum of about 10 steps.

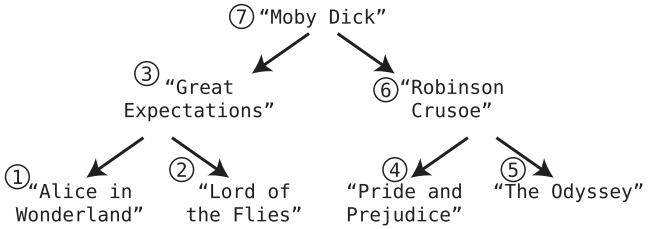
3. The greatest value within a binary search tree will always be the bottom rightmost node. We can find it by recursively following each node's right child until we hit the bottom:

```
def max(node):
    if node.rightChild:
        return max(node.rightChild)
    else:
        return node.value
```

4. Here is the order for preorder traversal:



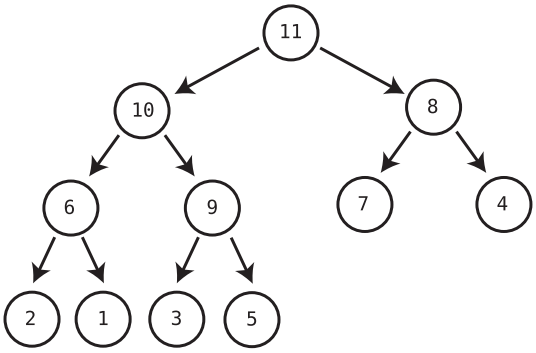
5. Here is the order for postorder traversal:



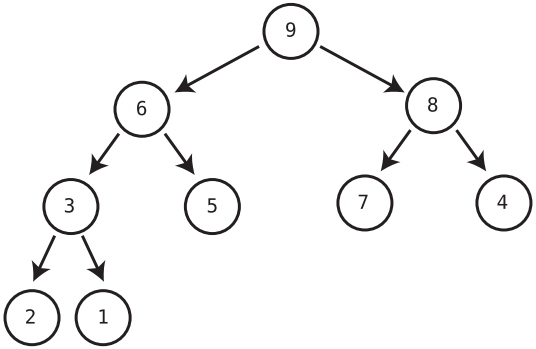
Chapter 16

These are the solutions to the exercises found in the section, Exercises, on page 303.

1. After inserting an 11, the heap would look like this:



2. After deleting the root node, the heap would look like this:



3. The numbers would be in perfect descending order. (This is for a max-heap. For a min-heap, they'd be in *ascending* order.)

Do you realize what this means? It means you've just discovered another sorting algorithm!

Heapsort is a sorting algorithm that inserts all the values into a heap, and then pops each one. As you can see from this exercise, the values always end up in sorted order.

Like Quicksort, Heapsort is O(N log N). This is because we need to insert N values into the heap, and each insertion takes log N steps.

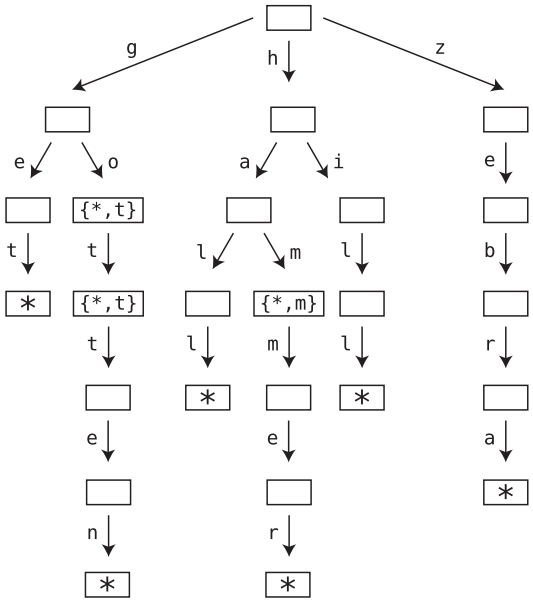
While there are fancier versions of Heapsort that try to maximize its efficiency, this is the basic idea.

Chapter 17

These are the solutions to the exercises found in the section, Exercises, on page 329. The solutions provided here are in Python, but you can find the solutions in JavaScript and Ruby in the code download.[11]

1. This trie stores the words: "tag," "tan," "tank," "tap," "today," "total," "we," "well," and "went."

2. Here is a trie that stores the words "get," "go," "got," "gotten," "hall," "ham," "hammer," "hill," and "zebra":



11. https://pragprog.com/titles/jwdsal2/source_code

3. The following code starts at the trie's node and iterates over each of its children. For each child, it prints the key and then recursively calls itself on the child node:

```
def traverse(self, node=None):
  currentNode = node or self.root

  for key, childNode in currentNode.children.items():
    print(key)
    if key != "*":
      self.traverse(childNode)
```

4. Our autocorrect implementation is a combination of the search and collectAllWords functions:

```
def autocorrect(self, word):
  currentNode = self.root
  # Keep track of how much of the user's word we've found
  # in the trie so far. We'll need to concatenate this with
  # the best suffix we can find in the trie.
  wordFoundSoFar = ""

  for char in word:
    # If the current node has child key with current character:
    if currentNode.children.get(char):
      wordFoundSoFar += char
      # Follow the child node:
      currentNode = currentNode.children.get(char)
    else:
      # If the current character isn't found among
      # the current node's children, collect all the suffixes that
      # descend from the current node and get the first one.
      # We concatenate the suffix with the prefix we've found so
      # far to suggest the word the user meant to type in:
      return wordFoundSoFar + self.collectAllWords(currentNode)[0]

  # If the user's word is found in the trie:
  return word
```

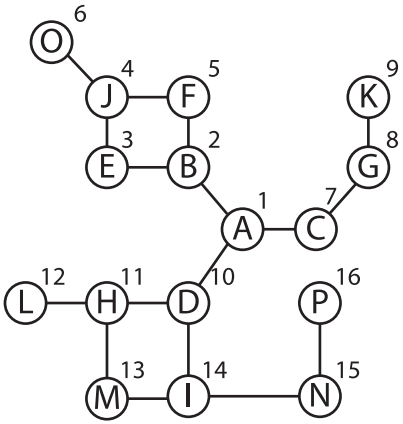
The basic approach is that we first search the trie to find as much of the prefix as we can. When we hit a dead end, instead of just returning None (as the search function does), we call collectAllWords on the current node to collect all the suffixes that stem from that node. We then use the first suffix of the array and concatenate it with the prefix to suggest a new word to the user.

Chapter 18

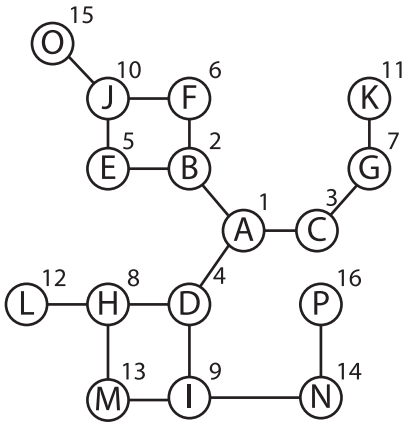
These are the solutions to the exercises found in the section, Exercises, on page 384. The solutions provided here are in Ruby, but you can find the solutions in JavaScript and Python in the code download.[12]

1. If a user is browsing "nails," the website will recommend "nail polish," "needles," "pins," and "hammer."

2. The order of depth-first search would be A-B-E-J-F-O-C-G-K-D-H-L-M-I-N-P, as seen in the following image:



3. The order of breadth-first search would be A-B-C-D-E-F-G-H-I-J-K-L-M-N-O-P, as seen in the following image:



12. https://pragprog.com/titles/jwdsal2/source_code

4. Following is an implementation of breadth-first search:

```
def bfs(starting_vertex, search_value, visited_vertices={})
  queue = Queue.new

  visited_vertices[starting_vertex.value] = true
  queue.enqueue(starting_vertex)

  while queue.read
    current_vertex = queue.dequeue

    return current_vertex if current_vertex.value == search_value

    current_vertex.adjacent_vertices.each do |adjacent_vertex|
      if !visited_vertices[adjacent_vertex.value]
        visited_vertices[adjacent_vertex.value] = true
        queue.enqueue(adjacent_vertex)
      end
    end
  end

  return nil
end
```

5. To find the shortest path in an unweighted graph, we are going to use breadth-first search. The main feature of breadth-first search is that it stays as close to the starting vertex as long as possible. This feature will serve as the key to finding the shortest path.

   Let's apply this to our social-networking example. Because breadth-first search stays close to Idris as long as possible, we'll end up finding Lina first through the shortest possible path. Only later on in the search would we end up finding Lina through longer paths. In fact, we can even stop the search as soon we as find Lina. (Our implementation that follows doesn't end early, but you can modify it to do so.)

   As we visit each vertex for the first time, then, we know that the *current* vertex is *always* part of the shortest path from the *starting* vertex to the *vertex* we're visiting. (Remember, in BFS, the current vertex and the vertex we're visiting aren't necessarily the same.)

   For example, when we visit Lina for the first time, Kamil will be the current vertex. This is because in BFS, we'll get to Lisa first through Kamil before we get to her through Sasha. When we do visit Lina (through Kamil), we can store in a table that the shortest path from Idris to Lina will be through Kamil. This table is similar to the `cheapest_previous_stopover_city_table` from Dijkstra's algorithm.

In fact, whenever we visit *any* vertex, the shortest path from Idris to that vertex will be through the current vertex. We'll store *all* of this data in the table called previous_vertex_table.

Finally, we can then use this data to work backward from Lina to Idris to build the precise shortest path between the two of them.

Here is our implementation:

```
def find_shortest_path(first_vertex, second_vertex, visited_vertices={})
  queue = Queue.new

  # As in Dijkstra's algorithm, we keep track in a table of each vertex
  # immediately preceding vertex.
  previous_vertex_table = {}

  # We employ breadth-first search:
  visited_vertices[first_vertex.value] = true
  queue.enqueue(first_vertex)

  while queue.read
    current_vertex = queue.dequeue
    current_vertex.adjacent_vertices.each do |adjacent_vertex|
      if !visited_vertices[adjacent_vertex.value]
        visited_vertices[adjacent_vertex.value] = true
        queue.enqueue(adjacent_vertex)

        # We store in the previous_vertex table the adjacent_vertex
        # as the key, and the current_vertex as the value. This
        # indicates that the current_vertex is the immediately
        # preceding vertex that leads to the adjacent_vertex.
        previous_vertex_table[adjacent_vertex.value] =
          current_vertex.value
      end
    end
  end

  # As in Dijkstra's algorithm, we work backwards through the
  # previous_vertex_table to build the shortest path;
  shortest_path = []
  current_vertex_value = second_vertex.value

  while current_vertex_value != first_vertex.value
    shortest_path << current_vertex_value
    current_vertex_value = previous_vertex_table[current_vertex_value]
  end
  shortest_path << first_vertex.value
  return shortest_path.reverse
end
```

Chapter 19

These are the solutions to the exercises found in the section, Exercises, on page 395. The solutions provided here are in JavaScript, but you can find the solutions in Python and Ruby in the code download.[13]

1. The space complexity is $O(N^2)$. This is because the function creates the array called `collection`, which will end up holding N^2 strings.

2. This implementation takes up O(N) space, as we create a `newArray` containing N items.

3. The following implementation uses this algorithm: we swap the first item with the last item in place. Then, we swap the second item with the second-to-last item in place. We then proceed to swap the third item with the third-to-last item in place, and so on. Since everything is done in place and we don't create any new data, this has a space complexity of O(1).

```
function reverse(array) {
  for (let i = 0; i < array.length / 2; i++) {
    // Newer JS way to swap values in an array:
    [array[i], array[(array.length - 1) - i]] =
    [array[(array.length - 1) - i], array[i]];

    // Alternate, old-fashioned way of swapping:

    // let temp = array[(array.length - 1) - i];
    // array[(array.length - 1)- i] = array[i];
    // array[i] = temp;
  }

  return array;
}
```

   (While we may create a temporary variable to accomplish each swap, we never have more than that one piece of data at any given time throughout the algorithm.)

4. Here is the completed table:

Version	Time Complexity	Space Complexity
Version #1	O(N)	O(N)
Version #2	O(N)	O(1)
Version #3	O(N)	O(N)

   All three versions run for as many steps as there are numbers in the array, so the time complexity is O(N) for all of them.

13. https://pragprog.com/titles/jwdsal2/source_code

Version #1 creates a brand-new array to store the doubled numbers. This array will have the same length as the original array, so takes up O(N) space.

Version #2 modifies the original array in place, so takes up zero extra space. This is expressed as O(1).

Version #3 also modifies the original array in place. However, since the function is recursive, the call stack at its peak will have N calls on it, taking up O(N) space.

Chapter 20

These are the solutions to the exercises found in the section, Exercises, on page 434. The solutions provided here are in Ruby, but you can find the solutions in JavaScript and Python in the code download.[14]

1. We can optimize this algorithm if we ask ourselves, "If I could magically find a desired piece of information in O(1) time, can I make my algorithm faster?"

   Specifically, as we iterate over one array, we'd want to "magically" look up that athlete from the other array in O(1) time. To accomplish this, we can first transform one of the arrays into a hash table. We'll use the full name (that is, the first *and* last name) as the key, and `true` (or any arbitrary item) as the value.

   Once we've turned one array into this hash table, we then iterate over the other array. As we encounter each athlete, we do an O(1) lookup in the hash table to see if that athlete already plays the other sport. If they do, we add that athlete to our `multisport_athletes` array, which we return at the end of the function.

   Here is the code for this approach:

```
def find_multisport_athletes(array_1, array_2)
  hash_table = {}
  multisport_athletes = []

  array_1.each do |athlete|
    hash_table[athlete[:first_name] + " " + athlete[:last_name]] = true
  end
```

14. https://pragprog.com/titles/jwdsal2/source_code

```
  array_2.each do |athlete|
    if hash_table[athlete[:first_name] + " " + athlete[:last_name]]
      multisport_athletes << athlete[:first_name] +
        " " + athlete[:last_name]
    end
  end

  return multisport_athletes
end
```

This algorithm is O(N + M) since we iterate through each set of players just once.

2. For this algorithm, generating examples to find a pattern will be immensely helpful.

   Let's take an array that has six integers and see what would happen if we removed a different integer each time:

```
[1, 2, 3, 4, 5, 6] : missing 0: sum = 21
[0, 2, 3, 4, 5, 6] : missing 1: sum = 20
[0, 1, 3, 4, 5, 6] : missing 2: sum = 19
[0, 1, 2, 4, 5, 6] : missing 3: sum = 18
[0, 1, 2, 3, 5, 6] : missing 4: sum = 17
[0, 1, 2, 3, 4, 6] : missing 5: sum = 16
```

   Hmm. When we remove the 0, the sum is 21. When we remove the 1, the sum is 20. And when we remove the 2, the sum is 19, and so on. This definitely seems like a pattern!

   Before we go further, let's call the 21 in this case the "full sum." This is the sum of the array when it's just missing the 0.

   If we analyze these cases carefully, we'll see that the sum of any array is less than the "full sum" by the *amount of the missing number*. For example, when we're missing the 4, the sum is 17, which is four less than 21. And when we're missing the 1, the sum is 20, which is one less than 21.

   So, we can begin our algorithm by calculating what the full sum is. We can then subtract the actual sum from the full sum, and that will be our missing number.

   Here is the code for this:

```
def find_missing_number(array)
  # Calculate the full sum (which is the sum if only the 0 were missing):
  full_sum = 0
  (1..array.length).each do |n|
    full_sum += n
  end
```

```
  # Calculate the CURRENT sum:
  current_sum = 0

  array.each do |n|
    current_sum += n
  end
  # The difference between the two sums will be the missing number:
  return full_sum - current_sum
end
```

This algorithm is O(N). It takes N steps to calculate the full sum, and then another N steps to calculate the actual sum. This is 2N steps, which reduces to O(N).

3. We can make this function much faster if we use a greedy algorithm. (Perhaps this shouldn't be a surprise given that our code is trying to make the greatest possible profit on stocks.)

   To make the most profit, we want to buy as low as possible and sell as high as possible. Our greedy algorithm begins by assigning the very first price to be the buy_price. We then iterate over all the prices, and as soon as we find a lower price, we make it the new buy_price.

   Similarly, as we iterate over the prices, we check how much profit we'd make if we sold at that price. This is calculated by subtracting the buy_price from the current price. In good greedy fashion, we save this profit in a variable called greatest_profit. As we iterate through all the prices, whenever we find a *greater* profit, we turn that into the greatest_profit.

   By the time we're done looping through the prices, the greatest_profit will hold the greatest possible profit we can make by buying and selling the stock one time.

   Here's the code for our algorithm:

```
def find_greatest_profit(array)
  buy_price = array[0]
  greatest_profit = 0

  array.each do |price|
    potential_profit = price - buy_price

    if price < buy_price
      buy_price = price
    elsif potential_profit > greatest_profit
      greatest_profit = potential_profit
    end
  end

  return greatest_profit
end
```

Because we iterate over the N prices just once, our function takes O(N) time. Not just did we make a lot of money, but we made it fast.

4. This is another algorithm where generating examples to find a pattern will be the key to optimizing it.

   As stated in the exercise, it's possible for the greatest product to be a result of negative numbers. Let's look at various examples of arrays and their greatest products formed by two numbers:

```
[-5, -4, -3, 0, 3, 4] -> Greatest product: 20 (-5 * -4)
[-9, -2, -1, 2, 3, 7] -> Greatest product: 21 (3 * 7)
[-7, -4, -3, 0, 4, 6] -> Greatest product: 28 (-7 * -4)
[-6, -5, -1, 2, 3, 9] -> Greatest product: 30 (-6 * -5)
[-9, -4, -3, 0, 6, 7] -> Greatest product: 42 (6 * 7)
```

   Seeing all these cases may help us realize that the greatest product can only be formed by either the greatest two numbers or the *lowest* two (negative) numbers.

   With this in mind, we should design our algorithm to keep track of these four numbers:

   - The greatest number
   - The second-to-greatest number
   - The lowest number
   - The second-to-lowest number

   We can then compare the product of the two greatest numbers versus the product of the two lowest numbers. And whichever product is greater is the greatest product in the array.

   Now, how do we find the greatest two numbers and the lowest two numbers? If we sorted the array, that would be easy. But that's still O(N log N), and the instructions say that we can achieve O(N).

   In fact, we can find all four numbers in *a single pass* through the array. It's time to get greedy again.

   Here is the code, followed by its explanation:

```
def greatest_product(array)
  greatest_number = -Float::INFINITY
  second_to_greatest_number = -Float::INFINITY

  lowest_number = Float::INFINITY
  second_to_lowest_number = Float::INFINITY
```

```
  array.each do |number|
    if number >= greatest_number
      second_to_greatest_number = greatest_number
      greatest_number = number
    elsif number > second_to_greatest_number
      second_to_greatest_number = number
    end

    if number <= lowest_number
      second_to_lowest_number = lowest_number
      lowest_number = number
    elsif number > lowest_number &&
      number < second_to_lowest_number
      second_to_lowest_number = number
    end
  end

  greatest_product_from_two_highest =
    greatest_number * second_to_greatest_number

  greatest_product_from_two_lowest =
    lowest_number * second_to_lowest_number

  if greatest_product_from_two_highest > greatest_product_from_two_lowest
    return greatest_product_from_two_highest
  else
    return greatest_product_from_two_lowest
  end
end
```

Before we begin our loop, we set the greatest_number and second_to_greatest_number to be *negative* infinity. This ensures they start out *lower* than any number currently in the array.

We then iterate over each number. If the current number is greater than the greatest_number, we greedily turn the current number into the new greatest_number. If we've already found a second_to_greatest_number, we reassign the second_to_greatest_number to be whatever the greatest_number was before we reached the current number. This ensures the second_to_greatest_number will indeed be the second-to-greatest number.

If the current number we're iterating over is less than the greatest_number but greater than then second_to_greatest_number, we update the second_to_greatest_number to be the current number.

We follow this very same process to find the lowest_number and the second_to_lowest_number.

Once we've found all four numbers, we compute the products from the two highest numbers and the products of the two lowest numbers and return whichever product is greater.

5. The key to optimizing this algorithm is the fact that we're sorting a finite number of values. Specifically, there are only 21 types of temperature readings that we may find in this array, from:

```
97.0, 97.1, 97.2, 97.3 ... 98.7, 98.8, 98.9, 99.0
```

Let's return to the example array mentioned in the exercise instructions:

```
[98.6, 98.0, 97.1, 99.0, 98.9, 97.8, 98.5, 98.2, 98.0, 97.1]
```

If were to imagine our array of temperatures as a *hash table*, we can store each temperature as a key and the number of occurrences as the value. This would look like something like this:

```
{98.6 => 1, 98.0 => 2, 97.1 => 2, 99.0 => 1,
 98.9 => 1, 97.8 => 1, 98.5 => 1, 98.2 => 1}
```

With this in mind, we can run a loop that runs from 97.0 up to 99.0 and *checks the hash table* for how many occurrences of that temperature there are. Each of these lookups take just O(1) time.

Then, we use that number of occurrences to populate a new array. Because our loop is set to go up from 97.0 to 99.0, our array will end up in perfect ascending order.

Here's the code for this:

```
def sort_temperatures(array)
  hash_table = {}

  # Populate hash table with occurrences of temperatures:
  array.each do |temperature|
    if hash_table[temperature]
      hash_table[temperature] += 1
    else
      hash_table[temperature] = 1
    end
  end

  sorted_array = []

  # We multiply the temperature by 10 on the outset
  # so that we can increment the temperature by a whole number
  # during the loop, and avoid floating-point math errors:
  temperature = 970

  # Run loop from 970 up to 990
  while temperature <= 990
```

```
    # If the hash table contains our the current temperature:
    if hash_table[temperature / 10.0]

      # Populate the sorted_array with as many occurences
      # of the current temperature that there are:
      hash_table[temperature / 10.0].times do
        sorted_array << temperature / 10.0
      end
    end

    temperature += 1
  end

  return sorted_array
end
```

You'll note that for the loop, we deal with the temperatures multiplied by 10. That's only to avoid floating arithmetic issues, because incrementing a variable by 0.1 in a loop causes weird things to happen, and the code doesn't work correctly.

Let's now analyze the efficiency of this algorithm. We take N steps to create the hash table. We then run a loop 21 times for all possible temperatures from 97.0 up to 99.0.

In each round of this loop, we run a nested loop to populate the sorted_array with the temperatures. However, this inner loop will *never end up running more times than the N temperatures.* This is because the inner loop only runs one time for each temperature in the original array.

Thus, we have N steps to create the hash table, 21 steps for the outer loop, and N steps for the inner loop. This is 2N + 21, which is reduced to a beautiful O(N).

6. This optimization employs the most brilliant use of magical lookups that I've ever seen.

   Imagine we're iterating over the array of numbers and we encounter a 5. Let's ask ourselves the magical lookup question: "If I could magically find a desired piece of information in O(1) time, can I make my algorithm faster?"

   Well, to determine whether the 5 is part of the longest consecutive sequence, we'd want to know whether there's a 6 in the array. We'd also want to know if there's a 7, and an 8, and so on.

   Now, we can actually achieve each of those lookups in O(1) time if we first store all the numbers from our array in a hash table. That is, the array,

[10, 5, 12, 3, 55, 30, 4, 11, 2] could look like this if we moved the data to a hash table:

```
{10 => true, 5 => true, 12 => true, 3 => true, 55 => true,
 30 => true, 4 => true, 11 => true, 2 => true}
```

In this case, if we were to encounter the 2, we can then run a loop that keeps checking for the next number in the hash table. If it finds it, we increase the length of the current sequence by one. The loop repeats this process until it can't find the next number in the sequence. Each of these lookups take just one step.

However, you may ask the following question: how does this help? Imagine our array is [6, 5, 4, 3, 2, 1]. When we iterate over the 6, we'll find that there isn't a sequence that builds up from there. When we reach the 5, we'll find the sequence 5-6. When we reach the 4, we'll find the sequence 4-5-6. When we reach the 3, we'll find the sequence 3-4-5-6, and so on. We'll still end up going through about N^2 / 2 steps finding all those sequences.

The answer to this is: we'll only start building a sequence if the current number is the *bottom number* of the sequence. So, we won't build 4-5-6 when there's a 3 in the array.

But how do we know if the current number is the bottom of a sequence? By doing a magical lookup!

How? Before running a loop to find a sequence, we'll do an O(1) lookup of the hash table to check whether there's a number *that's 1 less than current number.* So, if the current number is 4, we'll first check to see whether there's a 3 in the array. If there is, we won't bother to build a sequence. We only want to build a sequence starting from the bottom number of that sequence; otherwise we have redundant steps.

Here's the code for this:

```
def longest_sequence_length(array)
  hash_table = {}
  greatest_sequence_length = 0

  # Populate hash table with the numbers as keys:
  array.each do |number|
    hash_table[number] = true
  end

  # Iterate over each number in the array:
  array.each do |number|
```

```
    # If the current number is the first in the sequence,
    # (that is, there isn't a number one lower than it):
    if !hash_table[number - 1]

      # We start counting the length of the current sequence,
      # starting with the current number. Since this is the
      # first number of the sequence, the length of the sequence
      # is currently 1:
      current_sequence_length = 1

      # We establish a current_number for use in the
      # upcoming while loop:
      current_number = number

      # Run a while loop as long as there is
      # a next number in the sequence:
      while hash_table[current_number + 1]

        # Move on to the next number in the sequence:
        current_number += 1

        # Increase the length of the sequence by 1:
        current_sequence_length += 1

        # Greedily keep track of the greatest sequence length:
        if current_sequence_length > greatest_sequence_length
          greatest_sequence_length = current_sequence_length
        end
      end
    end
  end

  return greatest_sequence_length
end
```

In this algorithm, we take N steps to build the hash table. We take another N steps to iterate through the array. And we take about another N steps looking up numbers in the hash table to build the different sequences. All in all, this is about 3N, which is reduced to O(N).

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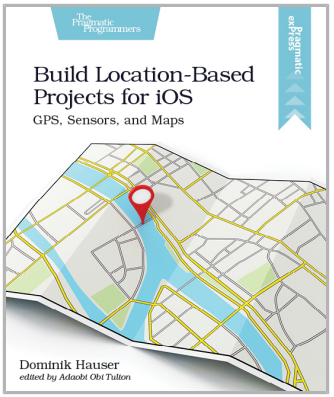
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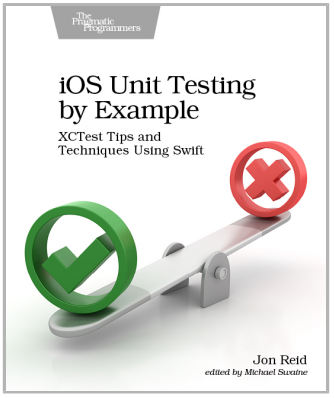
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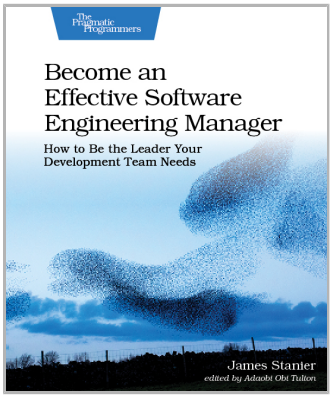
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Software startups make global headlines every day. As technology companies succeed and grow, so do their engineering departments. In your career, you'll may suddenly get the opportunity to lead teams: to become a manager. But this is often uncharted territory. How do you decide whether this career move is right for you? And if you do, what do you need to learn to succeed? Where do you start? How do you know that you're doing it right? What does "it" even mean? And isn't management a dirty word? This book will share the secrets you need to know to manage engineers successfully.

James Stanier
(396 pages) ISBN: 9781680507249. $45.95
https://pragprog.com/book/jsengman

Build Websites with Hugo

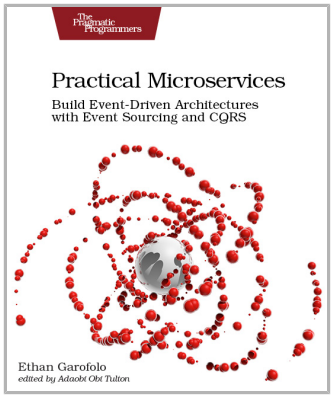


Rediscover how fun web development can be with Hugo, the static site generator and web framework that lets you build content sites quickly, using the skills you already have. Design layouts with HTML and share common components across pages. Create Markdown templates that let you create new content quickly. Consume and generate JSON, enhance layouts with logic, and generate a site that works on any platform with no runtime dependencies or database. Hugo gives you everything you need to build your next content site and have fun doing it.

Brian P. Hogan
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Practical Microservices

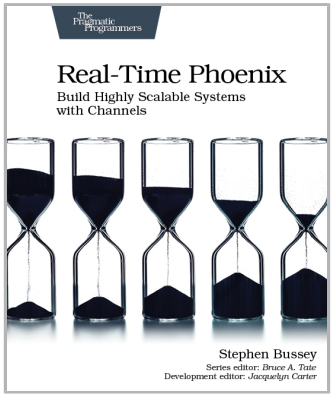
MVC and CRUD make software easier to write, but harder to change. Microservice-based architectures can help even the smallest of projects remain agile in the long term, but most tutorials meander in theory or completely miss the point of what it means to be microservice based. Roll up your sleeves with real projects and learn the most important concepts of evented architectures. You'll have your own deployable, testable project and a direction for where to go next.



Ethan Garofolo
(290 pages) ISBN: 9781680506457. $45.95
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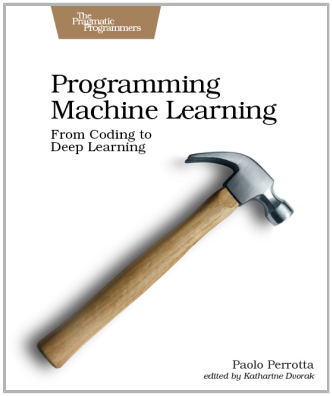
Real-Time Phoenix

Give users the real-time experience they expect, by using Elixir and Phoenix Channels to build applications that instantly react to changes and reflect the application's true state. Learn how Elixir and Phoenix make it easy and enjoyable to create real-time applications that scale to a large number of users. Apply system design and development best practices to create applications that are easy to maintain. Gain confidence by learning how to break your applications before your users do. Deploy applications with minimized resource use and maximized performance.



Stephen Bussey
(326 pages) ISBN: 9781680507195. $45.95
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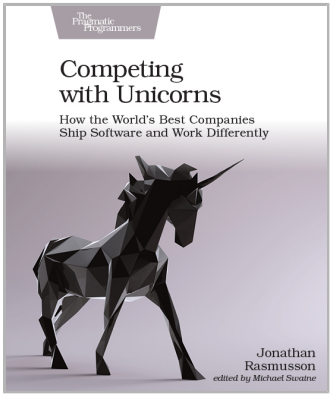
Programming Machine Learning



You've decided to tackle machine learning — because you're job hunting, embarking on a new project, or just think self-driving cars are cool. But where to start? It's easy to be intimidated, even as a software developer. The good news is that it doesn't have to be that hard. Master machine learning by writing code one line at a time, from simple learning programs all the way to a true deep learning system. Tackle the hard topics by breaking them down so they're easier to understand, and build your confidence by getting your hands dirty.

Paolo Perrotta
(340 pages) ISBN: 9781680506600. $47.95
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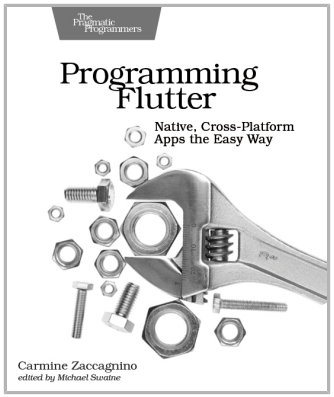
Competing with Unicorns



Today's tech unicorns develop software differently. They've developed a way of working that lets them scale like an enterprise while working like a startup. These techniques can be learned. This book takes you behind the scenes and shows you how companies like Google, Facebook, and Spotify do it. Leverage their insights, so your teams can work better together, ship higher-quality product faster, innovate more quickly, and compete with the unicorns.

Jonathan Rasmusson
(138 pages) ISBN: 9781680507232. $26.95
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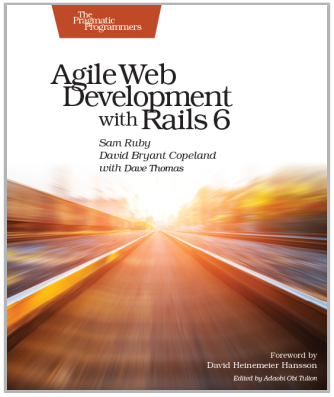
Programming Flutter



Develop your next app with Flutter and deliver native look, feel, and performance on both iOS and Android from a single code base. Bring along your favorite libraries and existing code from Java, Kotlin, Objective-C, and Swift, so you don't have to start over from scratch. Write your next app in one language, and build it for both Android and iOS. Deliver the native look, feel, and performance you and your users expect from an app written with each platform's own tools and languages. Deliver apps fast, doing half the work you were doing before and exploiting powerful new features to speed up development. Write once, run anywhere.

Carmine Zaccagnino
(368 pages) ISBN: 9781680506952. $47.95
https://pragprog.com/book/czflutr

Agile Web Development with Rails 6



Learn Rails the way the Rails core team recommends it, along with the tens of thousands of developers who have used this broad, far-reaching tutorial and reference. If you're new to Rails, you'll get step-by-step guidance. If you're an experienced developer, get the comprehensive, insider information you need for the latest version of Ruby on Rails. The new edition of this award-winning classic is completely updated for Rails 6 and Ruby 2.6, with information on processing email with Action Mailbox and managing rich text with Action Text.

Sam Ruby and David Bryant Copeland
(494 pages) ISBN: 9781680506709. $57.95
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